



U.S. Department
of Transportation

**National Highway
Traffic Safety
Administration**



DOT HS 813 707

August 2025

Crash Report Sampling System Analytical User's Manual, 2016-2023

DISCLAIMER

This publication is distributed by the U.S. Department of Transportation, National Highway Traffic Safety Administration, in the interest of information exchange. The opinions, findings, and conclusions expressed in this publication are those of the authors and not necessarily those of the Department of Transportation or the National Highway Traffic Safety Administration. The United States Government assumes no liability for its contents or use thereof. If trade or manufacturers' names or products are mentioned, it is because they are considered essential to the object of the publication and should not be construed as an endorsement. The United States Government does not endorse products or manufacturers.

Suggested APA Format Citation:

National Center for Statistics and Analysis. (2025, August). *Crash Report Sampling System analytical user's manual, 2016-2023* (Report No. DOT HS 813 707). National Highway Traffic Safety Administration.

Technical Report Documentation Page

1. Report No. DOT HS 813 707	2. Government Accession No.	3. Recipient's Catalog No.	
4. Title and Subtitle Crash Report Sampling System Analytical User's Manual, 2016-2023		5. Report Date August 2025	
		6. Performing Organization Code	
7. Author National Center for Statistics and Analysis		8. Performing Organization Report No. DOT-VNTSC-NHTSA-xx- xx	
9. Performing Organization Name and Address National Center for Statistics and Analysis National Highway Traffic Safety Administration 1200 New Jersey Avenue SE Washington, DC 20590		10. Work Unit No. (TRAI5)	
		11. Contract or Grant No.	
12. Sponsoring Agency Name and Address National Highway Traffic Safety Administration 1200 New Jersey Avenue SE Washington, DC 20590		13. Type of Report and Period Covered	
		14. Sponsoring Agency Code	
15. Supplementary Notes			
16. Abstract This is the updated and revised Crash Report Sampling System Analytical User's Manual for the period 2016 to 2023.			
17. Key Words CRSS; Crash Report Sampling System; analytical user's manual; 2016-2023		18. Distribution Statement Document is available to the public from the DOT, National Highway Traffic Safety Administration, National Center for Statistics and Analysis, https://crashstats.nhtsa.dot.gov .	
19. Security Classif. (of this report) Unclassified	20. Security Classif. (of this page) Unclassified	21 No. of Pages 444	22. Price

Form DOT F 1700.7 (8-72)

Reproduction of completed page authorized

Table of Contents

Introduction.....	1
New in 2023 CRSS	2
New and Noteworthy	2
Data Elements With Changes	2
Summary of SAS Naming Changes.....	7
CRSS Operations	8
CRSS Sample Design	9
National Estimates	12
CRSS Imputation	13
CRSS SAS Data Files.....	18
CRSS Data Element List	23
Data Element Definitions and Codes.....	34
Key Data Elements	35
The ACCIDENT Data File	47
The VEHICLE Data File	82
The PERSON Data File	179
The PARKWORK Data File.....	206
The PBTYPE Data File.....	266
The CEVENT Data File.....	293
The VEVENT Data File	301
The VSOE Data File	309
The CRASHRF Data File	314
The WEATHER Data File.....	317
The VEHICLESF Data File.....	319
The PVEHICLESF Data File.....	321
The DRIVERRF Data File.....	323
The DAMAGE Data File.....	326
The DISTRACT Data File.....	328
The DRIMPAIR Data File.....	331
The FACTOR Data File.....	333
The MANEUVER Data File.....	335
The VIOLATN Data File.....	337
The VISION Data File.....	342
The PERSONRF Data File	344
The NMCRAASH Data File.....	347
The NMDISTRACT Data File	350
The NMIMPAIR Data File	353
The NMPRIOR Data File	355
The SAFETYEQ Data File.....	357
The VPICDECODE Data File	365
The VPICTRAILERDECODE Data File	366

References.....	367
Appendix A: Crash Type Diagrams.....	A-1
Crash Type Diagram.....	A-2
Crash Type Configuration Diagram (2023).....	A-3
Appendix B: Rules for Derived Data Elements.....	B-1
Crash Level Counts.....	B-2
Crash and Vehicle Level Derived Data Elements	B-3
Appendix C: Analytical Classification of Select CRSS Data Elements	C-1
Appendix D: Auxiliary Data Files	D-1
Appendix E: Summary Statistics.....	E-1
Appendix F: Standard Errors	F-1
Appendix G: Special Notes for Analysts.....	G-1
Appendix H: Notable Changes	H-1

Introduction

One of the primary objectives of the National Highway Traffic Safety Administration (NHTSA) is to reduce the human toll and property damage that motor vehicle traffic crashes inflict on our society. Crashes each year result in thousands of lives lost, hundreds of thousands of injured victims, and billions of dollars in property damage. Accurate data are required to support the development, implementation, and assessment of highway safety programs aimed at reducing this toll. NHTSA uses data from many sources, including the Crash Report Sampling System (CRSS). CRSS is a sample of police-reported crashes involving all types of motor vehicles, pedestrians, and cyclists, ranging from property damage-only crashes to those that result in fatalities. CRSS is used to estimate the overall crash picture, identify highway safety problem areas, measure trends, drive consumer information initiatives, and form the basis for cost and benefit analyses of highway safety initiatives and regulations.

CRSS obtains its data from a nationally representative probability sample selected from the more than 6 million police-reported crashes that occur annually. Although various sources suggest that there are many more crashes that are not reported to the police, the majority of these unreported crashes involve only minor property damage and no significant personal injury. By restricting attention to police-reported crashes, CRSS concentrates on those crashes of greatest concern to the highway safety community and the general public.

This multi-year analytical user's manual provides documentation on the evolution of coding practices of CRSS from 2016 to 2023. The manual will continue to grow each year and present the historical coding of CRSS from inception through present. It includes documentation on the data elements that are contained in CRSS and other useful information that will enable the users to become familiar with the data system. The FARS/CRSS Coding and Validation Manual provides more detailed definitions and coding rules for each data element and attribute. NHTSA's National Center for Statistics and Analysis (NCSA) publishes these manuals for each year of data collection, and they are available at [NCSA Publications — Manuals and Documentation — CRSS](#).

The compilation of CRSS data is a NHTSA priority. These data store valuable information that will be preserved over time and are available for present and future use. This analytical user's manual should help improve the usefulness and accessibility of the data. With the exception of personal notes, there is no reason to keep older versions of this reference manual. All information in earlier editions has been retained in this newer version.

New in 2023 CRSS

New and Noteworthy

The Analytical User's Manual is updated annually to reflect necessary revisions and ensure quality data collection and analysis. CRSS data elements evolve based on any number of factors including the needs of end users. Changes are made with careful consideration and collaboration among key stakeholders. Below are the notable changes, challenges, reclassifications, or other issues the analyst should be aware of for this year.

Addition of Crash Type Configuration

In 2023, the data element "Crash Type" was condensed and made available as "Crash Type Configuration" in addition to the existing "Crash Type." For graphic descriptions of the condensed possible values see [Appendix A: PC23A Crash Type Configuration Diagram](#). For graphic descriptions of the existing "Crash Type" see [Appendix A: PC23 Crash Type Diagram](#).

Data Elements With Changes

Below is a list of CRSS data elements that have substantial changes for 2023. Changes are denoted in ***bold/italics*** for additions and strikethrough for deletions. Additional detailed information on each data element can be found in the FARS/CRSS Coding and Validation Manual. The NCSA publishes these manuals for each year of data collection and they can be found at [NCSA Publications — Manuals and Documentation](#).

Data Element ID	Data Element Name	SAS Table.NAME	Comments
C32	Related Factors— Crash Level	CrashRF.CRASHRF	- Removed attribute: 013 (Aggressive Driving/Road Rage by Non-Contact Vehicle Driver) - New attributes: 102 (Aggressive Driving by Non-Contact Vehicle Driver) 103 (Road Rage by Non-Contact Vehicle Driver)
D24	Related Factors— Driver Level	DriverRF.DRIVERRF	- Removed attributes: 008 (Aggressive Driving/Road Rage)

Data Element ID	Data Element Name	SAS Table.NAME	Comments
			○ <i>037 (Police Pursuing this Driver or Police Officer in Pursuit)</i> • New attributes: ○ <i>100 (Using a Belt-Positioning Device or Other)</i> ○ <i>102 (Aggressive Driving)</i> ○ <i>103 (Road Rage)</i> ○ <i>104 (Police Pursuing This Driver)</i> ○ <i>105 (Police Officer in Pursuit)</i>
PC23A	<i>Crash Type Configuration</i>	<i>Vehicle.ACC_CONFIG</i>	• Added new element • New attributes: ○ <i>101 (Right Roadside Departure)</i> ○ <i>102 (Left Roadside Departure)</i> ○ <i>103 (Struck Object While Moving Forward)</i> ○ <i>201 (Rear End, Trailing Vehicle)</i> ○ <i>202 (Rear End, Lead Vehicle)</i> ○ <i>203 (Rear End, Other or Unknown)</i> ○ <i>204 (Forward Impact, Frontal Impact After Maneuver)</i> ○ <i>205 (Forward Impact, Rear End Impact After Maneuver)</i> ○ <i>206 (Forward Impact, Other or Unknown)</i> ○ <i>207 (Sideswipe, Angle, Vehicle on Left)</i> ○ <i>208 (Sideswipe, Angle, Vehicle on Right)</i> ○ <i>209 (Sideswipe, Angle, Other or Unknown)</i>

Data Element ID	Data Element Name	SAS Table.NAME	Comments
			○ <i>301 (Lateral Move [Left/Right], Head-On, Sideswipe, or Angle)</i> ○ <i>302 (Lateral Move [Going Straight], Head-On, Sideswipe, or Angle)</i> ○ <i>303 (Lateral Move, Other or Unknown)</i> ○ <i>304 (Forward Impact After Maneuver, Departed Lane)</i> ○ <i>305 (Forward Impact After Maneuver, Remained in Lane)</i> ○ <i>306 (Forward Impact After Maneuver, Other or Unknown)</i> ○ <i>401 (Turn Across Path, Initial Opposite Directions [Left/Right])</i> ○ <i>402 (Turn Across Path, Initial Opposite Directions [Going Straight])</i> ○ <i>403 (Turn Across Path, Initial Same Directions [Turning Right])</i> ○ <i>404 (Turn Across Path, Initial Same Directions [Going Straight, Other Vehicle Turning Right])</i> ○ <i>405 (Turn Across Path, Initial Same Directions [Turning Left])</i> ○ <i>406 (Turn Across Path, Initial Same Directions [Going Straight, Other Vehicle Turning Left])</i> ○ <i>407 (Turn Across Path, Other or Unknown)</i> ○ <i>408 (Turn Into Path, Turn into Same Direction [Turning Left])</i>

Data Element ID	Data Element Name	SAS Table.NAME	Comments
			○ <i>409 (Turn Into Path, Turn into Same Direction [Going Straight, Other Vehicle Turning Left])</i> ○ <i>410 (Turn Into Path, Turn into Same Direction [Turning Right])</i> ○ <i>411 (Turn Into Path, Turn into Same Direction [Going Straight, Other Vehicle Turning Right])</i> ○ <i>412 (Turn Into Path, Turn into Opposite Directions [Turning Right])</i> ○ <i>413 (Turn Into Path, Turn into Opposite Directions [Going Straight, Other Vehicle Turning Right])</i> ○ <i>414 (Turn Into Path, Turn into Opposite Directions [Turning Left])</i> ○ <i>415 (Turn Into Path, Turn Into Opposite Directions [Going Straight, Other Vehicle Turning Left])</i> ○ <i>416 (Turn into Path, Other or Unknown)</i> ○ <i>501 (Straight Paths, Striking from the Right)</i> ○ <i>502 (Straight Paths, Struck on the Right)</i> ○ <i>503 (Straight Paths, Striking from the Left)</i> ○ <i>504 (Straight Paths, Struck on the Left)</i> ○ <i>505 (Straight Paths, Other or Unknown)</i> ○ <i>000 (No Impact)</i> ○ <i>992 (Backing Vehicle)</i>

Data Element ID	Data Element Name	SAS Table.NAME	Comments
			○ 993 (Other Vehicle) ○ 998 (Other Crash Type) ○ 999 (Unknown Crash Type)
P24	Related Factors— Person (MV Occupant) Level	PersonRF.PERSONRF	• New attributes: ○ 104 (Using a Belt-Positioning Device or Other) ○ 105 (Paraplegic or in a Wheelchair)
NM13	Non-Motorist Action/Circumstances	NMPrior.NMACTION	• Revised attribute labels: ○ 01 (Going To or From School {Pre-K-12})
NM26	Related Factors— Person (Not a MV Occupant) Level	PersonRF.PERSONRF	• New attributes: ○ 102 (Motor Vehicle Occupant in Prior Crash) ○ 103 (Road Rage)

Summary of SAS Naming Changes

Data Element ID	2022 SAS Name	New 2023 SAS Name	Data Element Name
<i>PC23A</i>	<i>N/A</i>	<i>ACC_CONFIG</i>	<i>Crash Type Configuration</i>

The data elements in ***bold/italics*** are new to 2023 CRSS.

The data elements in *italics* are changed in 2023 CRSS.

CRSS Operations

CRSS obtains its data from a nationally representative probability sample selected from the more than 6 million police-reported crashes that occur annually. To be eligible for the CRSS sample, a crash report must be completed by the police; it must involve at least one motor vehicle traveling on a trafficway; and the crash must result in property damage, injury, or death.

These crash reports are chosen from 60 selected sites across the United States that reflect the geography, population, miles driven, and crashes in the United States. CRSS data collectors review crash reports from hundreds of law enforcement agencies within the sites, systematically sampling tens of thousands of crash reports each year. The collectors obtain copies of the selected crash reports and send them to a central location for coding. No other data is collected beyond that in the selected crash reports.

Trained personnel interpret and code data directly from the crash reports into an electronic data file. Approximately 120 data elements are coded into a common format. After coding, quality checks are performed on the data to ensure validity and consistency. When these are completed, CRSS data files and coding documentation become publicly available.

CRSS data are also used to respond to requests from the international and national highway safety communities, State and local governments, the Congress, Federal agencies, research organizations, industry, the media, and the public.

CRSS Sample Design

Beginning 2016, as part of the effort to modernize NHTSA's data collection system, NCSA designed two new national probability-based crash sampling systems—CRSS to replace the NASS GES and CISS to replace the CDS. CRSS was designed completely independent of GES or CISS. CRSS has the same scope as GES: all police reported motor vehicle crashes that occur on a trafficway. The source of the information for CRSS continues solely to be the police crash report.

The CRSS police crash report sample is selected in multiple stages to produce a nationally representative probability sample since nationwide direct selection is infeasible. A brief description of the selection process at each of the three stages is given below.

1st Stage— PSU Sample: At the first stage, 3,117 counties in the country were grouped into 707 primary sampling units (PSU). U.S. Territories, some remote areas in Alaska, and small islands in Hawaii were excluded. A CRSS PSU is either a county or a group of counties. The 707 PSUs in the PSU frame were stratified into 50 strata by the four Census regions, urbanicity, vehicle miles traveled, total number of crashes, total truck miles traveled, and road miles. First, 101 PSUs were selected using a stratified probability-proportional-to-size (PPS) sampling method. Then a sequence of sub-samples was selected from the original 101 PSU sample and strata were collapsed if necessary. This produced a sequence of nested PSU samples with different sample sizes selected from the collapsed strata. This sequence of nested PSU samples provides NHTSA flexibility to change and scale the PSU sample size in the future without reselecting the sample. Therefore, the final PSU sample was the result of a multiphase sampling mechanism in which the PSU selection probability is still approximately PPS.

2nd Stage – PJ Sample: The secondary sampling units (SSU) are police jurisdictions (PJs) or groups of police jurisdictions. Within each selected PSU, PJs were stratified into three strata by their measure of size (MOS) that is a combination of crash counts in six categories of interest. A Pareto sampling method, an approximation to the PPS sampling method, was used to select PJ samples from each PJ strata. This method reduces the potential of changes to the existing PJ sample when a new PJ sample has to be selected because of PJ frame changes.

For some PSUs, NHTSA receives police crash reports (i.e., tertiary sampling unit) in electronic format through the Electronic Data Transfer (EDT) system. In 2023 there were 21 EDT PSUs. In an EDT PSU, police crash reports can be accessed directly from all PJs in an EDT PSU. Therefore, all PJs within an EDT PSU are combined into one consolidated pseudo PJ and the pseudo PJ is selected with certainty. This leads to a reduction in the number of selected PJs. NHTSA expects more PSUs to provide crash reports through the EDT system in the future.

3rd Stage – Police Crash Report Sample: The tertiary sampling units (TSU) are the police crash reports. The CRSS data collectors periodically obtain police crash reports from each selected PJ. During each collection, all new police crash reports since the last collection are sequentially stratified into nine police crash report strata (see Table 1 below). These nine strata were formed based on the results of NHTSA's internal and public data need assessments. The stratification allows NHTSA to over-sample in Strata 2-6.

From each stratum, a systematic sampling method is used to select the police crash report sample. The sampling intervals are determined in such a way that the final weights are

approximately equal for all the police crash reports in the same stratum with the ultimate aim of reducing the sampling variance for the domain estimates. The target annual sample size is approximately 50,000 police crash reports.

Table 1. CRSS Police Crash Report Domain Definition, Target Sample Allocation, and Population Distribution

Stratum	Description (Hierarchical Structure)	Target Percent of Sample	Estimated Percent of Population¹
2	Crashes with killed or injured pedestrian	9%	2.1%
3	Crashes with killed or injured motorcycle or moped occupant	6%	1.3%
4	LMY passenger vehicle crashes with killed or incapacitated occupant	4%	0.3%
5	NLMY passenger vehicle crashes with killed or incapacitated occupant	7%	1.4%
6	LMY passenger vehicle crashes with injured occupant	14%	5.7%
7	Crashes involving medium or heavy truck or bus	6%	8.7%
8	NLMY passenger vehicle crashes with injured occupant	12%	14.1%
9	LMY passenger vehicle crashes AND no one is killed or injured	22%	24.9%
10	Crashes not in strata 2-9	20%	41.4%
Late Model Year (LMY) passenger vehicle: ≤ 4 years old, Non-Late Model Year (NLMY) passenger vehicle: ≥ 5 years old			

¹ Estimated percentage of population is based on 2023 CRSS estimates.

In 2023, 61 PSUs were selected, and 60 PSUs cooperated. From the 60 cooperating PSUs, 289 PJs were selected, and 275 PJs cooperated. A total of 50,241 PCRs were selected from the cooperating PJs. Among them, 21 PCRs were non-responding (i.e., PCRs with missing or unreadable pages), 25 PCRs were duplicates, and 92 PCRs were out-of-scope. Therefore, the 2023 CRSS analysis files were created based on data collected from 50,103 PCRs. Weight adjustments were made to mitigate the potential bias caused by the non-responding (i.e., non-cooperating) PSUs, PJs, PCRs. In addition, weights were adjusted for the duplicate PCRs.

Table 2 below shows the sample sizes of the three stages for each sample year of CRSS.

Table 2. CRSS Sample Sizes

Year	PSU		PJ		PCR				
	Sampled	Cooperated	Sampled ¹	Cooperated	Sampled	Non-Responding	Duplicates	Out-of-Scope	Coded
2016	60	53	394 ²	337	47,872	80	140	1,141	46,511
2017	61	60	397	393	55,274	17	26	262	54,969
2018	61	60	396	389	48,644	6	11	184	48,443
2019	61	60	326	313	54,507	11	14	73	54,409
2020	61	60	301	288	54,883	13	30	95	54,745
2021	61	60	295	282	54,389	63	17	109	54,200
2022	61	59	295 ³	275	54,148	28	38	127	53,955
2023	61	60	289	275	50,241	21	25	92	50,103

¹ In general, sampled PJs are selected from the cooperating PSUs.

² In 2016, PJs were selected from the 60 sampled PSUs at the beginning of the sampling year. During the year, 7 PSUs were determined to be non-cooperating. The number of sampled PJs includes 44 PJs from 7 non-cooperating PSUs.

³ In 2022, PJs were selected from the 60 cooperating PSUs at the beginning of the sampling year. During the year, one more PSU was determined to be non-cooperating. The number of sample PJs includes 6 PJs from the additional non-responding PSU.

Please refer to the NHTSA Technical Reports [Crash Report Sampling System: Sample Design and Weighting](#) and [Crash Report Sampling System: Design Overview, Analytic Guidance, and FAQs](#) for more in-depth discussions of the CRSS sample design.

National Estimates

The CRSS police crash report sample is a complex multi-stage, stratified sample with unequal selection probabilities. Estimates from CRSS data must be properly weighted to ensure unbiased and robust estimates. CRSS weights were created using the following steps:

1. Calculate base weights—the inverse of selection probabilities—at all three stages (PSU, PJ, and Police Crash Report) to reflect the unequal selection probabilities.
2. Adjust the base weights for non-response at all three stages to correct potential non-response bias.
3. Adjust the weights for duplicate crashes that were identified post sampling.
4. Calibrate PJ and Police Crash Report weights using the PSU level total crash report stratum counts to further correct potential non-response bias and coverage bias (Only in 2016 and 2017).
5. Calibrate case weights by benchmarking Census resident population counts and FARS crash counts.

The final CRSS weight variable that incorporates the above steps is called WEIGHT in the CRSS analysis file. Please refer to the NHTSA Technical Report [Crash Report Sampling System: Sample Design and Weighting](#) for a more in-depth discussion on the CRSS weighting procedure.

Complex sample design features employed in CRSS data collection should be considered in analysis of the CRSS data. Treating the CRSS sample as a simple random sample in estimation may cause severe bias to both point estimates and standard error estimates. Specialized computer software for complex survey data analysis, such as SAS PROC SURVEY procedures and SUDAAN procedures, should be used for CRSS data analysis along with proper design statements. Because of the low PSU level sampling rates, the CRSS PSU sample can be treated as a with-replacement sample with unequal selection probabilities. This simplifies the variance estimation.

In the CRSS analysis file, the variable PSUSTRAT defines the PSU strata, and PSU_VAR identifies sampled PSUs for variance estimation. Also, certainty PSU is treated as a stratum in PSUSTRAT. The PJs selected in the certainty PSU are treated as PSUs in PSU_VAR.

CRSS is designed for national level estimates. For other smaller analysis domains, the point estimates may have large standard errors.

Please refer to the NHTSA Technical Report [Crash Report Sampling System: Design Overview, Analytic Guidance, and FAQs](#) for more detailed information on CRSS estimation and examples.

CRSS Imputation

CRSS data are obtained either directly from an item on the police crash report or by interpreting the information provided in the crash report through a review of the crash diagrams, the police officer's written summary of the crash, or combinations of data elements on the report. During this process of data acquisition, some records of the data elements are found missing or entered as "unknown" or "not reported" resulting in incomplete data for analysis. To offer more complete CRSS data for analysis, NHTSA imputes selected data elements from the Accident, Vehicle, and Person files as follows:

- ***Accident data file:*** Alcohol Involved in Crash, Atmospheric Conditions, Crash Date (Day of Week), Crash time (Hour), Crash Time (Minute), First Harmful Event, Light Condition, Manner of Collision, Maximum Injury Severity in Crash, Number of Injured in Crash, Relation to Junction – Within Interchange Area, Relation to Junction – Specific Location;
- ***Vehicle data file:*** Areas of Impact – Initial Contact Point, Driver Drinking in Vehicle, Number of Injured in Vehicle, Maximum Injury Severity in Vehicle, Most Harmful Event, Vehicle Model Year, Movement Prior to Critical Event;
- ***Person data file:*** Age, Police Reported Alcohol Involvement, Ejection, Injury Severity, Seating Position, Sex.

The above data elements are consistent with the ones imputed in the corresponding three files of NASS GES data from 2010 to 2015. More details about GES data imputation in 2015 and earlier years are available in the [1988-2015 NASS GES Analytical User's Manual](#).

The imputation process for CRSS data imputes a single value for each unknown or not reported value. In other words, instead of filling in an unknown or not reported value with a set of plausible values, a single estimated value is used to replace the unknown or not reported value. The procedure is a multivariate imputation of each selected data element by means of its covariates. If this process produces inconsistent imputed values, a separate univariate imputation is conducted to impute the inconsistent imputed values. In the case of "Body Type," however, imputation is done by univariate imputation only. Starting in 2021, NHTSA will no longer impute body type. See [Appendix H: Changes to Imputed Elements](#) for more information.

The multivariate imputation is carried out by sequential regression modeling in which logistic regression models estimate unknown or not reported values for the categorical data elements, and linear regression models for the continuous data elements. In each case, the stepwise regression algorithm automatically selects the covariates and computes the imputed (predicted) values of the data element. This process is done using the SAS callable software "IVEware" developed at the University of Michigan (<http://www.isr.umich.edu/src/smp/ive/>). This multivariate imputation procedure may produce imputed values inconsistent with other observed values or may terminate prematurely because of the number of iterations or other convergence criteria provisioned in the software. Then the univariate imputation procedure will be used to impute the inconsistent values or the remaining unknown or not reported values. All data elements, except "Body Type," are first imputed by the multivariate regression method.

It should be noted that the data elements produced by the imputation do not replace the originals; all original data elements are kept intact in the CRSS data files. Rather, new imputed data elements are created from the original data elements having each unknown or not reported value substituted by the estimated value. The imputed data elements, identified by the suffix `_IM` (e.g. `AGE_IM`, `WEATHER_IM` for the data elements `AGE` and `WEATHER`, respectively) are added as additional data elements to their respective files. It is also worth noting that:

- the imputed maximum severity `MAXSEV_IM` and imputed number of injured `NO_INJ_IM` at the accident level are derived from `INJSEV_IM` which contains the imputed values of the Injury Severity at the person level;
- the imputed maximum severity `MXVSEV_IM` and imputed number of injured `NUMINJ_IM` at the vehicle level are derived from `INJSEV_IM` that contains the imputed values of the Injury Severity at the person level;
- the imputed police reported alcohol involvement `ALCHL_IM` at the accident level is derived from `PERALCH_IM` that contains the imputed values of alcohol involvement at the person level;
- the imputed police reported alcohol involvement `V_ALCH_IM` at the vehicle level is derived from `PERALCH_IM` that contains the imputed values of alcohol involvement at the person level.

Overall, the CRSS imputation process employs IVEware software and several other programs written in SAS. Some text files input to this software provide additional controls to accurately and efficiently obtain the best estimates of the unknown or not reported values. In addition, the process makes provision for edit and consistency checks on the data to avoid any implausible value that might have been predicted by the applicable regression models. Please refer to the NHTSA Technical Report [Crash Report Sampling System: Imputation](#) for a more in-depth discussion on the CRSS imputation procedure.

The table below shows the SAS names and the corresponding SAS labels of the selected data elements for both the original and imputed versions for the Accident, Vehicle, and Person files.

Data Elements and Their Imputed Counterparts – SAS Names and Labels

SAS Data File	Data Element		<u>Imputed</u> Data Element	
	SAS Name	SAS Label	SAS Name	SAS Label
<i>Accident</i>				
Accident	ALCOHOL	Alcohol Involved in Crash	ALCHL_IM	Imputed Alcohol Involved in Crash
Accident	DAY_WEEK	Crash Date (Day of Week)	WKDY_IM	Imputed Day of Week
Accident	HARM_EV	First Harmful Event	EVENT1_IM	Imputed First Harmful Event
Accident	HOUR	Crash Time (Hour)	HOUR_IM	Imputed Hour

SAS Data File	Data Element		<u>Imputed</u> Data Element	
	SAS Name	SAS Label	SAS Name	SAS Label
Accident	LGT_COND	Light Condition	LGTCOIM	Imputed Light Condition
Accident	MINUTE	Crash Time (Minute)	MINUTE_IM	Imputed Minute
Accident	MAN_COLL	Manner of Collision	MANCOL_IM	Imputed Manner of Collision
Accident	MAX_SEV	Maximum Injury Severity in Crash	MAXSEV_IM	Imputed Maximum Injury Severity in Crash
Accident	NUM_INJ	Number Injured in Crash	NO_INJ_IM	Imputed Number Injured in Crash
Accident	RELJCT1	Relation to Junction – Within Interchange Area	RELJCT1_IM	Imputed Relation to Junction – Within Interchange Area
Accident	RELJCT2	Relation to Junction – Specific Location	RELJCT2_IM	Imputed Relation to Junction – Specific Location
Accident	WEATHER	Atmospheric Conditions	WEATHR_IM	Imputed Atmospheric Conditions
<i>Vehicle</i>				
Vehicle	IMPACT1	Area of Impact – Initial Contact Point	IMPACT1_IM	Imputed Area of Impact – Initial Contact Point
Vehicle	VEH_ALCH	Driver Drinking in Vehicle	V_ALCH_IM	Imputed Driver Drinking in Vehicle
Vehicle	MAX_VSEV	Maximum Injury Severity in Vehicle	MXVSEV_IM	Imputed Maximum Injury Severity in Vehicle
Vehicle	MOD_YEAR	Vehicle Model Year	MDLYR_IM	Imputed Vehicle Model Year
Vehicle	P_CRASH1	Pre-Event Movement	PCRASH1_IM	Imputed Pre-Event Movement

SAS Data File	Data Element		<u>Imputed</u> Data Element	
	SAS Name	SAS Label	SAS Name	SAS Label
Vehicle	M_HARM	Most Harmful Event	VEVENT_IM	Imputed Most Harmful Event
Vehicle	NUM_INJV	Number Injured in Vehicle	NUMINJ_IM	Imputed Number Injured in Vehicle
<i>Person</i>				
Person	AGE	Age	AGE_IM	Imputed Age
Person	EJECTION	Ejection	EJECT_IM	Imputed Ejection
Person	INJ_SEV	Injury Severity	INJSEV_IM	Imputed Injury Severity
Person	DRINKING	Police Reported Alcohol Involvement	PERALCH_IM	Imputed Police Rep. Alcohol Inv.
Person	SEAT_POS	Seating Position	SEAT_IM	Imputed Seating Position
Person	SEX	Sex	SEX_IM	Imputed Sex

The following table shows percentages of “Not Reported” and “Reported as Unknown” values for the selected data elements for the Accident, Vehicle, and Person files in CRSS 2023 data.

Data Elements and Percentages of Unknown and Not Reported Values

SAS Data File	Data Element		Unknown/ Not Reported Percentage
	SAS Name	SAS Label	
<i>Accident</i>			
Accident	ALCOHOL	Alcohol Involved in Crash	39.2%
Accident	DAY_WEEK	Crash Date (Day of Week)	0.0%
Accident	HARM_EV	First Harmful Event	0.1%
Accident	HOURL	Crash Time (Hour)	0.7%
Accident	LGT_COND	Light Condition	0.4%
Accident	MINUTE	Crash Time (Minute)	0.7%
Accident	MAN_COLL	Manner of Collision	0.3%
Accident	MAX_SEV	Maximum Injury Severity in Crash	1.9%
Accident	NUM_INJ	Number Injured in Crash	1.9%

SAS Data File	Data Element		Unknown/ Not Reported Percentage
	SAS Name	SAS Label	
Accident	RELJCT1	Relation to Junction – Within Interchange Area	0.1%
Accident	RELJCT2	Relation to Junction – Specific Location	0.2%
Accident	WEATHER	Atmospheric Conditions	3.5%
<i>Vehicle</i>			
Vehicle	IMPACT1	Area of Impact – Initial Contact Point	1.8%
Vehicle	VEH_ALCH	Driver Drinking in Vehicle	34.1%
Vehicle	MAX_VSEV	Maximum Injury Severity in Vehicle	4.5%
Vehicle	MOD_YEAR	Vehicle Model Year	4.8%
Vehicle	P_CRASH1	Pre-Event Movement	1.6%
Vehicle	M_HARM	Most Harmful Event	0.1%
Vehicle	NUM_INJV	Number Injured in Vehicle	4.5%
<i>Person</i>			
Person	AGE	Age	6.4%
Person	EJECTION	Ejection	3.5%
Person	INJ_SEV	Injury Severity	3.8%
Person	DRINKING	Police Reported Alcohol Involvement	48.3%
Person	SEAT_POS	Seating Position	1.9%
Person	SEX	Sex	4.8%

CRSS SAS Data Files

CRSS data are made available to the public in Statistical Analysis System (SAS) data files as well as comma-separated values (CSV) files. For the current data collection year, there are 28 data files. The current data files are: Accident, Vehicle, Person, Parkwork, Vpicdecode, Vpictrailerdecode, Pbtype, Cevent, Vevent, Vsoe, Crashrf, Weather, Vehiclesf, Pvehiclesf, Driverrf, Damage, Distract, DrimPAIR, Factor, Maneuver, Violatn, Vision, Personrf, Nmcrash, Nmdistract, NmimPAIR, Nmprior, and Safetyeq data files. Seventeen of these data files contain only one data element and the coder can code more than one response for these elements (i.e., “select all that apply”); thus, there is a record for each response. These data files are: Crashrf, Weather, Vehiclesf, Pvehiclesf, Driverrf, Damage, Distract, DrimPAIR, Factor, Maneuver, Violatn, Vision, Personrf, Nmcrash, Nmdistract, NmimPAIR, and Nmprior. Two data files, Vpicdecode and Vpictrailerdecode, contain elements derived from the vehicle’s and trailer’s VIN, respectively. Details on these elements are found in a separate manual, the *Product Information Catalog and Vehicle Listing (vPIC) Analytical User’s Manual*, found in the [NCSA Publications — Manuals and Documentation](#) section of NHTSA’s website.

The data files are presented with their data elements in the Data Elements Definitions and Codes section. For each of the data elements a brief definition is provided along with any additional information that could assist analyses. SAS names and values are also provided for the data elements. Discontinued data elements are moved to the end of the data file.

The SAS data files and years of availability are:

- ***Accident*** – (2016-current): This data file contains information about crash characteristics and environmental conditions at the time of the crash. There is one record per crash.
- ***Vehicle*** – (2016-current): This data file contains information describing the motor vehicles in-transport and the drivers of motor vehicles in-transport who are involved in the crash. There is one record per motor vehicle in-transport. Parked and working vehicle information is in the Parkwork data file.
- ***Person*** – (2016-current): This data file contains information describing all people involved in the crash including motorists (i.e., drivers and passengers of motor vehicles in-transport) and non-motorists (e.g., pedestrians, pedalcyclists, and occupants of motor vehicles not in-transport). It provides information such as age, sex, vehicle occupant restraint use, and injury severity. There is one record per person.
- ***Parkwork*** – (2016-current): This data file contains information about parked and working vehicles that were involved in CRSS crashes. A parked vehicle is a motor vehicle that is stopped off the roadway. A working vehicle is a motor vehicle involved in trafficway maintenance, construction, or utility activities. It excludes vehicles performing private maintenance, construction, or utility activities. Data users are strongly advised to consult the annual FARS/CRSS Coding and Validation Manuals for a detailed description. There is one record per parked/working vehicle.
- ***Vpicdecode*** – (2016-current): This data file contains vehicle features and specifications based on the vehicle’s VIN that is decoded using NHTSA’s Product Information Catalog and Vehicle Listing, known as vPIC. There is one record per vehicle. First released in

2020, NHTSA also provided this data file for the previous 4 years and plans to release more previous years in the future.

- ***Vpictrailerdecode*** – (2016-current): This data file contains trailer features and specifications based on the trailer’s VIN that is decoded using NHTSA’s Product Information Catalog and Vehicle Listing, known as vPIC. There is one record per trailer. First released in 2020, NHTSA also provided this data file for the previous 4 years and plans to release more previous years in the future.
- ***Pbtype*** – (2016-current): This data file contains information about crashes between motor vehicles and pedestrians, people on personal conveyances, and bicyclists. Data from the crash are entered into the Pedestrian and Bicycle Crash Analysis Tool (PBCAT). The output fields from PBCAT, including the pre-crash actions of the parties involved (crash type), are included in this data file. There is one record for each pedestrian, bicyclist, or person on a personal conveyance.
- ***Cevent*** – (2016-current): This data file contains information for all of the qualifying events (both harmful and non-harmful) that occurred in the crash. It details the chronological sequence of events resulting from an unstabilized situation that constitutes a motor vehicle traffic crash. There is one record per event. Included in each record is a description of the event or object contacted (e.g., ran off road-right, crossed center line, guardrail, parked motor vehicle), the vehicles involved, and the vehicles’ areas of impact.
- ***Vevent*** – (2016-current): This data file contains the sequence of events for each motor vehicle in-transport involved in the crash. This data file has the same data elements as the Cevent data file. In addition, this data file has a data element that records the sequential event number for each vehicle (VEVENTNUM). There is one record for each event for each motor vehicle in-transport.
- ***Vsoe*** – (2016-current): This data file contains the sequence of events for each motor vehicle in-transport involved in the crash. This data file has a subset of the data elements contained in the Vevent data file (it is a simplified Vevent data file). There is one record for each event for each motor vehicle in-transport.
- ***Weather*** – (2020-current): This data file contains information describing the atmospheric conditions at the time of the crash. There is one record per condition and at least one record for each crash.
- ***Crashrf*** – (2020-current): This data file contains factors related to the crash based on a list of unusual conditions and special circumstances. Each factor is a separate record and there is at least one record for each crash.
- ***Vehiclesf*** – (2020-current): This data file contains factors related to the motor vehicles in-transport involved in the crash based on a list of special circumstances. There is one record per factor and at least one record for each motor vehicle in-transport.
- ***Pvehiclesf*** – (2020-current): This data file contains factors related to parked and working vehicles involved in CRSS crashes based on a list of special circumstances. There is one record per factor and at least one record for each parked and working vehicle.

- ***Driverrf*** – (2020-current): This data file contains factors related to the drivers of motor vehicles in-transport involved in the crash based on a list of driver conditions, unusual situations, and special circumstances. There is one record per factor and at least one record for each driver.
- ***Damage*** – (2016-current): This data file contains information about all of the areas on this vehicle that were damaged in the crash. There is one record per damaged area.
- ***Distract*** – (2016-current): This data file contains information about driver distractions. Each distraction is a separate record. There is at least one record for each driver of a motor vehicle in-transport.
- ***Drimpair*** – (2016-current): This data file contains information about physical impairments of drivers of motor vehicles. There is one record per impairment, and there is at least one record for each driver of a motor vehicle in-transport.
- ***Factor*** – (2016-current): This data file contains information about vehicle circumstances that may have contributed to the crash. Each factor is a separate record. There is at least one record per motor vehicle in-transport.
- ***Maneuver*** – (2016-current): This data file contains information about actions taken by the driver to avoid something or someone in the road. Each maneuver is a separate record. There is at least one record per motor vehicle in-transport.
- ***Violatn*** – (2016-current): This data file contains information about violations that were charged to drivers. Each violation is a separate record. There is at least one record per motor vehicle in-transport.
- ***Vision*** – (2016-current): This data file contains information about circumstances that may have obscured the driver's vision. Each obstruction is a separate record. There is at least one record per motor vehicle in-transport.
- ***Personrf*** – (2020-current): This data file contains factors related to each person, occupants, and non-occupants involved in the crash based on a list of unusual situations and special circumstances. There is one record per factor and at least one record for each person.
- ***Nmcrash*** – (2016-current): This data file contains information about contributing circumstances or improper actions of people who are not occupants of motor vehicles (e.g., pedestrians and bicyclists) noted on the police report. There is one record per action, and there is at least one record for each person who is not an occupant of a motor vehicle.
- ***Nmdistract*** – (2019-current): This data file contains information about non-motorist distractions. Each distraction is a separate record. There is at least one record for each person who is not an occupant of a motor vehicle.
- ***Nmimpair*** – (2016-current): This data file contains information about physical impairments of people who are not occupants of motor vehicles. There is one record per

impairment and there is at least one record for each person who is not an occupant of a motor vehicle.

- ***Nmprior*** – (2016-current): This data file contains information about the actions of people who are not occupants of motor vehicles (e.g., pedestrians and bicyclists) at the time of their involvement in the crash. There is one record per action and there is at least one record for each person who is not an occupant of a motor vehicle.
- ***Safetyeq*** – (2016-current): This data file contains information about safety equipment used by people who are not occupants of motor vehicles. In 2016 the file contains a record for each type of safety equipment used by a person who is not an occupant of a motor vehicle. From 2017 onward the file contains six safety equipment data elements and only one record for each person who is not an occupant of a motor vehicle.

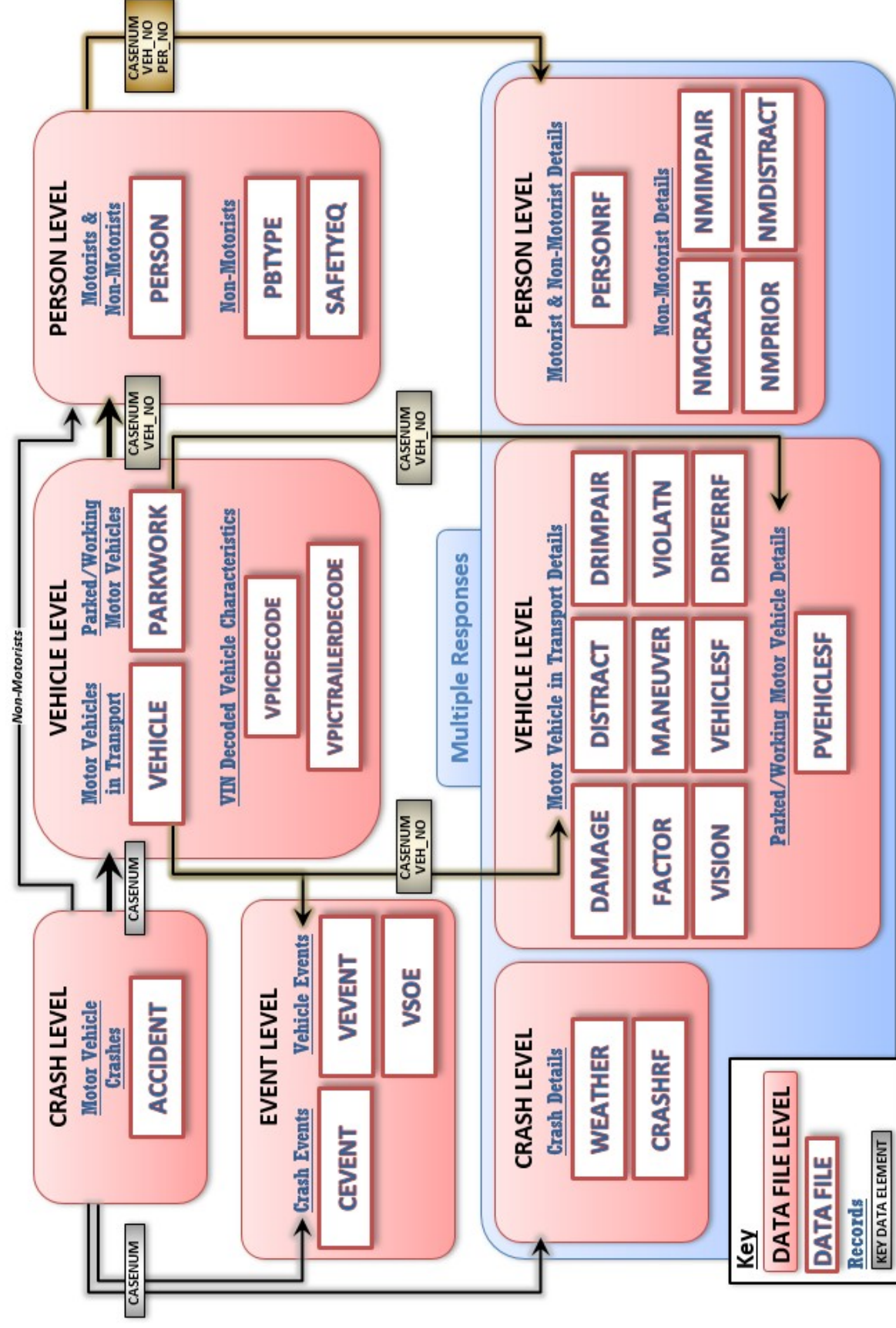


Figure 1. CRSS Schema Diagram

CRSS Data Element List

The following lists all SAS data elements with their SAS data file locations. Data elements that do not have a Data Element ID at the left side of the list have been discontinued (with the exception of some key data elements).

Data Element List

Key Data Elements

Data Element ID	Data Element Name	SAS Name
	Case Number	CASENUM
	Primary Sampling Unit (PSU)	PSU
	Primary Sampling Unit for Variance Estimation	PSU_VAR
	Primary Sampling Unit Stratum	PSUSTRAT
	Region of the Country	REGION
	Urbanicity	URBANICITY
C34	Stratum	STRATUM
C35	Police Jurisdiction (PJ)	PJ
	Case Weight	WEIGHT
V3/D3/PC3/P3/NM4	Vehicle Number	VEH_NO
P4/NM3	Person Number	PER_NO
C18	Event Number	EVENTNUM
C18	Vehicle Event Number	VEVENTNUM

The ACCIDENT Data File

Data Element ID	Data Element Name	SAS Name
C3	Number of Forms Submitted for Persons Not in MV	PEDS
C3A	Number of Persons Not in Motor Vehicles In-Transport (MVIT)	PERNOTMVIT
C4	Number of Vehicle Forms Submitted	VE_TOTAL
C4A	Number of Motor Vehicles In-Transport (MVIT)	VE_FORMS
C4B	Number of Parked/Working Vehicles	PVH_INVL
C5A	Number of Persons in Motor Vehicles In-Transport (MVIT)	PERMVIT
C8A	Month of Crash	MONTH
C8C	Day of Week	WKDY_IM
C8CI	Imputed Day of Week	DAY_WEEK
C8D	Year of Crash	YEAR

CRSS Data Element List

Data Element ID	Data Element Name	SAS Name
C9A	Hour of Crash	HOUR
C9AI	Imputed Hour of Crash	HOUR_IM
C9B	Minute of Crash	MINUTE
C9BI	Imputed Minute of Crash	MINUTE_IM
C19	First Harmful Event	HARM_EV
C19I	Imputed First Harmful Event	EVENT1_IM
C20	Manner of Collision of the First Harmful Event	MAN_COLL
C20I	Imputed Manner of Collision of the First Harmful Event	MANCOL_IM
C21A	Relation to Junction—Within Interchange Area	RELJCT1
C21AI	Imputed Relation to Junction—Within Interchange Area	RELJCT1_IM
C21B	Relation to Junction—Specific Location	RELJCT2
C21BI	Imputed Relation to Junction—Specific Location	RELJCT2_IM
C22	Type of Intersection	TYP_INT
C23	Relation to Trafficway	REL_ROAD
C24	Work Zone	WRK_ZONE
C25	Light Condition	LGT_COND
C25I	Imputed Light Condition	LGTCON_IM
C26	Atmospheric Conditions	WEATHER
C26I	Imputed Atmospheric Conditions	WEATHR_IM
C27	School Bus Related	SCH_BUS
C33	Interstate Highway	INT_HWY
C90	Maximum Injury Severity in Crash	MAX_SEV
C90I	Imputed Maximum Injury Severity in Crash	MAXSEV_IM
C91	Number Injured in Crash	NUM_INJ
C91I	Imputed Number Injured in Crash	NO_INJ_IM
C92	Alcohol Involved in Crash	ALCOHOL
C92I	Imputed Alcohol Involved in Crash	ALCHL_IM
	Atmospheric Conditions (discontinued)	WEATHER1
	Atmospheric Conditions (discontinued)	WEATHER2
	Related Factors—Crash Level (discontinued)	CF1
	Related Factors—Crash Level (discontinued)	CF2
	Related Factors—Crash Level (discontinued)	CF3

The VEHICLE Data File

Data Element ID	Data Element Name	SAS Name
V4	Number of Occupants	NUMOCCS
V5	Unit Type	UNITTYPE
V6	Hit-and-Run	HIT_RUN
V9	Vehicle Identification Number (VIN)	VIN
V10	Vehicle Model Year	MOD_YEAR
V10I	Imputed Vehicle Model Year	MDLYR_IM
V11	vPIC Make	VPICMAKE
V12	vPIC Model	VPICMODEL
V13	vPIC Body Class	VPICBODYCLASS
V14	NCSA Make	MAKE
V15	NCSA Model	MODEL
V16	NCSA Body Type	BODY_TYP
V17	Final Stage Body Class	ICFINALBODY
V18	Power Unit Gross Vehicle Weight Rating (GVWR)	GVWR_FROM
V18	Power Unit Gross Vehicle Weight Rating (GVWR)	GVWR_TO
V19	Vehicle Trailing	TOW_VEH
V20	Trailer Vehicle Identification Number	TRLR1VIN
V20	Trailer Vehicle Identification Number	TRLR2VIN
V20	Trailer Vehicle Identification Number	TRLR3VIN
V21	Trailer Gross Vehicle Weight Rating (GVWR)	TRLR1GVWR
V21	Trailer Gross Vehicle Weight Rating (GVWR)	TRLR2GVWR
V21	Trailer Gross Vehicle Weight Rating (GVWR)	TRLR3GVWR
V22	Jackknife	J_KNIFE
V23	Motor Carrier Identification Number (MCID)	MCARR_ID
V23A	MCID Issuing Authority	MCARR_I1
V23B	MCID Identification Number	MCARR_I2
V24	Vehicle Configuration	V_CONFIG
V25	Cargo Body Type	CARGO_BT
V26A/HM1	Hazardous Materials Involvement	HAZ_INV
V26B/HM2	Hazardous Materials Placard	HAZ_PLAC
V26C/HM3	Hazardous Material Identification Number	HAZ_ID
V26D/HM4	Hazardous Material Class Number	HAZ_CNO
V26E/HM5	Release of Hazardous Material from the Cargo Compartment	HAZ_REL
V27	Bus Use	BUS_USE

CRSS Data Element List

Data Element ID	Data Element Name	SAS Name
V28	Special Use	SPEC_USE
V29	Emergency Motor Vehicle Use	EMER_USE
V30	Travel Speed	TRAV_SP
V31	Vehicle Underride/Override	UNDEROVERRIDE
V32	Rollover	ROLLOVER
V33	Location of Rollover	ROLINLOC
V34A	Areas of Impact—Initial Contact Point	IMPACT1
V34AI	Imputed Areas of Impact—Initial Contact Point	IMPACT1_IM
V35	Extent of Damage	DEFORMED
V36	Vehicle Towed	TOWED
V38	Most Harmful Event	M_HARM
V38I	Imputed Most Harmful Event	VEVENT_IM
V39	Fire Occurrence	FIRE_EXP
V40A	Automation System or Systems Present in Vehicle	ADS_PRES
V40B	Highest Automation System Level Present in Vehicle	ADS_LEV
V40C	Highest Automation System Level Engaged at Time of Crash	ADS_ENG
V90	Maximum Injury Severity in Vehicle	MAX_VSEV
V90I	Imputed Maximum Injury Severity in Vehicle	MXVSEV_IM
V91	Number Injured in Vehicle	NUM_INJV
V91I	Imputed Number Injured in Vehicle	NUMINJ_IM
V92	Driver Drinking in Vehicle	VEH_ALCH
V92I	Imputed Driver Drinking in Vehicle	V_ALCH_IM
V100	NCSA Make Model Combined	MAK_MOD
D4	Driver Presence	DR_PRES
D6	Driver's Zip Code	DR_ZIP
D22	Speeding Related	SPEEDREL
PC5	Trafficway Description	VTRAFWAY
PC6	Total Lanes in Roadway	VNUM_LAN
PC7	Speed Limit	VSPD_LIM
PC8	Roadway Alignment	VALIGN
PC9	Roadway Grade	VPROFILE
PC11	Roadway Surface Conditions	VSURCOND
PC12	Traffic Control Device	VTRAFCON
PC13	Traffic Control Device Functioning	VTCONT_F
PC17	Pre-Event Movement (Prior to Recognition of Critical Event)	P_CRASH1

CRSS Data Element List

Data Element ID	Data Element Name	SAS Name
PC17I	Imputed Pre-Event Movement (Prior to Recognition of Critical Event)	PCRASH1_IM
PC19	Critical Event—Precrash	P_CRASH2
PC20	Attempted Avoidance Maneuver	P_CRASH3
PC21	Pre-Impact Stability	PCRASH4
PC22	Pre-Impact Location	PCRASH5
PC23	Crash Type	ACC_TYPE
PC23A	Crash Type Configuration	ACC_CONFIG
	Gross Vehicle Weight Rating (discontinued)	GVWR
	Imputed Hit and Run (discontinued)	HITRUN_IM
	Imputed NCSA Body Type (discontinued)	BDYTYP_IM
	Related Factors—Driver Level (discontinued)	DR_SF1
	Related Factors—Driver Level (discontinued)	DR_SF2
	Related Factors—Driver Level (discontinued)	DR_SF3
	Related Factors—Driver Level (discontinued)	DR_SF4
	Related Factors—Vehicle Level (discontinued)	VEH_SC1
	Related Factors—Vehicle Level (discontinued)	VEH_SC2

The PERSON Data File

Data Element ID	Data Element Name	SAS Name
P5/NM5	Age	AGE
P5/NM5I	Imputed Age	AGE_IM
P6/NM6	Sex	SEX
P6/NM6I	Imputed Sex	SEX_IM
P7/NM7	Person Type	PER_TYP
P8/NM10	Injury Severity	INJ_SEV
P8/NM10I	Imputed Injury Severity	INJSEV_IM
P9	Seating Position	SEAT_POS
P9I	Imputed Seating Position	SEAT_IM
P10A	Restraint System Use	REST_USE
P10B	Indication of Restraint System Misuse	REST_MIS
P11A	Helmet Use	HELM_USE
P11B	Indication of Helmet Misuse	HELM_MIS
P12	Air Bag Deployed	AIR_BAG
P13	Ejection	EJECTION
P13I	Imputed Ejection	EJECT_IM
P16/NM18	Police Reported Alcohol Involvement	DRINKING

CRSS Data Element List

Data Element ID	Data Element Name	SAS Name
P16/NM18I	Imputed Police Reported Alcohol Involvement	PERALCH_IM
P17A/NM19A	Alcohol Test Status	ALC_STATUS
P17B/NM19B	Alcohol Test Type	ATST_TYP
P17C/NM19C	Alcohol Test Result	ALC_RES
P18/NM20	Police Reported Drug Involvement	DRUGS
P20/NM22	Transported to First Medical Facility By	HOSPITAL
NM4	Vehicle Number of Motor Vehicle Striking Non-Motorist	STR_VEH
NM8	Non-Motorist Device Type	DEVTYPE
NM9	Non-Motorist Device Motorization	DEVMOTOR
NM12	Non-Motorist Location at Time of Crash	LOCATION
	Drug Test Result (discontinued)	DRUGRES1
	Drug Test Result (discontinued)	DRUGRES2
	Drug Test Result (discontinued)	DRUGRES3
	Drug Test Status (discontinued)	DSTATUS
	Drug Test Type (discontinued)	DRUGTST1
	Drug Test Type (discontinued)	DRUGTST2
	Drug Test Type (discontinued)	DRUGTST3
	Related Factors—Person Level (discontinued)	P_SF1
	Related Factors—Person Level (discontinued)	P_SF2
	Related Factors—Person Level (discontinued)	P_SF3

The PARKWORK Data File

Data Element ID	Data Element Name	SAS Name
C4A	Number of Motor Vehicles In-Transport (MVIT)	PVE_FORMS
C8A	Month of Crash	PMONTH
C9A	Hour of Crash	PHOUR
C9B	Minute of Crash	PMINUTE
C19	First Harmful Event	PHARM_EV
C20	Manner of Collision of the First Harmful Event	PMAN_COLL
V4	Number of Occupants	PNUMOCCS
V5	Unit Type	PTYPE
V6	Hit-and-Run	PHIT_RUN
V9	Vehicle Identification Number (VIN)	PVIN

CRSS Data Element List

Data Element ID	Data Element Name	SAS Name
V10	Vehicle Model Year	PMODYEAR
V11	vPIC Make	PVPICMAKE
V12	vPIC Model	PVPICMODEL
V13	vPIC Body Class	PVPICBODYCLASS
V14	NCSA Make	PMAKE
V15	NCSA Model	PMODEL
V16	NCSA Body Type	PBODYTYP
V17	Final Stage Body Class	PICFINALBODY
V18	Power Unit Gross Vehicle Weight Rating (GVWR)	PGVWR_FROM
V18	Power Unit Gross Vehicle Weight Rating (GVWR)	PGVWR_TO
V19	Vehicle Trailing	PTRAILER
V20	Trailer Vehicle Identification Number	PTRLR1VIN
V20	Trailer Vehicle Identification Number	PTRLR2VIN
V20	Trailer Vehicle Identification Number	PTRLR3VIN
V21	Trailer Gross Vehicle Weight Rating (GVWR)	PTRLR1GVWR
V21	Trailer Gross Vehicle Weight Rating (GVWR)	PTRLR2GVWR
V21	Trailer Gross Vehicle Weight Rating (GVWR)	PTRLR3GVWR
V23	Motor Carrier Identification Number (MCID)	PMCARR_ID
V23A	MCID Issuing Authority	PMCARR_I1
V23B	MCID Identification Number	PMCARR_I2
V24	Vehicle Configuration	PV_CONFIG
V25	Cargo Body Type	PCARGTYP
V26A/HM1	Hazardous Materials Involvement	PHAZ_INV
V26B/HM2	Hazardous Materials Placard	PHAZPLAC
V26C/HM3	Hazardous Material Identification Number	PHAZ_ID
V26D/HM4	Hazardous Material Class Number	PHAZ_CNO
V26E/HM5	Release of Hazardous Material from the Cargo Compartment	PHAZ_REL
V27	Bus Use	PBUS_USE
V28	Special Use	PSP_USE
V29	Emergency Motor Vehicle Use	PEM_USE
V31	Vehicle Underride/Override	PUNDEROVERRIDE
V34A	Areas of Impact—Initial Contact Point	PIMPACT1
V35	Extent of Damage	PVEH_SEV
V36	Vehicle Towed	PTOWED
V38	Most Harmful Event	PM_HARM
V39	Fire Occurrence	PFIRE

CRSS Data Element List

Data Element ID	Data Element Name	SAS Name
V100	NCSA Make Model Combined	PMMAK_MOD
	Gross Vehicle Weight Rating (discontinued)	PGVWR
	Related Factors—Vehicle Level (discontinued)	PVEH_SC1
	Related Factors—Vehicle Level (discontinued)	PVEH_SC2

The PBTYPE Data File

Data Element ID	Data Element Name	SAS Name
P5/NM5	Age	PBAGE
P6/NM6	Sex	PBSEX
P7/NM7	Person Type	PBPTYPE
NM11-PB27	Marked Crosswalk Present	PBCWALK
NM11-PB28	Sidewalk Present	PBSWALK
NM11-PB29	School Zone	PBSZONE
NM11-PB30	Crash Type—Pedestrian	PEDCTYPE
NM11-PB30B	Crash Type—Bicycle	BIKECTYPE
NM11-PB31	Crash Location—Pedestrian	PEDLOC
NM11-PB31B	Crash Location—Bicycle	BIKELOC
NM11-PB32	Pedestrian Position	PEDPOS
NM11-PB32B	Bicyclist Position	BIKEPOS
NM11-PB33	Pedestrian Initial Direction of Travel	PEDDIR
NM11-PB33B	Bicyclist Initial Direction of Travel	BIKEDIR
NM11-PB34	Motorist Initial Direction of Travel	MOTDIR
NM11-PB35	Motorist Maneuver	MOTMAN
NM11-PB36	Intersection Leg	PEDLEG
NM11-PB37	Pedestrian Scenario	PEDSNR
NM11-PB38	Crash Group – Pedestrian	PEDCGP
NM11-PB38B	Crash Group – Bicycle	BIKECGP

The CEVENT Data File

Data Element ID	Data Element Name	SAS Name
C18A	Vehicle Number (This Vehicle)	VNUMBER1
C18B	Area of Impact (This Vehicle)	AOI1
V37	Sequence of Events	SOE
C18C	Vehicle Number (Other Vehicle)	VNUMBER2
C18D	Area of Impact (Other Vehicle)	AOI2

The VEVENT Data File

Data Element ID	Data Element Name	SAS Name
C18A	Vehicle Number (This Vehicle)	VNUMBER1
C18B	Area of Impact (This Vehicle)	AOI1
V37	Sequence of Events	SOE
C18C	Vehicle Number (Other Vehicle)	VNUMBER2
C18D	Area of Impact (Other Vehicle)	AOI2

The VSOE Data File

Data Element ID	Data Element Name	SAS Name
C18E	Area of Impact	AOI
V37	Sequence of Events	SOE

The CRASHRF Data File

Data Element ID	Data Element Name	SAS Name
C32	Related Factors—Crash Level	CRASHRF

The WEATHER Data File

Data Element ID	Data Element Name	SAS Name
C26	Atmospheric Conditions	WEATHER

The VEHICLESF Data File

Data Element ID	Data Element Name	SAS Name
V41	Related Factors—Vehicle Level (Motor Vehicles in Transport)	VEHICLESF

The PVEHICLESF Data File

Data Element ID	Data Element Name	SAS Name
V41	Related Factors—Vehicle Level (Parked/Working Vehicles)	PVEHICLESF

The DRIVERRF Data File

Data Element ID	Data Element Name	SAS Name
D24	Related Factors—Driver Level	DRIVERRF

The DAMAGE Data File

Data Element ID	Data Element Name	SAS Name
V34B	Areas of Impact – Damaged Areas	DAMAGE

The DISTRACT Data File

Data Element ID	Data Element Name	SAS Name
PC16	Driver Distracted By	DRDISTRACT

The DRIMPAIR Data File

Data Element ID	Data Element Name	SAS Name
D23	Condition (Impairment) at Time of Crash— Driver	DRIMPAIR

The FACTOR Data File

Data Element ID	Data Element Name	SAS Name
PC4	Contributing Circumstances, Motor Vehicle	VEHICLECC

The MANEUVER Data File

Data Element ID	Data Element Name	SAS Name
PC15	Driver Maneuvered to Avoid	MANEUVER

The VIOLATN Data File

Data Element ID	Data Element Name	SAS Name
D21	Violations Charged	VIOLATION

The VISION Data File

Data Element ID	Data Element Name	SAS Name
PC14	Driver's Vision Obscured By	VISION

The PERSONRF Data File

Data Element ID	Data Element Name	SAS Name
P24/NM26	Related Factors—Person Level	PERSONRF

The NMCRASH Data File

Data Element ID	Data Element Name	SAS Name
NM14	Non-Motorist Contributing Circumstances	NMCC

The NMDISTRACK Data File

Data Element ID	Data Element Name	SAS Name
NM15	Non-Motorist Distracted By	NMDISTRACK

The NMIMPAIR Data File

Data Element ID	Data Element Name	SAS Name
NM17	Condition (Impairment) at Time of Crash—Non-Motorist	NMIMPAIR

The NMPRIOR Data File

Data Element ID	Data Element Name	SAS Name
NM11	Non-Motorist Action/Circumstances	NMACTION

The SAFETYQ Data File

Data Element ID	Data Element Name	SAS Name
NM16A	Non-Motorist Helmet Use	NMHELMET
NM16B	Non-Motorist Use of Protective Pads	NMPROPAD
NM16C	Non-Motorist Use of Other Protective Safety Equipment	NMOTHPRO
NM16D	Non-Motorist Use of Reflective Clothing/Carried Item	NMREFCLO
NM16E	Non-Motorist Use of Lighting	NMLIGHT
NM16F	Non-Motorist Use of Other Preventive Safety Equipment	NMOTHPRE
	Non-Motorist Safety Equipment Use (discontinued)	MSAFEQMT

Data Element Definitions and Codes

This section represents the majority of the manual. It provides information on each data element, including definitions, SAS names, attribute codes and attribute labels. Over the years changes have been made to the data collected. Some data elements have been dropped, new ones added, and attribute codes of individual data elements have changed. Element changes and the years for which individual attributes are available are shown for each data element.

For a detailed description of each data element including coding instructions and attribute definitions, see the FARS/CRSS Coding and Validation Manual. The Coding Manual is published for each year of data collection and is available at:

[NCSA Publications — Manuals and Documentation — CRSS.](#)

Additionally, a SAS program (format[YY].sas) and SAS catalog (formats.sas7bcat) are provided with the data files each year for applying the labels and formats described in this section to the current year's attributes.

The data elements in this section are listed under the data file in which they are stored. Some data elements are provided in more than one data file to facilitate analyses. For example, Month of Crash (MONTH) is a crash level data element but for convenience it is also provided in the Vehicle, Parkwork, and Person files. For such elements, they are listed under the primary data file only.

All data elements are numeric except the following that are character.

- V9 Vehicle Identification Number (VIN, PVIN) *[12 characters]*
- V20 Trailer Vehicle Identification Number (TRLR1VIN *[12 characters]*, TRLR2VIN *[12 characters]*, TRLR3VIN *[12 characters]*)
- V23 and V23B Motor Carrier ID (MCARR_ID) *[11 characters]*, (MCARR_I2) *[9 characters]*
- D6 Driver's ZIP Code (DR_ZIP) *[5 characters]*
- NM9-PB37 Pedestrian Scenario (PEDSNR) *[10 characters]*

Key Data Elements

All of the data files contain the following nine crash level data elements.

Case Number

Definition

This data element is the unique case number assigned to each crash. It appears on each data file and is used to merge information from the data files together.

Additional Information

This data element is assigned by the data entry system to each crash and is the unique identifier for the crash within the year. It is used as the key, when any two of these files from the same year are merged.

SAS Name

CASENUM

Attribute Codes

2016-Later	
xx	Case Number

Primary Sampling Unit (PSU)

Definition

This data element identifies the general geographic location from where the police report was sampled. A PSU is either a large central city, a county surrounding a city, or a group of counties.

Additional Information

See the section [CRSS Sample Design](#) for more information.

SAS Name

PSU

Attribute Codes

2016-Later	
10-83	CRSS Primary Sampling Unit Number

Primary Sampling Unit for Variance Estimation

Definition

This data element provides the PSU identifier to be used for variance estimation.

Additional Information

See [Appendix B: Rules for Derived Data Elements](#) for an explanation of this data element and how it is derived.

SAS Name

PSU_VAR

Attribute Codes

2016-Later	
	10 to 206

Primary Sampling Unit Stratum

Definition

The PSUs are grouped into strata to reflect the first stage of the sample selection. This data element is used by statistical software packages that use complex sample design for calculating variances, such as SUDAAN and SAS V9.

Additional Information

SAS Name

PSUSTRAT

Attribute Codes

2016-Later	
	1 to 25

Region of the Country

Definition

This data element identifies the region of the country where the crash occurred.

Additional Information

This data element is derived based on the State in which the Primary Sampling Unit is located where the crash occurred.

See [Appendix B: Rules for Derived Data Elements](#) for an explanation of this data element and how it is derived.

SAS Name

REGION

Attribute Codes

2016-Later	
1	Northeast (PA, NJ, NY, NH, VT, RI, MA, ME, CT)
2	Midwest (OH, IN, IL, MI, WI, MN, ND, SD, NE, IA, MO, KS)
3	South (MD, DE, DC, WV, VA, KY, TN, NC, SC, GA, FL, AL, MS, LA, AR, OK, TX)
4	West (MT, ID, WA, OR, CA, NV, NM, AZ, UT, CO, WY, AK, HI)

Urbanicity

Definition

This data element describes whether the geographical area of the crash is essentially urban or rural. The area is considered urban if it has a population of 250,000 or greater, otherwise it is rural. Census population data is used to define urbanicity.

Additional Information

See [Appendix B: Rules for Derived Data Elements](#) for an explanation of this data element and how it is derived.

SAS Name

URBANICITY

Attribute Codes

2016-Later	
1	Urban
2	Rural

Stratum

Definition

This data element identifies the number of the category in which the police report was originally listed in the PARSE Program.

Additional Information

See [CRSS Sample Design](#) for more information.

SAS Name

STRATUM

Attribute Codes

2016-Later	
2	CRSS crashes involving at least one injured (A, B, C, or ISU) or Killed (K) person who was not in a motor vehicle (i.e., non-motorist). [Not a MV Occupant- Any Injury]
3	CRSS crashes not qualifying for Stratum 2 involving at least one injured (A, B, C, or ISU) or Killed (K) occupant of a motorcycle or moped. [Motorcycle- Any Injury]
4	CRSS crashes not qualifying for Strata 2 or 3 involving at least one occupant of a late model year (LMY) passenger vehicle who was injured with a Suspected Serious Injury (A) or Killed (K). [Late Model Year Passenger Vehicle- Serious Injury]
5	CRSS crashes not qualifying for Strata 2, 3, or 4 involving at least one occupant of a non-late model year (NLMY) passenger vehicle who was injured with a Suspected Serious Injury (A) or Killed (K). [Non-Late Model Year Passenger Vehicle- Serious Injury]
6	CRSS crashes not qualifying for Strata 2, 3, 4, or 5 involving at least one occupant of a late model year (LMY) passenger vehicle who was injured (B, C, or ISU). [Late Model Year Passenger Vehicle- Minor Injury]
7	CRSS crashes not qualifying for Strata 2, 3, 4, 5, or 6 involving at least one medium or heavy truck or bus (includes school bus, transit bus, and motor coach) with GVWR equal to or greater than 10,001 pounds. [Medium/Heavy Truck or Bus]
8	CRSS crashes not qualifying for Strata 2, 3, 4, 5, 6, or 7 involving at least one occupant of a non-late model year (NLMY) passenger vehicle who was injured (B, C, or ISU). [Non-Late Model Year Passenger Vehicle- Minor Injury]
9	CRSS crashes not qualifying for Strata 2, 3, 4, 5, 6, 7, or 8 involving at least one late model year (LMY) passenger vehicle AND no one in the crash was injured (A, B, C, or ISU) or Killed (K). [Late Model Year Passenger Vehicle- No Injuries in Crash]
10	CRSS crashes not qualifying for Strata 2, 3, 4, 5, 6, 7, 8, or 9. [Other]

Police Jurisdiction (PJ)**Definition**

This data element identifies the number of the police jurisdiction from which the police crash report was originally sampled.

Additional Information**SAS Name****PJ****Attribute Codes**

2016-Later	
46-4060	CRSS Police Jurisdiction Number

Case Weight**Definition**

This data element is used to produce national estimates from the data.

Additional Information

See the section [National Estimates](#) for more information.

SAS Name**WEIGHT**

All of the vehicle level data files contain the preceding accident level data elements as well as VEH_NO.

Vehicle Number

Definition

This data element is the consecutive number assigned to each vehicle in the case. This data element appears on each vehicle level data file and is used in conjunction with the CASENUM data element to merge information from vehicle level data files.

Additional Information

All vehicles (motor vehicles in-transport as well as parked/working vehicles) are sequentially ordered starting with 1.

SAS Name

VEH_NO

Attribute Codes

2016-Later	
0	Non-Motorist
1-999	Assigned Vehicle Number

All of the person level data files contain the preceding accident level and vehicle level data elements as well as PER_NO.

Person Number

Definition

This data element is the consecutive number assigned to each person in the case (i.e., each occupant, pedestrian, or non-motorists involved in the crash). This data element appears on each person level data file and is used in conjunction with the CASENUM data element (and sometimes the VEH_NO data element) to merge information from person level data files.

Additional Information

This data element is computer assigned. Each occupant of the vehicle is numbered and each non-occupant is numbered; in the case of a non-occupant the vehicle number is zero. The numbers for occupants are consecutive for each vehicle beginning with 1. Numbers are never skipped. Drivers do not have to be coded 1. Non-occupants are identified by vehicle number 0 and are numbered consecutively starting with 1 for each non-motorist. To get drivers see data element PER_TYP under Person Type.

SAS Name

PER_NO

Attribute Codes

2016-Later	
1-999	Assigned Person Number

The CEVENT and VEVENT data files contain the preceding crash level data elements as well as EVENTNUM.

Event Number

Definition

This data element is the consecutive number assigned to each harmful and non-harmful event in a crash in chronological order.

Additional Information

Qualifying events are those that involve a motor vehicle in-transport or an object set in motion by a motor vehicle in-transport.

SAS Name

EVENTNUM

Attribute Codes

2016-Later	
1-999	Event Number

The VEVENT and VSOE data files contain the preceding crash level data elements and VEH_NO as well as VEVENTNUM.

Vehicle Event Number

Definition

This data element is the consecutive number assigned to each harmful and non-harmful event for this vehicle in chronological order.

Additional Information

The vehicle's event number shows the chronological sequence of the qualifying harmful and non-harmful events involving a particular vehicle. Qualifying events are those that involve a motor vehicle in-transport or an object set in motion by a motor vehicle in-transport.

SAS Name

VEVENTNUM

Attribute Codes

2016-Later	
1-999	Vehicle Event Number

The ACCIDENT Data File

The Accident data file includes crash data. It contains the data elements CASENUM, PSU, PJ, STRATUM, PSUSTRAT, PSU_VAR, REGION, URBANICITY, and WEIGHT, which are described in the Key Data Elements at the beginning of the Data Element Definitions and Codes section. The Accident data file also contains the data elements on the following pages.

CASENUM is the unique case identifier for each record.

C3. Number of Forms Submitted for Persons Not in MV**Definition**

This data element is the number of Person Forms (Not a Motor Vehicle Occupant) that are applicable to this case (i.e., non-occupants).

Additional Information

This represents the number of forms created for people *not* in motor vehicles. Prior to 2020 it is the number of people in the crash where “Person Type” is in (4, 5, 6, 7, 8, 10, or 19). Starting in 2020 the attributes are in (4, 5, 6, 7, 10, 11, 12, 13, or 19).

Note: People where “Person Type” = 3 (Occupant of a Motor Vehicle Not In-Transport) are *not* included in this data element but are counted in C3A below.

SAS Name**PEDS****Attribute Codes**

2016-Later	
0-99	Number of Persons Not in Motor Vehicles

C3A. Number of Persons Not in Motor Vehicles In-Transport (MVIT)**Definition**

This data element is a count of the number of non-motorists in the crash. A non-motorist is defined as a pedestrian, a cyclist, an occupant of a motor vehicle not in-transport, a person riding a horse, an occupant of an animal drawn conveyance, person associated with non-motorist conveyance (e.g., baby carriage, skateboard, wheelchair), or an other non-motorist (e.g., person outside a trafficway, person in a house).

Additional Information

Prior to 2020, this data element is calculated as the count of all people in the crash where “Person Type” is in (3, 4, 5, 6, 7, 8, 10, or 19). Starting in 2020 the attributes are in (3, 4, 5, 6, 7, 10, 11, 12, 13, or 19).

SAS Name**PERNOTMVIT****Attribute Codes**

2016-Later	
0-98	Number of Persons Not in Motor Vehicles In-Transport

C4. Number of Vehicle Forms Submitted**Definition**

This data element is the number of contact motor vehicles that the officer reported on the police crash report as a unit involved in the crash.

Additional Information

This number represents all of the motor vehicles in the crash. This includes the vehicles in-transport that are documented in the Vehicle data file and the vehicles not in-transport that are documented in the Parkwork data file. This data element only appears in the Accident data file.

SAS Name**VE_TOTAL****Attribute Codes**

2016-Later	
1-999	Number of Vehicles in Crash

C4A. Number of Motor Vehicles In-Transport (MVIT)**Definition**

This data element is a count of the number of motor vehicles in-transport involved in the crash. Legally parked vehicles are not included.

Additional Information

This data element is derived as the count of all vehicles in the crash where “Unit Type” = 1. It is the number of records in the Vehicle data file.

This data element also appears in the Vehicle and Person data files, and in the Parkwork data file as PVE_FORMS.

SAS Name**VE_FORMS****Attribute Codes**

2016-Later	
1-999	Number of Vehicle Forms

C4B. Number of Parked/Working Vehicles**Definition**

This data element is a count of the number of parked and working motor vehicles involved in the crash.

Additional Information

This data element is derived as the count of all vehicles in the crash where “Unit Type” is in (2, 3, or 4). It is the number of records in the Parkwork data file.

Working vehicles include only vehicles involved in trafficway maintenance, construction, or utility activities. Vehicles performing private maintenance, construction, or utility activities are excluded.

SAS Name**PVH_INVL****Attribute Codes**

2016-Later	
0-999	Number of Parked/Working Vehicles in the Crash

C5A. Number of Persons in Motor Vehicles In-Transport (MVIT)**Definition**

This data element is a count of the number of motorists in the crash. A motorist is a driver, passenger or unknown occupant type of a motor vehicle in-transport.

Additional Information

This data element is derived as the count of all people in the crash where “Person Type” is in (1, 2, or 9).

Note: People where “Person Type” = 3 (Occupant of a Motor Vehicle Not In-Transport) are *not* included in this data element.

SAS Name**PERMVIT****Attribute Codes**

2016-Later	
0-999	Number of Persons in Motor Vehicles In-Transport

C8. Crash Date**C8A. Month of Crash****Definition**

This data element records the month in which the crash occurred.

Additional Information

This data element also appears in the Vehicle and Person data files and in the Parkwork data file as PMONTH.

SAS Name**MONTH****Attribute Codes**

2016-Later	
1	January
2	February
3	March
4	April
5	May
6	June
7	July
8	August
9	September
10	October
11	November
12	December

C8C. Day of Week**Definition**

This data element records the day of the week on which the crash occurred.

Additional Information

This data element is derived from the SAS Weekday function. The SAS Weekday function returns the day of the week from a date.

See [Appendix C: Analytical Classification of Select CRSS Data Elements](#) for the standard NCSA classifications for this data element.

SAS Name

DAY_WEEK

Attribute Codes

2016-Later	
1	Sunday
2	Monday
3	Tuesday
4	Wednesday
5	Thursday
6	Friday
7	Saturday
9	Unknown

C8CI. Imputed Day of Week**Definition**

This imputed data element has the same definition and data element values as Day of Week, excluding value 9 for unknown day of week.

Additional Information

See the [CRSS Imputation](#) section of this manual.

SAS Name

WKDY_IM

C8D. Year of Crash**Definition**

This data element records the year in which the crash occurred.

Additional Information**SAS Name****YEAR****Attribute Codes**

2016-Later	
xxxx	Year of the Crash

C9. Crash Time**C9A. Hour of Crash****Definition**

This data element records the hour at which the crash occurred.

Additional Information

Military time is used. Noon is coded as "12." Midnight is coded as HOUR=0 and MINUTE=0. Hour is coded 0 for 1 minute after midnight to 59 minutes after midnight.

See [Appendix C: Analytical Classification of Select CRSS Data Elements](#) for the standard NCSA classifications for this data element.

This data element also appears in the Vehicle and Person data files and in the Parkwork data file as PHOUR.

SAS Name**HOURL****Attribute Codes**

2016-Later	
0-23	Hour
99	Unknown

C9AI. Imputed Hour of Crash**Definition**

This imputed data element has the same definition and data element values as Hour of the Crash, excluding value 99 for unknown hour.

Additional Information

See the [CRSS Imputation](#) section of this manual.

SAS Name**HOURL_IM**

C9B. Minute of Crash**Definition**

This data element records the minutes after the hour at which the crash occurred.

Additional Information

This data element also appears in the Vehicle and Person data files and in the Parkwork data file as PMINUTE.

SAS Name**MINUTE****Attribute Codes**

2016-Later	
0-59	Minute
99	Unknown

C9BI. Imputed Minute of Crash**Definition**

This imputed data element has the same definition and data element values as Minute of the Crash, excluding value 99 for unknown minutes.

Additional Information

See the [CRSS Imputation](#) section of this manual.

SAS Name**MINUTE_IM**

C19. First Harmful Event

Definition

This data element describes the first injury- or damage-producing event of the crash.

Additional Information

“First Harmful Event” applies to the crash. “Most Harmful Event” (M_HARM) applies to the vehicle. “First Harmful Event,” “Most Harmful Event,” and the “Sequence of Events” data elements have the same harmful event attributes. “Sequence of Events” also has non-harmful event attributes.

This data element is derived from the “Sequence of Events” data element as the first value that is not between codes 60 and 79 (non-harmful events). See [Appendix B: Rules for Derived Data Elements](#) for an explanation of this data element and how it is derived.

This data element also appears in the Vehicle and Person data files and in the Parkwork data file as PHARM_EV.

SAS Name

HARM_EV

Attribute Codes

Non-Collision Harmful Events

2016	2017	2018-Later	
1	1	1	Rollover/Overturn
2	2	2	Fire/Explosion
3	3	3	Immersion or Partial Immersion
4	4	4	Gas Inhalation
5	5	5	Fell/Jumped From Vehicle
6	6	6	Injured in Vehicle (Non-Collision)
7	7	7	Other Non-Collision
16	16	16	Thrown or Falling Object
44	44	44	Pavement Surface Irregularity (Ruts, Potholes, Grates, etc.)
51	51	51	Jackknife (Harmful to This Vehicle)
72	72	--	Cargo/Equipment Loss or Shift (Harmful to This Vehicle)
--	--	72	Cargo/Equipment Loss, Shift, or Damage (Harmful)

Collision with Motor Vehicle In-Transport

2016	2017	2018-Later	
12	12	12	Motor Vehicle In-Transport
54	54	54	Motor Vehicle In-Transport Strikes or Is Struck by Cargo, Persons or Objects Set-in-Motion From/by Another Motor Vehicle In-Transport
55	55	55	Motor Vehicle in Motion Outside the Trafficway

Collision with Object Not Fixed

2016	2017	2018-Later	
8	8	8	Pedestrian
9	9	9	Pedalcyclist
10	10	10	Railway Vehicle
11	11	11	Live Animal
14	14	14	Parked Motor Vehicle
15	15	15	Non-Motorist on Personal Conveyance
18	18	18	Other Object Not Fixed
45	45	45	Working Motor Vehicle
49	49	49	Ridden Animal or Animal Drawn Conveyance
73	73	73	Object That Had Fallen From Motor Vehicle In-Transport
74	74	74	Road Vehicle on Rails
--	91	91	Unknown Object Not Fixed

Collision with Fixed Object

2016	2017	2018-Later	
17	17	17	Boulder
19	19	19	Building
20	20	20	Impact Attenuator/Crash Cushion
21	21	21	Bridge Pier or Support
23	23	23	Bridge Rail (Includes Parapet)
24	24	24	Guardrail Face
25	25	25	Concrete Traffic Barrier
26	26	26	Other Traffic Barrier
30	30	30	Utility Pole/Light Support
31	31	31	Post, Pole or Other Support
32	32	32	Culvert
33	33	33	Curb

2016	2017	2018-Later	
34	34	34	Ditch
35	35	35	Embankment
38	38	38	Fence
39	39	39	Wall
40	40	40	Fire Hydrant
41	41	41	Shrubbery
42	42	42	Tree (Standing Only)
43	43	43	Other Fixed Object
46	46	46	Traffic Signal Support
48	48	48	Snow Bank
50	50	50	Bridge Overhead Structure
52	52	52	Guardrail End
53	53	53	Mail Box
57	57	57	Cable Barrier
58	58	58	Ground
59	59	59	Traffic Sign Support
--	93	93	Unknown Fixed Object

--	--	98	Harmful Event, Details Not Reported (Since 2019)
99	99	--	Unknown
--	--	99	Reported as Unknown

C19I. Imputed First Harmful Event

Definition

This imputed data element has the same definition as First Harmful Event, excluding values 98 and 99 for not reported and unknown first harmful events.

Additional Information

See the [CRSS Imputation](#) section of this manual.

SAS Name

EVENT1_IM

C20. Manner of Collision of the First Harmful Event

Definition

This data element describes the orientation of two motor vehicles in-transport when they are involved in the “First Harmful Event” of a collision crash. If the “First Harmful Event” is not a collision between two motor vehicles in-transport, it is classified as such.

Additional Information

Prior to 2019 this data element’s name was “Manner of Collision.”

This data element also appears in the Vehicle and Person data files and in the Parkwork data file as PMAN_COLL.

SAS Name

MAN_COLL

Attribute Codes

2016-2017	2018	2019-Later	
0	0	--	Not Collision With Motor Vehicle In-Transport
--	--	0	First Harmful Event Was Not a Collision With Motor Vehicle In-Transport
1	1	1	Front-to-Rear
2	2	2	Front-to-Front
6	6	6	Angle
7	7	7	Sideswipe – Same Direction
8	8	8	Sideswipe – Opposite Direction
9	9	9	Rear-to-Side
10	10	10	Rear-to-Rear
11	11	11	Other
98	98	98	Not Reported
99	--	--	Unknown
--	99	99	Reported as Unknown

C20I. Imputed Manner of Collision of the First Harmful Event

Definition

This imputed data element has the same definition and data element values as “**Manner of Collision of the First Harmful Event**,” excluding value 99 for unknown manner of collision and value 98 for not reported manner of collision.

Additional Information

See the [CRSS Imputation](#) section of this manual.

SAS Name

MANCOL_IM

C21. Relation to Junction**C21A. Relation to Junction—Within Interchange Area****Definition**

This data element identifies the crash's location with respect to presence in an interchange area. The coding of this data element is done in two sub-fields (see also C21B) and is based on the location of the "First Harmful Event" of the crash.

Additional Information**SAS Name****RELJCT1****Attribute Codes**

2016-2017	2018-Later	
0	0	No
1	1	Yes
8	8	Not Reported
9	--	Unknown
--	9	Reported as Unknown

C21AI. Imputed Relation to Junction—Within Interchange Area**Definition**

This imputed data element has the same definition and data element values as Relation to Junction – Within Interchange Area excluding value 8 for not reported and 9 for unknown.

Additional Information

See the [CRSS Imputation](#) section of this manual.

This imputed data element was discontinued in 2019 and added back in 2020.

SAS Name**RELJCT1_IM**

C21B. Relation to Junction—Specific Location

Definition

This data element identifies the crash's location with respect to presence in or proximity to components typically in junction or interchange areas. The coding of this data element is done in two sub-fields (see also C21A) and is based on the location of the "First Harmful Event" of the crash.

Additional Information

SAS Name

RELJCT2

Attribute Codes

2016-2017	2018-Later	
1	1	Non-Junction
2	2	Intersection
3	3	Intersection Related
4	4	Driveway Access
5	5	Entrance/Exit Ramp Related
6	6	Railway Grade Crossing
7	7	Crossover Related
8	8	Driveway Access Related
16	16	Shared-Use Path Crossing
17	17	Acceleration/Deceleration Lane
18	18	Through Roadway
19	19	Other Location Within Interchange Area
20	20	Entrance/Exit Ramp
98	98	Not Reported
99	--	Unknown
--	99	Reported as Unknown

See [Analysis of Pedestrian and Bicycle Crashes Around Intersections](#) for guidance on analyzing Pedestrian/Bicyclist crash locations.

C21BI. Imputed Relation to Junction—Specific Location**Definition**

This imputed data element has the same definition and data element values as Relation to Junction – Specific Location, excluding value 98 for not reported and 99 for unknown.

Additional Information

See the [CRSS Imputation](#) section of this manual.

SAS Name**RELJCT2_IM**

C22. Type of Intersection

Definition

This data element identifies and allows separation of various intersection types.

Additional Information

SAS Name

TYP_INT

Attribute Codes

2016-2017	2018-2019	2020-Later	
1	1	1	Not an Intersection
2	2	2	Four-Way Intersection
3	3	3	T-Intersection
4	4	4	Y-Intersection
5	5	5	Traffic Circle
6	6	6	Roundabout
7	7	7	Five-Point, or More
10	10	10	L-Intersection
--	--	11	Other Intersection Type
98	98	98	Not Reported
99	--	--	Unknown
--	99	99	Reported as Unknown

C23. Relation to Trafficway

Definition

This data element identifies the location of the crash as it relates to its position within or outside the trafficway based on the “First Harmful Event.”

Additional Information

SAS Name

REL_ROAD

Attribute Codes

2016-2017	2018-Later	
1	1	On Roadway
2	2	On Shoulder
3	3	On Median
4	4	On Roadside
5	5	Outside Trafficway
6	6	Off Roadway – Location Unknown
7	7	In Parking Lane/Zone
8	8	Gore
10	10	Separator
11	11	Continuous Left Turn Lane
--	12	Pedestrian Refuge Island or Traffic Island
98	98	Not Reported
99	--	Unknown
--	99	Reported as Unknown

C24. Work Zone

Definition

This data element identifies a motor vehicle traffic crash in which the first harmful event occurs within the boundaries of a work zone or on an approach to or exit from a work zone, resulting from an activity, behavior, or control related to the movement of the traffic units through the work zone.

Additional Information

This data element identifies a “Work Zone Accident” as defined in ANSI D16.1, 7th Edition. If the crash qualifies as a "Work Zone Accident" then the type of work activity is identified. Use of the codes does not imply that the crash was caused by the construction, maintenance, or work activity.

SAS Name

WRK_ZONE

Attribute Codes

2016-Later	
0	None
1	Construction
2	Maintenance
3	Utility
4	Work Zone, Type Unknown

C25. Light Condition**Definition**

This data element records the type/level of light that existed at the time of the crash as indicated in the police crash report.

Additional Information**SAS Name****LGT_COND****Attribute Codes**

2016-2017	2018-Later	
1	1	Daylight
2	2	Dark – Not Lighted
3	3	Dark – Lighted
4	4	Dawn
5	5	Dusk
6	6	Dark – Unknown Lighting
7	7	Other
8	8	Not Reported
9	--	Unknown
--	9	Reported as Unknown

C25I. Imputed Light Condition**Definition**

This imputed data element has the same definition and data element values as **Light Condition**, excluding value 9 for unknown light condition and value 8 for not-reported light condition.

Additional Information

See the [CRSS Imputation](#) section of this manual.

SAS Name**LGTCN_IM**

C26. Atmospheric Conditions

Definition

This derived data element records the prevailing atmospheric conditions that existed at the time of the crash as indicated in the police crash report.

Additional Information

Prior to 2020 this data element was derived from up to two conditions that could be selected, WEATHER1 and WEATHER2 based on a hierarchy. The two coded data elements were discontinued after 2019 and moved to the Discontinued Accident Data Elements at the end of the Accident Data File section.

Beginning in 2020 all applicable atmospheric conditions are selected and stored in the Weather data file and this data element is derived from those multiple responses using the same hierarchy.

See [Appendix B: Rules for Derived Data Elements](#) for an explanation of how this data element is derived.

SAS Name

WEATHER

Attribute Codes

2016-2017	2018-Later	
1	1	Clear
2	2	Rain
3	3	Sleet or Hail
4	4	Snow
5	5	Fog, Smog, Smoke
6	6	Severe Crosswinds
7	7	Blowing Sand, Soil, Dirt
8	8	Other
10	10	Cloudy
11	11	Blowing Snow
12	12	Freezing Rain or Drizzle
98	98	Not Reported
99	--	Unknown
--	99	Reported as Unknown

C26I. Imputed Atmospheric Conditions**Definition**

This imputed data element has the same definition and data element values as **Atmospheric Conditions**, excluding value 99 for unknown atmospheric conditions and value 98 for not reported atmospheric conditions.

Additional Information

See the [CRSS Imputation](#) section of this manual.

SAS Name

WEATHR_IM

C27. School Bus Related**Definition**

This data element identifies if a school bus, or motor vehicle functioning as a school bus, is related to the crash.

Additional Information

The number of school bus related crashes may not equal the number of crashes with school buses involved. For example, if a vehicle goes around a stopped school bus and hits a pedestrian, the school bus usually will not be coded, but the crash is school bus related.

This data element also appears in the Person data file.

SAS Name**SCH_BUS****Attribute Codes**

2016-Later	
0	No
1	Yes

C33. Interstate Highway**Definition**

This data element identifies whether the crash occurred on an interstate highway. Interstate highway is a Federal Highway Administration classification.

Additional Information**SAS Name****INT_HWY****Attribute Codes**

2016-Later	
0	No
1	Yes
9	Unknown

C90. Maximum Injury Severity in Crash

Definition

This data element records the single most severe injury of all people involved in the crash, and is derived from “Injury Severity” in the Person data file.

Additional Information

The following order of severity is used.

- **4-Fatal**
- **3-Suspected Serious Injury**
- **2-Suspected Minor Injury**
- **1-Possible Injury**
- **5-Injured, Unknown Severity**
- **0-No Apparent Injury**
- **6-Died Prior**
- **9-Unknown/Not Reported**
- **8-No Person Involved in Crash**

See [Appendix B: Rules for Derived Data Elements](#) for an expanded explanation of this data element and how it is derived.

SAS Name

MAX_SEV

Attribute Codes

2016-Later	
0	No Apparent Injury
1	Possible Injury
2	Suspected Minor Injury
3	Suspected Serious Injury
4	Fatal
5	Injured, Severity Unknown
6	Died Prior to Crash
8	No Person Involved in Crash
9	Unknown/Not Reported

C90I. Imputed Maximum Injury Severity in Crash**Definition**

This imputed data element has the same definition and data element values as **Maximum Injury Severity in Crash**, excluding value 9 for unknown maximum injury severity.

Additional Information

See the [CRSS Imputation](#) section of this manual.

This data element is derived from “Imputed Injury Severity” in the Person data file.

SAS Name**MAXSEV_IM**

C91. Number Injured in Crash

Definition

This data element records the number of people injured in the crash and is derived by counting all people with “Injury Severity” of (1, 2, 3, 4, or 5) in the crash. This count includes fatally injured occupants.

Additional Information

See [Appendix B: Rules for Derived Data Elements](#) for an expanded explanation of this data element and how it is derived.

SAS Name

NUM_INJ

Attribute Codes

2016-Later	
0	No Person Injured/Property Damage Only Crash
x	Number of Known Injured
98	No Person Involved in the Crash
99	All Persons in Crash Are Unknown if Injured

C91I. Imputed Number Injured in Crash

Definition

This imputed data element has the same definition and data element values as **Number Known Injured in Crash**, excluding value 99 for unknown number injured, which is imputed, and the attribute code 98, which is converted to code 0.

Additional Information

See the [CRSS Imputation](#) section of this manual.

This data element is derived from “Imputed Injury Severity” in the Person data file.

SAS Name

NO_INJ_IM

C92. Alcohol Involved in Crash

Definition

This data element records alcohol use for drivers, pedestrians, cyclists and other types of non-motorists (except occupants of motor vehicles not in-transport) involved in the crash. The data element is derived from “Police Reported Alcohol Involvement” in the Person data file.

Additional Information

8 (No Applicable Person) is coded if the crash involved only passengers of motor vehicles in-transport, occupants of motor vehicles not in-transport or unknown occupant types who are in a motor vehicle in-transport where there is no driver present.

See [Appendix B: Rules for Derived Data Elements](#) for an expanded explanation of this data element and how it is derived.

SAS Name

ALCOHOL

Attribute Codes

2016-Later	
1	Alcohol Involved
2	No Alcohol Involved
8	No Applicable Person
9	Unknown

C92I. Imputed Alcohol Involved in Crash

Definition

This data element has the same definition and data element values as **Alcohol Involved in Crash**, excluding value 9 for unknown alcohol involvement, which is imputed, and the value 8, which is converted to attribute code 2.

Additional Information

See the [CRSS Imputation](#) section of this manual.

This imputed data element is derived from “Imputed Police Reported Alcohol Involvement” in the Person data file.

SAS Name

ALCHL_IM

Discontinued ACCIDENT Data Elements

Atmospheric Conditions (discontinued)

Definition

This data element records the prevailing atmospheric conditions that existed at the time of the crash as indicated in the police crash report.

Additional Information

This data element identifies up to two values. If more than two atmospheric conditions were reported, the two conditions that most affect visibility were selected. Accident.WEATHER1 and Accident.WEATHER2 were the coded data elements, and Accident.WEATHER was derived from these two.

The two coded data elements were discontinued after 2019. Beginning in 2020 all applicable atmospheric conditions are selected and stored in the Weather data file. Only the derived data element WEATHER is still stored in the Accident data file. It is now derived from the multiple responses in the Weather data file using the same hierarchy.

SAS Name

WEATHER1, WEATHER2

Attribute Codes

2016-2017	2018-2019	
0	0	No Additional Atmospheric Conditions
1	1	Clear
2	2	Rain
3	3	Sleet or Hail
4	4	Snow
5	5	Fog, Smog, Smoke
6	6	Severe Crosswinds
7	7	Blowing Sand, Soil, Dirt
8	8	Other
10	10	Cloudy
11	11	Blowing Snow
12	12	Freezing Rain or Drizzle
98	98	Not Reported
99	--	Unknown
--	99	Reported as Unknown

Related Factors—Crash Level (discontinued)**Definition**

This data element records factors related to the crash expressed in the case materials.

Additional Information

There are also vehicle level related factors in the Vehicle data file, VEH_SC1 and VEH_SC2 and driver-related factors, also in the Vehicle data file, namely DR_SF1, DR_SF2, DR_SF3, and DR_SF4. In addition there are person-related factors P_SF1, P_SF2, and P_SF3 in the Person data file.

The CRSS coder may have used any of the three data elements to code a related factor. One must test all three data elements to insure that the selected related factor is included.

Beginning in 2020 this data element was no longer collected at the Accident level. It is now collected in the Crashrf data file as CRASHRF.

SAS Name

CF1, CF2, CF3

Attribute Codes

2016-2017	2018	2019-Later	
0	0	0	None
3	3	3	Other Maintenance or Construction-Created Condition
5	5	5	Surface Under Water
7	7	7	Surface Washed out (Caved in, Road Slippage)
--	12	12	Distracted Driver of a Non-Contact Vehicle
13	13	13	Aggressive Driving/Road Rage by Non-Contact Vehicle Driver
14	14	14	Motor Vehicle Struck By Falling Cargo or Something That Came Loose From or Something That Was Set in Motion By a Vehicle
15	15	15	Non-Occupant Struck By Falling Cargo, or Something Came Loose From or Something That Was Set in Motion By a Vehicle
16	16	16	Non-Occupant Struck Vehicle
17	17	17	Vehicle Set in Motion By Non-Driver
19	19	19	Recent Previous Crash Scene Nearby
20	20	20	Police-Pursuit-Involved
21	21	21	Within Designated School Zone
23	23	23	Indication of a Stalled/Disabled Vehicle

2016-2017	2018	2019-Later	
24	24	24	Unstabilized Situation Began and All Harmful Events Occurred off the Roadway
25	25	25	Toll Booth/Plaza Related
26	26	--	Backup Due to Prior Non-Recurring Incident
--	--	26	Prior Non-Recurring Incident
27	27	27	Backup Due to Prior Crash
28	28	--	Backup Due to Regular Congestion
--	--	28	Regular Congestion
--	--	30	Obstructed Crosswalks
--	--	31	Related to a Bus Stop
99	--	--	Unknown
--	99	99	Reported as Unknown

The VEHICLE Data File

The Vehicle data file includes motor vehicle in-transport data as well as driver and precrash data. It contains the data elements CASENUM, PSU, PJ, STRATUM, PSUSTRAT, PSU_VAR, REGION, URBANICITY, WEIGHT, and VEH_NO, which are described in the Key Data Elements at the beginning of the Data Element Definitions and Codes section. The Vehicle data file also contains the data elements on the following pages.

CASENUM and VEH_NO are the unique identifiers for each record. CASENUM should be used to merge the Vehicle data file with the Accident data file. CASENUM and VEH_NO should be used to merge the Vehicle data file with other vehicle level data files and the Person data file.

V4. Number of Occupants**Definition**

This data element is a count of the number of occupants in this vehicle.

Additional Information

This data element also appears in the Parkwork data file as PNUMOCCS.

SAS Name**NUMOCCS****Attribute Codes**

2016-Later	
0	None
1-98	Number of Occupants
99	Unknown

V5. Unit Type

Definition

This data element identifies the type of unit that applies to this motor vehicle at the time it became an involved vehicle in the crash and was reported as a unit on the police crash report.

Additional Information

This data element also appears in the Parkwork data file as PTYPE. The valid attributes for PTYPE are:

- 2 Motor Vehicle Not In-Transport Within the Trafficway
- 3 Motor Vehicle Not In-Transport Outside the Trafficway
- 4 Working Motor Vehicle (Highway Construction, Maintenance, Utility Only)

SAS Name

UNITTYPE

Attribute Codes

2016-Later	
1	Motor Vehicle In-Transport (Inside or Outside the Trafficway)

V6. Hit-and-Run**Definition**

This data element identifies whether this vehicle was a contact vehicle in the crash that did not stop to render aid (this can include drivers who flee the scene on foot). Hit and run is coded when a motor vehicle in-transport, or its driver, departs from the scene; motor vehicles not in-transport are excluded. It does not matter whether the hit-and-run vehicle was striking or struck.

Additional Information

This data element also appears in the Parkwork data file as PHIT_RUN.

SAS Name**HIT_RUN****Attribute Codes**

2016-2017	2018-2019	2020-Later	
0	0	0	No
1	1	1	Yes
9	--	--	Unknown
--	9	--	Reported as Unknown

V9. Vehicle Identification Number (VIN)

Definition

This data element records the vehicle identification number (VIN) of this vehicle assigned by the vehicle manufacturer. The VIN contains information on the vehicle such as: manufacturer, model year, model, body type, restraint type, etc.

Additional Information

The vehicle manufacturers use the VIN to describe certain characteristics of a vehicle and to assign a serial number to the vehicle.

Prior to 2018 if a character of the VIN is missing or undecipherable, the VIN length will be less than 12 characters. Starting in 2018 an asterisk (*) is used for missing or undecipherable VIN characters. Prior to 2020 the Data Element ID was V13.

This data element also appears in the Parkwork data file as PVIN.

SAS Name

VIN

Attribute Codes

2016-2017	2018-2020	2021-Later	
000000000000	000000000000	--	No VIN Required
--	--	000000000000	No VIN Required, Not a Vehicle for Road Use
xxxxxxxxxxxx	xxxxxxxxxxxx	xxxxxxxxxxxx	First 12 Characters of the VIN
888888888888	888888888888	888888888888	Not Reported
999999999999	--	--	Unknown
--	999999999999	999999999999	Reported as Unknown
--	*	*	VIN Character Missing or Not Decipherable

V10. Vehicle Model Year**Definition**

This data element identifies the manufacturer's model year of this vehicle.

Additional Information

Prior to 2020 the Data Element ID was V12.

This data element also appears in the Person data file and in the Parkwork data file as PMODYEAR.

SAS Name**MOD_YEAR****Attribute Codes**

2016-Later	
xxxx	Actual Model Year
9998	Not Reported
9999	Unknown

V10I. Imputed Vehicle Model Year**Definition**

This imputed data element has the same definition and data element values as Model Year, excluding value 9999 for unknown model year and value 9998 for not reported.

Additional Information

Prior to 2020 the Data Element ID was V12I.

See the [CRSS Imputation](#) section of this manual.

SAS Name**MDLYR_IM**

V11. vPIC Make

Definition

This element identifies the Make (manufacturer brand name) of this vehicle as per NHTSA vPIC submissions.

Additional Information

Refer to [Addition of VIN-Decoded Data](#) for more details. For more information on NHTSA's Product Information Catalog and Vehicle Listing (vPIC), go to <https://vpic.nhtsa.dot.gov/>.

A complete listing of vPIC Makes can be downloaded using the following URL:
<https://vpic.nhtsa.dot.gov/api/vehicles/getallmakes?format=csv>.

The vPIC Make Name (make_name) and vPIC Make ID (make_id) in the listing can be used to download the vPIC Models for a particular vPIC Make. (See [vPIC Model](#) for more details.)

This data element also appears in the Person data file and in the Parkwork data file as PVPICMAKE.

SAS Name

VPICMAKE

Attribute Codes

2020- Later	
xxxxx	Actual Make (up to five digits)
99997	Other
99998	Not Reported
99999	Unknown

V12. vPIC Model

Definition

This element identifies the Model of this vehicle using NHTSA's VIN decoder application, vPIC.

Additional Information

Refer to [Addition of VIN-Decoded Data](#) for more details. For more information on NHTSA's Product Information Catalog and Vehicle Listing (vPIC), go to <https://vpic.nhtsa.dot.gov/>.

A complete listing of vPIC Models for a particular vPIC Make can be downloaded using the following URLs as a guide. The first uses vPIC Make ID (make_id) as a search parameter and the second uses vPIC Make Name (make_name). (See [vPIC Make](#) for obtaining vPIC Make Names and IDs.)

- Replace * in the URL with vPIC Make ID:
https://vpic.nhtsa.dot.gov/api/vehicles/GetModelsForMakeId/*?format=csv
- Replace * in the URL with vPIC Make Name:
https://vpic.nhtsa.dot.gov/api/vehicles/getmodelsformake/*?format=csv

Example 1: Use the following URLs to download all the Models for Buick.

Use **Buick** Make ID **468** as parameter:

<https://vpic.nhtsa.dot.gov/api/vehicles/GetModelsForMakeId/468?format=csv>

Use the Make Name “**Buick**” as parameter:

<https://vpic.nhtsa.dot.gov/api/vehicles/getmodelsformake/Buick?format=csv>

Example 2: Use the following URLs to download all the Models for Toyota.

Use **Toyota** Make ID **448** as parameter:

<https://vpic.nhtsa.dot.gov/api/vehicles/GetModelsForMakeId/448?format=csv>

Use the Make Name “**Toyota**” as parameter:

<https://vpic.nhtsa.dot.gov/api/vehicles/getmodelsformake/Toyota?format=csv>

This data element also appears in the Person data file and in the Parkwork data file as PVPICMODEL.

SAS Name

VPICMODEL

Attribute Codes

2020-Later	
xxxxx	Actual Model (up to five digits)
99997	Other

2020-Later	
99998	Not Reported
99999	Unknown

V13. vPIC Body Class

Definition

This element identifies a classification of this vehicle based on its general body configuration, size, shape, doors, etc. as defined by the manufacturer.

Additional Information

Refer to [Addition of VIN-Decoded Data](#) for more details. For more information on NHTSA's Product Information Catalog and Vehicle Listing (vPIC), go to <https://vpic.nhtsa.dot.gov/>.

See [Appendix C: Analytical Classification of Select CRSS Data Elements](#) for standard classifications for this data element.

Attributes with an asterisk (*) must have a finished body class for an incomplete vehicle captured under Final Stage Body Class. Other attributes may have a Final Stage Body Class if VIN decoding indicates that the vehicle is manufactured as an incomplete vehicle.

This data element also appears in the Person data file and in the Parkwork data file as PVPICBODYCLASS.

SAS Name

VPICBODYCLASS

Attribute Codes

2020	2021-Later	
1	1	Convertible/Cabriolet
2	2	Minivan
3	3	Coupe
4	4	Low Speed Vehicle (LSV)/Neighborhood Electric Vehicle (NEV)
5	5	Hatchback/Liftback/Notchback
6	6	Motorcycle -Standard
7	7	Sport Utility Vehicle (SUV)/Multi-Purpose Vehicle (MPV)
8	8	Crossover Utility Vehicle (CUV)
9	9	Van
10	10	Roadster
11	11	Truck
12	12	Motorcycle -Scooter
13	13	Sedan/Saloon
15	15	Wagon
16	16	Bus
60	60	Pickup

2020	2021-Later	
62	62	Incomplete -Cutaway*
63	63	Incomplete -Chassis Cab (Single Cab)*
64	64	Incomplete -Glider*
65	65	Incomplete*
66	66	Truck-Tractor
67	67	Incomplete -Stripped Chassis*
68	68	Streetcar/Trolley
69	69	Off-Road Vehicle -All Terrain Vehicle (ATV) (Motorcycle-Style)
70	70	Incomplete -Chassis Cab (Double Cab)*
71	71	Incomplete -School Bus Chassis*
72	72	Incomplete -Commercial Bus Chassis*
73	73	Bus -School Bus
74	74	Incomplete -Chassis Cab (Number of Cab Unknown)*
75	75	Incomplete -Transit Bus Chassis*
76	76	Incomplete -Motor Coach Chassis*
77	77	Incomplete -Shuttle Bus Chassis*
78	78	Incomplete -Motor Home Chassis*
80	80	Motorcycle -Sport
81	81	Motorcycle -Touring/Sport Touring
82	82	Motorcycle -Cruiser
83	83	Motorcycle -Trike
84	84	Off-Road Vehicle -Dirt Bike/Off-Road
85	85	Motorcycle -Dual Sport/Adventure/Supernoto/On/Off-Road
86	86	Off-Road Vehicle -Enduro (off-road long-distance racing)
87	87	Motorcycle -Small/Minibike
88	88	Off-Road Vehicle -Go Kart
90	90	Motorcycle -Side Car
94	94	Motorcycle -Custom
95	95	Cargo Van
97	97	Off-Road Vehicle -Snowmobile
98	98	Motorcycle -Street
100	100	Motorcycle -Enclosed Three Wheeled/Enclosed Autocycle
103	103	Motorcycle -Unenclosed Three Wheeled/Open Autocycle
104	104	Motorcycle -Moped
105	105	Off-Road Vehicle -Recreational Off-Road Vehicle (ROV)
107	107	Incomplete -Bus Chassis*
108	108	Motorhome
109	109	Motorcycle -Cross Country

2020	2021-Later	
110	110	Motorcycle -Underbone
111	111	Step Van/Walk-in Van
112	112	Incomplete -Commercial Chassis*
113	113	Off-Road Vehicle -Motocross (Off-Road Short-Distance, Closed-Track Racing)
114	114	Motorcycle -Competition
117	117	Limousine
119	119	Sport Utility Truck (SUT)
124	124	Off-Road Vehicle -Golf Cart
125	125	Motorcycle -Unknown Body Type
126	126	Off-Road Vehicle -Farm Equipment
127	127	Off-Road Vehicle -Construction Equipment
--	128	Ambulance
--	129	Street Sweeper
--	130	Fire Apparatus
996	996	Motorized Bicycle (discontinued in 2022)
997	997	Other
998	998	Not Reported
999	999	Unknown

V14. NCSA Make**Definition**

This data element identifies the make (manufacturer) of this vehicle by NCSA historically.

Additional Information

Prior to 2020 this data element's name was "Vehicle Make" and the Data Element ID was V9.

This data element also appears in the Person data file and in the Parkwork data file as PMAKE.

SAS Name

MAKE

Attribute Codes

2016-Later	
1	American Motors
2	Jeep/Kaiser-Jeep/Willys-Jeep
3	AM General
6	Chrysler
7	Dodge
8	Imperial
9	Plymouth
10	Eagle
12	Ford
13	Lincoln
14	Mercury
18	Buick/Opel
19	Cadillac
20	Chevrolet
21	Oldsmobile
22	Pontiac
23	GMC
24	Saturn
25	Grumman
26	Coda
29	Other Domestic Manufacturers
	Avanti
	Checker
	DeSoto

2016-Later	
	Excalibur
	Hudson
	Packard
	Panoz
	Saleen
	Studebaker
	Stutz
	Tesla
30	Volkswagen
31	Alfa Romeo
32	Audi
33	Austin/Austin-Healey
34	BMW
35	Datsun/Nissan
36	Fiat
37	Honda
38	Isuzu
39	Jaguar
40	Lancia
41	Mazda
42	Mercedes-Benz
43	MG
44	Peugeot
45	Porsche
46	Renault
47	Saab
48	Subaru
49	Toyota
50	Triumph
51	Volvo
52	Mitsubishi
53	Suzuki
54	Acura
55	Hyundai
56	Merkur
57	Yugo
58	Infiniti
59	Lexus

2016-Later	
60	Daihatsu
61	Sterling
62	Land Rover
63	Kia
64	Daewoo
65	Smart
67	Scion
69	Other Imports
	Aston Martin
	Bentley
	Bertone
	Bricklin
	Bugatti
	Caterham
	Citroën
	DeLorean
	Desta
	Ferrari
	Fisker
	Gazelle
	Hillman
	Jensen
	Koenigsegg
	Lada
	Lamborghini
	Lotus
	Mahindra
	Maserati
	Maybach
	McLaren
	Mini Cooper
	Morgan
	Morris
	Reliant (British)
	Rolls-Royce
	Simca
	Singer
	Spyker

2016-Later		
		Sunbeam
		TVR
70	BSA	
71	Ducati	
72	Harley-Davidson	
73	Kawasaki	
74	Moto Guzzi	
75	Norton	
76	Yamaha	
78	Other Make Moped	
79	Other Make Motored Cycle	
80	Brockway	
81	Diamond Reo/Reo	
82	Freightliner/White	
83	FWD	
84	International Harvester/Navistar	
85	Kenworth	
86	Mack	
87	Peterbilt	
88	Iveco/Magirus	
89	White/Autocar, White/GMC	
90	Bluebird	
91	Eagle Coach	
92	Gillig	
93	MCI	
94	Thomas Built	
97	Not Reported	
98	Other Make	
		Auto-Union-DKW
		Carpenter
		Collins Bus
		DINA
		Divco
		Hino
		Meyers Motors
		Mid Bus
		Neoplan
		Orion

2016-Later		
		Oshkosh
		Scania
		Sterling
		Think
		UD
		Van Hool
		Western Star
99	Unknown Make	

V15. NCSA Model**Definition**

This data element identifies the NCSA model of this vehicle within a given NCSA make.

Additional Information

Prior to 2020 this data element's name was "Vehicle Model" and the Data Element ID was V10.

This data element also appears in the Person data file and in the Parkwork data file as PMODEL.

SAS Name**MODEL****Attribute Codes**

2016-Later	
	See the current FARS/CRSS Coding and Validation Manual for vehicle model codes.

V16. NCSA Body Type

Definition

This data element identifies a classification of this vehicle based on its general body configuration, size, shape, doors, etc. as defined by NCSA.

Additional Information

Prior to 2020 this data element's name was "Body Type" and the Data Element ID was V11.

See [Appendix C: Analytical Classification of Select CRSS Data Elements](#) for the standard NCSA classifications for this data element.

This data element also appears in the Person data file and in the Parkwork data file as PBODYTYP.

SAS Name

BODY_TYP

Attribute Codes

Automobiles

2016	2017-2019	2020-Later	
1	1	1	Convertible (Excludes Sun-Roof, T-Bar)
2	2	2	2-Door Sedan, Hardtop, Coupe
3	3	3	3-Door/2-Door Hatchback
4	4	4	4-Door Sedan, Hardtop
5	5	5	5-Door/4-Door Hatchback
6	6	6	Station Wagon (Excluding Van-and Truck-Based)
7	7	7	Hatchback, Number of Doors Unknown
8	8	8	Sedan/Hardtop, Number of Doors Unknown
9	9	9	Other or Unknown Automobile Type
17	17	17	3-Door Coupe

Automobile Derivatives

2016	2017-2019	2020-Later	
10	10	10	Auto Based Pickup (Includes El Camino, Caballero, Ranchero, SSR, G8-ST, Baha, Brat, and Rabbit Pickup)
11	11	11	Auto Based Panel (Cargo Station Wagon, Auto-Based Ambulance/Hearse)

2016	2017-2019	2020-Later	
12	12	12	Large Limousine (More Than Four Side Doors or Stretched Chassis)
13	13	13	Three Wheel Automobile or Automobile Derivative

Utility Vehicles

2016	2017-2019	2020-Later	
14	14	14	Compact Utility (ANSI D.16 Utility Vehicle Categories “Small” and “Midsize”)
15	15	15	Large Utility (ANSI D.16 Utility Vehicle Categories “Full Size” and “Large”)
16	16	16	Utility Station Wagon
19	19	19	Utility Vehicle, Unknown Body Type

Van-Based Light Trucks (GVWR ≤ 10,000 lbs)

2016	2017-2019	2020-Later	
20	20	20	Minivan
21	21	21	Large Van – Includes Van-Based Buses
22	22	22	Step Van or Walk-in Van (GVWR ≤ 10,000 lbs)
28	28	28	Other Van Type
29	29	29	Unknown Van Type

Light Conventional Trucks (Pickup style cab, GVWR ≤ 10,000 lbs)

2016	2017-2019	2020-Later	
30	--	--	Compact Pickup (S-10, LUV, Ram 50, Rampage, Courier, Ranger, S-5, Pup, Mazda Pickup, Mitsubishi Truck, Datsun/Nissan Pickup, Arrow Pickup, Scamp, Toyota Pickup, VW Pickup, D50, Colt P/U, T-10, S-15, T-15, Ram 100, Dakota, Sonoma)
31	--	--	Standard Pickup (C10-C35, Jeep P/U, Comanche, Ram P/U, K10-K35, D100-D350, W100-350, F100F-350, R100-500, R10-R35, V10-35, Silverado, Sierra, T100)
32	32	32	Pickup With Slide-in Camper (2016-2017 Only)
33	33	33	Convertible Pickup
--	34	34	Light Pickup
39	39	39	Unknown (Pickup Style) Light Conventional Truck

Other Light Trucks (GVWR ≤ 10,000 lbs)

2016	2017-2019	2020-Later	
40	40	40	Cab Chassis Based (Included Rescue Vehicle, Light Stake, Dump, and Tow Truck)
41	41	41	Truck-Based Panel
45	45	45	Other Light Conventional Truck Type
48	48	48	Unknown Light Truck Type
49	49	49	Unknown Light Vehicle Type (Automobile, Utility, Van, or Light Truck)

Buses (excludes van based buses with a GVWR ≤ 10,000 lbs)

2016	2017-2019	2020-Later	
50	50	50	School Bus (Designed to Carry Students, Not Cross Country or Transit)
51	51	51	Cross Country/Intercity Bus (i.e., Greyhound)
52	52	52	Transit Bus (City Bus)
55	55	55	Van-Based Bus (GVWR Greater Than 10,000 lbs)
58	58	58	Other Bus Type
59	59	59	Unknown Bus Type

Medium/Heavy Trucks (GVWR > 10,000 lbs)

2016	2017-2019	2020-Later	
60	60	60	Step Van (GVWR Greater Than 10,000 lbs)
61	61	61	Single-Unit Straight Truck or Cab-Chassis (GVWR Range 10,001 to 19,500 lbs)
62	62	62	Single-Unit Straight Truck or Cab-Chassis (GVWR Range 19,501 to 26,000 lbs)
63	63	63	Single-Unit Straight Truck or Cab-Chassis (GVWR Greater Than 26,000 lbs)
64	64	64	Single Unit Straight Truck or Cab-Chassis (GVWR Unknown)
66	66	66	Truck-Tractor (Cab Only, or With Any Number of Trailing Units; Any Weight)
67	67	67	Medium/Heavy Pickup (GVWR > 10,000 lbs)
71	71	71	Unknown if Single-Unit or Combination-Unit Medium Truck (GVWR Range 10,001 to 26,000 lbs)
72	72	72	Unknown if Single-Unit or Combination-Unit Heavy Truck (GVWR Greater Than 26,000 lbs)

2016	2017-2019	2020-Later	
78	78	78	Unknown Medium/Heavy Truck Type
79	79	79	Unknown Truck Type (Light/Medium/Heavy)

Motor Homes

2016	2017-2019	2020-Later	
42	42	--	Light Truck-Based Motor Home (Chassis-Mounted)
--	--	42	Light Vehicle-Based Motor Home (Chassis-Mounted)
65	65		Medium/Heavy Truck-Based Motor Home
--	--	65	Medium/Heavy Vehicle-Based Motor Home
73	73	--	Camper or Motor Home, Unknown Truck Type
--	--	73	Camper or Motor Home, Unknown GVWR

Motored Cycles, Mopeds, All-Terrain Vehicles, All-Terrain cycles

2016	2017-2019	2020-Later	
80	--	--	Motorcycle
--	80	80	Two Wheel Motorcycle (Excluding Motor Scooters)
81	--	--	Moped (Motorized Bicycle)
--	81	81	Moped (Since 2022)
82	--	--	Three-Wheeled Motorcycle or Moped
--	82	82	Three-Wheel Motorcycle (2 Rear Wheels)
83	--	--	Off-Road Motorcycle (2-Wheel)
--	83	83	Off-Road Motorcycle
--	84	84	Motor Scooter
--	85	85	Unenclosed Three-Wheel Motorcycle/Unenclosed Autocycle (1 Rear Wheel)
--	86	86	Enclosed Three-Wheel Motorcycle/Enclosed Autocycle (1 Rear Wheel)
--	87	87	Unknown Three-Wheel Motorcycle Type
88	--	--	Other Motored Cycle Type (Minibike, Motor Scooter, Pocket Motorcycles, Pocket Bikes)
--	88	88	Other Motored Cycle Type (Minibikes, Pocket Motorcycles, "Pocket Bikes")
89	89	89	Unknown Motored Cycle Type
90	90	90	ATV (All-Terrain Vehicle)/ATC (All-Terrain Cycle)

Other Vehicles

2016	2017-2019	2020-Later	
91	91	91	Snowmobile
92	92	92	Farm Equipment Other Than Trucks
93	93	93	Construction Equipment Other Than Trucks (Includes Graders)
94	94	94	Low Speed Vehicle (LSV)/Neighborhood Electric Vehicle (NEV)
95	95	95	Golf Cart
--	96	96	Recreational Off-Highway Vehicle (ROV)
97	97	97	Other Vehicle Type (Includes Go-Cart, Fork-Lift, City Street Sweeper)
98	98	98	Not Reported
99	99	99	Unknown Body Type

V17. Final Stage Body Class

Definition

This element captures the completed/finished body class for an Incomplete Vehicle. An incomplete vehicle is completed by a final stage manufacturer. The intent of this data element is to capture the body class for incomplete vehicles when they are finished for road use.

Additional Information

This data element is only applicable to incomplete vehicles, and the attributes are a subset of the vPIC Body Class attributes. Information captured in this data element is based on the police crash report.

This data element also appears in the Person data file and in the Parkwork data file as PICFINALBODY.

SAS Name

ICFINALBODY

Attribute Codes

2020	2021-Later	
0	0	Not Applicable
2	2	Minivan
4	4	Low-Speed Vehicle (LSV)
7	7	Sport Utility Vehicle (SUV)/Multi-Purpose Vehicle (MPV)
8	8	Crossover Utility Vehicle (CUV)
9	9	Van
11	11	Truck
15	15	Wagon
16	16	Bus
60	60	Pickup
66	66	Truck-Tractor
68	68	Streetcar/Trolley
73	73	Bus-School Bus
95	95	Cargo Van
108	108	Motorhome
111	111	Step Van/Walk-in Van
117	117	Limousine
119	119	Sport Utility Truck
--	128	Ambulance
--	129	Street Sweeper

2020	2021-Later	
--	130	Fire Apparatus
997	997	Other
998	998	Not Reported
999	999	Unknown

V18. Power Unit Gross Vehicle Weight Rating (GVWR)

Definition

This element identifies the range of gross vehicle weight rating of the power unit as identified by the manufacturer through the vehicle's VIN submission. GVWR_FROM defines the lowest value and GVWR_TO defines the highest value for the range of the GVWR specified by the manufacturer as the recommended loaded weight for a vehicle.

Additional Information

Refer to [Addition of VIN-Decoded Data](#) for more details. These data elements also appear in the Parkwork data file as PGVWR_FROM and PGVWR_TO.

SAS Name

GVWR_FROM, GVWR_TO

Attribute Codes

2020-Later	
11	Class 1: 6,000 lbs or less (2,722 kg or less)
12	Class 2: 6,001 - 10,000 lbs (2,722 - 4,536 kg)
13	Class 3: 10,001 - 14,000 lbs (4,536 - 6,350 kg)
14	Class 4: 14,001 - 16,000 lbs (6,350 - 7,258 kg)
15	Class 5: 16,001 - 19,500 lbs (7,258 - 8,845 kg)
16	Class 6: 19,501 - 26,000 lbs (8,845 - 11,794 kg)
17	Class 7: 26,001 - 33,000 lbs (11,794 - 14,969 kg)
18	Class 8: 33,001 lbs and above (14,969 kg and above)
98	Not Reported
99	Reported as Unknown

V19. Vehicle Trailing

Definition

This data element identifies whether this vehicle had any attached trailing units or was towing another motor vehicle.

Additional Information

Trailing unit applies to any device connected to a motor vehicle by a hitch, including tractor-trailer combinations, a single-unit truck pulling a trailer (truck trailer), a boat trailer hitched onto a motor vehicle, etc.

Prior to 2020 the Data Element ID was V14.

This data element also appears in the Person data file and in the Parkwork data file as PTRAILER.

SAS Name

TOW_VEH

Attribute Codes

2016-2021	2022-Later	
0	--	No Trailing Units
--	0	No Trailers
1	--	Yes, One Trailing Unit
--	1	One Trailer
2	--	Yes, Two Trailing Units
--	2	Two Trailers
3	--	Yes, Three or More Trailing Units
--	3	Three or More Trailers
4	--	Yes, Number of Trailing Units Unknown
--	4	Yes, Number of Trailers Unknown
5	5	Vehicle Towing Another Motor Vehicle – Fixed Linkage
6	6	Vehicle Towing Another Motor Vehicle – Non-Fixed Linkage
--	7	Trailing Unit Other than a Trailer or Another Motor Vehicle
9	9	Unknown

V20. Trailer Vehicle Identification Number

Definition

This data element records the vehicle identification number (VIN) of any trailing units of a combination vehicle.

Additional Information

Prior to 2018 if a character of the VIN is missing or undecipherable, the VIN length will be less than 12 characters. Starting in 2018 an asterisk (*) is used for missing or undecipherable VIN characters. Prior to 2020 the Data Element ID was V15.

These data elements also appear in the Parkwork data file as PTRLR1VIN, PTRLR2VIN, and PTRLR3VIN.

SAS Name

TRLR1VIN, TRLR2VIN, TRLR3VIN

Attribute Codes

2016-2017	2018-2020	2021-Later	
000000000000	000000000000	--	No VIN Required
--	--	000000000000	No VIN Required, Not a Vehicle for Road Use
xxxxxxxxxxxx	xxxxxxxxxxxx	xxxxxxxxxxxx	First 12 Characters of the VIN
777777777777	777777777777	777777777777	No Trailing Units
888888888888	888888888888	888888888888	Not Reported
999999999999	--	--	Unknown
--	999999999999	999999999999	Reported as Unknown
--	*	*	VIN Character Missing or Not Decipherable

V21. Trailer Gross Vehicle Weight Rating (GVWR)

Definition

This element identifies the gross vehicle weight rating of any trailing units as identified by the manufacturer in the vehicle's VIN.

Additional Information

These data elements also appear in the Parkwork data file as PTRLR1GVWR, PTRLR2GVWR, and PTRLR3GVWR.

SAS Name

TRLR1GVWR, TRLR2GVWR, TRLR3GVWR

Attribute Codes

2020-Later	
0	No Trailer GVWR Required
11	Class 1: 6,000 lbs or less (2,722 kg or less)
12	Class 2: 6,001 - 10,000 lbs (2,722 - 4,536 kg)
13	Class 3: 10,001 - 14,000 lbs (4,536 - 6,350 kg)
14	Class 4: 14,001 - 16,000 lbs (6,350 - 7,258 kg)
15	Class 5: 16,001 - 19,500 lbs (7,258 - 8,845 kg)
16	Class 6: 19,501 - 26,000 lbs (8,845 - 11,794 kg)
17	Class 7: 26,001 - 33,000 lbs (11,794 - 14,969 kg)
18	Class 8: 33,001 lbs and above (14,969 kg and above)
77	No Trailing Units
98	Not Reported
99	Reported as Unknown

V22. Jackknife**Definition**

This data element identifies whether this vehicle experienced a jackknife anytime during the unstabilized situation.

Additional Information

Jackknife applies to a condition that occurs to a “semi” truck (i.e., cab and one or more trailers) while in motion. The condition reflects a loss of control of the truck by the driver in which the trailer yaws more than 15° from its normal straight-line path behind the cab. If the final resting configuration of the vehicle is in the jackknife position, it does not necessarily mean that the vehicle has jackknifed (such as, a crash occurring while the vehicle is backing up or parking).

Prior to 2020 the Data Element ID was V16.

SAS Name

J_KNIFE

Attribute Codes

2016-Later	
0	Not an Articulated Vehicle
1	No
2	Yes, First Event
3	Yes, Subsequent Event

V23. Motor Carrier Identification Number (MCID)

Definition

This data element records the issuing authority and motor carrier identification number (if applicable) to this vehicle.

Additional Information

This 11-character data element is the combination of two data elements, the two-digit “Motor Carrier Issuing Authority” code (MCARR_I1) followed by the nine-character “Identification Number” (MCARR_I2).

The Carrier Identification Number is found only on vehicles of interstate for-hire or private carriers in the transportation business. It is the unique number assigned to the Carrier by the United States Department of Commerce Commission, or the State. The number can be either a U.S. DOT number (on interstate private carriers) or an ICC MC number (interstate for-hire carriers). Collected only for buses and trucks over 4,500 kg GVWR, this data element is applicable to the following vehicles:

- Medium/Heavy Trucks: vehicles with two axles/six tires and/or gross weight greater than 10,000 pounds.
- Buses with 16 or more seats (including the driver)
- Trucks and Vans of any size carrying hazardous cargo.
- Light commercial trucks pulling a trailer with gross combination weight rating (GCWR) greater than 10,000 pounds.

Prior to 2020 the Data Element ID was V17.

This data element also appears in the Parkwork data file as PMCARR_ID.

SAS Name

MCARR_ID

Attribute Codes

2016-Later	
00000000000	Not Applicable
xxxxxxxxxxx	11-Character (Combination of MCARR_I1 followed by MCARR_I2)
77777777777	Not Reported
88888888888	None
99999999999	Unknown (Reported as Unknown, 2018-2019)

V23A. MCID Issuing Authority**Definition**

This data element records the issuing authority if applicable to this vehicle.

Additional Information

This data element is only applicable for the following vehicles:

- Medium/Heavy Trucks: vehicles with two axles/six tires and/or gross weight greater than 10,000 pounds.
- Buses with 16 or more seats (including the driver)
- Trucks and Vans of any size carrying hazardous cargo.
- Light commercial trucks pulling a trailer with gross combination weight rating (GCWR) greater than 10,000 pounds.

See the current [FARS/CRSS Coding and Validation Manual](#) for state codes.

Prior to 2020 the Data Element ID was V17A.

This data element also appears in the Parkwork data file as PMCARR_I1.

SAS Name

MCARR_I1

Attribute Codes

2016-Later	
0	Not Applicable
1-56	State Code
57	U.S. DOT
58	MC/MX (ICC)
77	Not Reported
88	None
95	Canada
96	Mexico
99	Unknown (Reported as Unknown, 2018-2019)

V23B. MCID Identification Number**Definition**

This data element records the motor carrier identification number if applicable to this vehicle.

Additional Information

The Carrier Identification Number is found only on vehicles of interstate for-hire or private carriers in the transportation business. It is the unique number assigned to the Carrier by the United States Department of Commerce Commission, or the State. The number can be either a U.S. DOT number (on interstate private carriers) or an ICC MC number (interstate for-hire carriers). Collected only for buses and trucks over 4,500 kg GVWR, this data element is applicable to the following vehicles:

- Medium/Heavy Trucks: vehicles with two axles/six tires and/or gross weight greater than 10,000 pounds.
- Buses with 16 or more seats (including the driver).
- Trucks and Vans of any size carrying hazardous cargo.
- Light commercial trucks pulling a trailer with gross combination weight rating (GCWR) greater than 10,000 pounds.

Prior to 2020 the Data Element ID was V17B.

This data element also appears in the Parkwork data file as PMCARR_I2.

SAS Name**MCARR_I2****Attribute Codes**

2016-Later	
000000000	Not Applicable
xxxxxxxxx	Actual 9-Digit Number
777777777	Not Reported
888888888	None
999999999	Unknown (Reported as Unknown, 2018-2019)

V24. Vehicle Configuration

Definition

This data element describes the general configuration of this vehicle if applicable.

Additional Information

Not Applicable is used for automobiles, motorcycles, passenger vans (with less than nine seats, including driver) and single-unit light trucks or cargo vans (10,000 lbs or less GVWR), not carrying hazardous cargo.

Prior to 2020 the Data Element ID was V19.

This data element also appears in the Parkwork data file as PV_CONFIG.

SAS Name

V_CONFIG

Attribute Codes

2016-2020	2021-Later	
0	0	Not Applicable
1	1	Single-Unit Truck (2 Axles and GVWR More Than 10,000 lbs)
2	2	Single-Unit Truck (3 or More Axles)
4	4	Truck Pulling Trailer(s)
5	5	Truck Tractor (Bobtail)
6	6	Truck Tractor/Semi-Trailer
7	7	Truck Tractor/Double
8	8	Truck Tractor/Triple
10	10	Vehicle 10,000 lbs or Less Placarded for Hazardous Materials
19	--	Truck More Than 10,000 lbs, Cannot Classify
--	19	Vehicle More Than 10,000 lbs, Other
20	20	Bus/Large Van (Seats for 9-15 Occupants, Including Driver)
21	21	Bus (Seats for More Than 15 Occupants, Including Driver)
--	88	Qualifying Vehicle, Unknown Configuration
99	99	Unknown (Reported as Unknown, 2018-2019)

V25. Cargo Body Type

Definition

This data element describes the primary cargo carrying capability of this vehicle if applicable.

Additional Information

Passenger vehicles and light trucks that display a hazardous cargo placard are coded “No Cargo Body,” as are medium/heavy trucks with no cargo carrying capability. “Not Applicable” is coded only for passenger vehicles and light trucks and vans that do not display a hazardous cargo placard.

Prior to 2020 the Data Element ID was V20.

This data element also appears in the Parkwork data file as PCARGTYP.

SAS Name

CARGO_BT

Attribute Codes

2016-Later	
0	Not Applicable
1	Van/Enclosed Box
2	Cargo Tank
3	Flatbed
4	Dump
5	Concrete Mixer
6	Auto Transporter
7	Garbage/Refuse
8	Grain/Chips/Gravel
9	Pole-Trailer
10	Log
11	Intermodal Container Chassis
12	Vehicle Towing Another Motor Vehicle
22	Bus
96	No Cargo Body Type
97	Other
98	Unknown Cargo Body Type
99	Unknown (Reported as Unknown, 2018-2019)

V26A/HM1. Hazardous Materials Involvement**Definition**

This data element identifies whether this vehicle was carrying hazardous materials.

Additional Information

Prior to 2020 the Data Element ID was V21A/HM1.

This data element also appears in the Parkwork data file as PHAZ_INV.

SAS Name

HAZ_INV

Attribute Codes

2016-Later	
1	No
2	Yes

V26B/HM2. Hazardous Materials Placard**Definition**

This data element identifies the presence of hazardous materials for this vehicle and whether this vehicle displayed a hazardous materials placard.

Additional Information

Prior to 2020 the Data Element ID was V21B/HM2.

This data element also appears in the Parkwork data file as PHAZPLAC.

SAS Name**HAZ_PLAC****Attribute Codes**

2016-Later	
0	Not Applicable
1	No
2	Yes
8	Not Reported

V26C/HM3. Hazardous Material Identification Number**Definition**

This data element identifies the four-digit hazardous material identification number for this vehicle.

Additional Information

Prior to 2020 the Data Element ID was V21C/HM3.

This data element also appears in the Parkwork data file as PHAZ_ID.

SAS Name**HAZ_ID****Attribute Codes**

2016-Later	
0	Not Applicable
xxxx	Actual 4-Digit Number
8888	Not Reported

V26D/HM4. Hazardous Material Class Number**Definition**

This data element identifies the single-digit hazardous material class number for this vehicle.

Additional Information

Prior to 2020 the Data Element ID was V21D/HM4.

This data element also appears in the Parkwork data file as PHAZ_CNO.

SAS Name**HAZ_CNO****Attribute Codes**

2016-Later	
0	Not Applicable
1	Explosives
2	Gases
3	Flammable/Combustible Liquid
4	Flammable Solid, Spontaneously Combustible, and Dangerous When Wet
5	Oxidizer and Organic Peroxide
6	Poison and Poison Inhalation Hazard
7	Radioactive
8	Corrosive
9	Miscellaneous
88	Not Reported

V26E/HM5. Release of Hazardous Material from the Cargo Compartment**Definition**

This data element identifies whether any hazardous cargo was released from the cargo tank or compartment of this vehicle.

Additional Information

Prior to 2020 the Data Element ID was V21E/HM5.

This data element also appears in the Parkwork data file as PHAZ_REL.

SAS Name**HAZ_REL**

2016-Later	
0	Not Applicable
1	No
2	Yes
8	Not Reported

V27. Bus Use**Definition**

This data element describes the common type of bus service this vehicle was being used as at the time of the crash or the primary use for the bus if not in service at the time of the crash.

Additional Information

Prior to 2020 the Data Element ID was V22.

This data element also appears in the Parkwork data file as PBUS_USE.

SAS Name

BUS_USE

Attribute Codes

2016-2017	2018-2021	2022-Later	
0	0	0	Not a Bus
1	1	1	School
4	4	4	Intercity
5	5	5	Charter/Tour
6	6	6	Transit/Commuter
7	7	7	Shuttle
8	8	8	Modified for Personal/Private Use
--	--	97	Bus, Unknown Use
98	98	98	Not Reported
99	--	--	Unknown
--	99	99	Reported as Unknown

V28. Special Use

Definition

This data element identifies any special use associated with this vehicle at the time of the crash.

Additional Information

All military vehicles are classified as "4" even if they are police, ambulance, or fire trucks.

Prior to 2020 the Data Element ID was V23.

This data element also appears in the Person data file and in the Parkwork data file as PSP_USE.

SAS Name

SPEC_USE

Attribute Codes

2016-2018	2019	2020	2021-Later	
0	0	0	--	No Special Use
--	--	--	0	No Special Use Noted
1	1	1	1	Taxi
2	2	2	2	Vehicle Used for School Transport
3	3	3	3	Vehicle Used as Other Bus
4	4	4	4	Military
5	5	5	5	Police
6	6	6	6	Ambulance
7	7	7	7	Fire Truck
8	8	8	8	Non-Transport Emergency Services Vehicle
--	10	10	10	Safety Service Patrols – Incident Response
--	11	11	11	Other Incident Response
--	12	12	12	Towing – Incident Response
13	--	--	--	Incident Response
--	--	19	19	Motor Vehicle Used for Vehicle Sharing Mobility
--	20	--	--	Vehicle Used for Electronic Ride-Hailing (Transportation Network Company)
--	--	20	20	Motor Vehicle Used for Electronic Ride-Hailing
--	21	21	21	Mail Carrier
--	22	22	22	Public Utility
--	23	23	23	Rental Truck Over 10,000 lbs
--	24	24	24	Truck Operating With Crash Attenuator Equipment

2016-2018	2019	2020	2021-Later	
98	98	98	--	Not Reported
99	--	--	--	Unknown
99	99	99	99	Reported as Unknown (Since 2018)

V29. Emergency Motor Vehicle Use

Definition

This data element identifies whether this vehicle was engaged in emergency use. Emergency Motor Vehicle Use indicates operation of any motor vehicle that is legally authorized by a government authority to respond to emergencies with or without the use of emergency warning equipment, such as a police vehicle, fire truck or ambulance while actually engaged in such response.

Additional Information

Prior to 2020 the Data Element ID was V24.

This data element also appears in the Person data file and in the Parkwork data file as PEM_USE.

SAS Name

EMER_USE

Attribute Codes

2016-2017	2018-Later	
0	0	Not Applicable
2	2	Non-Emergency, Non-Transport
3	3	Non-Emergency Transport
4	4	Emergency Operation, Emergency Warning Equipment Not in Use
5	5	Emergency Operation, Emergency Warning Equipment in Use
6	6	Emergency Operation, Emergency Warning Equipment in Use Unknown
8	8	Not Reported
9	--	Unknown
--	9	Reported as Unknown

V30. Travel Speed**Definition**

This data element records the speed the vehicle was traveling prior to the occurrence of the crash as reported by the investigating officer.

Additional Information

Prior to 2020 the Data Element ID was V25.

SAS Name

TRAV_SP

Attribute Codes

2016-2017	2018-Later	
0	0	Stopped Motor Vehicle In-Transport
1-151	1-151	Reported Speed up to 151 mph
997	997	Speed Greater Than 151 mph
998	998	Not Reported
999	--	Unknown
--	999	Reported as Unknown

V31. Vehicle Underride/Override**Definition**

This data element indicates whether this vehicle experienced an underride or override with another vehicle during the crash.

Additional Information

See [Appendix H: Notable Changes](#) for more information on this data element. This data element also appears in the Parkwork data file as PUNDEROVERRIDE.

SAS Name**UNDEROVERRIDE****Attribute Codes**

2021-Later	
0	No Underride or Override
1	Underride
2	Override
7	Not Applicable
8	Not Reported
9	Reported as Unknown

V32. Rollover

Definition

This data element identifies this vehicle's involvement in a rollover or overturn during the crash. Rollover is defined as any vehicle rotation of 90° or more about any true longitudinal or lateral axis. Rollover can occur at any time during the crash.

Additional Information

Prior to 2020 the Data Element ID was V27.

Starting in 2022 this data element is derived from the "Sequence of Events" data element for each vehicle with an event equal to 01. See [Appendix B: Rules for Derived Data Elements](#) for an explanation of this data element and how it is derived.

This data element also appears in the Person data file.

SAS Name

ROLLOVER

Attribute Codes

2016-2021	2022-Later	
0	0	No Rollover
1	--	Rollover, Tripped By Object/Vehicle
2	--	Rollover, Untripped
--	3	Rollover
--	8	Not Applicable
9	--	Rollover, Unknown Type

V33. Location of Rollover**Definition**

This data element identifies the location of the trip point or start of this vehicle's roll.

Additional Information

Prior to 2020 the Data Element ID was V28.

SAS Name

ROLINLOC

Attribute Codes

2016-2021	2022-Later	
0	0	No Rollover
1	1	On Roadway
2	2	On Shoulder
3	3	On Median/Separator
4	4	In Gore
5	5	On Roadside
6	6	Outside of Trafficway
7	7	In Parking Lane/Zone
--	8	Not Applicable
9	9	Unknown

V34A. Areas of Impact—Initial Contact Point

Definition

This data element identifies the area on this vehicle that produced the first instance of injury to non-motorists or occupants of this vehicle, or that resulted in the first instance of damage to other property or to this vehicle.

Additional Information

This data element is derived from the crash events for the vehicle. It is the first recorded “Area of Impact (This Vehicle)” value for this vehicle. See [Appendix B: Rules for Derived Data Elements](#) for an explanation of this data element and how it is derived.

Prior to 2020 the Data Element ID was V29A.

This data element also appears in the Person data file and in the Parkwork data file as PIMPACT1.

SAS Name

IMPACT1

Attribute Codes

2016	2017	2018	2019-Later	
0	0	0	0	Non-Collision
1-12	1-12	1-12	1-12	Clock Points
13	13	13	13	Top
14	14	14	14	Undercarriage
18	18	18	18	Cargo/Vehicle Parts Set-in-Motion
19	19	19	--	Other Objects Set-in-Motion
--	--	--	19	Other Objects or Person Set-in-Motion
--	20	20	20	Object Set in Motion, Unknown if Cargo/Vehicle Parts or Other
61	61	61	61	Left
62	62	62	62	Left-Front Side
63	63	63	63	Left-Back Side
81	81	81	81	Right
82	82	82	82	Right-Front Side
83	83	83	83	Right-Back Side
98	98	98	98	Not Reported
99	99	--	--	Unknown
--	--	99	99	Reported as Unknown

V34AI. Imputed Areas of Impact—Initial Contact Point**Definition**

This imputed data element has the same definition and data element values as Initial Contact Point, excluding value 99 for unknown initial contact point and value 98 for not reported initial contact point.

Additional Information

Prior to 2020 the Data Element ID was V29AI.

See **the** [CRSS Imputation](#) section of this manual.

SAS Name

IMPACT1_IM

V35. Extent of Damage**Definition**

This data element records the amount of damage sustained by this vehicle as indicated on the police crash report based on an operational damage scale.

Additional Information

Prior to 2020 the Data Element ID was V30.

This data element also appears in the Parkwork data file as PVEH_SEV.

SAS Name

DEFORMED

Attribute Codes

2016-2017	2018-2021	2022-Later	
0	0	0	No Damage
2	2	2	Minor Damage
4	4	4	Functional Damage
6	6	6	Disabling Damage
--	--	7	Damage Reported, Extent Unknown
8	8	8	Not Reported
9	--	--	Unknown
--	9	9	Reported as Unknown

V36. Vehicle Towed**Definition**

This data element identifies whether the vehicle was towed from the scene of the crash.

Additional Information

Prior to 2022 this data element's name was "Vehicle Removal." In 2022 the attribute structure was revised to remove the multiple contracts (i.e., tow status and damage) and to simply indicate whether or not the vehicle was towed. Prior to 2020 the Data Element ID was V31.

See this data element in the Parkwork data file section for more information.

SAS Name**TOWED****Attribute Codes**

2016-2017	2018-2019	2020-2021	2022-Later	
2	2	2	--	Towed Due to Disabling Damage
3	3	--	--	Towed Not Due to Disabling Damage
--	--	3	--	Towed but Not Due to Disabling Damage
5	5	5	5	Not Towed
--	--	--	6	Towed
--	7	7	--	Towed, Unknown Reason
8	8	8	8	Not Reported
9	--	--	--	Unknown
--	9	9	9	Reported as Unknown

V38. Most Harmful Event

Definition

This data element describes the event that resulted in the most severe injury or, if no injury, the greatest property damage involving this vehicle.

Additional Information

“First Harmful Event” applies to the crash (HARM_EV). “Most Harmful Event” applies to the vehicle. “First Harmful Event,” “Most Harmful Event,” and the “Sequence of Events” data elements have the same harmful event attributes. “Sequence of Events” also has non-harmful event attributes.

Prior to 2020 the Data Element ID was V33.

This data element also appears in the Parkwork data file as PM_HARM.

SAS Name

M_HARM

Attribute Codes

Non-collision

2016	2017	2018-Later	
1	1	1	Rollover/Overturn
2	2	2	Fire/Explosion
3	3	3	Immersion or Partial Immersion
4	4	4	Gas Inhalation
5	5	5	Fell/Jumped From Vehicle
6	6	6	Injured in Vehicle (Non-Collision)
7	7	7	Other Non-Collision
16	16	16	Thrown or Falling Object
44	44	44	Pavement Surface Irregularity (Ruts, Potholes, Grates, etc.)
51	51	51	Jackknife (Harmful to This Vehicle)
72	72	--	Cargo/Equipment Loss or Shift (Harmful to This Vehicle)
--	--	72	Cargo/Equipment Loss, Shift, or Damage (Harmful)

Collision with Motor Vehicle in-Transport

2016	2017	2018-Later	
12	12	12	Motor Vehicle In-Transport
54	54	54	Motor Vehicle In-Transport Strikes or Is Struck by Cargo, Persons or Objects Set-in-Motion From/by Another Motor Vehicle In-Transport
55	55	55	Motor Vehicle in Motion Outside the Trafficway

Collision with Object Not Fixed

2016	2017	2018-Later	
8	8	8	Pedestrian
9	9	9	Pedalcyclist
10	10	10	Railway Vehicle
11	11	11	Live Animal
14	14	14	Parked Motor Vehicle
15	15	15	Non-Motorist on Personal Conveyance
18	18	18	Other Object Not Fixed
45	45	45	Working Motor Vehicle
49	49	49	Ridden Animal or Animal Drawn Conveyance
73	73	73	Object That Had Fallen From Motor Vehicle In-Transport
74	74	74	Road Vehicle on Rails
--	91	91	Unknown Object Not Fixed

Collision with Fixed Object

2016	2017	2018-Later	
17	17	17	Boulder
19	19	19	Building
20	20	20	Impact Attenuator/Crash Cushion
21	21	21	Bridge Pier or Support
23	23	23	Bridge Rail (Includes Parapet)
24	24	24	Guardrail Face
25	25	25	Concrete Traffic Barrier
26	26	26	Other Traffic Barrier
30	30	30	Utility Pole/Light Support
31	31	31	Post, Pole or Other Support
32	32	32	Culvert
33	33	33	Curb

2016	2017	2018-Later	
34	34	34	Ditch
35	35	35	Embankment
38	38	38	Fence
39	39	39	Wall
40	40	40	Fire Hydrant
41	41	41	Shrubbery
42	42	42	Tree (Standing Only)
43	43	43	Other Fixed Object
46	46	46	Traffic Signal Support
48	48	48	Snow Bank
50	50	50	Bridge Overhead Structure
52	52	52	Guardrail End
53	53	53	Mail Box
57	57	57	Cable Barrier
58	58	58	Ground
59	59	59	Traffic Sign Support
--	93	93	Unknown Fixed Object
--	--	98	Harmful Event, Details Not Reported (Since 2019)
99	99	--	Unknown
--	--	99	Reported as Unknown

V38I. Imputed Most Harmful Event.

Definition

This imputed data element has the same data element values as Most Harmful Event, excluding values 98 and 99 for not reported and unknown most harmful events.

Additional Information

Prior to 2020 the Data Element ID was V33I.

See the [CRSS Imputation](#) section of this manual.

SAS Name

VEVENT_IM

V39. Fire Occurrence**Definition**

This data element identifies whether a fire in any way related to the crash occurred in this vehicle.

Additional Information

Prior to 2020 the Data Element ID was V34.

This data element also appears in the Person data file and in the Parkwork data file as PFIRE.

SAS Name**FIRE_EXP****Attribute Codes**

2016-Later	
0	No or Not Reported
1	Yes

V40. Motor Vehicle Automated Driving System(s)**V40A. Automation System or Systems Present in Vehicle****Definition**

This data element indicates the presence of an Automation System or Systems in this vehicle.

Additional Information

An automation system is the hardware and software that are collectively capable of performing part of or all the dynamic driving task on a sustained basis. Automated Driving System (ADS), is used generically to describe any system capable of level 1-5 driving automation. For details regarding the collection of this element see [Appendix G: Special Notes for Analysts - Removal of Automated Driving Systems \(ADS\) Data Elements in CRSS](#).

Prior to 2020 the Data Element ID was V35A.

SAS Name**ADS_PRES****Attribute Codes**

2019-Later	
0	No
1	Yes
98	Not Reported
99	Reported as Unknown

V40B. Highest Automation System Level Present in Vehicle**Definition**

This data element indicates the highest level of automation present in this vehicle.

Additional Information

These systems do not have to be engaged in this vehicle at the time of the crash. For details regarding the collection of this element see [Appendix G: Special Notes for Analysts - Removal of Automated Driving Systems \(ADS\) Data Elements in CRSS](#).

Prior to 2020 the Data Element ID was V35B.

SAS Name

ADS_LEV

Attribute Codes

2019-Later	
0	Level 0 – No Automation
1	Level 1 – Driver Assistance Present
2	Level 2 – Partial Automation Present
3	Level 3 – Conditional Automation Present
4	Level 4 – High Automation Present
5	Level 5 – Full Automation Present
9	Automation Present, Level Unknown
98	Not Reported
99	Reported as Unknown

V40C. Highest Automation System Level Engaged at Time of Crash**Definition**

This data element indicates the highest level of automation that was known to have been engaged in this vehicle at the time of the crash. For details regarding the collection of this element see [Appendix G: Special Notes for Analysts - Removal of Automated Driving Systems \(ADS\) Data Elements in CRSS](#).

Additional Information

Prior to 2020 the Data Element ID was V35C.

SAS Name

ADS_ENG

Attribute Codes

2019-Later	
0	Level 0 – No Automation
1	Level 1 – Driver Assistance Engaged
2	Level 2 – Partial Automation Engaged
3	Level 3 – Conditional Automation Engaged
4	Level 4 – High Automation Engaged
5	Level 5 – Full Automation Engaged
6	Automation Systems Engaged, Level Unknown
9	Automation Systems Present, Unknown if Any Engaged
90	Automation Systems Present, Not Engaged
98	Not Reported
99	Reported as Unknown

V90. Maximum Injury Severity in Vehicle

Definition

This data element records the single most severe injury level reported for any occupant in this vehicle. This data element is derived by comparing “Injury Severity” from the Person data file for each occupant record in this vehicle. The following is the order of severity codes.

- 4-Fatal
- 3-Suspected Serious Injury
- 2-Suspected Minor Injury
- 1-Possible Injury
- 5-Injured, Unknown Severity
- 0-No Apparent Injury
- 6-Died Prior
- 9- Unknown/Not Reported
- 8-No Person in Vehicle

Additional Information

See [Appendix B: Rules for Derived Data Elements](#) for an expanded explanation of this data element and how it is derived.

SAS Name

MAX_VSEV

Attribute Codes

2016-Later	
0	No Apparent Injury
1	Possible Injury
2	Suspected Minor Injury
3	Suspected Serious Injury
4	Fatal
5	Injured, Severity Unknown
6	Died Prior to Crash
8	No Person in Vehicle
9	Unknown/Not Reported

V90I. Imputed Maximum Injury Severity in Vehicle**Definition**

This imputed data element has the same definition and data element values as Maximum Injury Severity in Vehicle, excluding value 9 for unknown maximum injury severity.

Additional Information

See the [CRSS Imputation](#) section of this manual.

The data element is derived from “Imputed Injury Severity” in the Person data file.

SAS Name**MXVSEV_IM**

V91. Number Injured in Vehicle

Definition

This data element records the number of people injured in the vehicle and is derived by counting all the people with “Injury Severity” of (1, 2, 3, 4, or 5) in a vehicle. This count includes fatally injured occupants.

Additional Information

See [Appendix B: Rules for Derived Data Elements](#) for an expanded explanation of this data element and how it is derived.

SAS Name

NUM_INJV

Attribute Codes

2016-Later	
0	No Person Injured in Vehicle
1-97	Actual Number
98	No Person in the Vehicle
99	All Persons in the Vehicle Are Unknown if Injured

V91I. Imputed Number Injured in Vehicle

Definition

This imputed data element has the same definition and data element values as “Number Injured in Vehicle,” excluding value 99 for unknown number injured, which is imputed, and the attribute code 98, which is converted to code 0.

Additional Information

See the [CRSS Imputation](#) section of this manual.

This data element is derived from “Imputed Injury Severity” in the Person data file.

SAS Name

NUMINJ_IM

V92. Driver Drinking in Vehicle

Definition

This data element records alcohol use by the driver of the vehicle. The data element is derived from “Police Reported Alcohol Involvement” in the Person data file.

Additional Information

See [Appendix B: Rules for Derived Data Elements](#) for an expanded explanation of this data element and how it is derived.

SAS Name

VEH_ALCH

Attribute Codes

2016-Later	
1	Alcohol Involved
2	No Alcohol Involved
8	No Driver Present/Unknown if Driver Present
9	Unknown

V92I. Imputed Driver Drinking in Vehicle

Definition

This data element has the same definition and data element values as Driver “Drinking in Vehicle,” excluding value 9 for unknown alcohol involvement, which is imputed, and value 8, which is converted to attribute code 2.

Additional Information

See **the** [CRSS Imputation](#) section of this manual.

This imputed data element is derived from “Imputed Police Reported Alcohol Involvement” in the Person data file.

SAS Name

V_ALCH_IM

V100. NCSA Make Model Combined**Definition**

This derived data element represents the five-digit combination of two data elements, the two-digit “NCSA Make” code (MAKE) followed by the three-digit “NCSA Model” code (MODEL).

Additional Information

Prior to 2020 this data element’s name was “Make Model Combined.”

This data element also appears in the Person data file and in the Parkwork data file as PMAK_MOD.

SAS Name**MAK_MOD****Attribute Codes**

2016-Later	
	See the current FARS/CRSS Coding and Validation Manual for vehicle make and model codes.

D4. Driver Presence**Definition**

This data element identifies whether a driver was present in this vehicle at the onset of the unstabilized situation.

Additional Information**SAS Name****DR_PRES****Attribute Codes**

2016-Later	
0	No Driver Present/Not Applicable
1	Yes
9	Unknown

D6. Driver's ZIP Code**Definition**

This data element records the ZIP Code of the driver's address as listed on the police crash report.

Additional Information**SAS Name****DR_ZIP****Attribute Codes**

2016-2019	2020-Later	
00000	00000	Not Resident of U.S. or Territories
xxxxx	xxxxx	Actual ZIP Code
99997	99997	No Driver Present/Unknown if Driver Present
--	99998	Not Reported
99999	--	Unknown
--	99999	Reported as Unknown

D22. Speeding Related**Definition**

This data element identifies if the driver was speeding and it was related to the crash as identified by law enforcement.

Additional Information**SAS Name****SPEEDREL****Attribute Codes**

2016-2017	2018-Later	
0	0	No
2	2	Yes, Racing
3	3	Yes, Exceeded Speed Limit
4	4	Yes, Too Fast for Conditions
5	5	Yes, Specifics Unknown
8	8	No Driver Present/Unknown if Driver Present
9	--	Unknown
--	9	Reported as Unknown

PC5. Trafficway Description

Definition

This data element identifies the attribute that best describes the trafficway flow just prior to this vehicle's critical precrash event.

Additional Information

SAS Name

VTRAFWAY

Attribute Codes

2016-2017	2018-2021	2022-Later	
0	0	0	Non-Trafficway or Driveway Access
1	1	1	Two-Way, Not Divided
2	2	2	Two-Way, Divided, Unprotected Median
3	3	3	Two-Way, Divided, Positive Median Barrier
4	4	4	One-Way Trafficway
5	5	5	Two-Way, Not Divided With a Continuous Left-Turn Lane
6	6	6	Entrance/Exit Ramp
--	--	7	Two-Way Divided, Unknown if Unprotected Median or Positive Median Barrier
8	8	8	Not Reported
9	--	--	Unknown
--	9	9	Reported as Unknown

PC6. Total Lanes in Roadway

Definition

This data element identifies the attribute that best describes the number of travel lanes just prior to this vehicle's critical precrash event.

Additional Information

The number of lanes refers to the number of lanes of a continuous cross-section of roadway. For example, a local roadway with one lane going north and one lane going south would be coded as two lanes. However, if a trafficway is a divided highway with two lanes going north, a median, and two lanes going south, then the number of lanes is coded as two. If a trafficway has two lanes going north immediately adjacent to two lanes going south, one continuous cross-section of roadway, then the number of lanes is coded as four. This data element can be used with the Trafficway Description data element VTRAFWAY to determine the trafficway geometry. For example: If (VNUM_LAN=2) AND (VTRAFWAY=1), then one has a two-lane roadway that is not physically divided, which is what most people think of as a two-lane road (i.e., one lane going in each direction).

If the roadway is a divided trafficway, the number of travel lanes counts only lanes in the direction of travel of the first harmful event. If the roadway is an undivided trafficway, the number of travel lanes are all the lanes regardless of their direction of travel.

SAS Name

VNUM_LAN

Attribute Codes

2016-2017	2018-Later	
0	0	Non-Trafficway or Driveway Access
1	1	One Lane
2	2	Two Lanes
3	3	Three Lanes
4	4	Four Lanes
5	5	Five Lanes
6	6	Six Lanes
7	7	Seven or More Lanes
8	8	Not Reported
9	--	Unknown
--	9	Reported as Unknown

PC7. Speed Limit**Definition**

This data element records the posted speed limit in miles per hour.

Additional Information**SAS Name****VSPD_LIM****Attribute Codes**

2016-2017	2018-Later	
0	0	No Statutory Limit/Non-Trafficway or Driveway Access
5-95	5-95	Speed Limit (in 5 mph Increments)
98	98	Not Reported
99	--	Unknown
--	99	Reported as Unknown

PC8. Roadway Alignment

Definition

This data element identifies the attribute that best represents the roadway alignment prior to this vehicle's critical precrash event.

Additional Information

SAS Name

VALIGN

Attribute Codes

2016-2017	2018-Later	
0	0	Non-Trafficway or Driveway Access
1	1	Straight
2	2	Curve Right
3	3	Curve Left
4	4	Curve – Unknown Direction
8	8	Not Reported
9	--	Unknown
--	9	Reported as Unknown

PC9. Roadway Grade**Definition**

This data element identifies the attribute that best represents the roadway grade prior to this vehicle's critical precrash event.

Additional Information**SAS Name****VPROFILE****Attribute Codes**

2016-2017	2018-Later	
0	0	Non-Trafficway or Driveway Access
1	1	Level
2	2	Grade, Unknown Slope
3	3	Hillcrest
4	4	Sag (Bottom)
5	5	Uphill
6	6	Downhill
8	8	Not Reported
9	--	Unknown
--	9	Reported as Unknown

PC11. Roadway Surface Conditions

Definition

This data element identifies the attribute that best represents the roadway surface condition prior to this vehicle's critical precrash event.

Additional Information

SAS Name

VSURCOND

Attribute Codes

2016-2017	2018-Later	
0	0	Non-Trafficway or Driveway Access
1	1	Dry
2	2	Wet
3	3	Snow
4	4	Ice/Frost
5	5	Sand
6	6	Water (Standing or Moving)
7	7	Oil
8	8	Other
10	10	Slush
11	11	Mud, Dirt, Gravel
98	98	Not Reported
99	--	Unknown
--	99	Reported as Unknown

PC12. Traffic Control Device

Definition

This data element identifies the attribute that best describes the traffic controls in the vehicle's environment just prior to this vehicle's critical precrash event.

Additional Information

If a vehicle is controlled by more than one device, the device coded is based on the following priority:

- 51 - Officer, Crossing Guard, Flagman, etc.
- The lowest numbered device shown below.
- 0 - No traffic control device.

SAS Name

VTRAFCON

Attribute Codes

2016-2017	2018-Later	
0	0	No Controls

Traffic Signals

2016-2017	2018-Later	
1	1	Traffic Control Signal (on Colors) Without Pedestrian Signal
2	2	Traffic Control Signal (on Colors) With Pedestrian Signal
3	3	Traffic Control Signal (on Colors) Not Known if Pedestrian Signal
4	4	Flashing Traffic Control Signal
7	7	Lane Use Control Signal
8	8	Other Highway Traffic Signal
9	9	Unknown Highway Traffic Signal

Regulatory Signs

2016-2017	2018-Later	
20	20	Stop Sign
21	21	Yield Sign
23	23	School Zone Sign/Device
28	28	Other Regulatory Sign
29	29	Unknown Regulatory Sign

Other Signs and Signals

2016-2017	2018-Later	
40	40	Warning Sign
50	50	Person
65	65	Railway Crossing Device
98	98	Other

Not Reported and Unknown

2016-2017	2018-Later	
97	97	Not Reported
99	--	Unknown
--	99	Reported as Unknown

PC13. Traffic Control Device Functioning**Definition**

This data element identifies the functionality of the traffic control device recorded for this vehicle in the data element “Traffic Control Device.”

Additional Information**SAS Name****VTCONT_F****Attribute Codes**

2016-2017	2018	2019-Later	
0	0	0	No Controls
1	1	1	Device Not Functioning
2	2	2	Device Functioning – Functioning Improperly
3	3	3	Device Functioning Properly
--	--	4	Device Not Functioning or Device Functioning Improperly, Specifics Unknown
8	8	8	Not Reported
9	--	--	Unknown
--	9	9	Reported as Unknown

PC17. Pre-Event Movement (Prior to Recognition of Critical Event)**Definition**

This data element identifies the attribute that best describes this vehicle's activity prior to the driver's realization of an impending critical event or just prior to impact if the driver took no action or had no time to attempt any evasive maneuvers.

Additional Information

These data elements were designed to identify: (1) what the vehicle was doing just prior to the critical precrash event, (2) what made the vehicle's situation critical, (3) what was the corrective action made, if any, to this critical situation, and what was the (4) location and (5) stability of the vehicle just prior to impact.

SAS Name**P_CRASH1****Attribute Codes**

2016-Later	
0	No Driver Present/Unknown if Driver Present
1	Going Straight
2	Decelerating in Road
3	Accelerating in Road
4	Starting in Road
5	Stopped in Roadway
6	Passing or Overtaking Another Vehicle
7	Disabled or "Parked" in Travel Lane
8	Leaving a Parking Position
9	Entering a Parking Position
10	Turning Right
11	Turning Left
12	Making a U-turn
13	Backing up (Other Than for Parking Position)
14	Negotiating a Curve
15	Changing Lanes
16	Merging
17	Successful Corrective Action to a Previous Critical Event
98	Other
99	Unknown

PC17I. Imputed Pre-Event Movement (Prior to Recognition of Critical Event)**Definition**

This imputed data element has the same definition and data element values as Movement Prior to Critical Event, excluding value 99 for unknown movement prior to critical event.

Additional Information

See the [CRSS Imputation](#) section of this manual.

SAS Name

PCRASH1_IM

PC19. Critical Event—Precrash

Definition

This data element identifies the attribute that best describes the critical event that made this crash imminent (i.e., something occurred that made the collision possible).

Additional Information

A critical event is coded for each vehicle and identifies the circumstances leading to the vehicle's first impact in the crash.

These data elements were designed to identify: (1) what the vehicle was doing just prior to the critical precrash event, (2) what made the vehicle's situation critical, (3) what was the corrective action made, if any, to this critical situation, and what was the (4) location and (5) stability of the vehicle just prior to impact.

SAS Name

P_CRASH2

Attribute Codes

This Vehicle Loss of Control Due to:

2016-2018	2019-Later	
1	1	Blow Out/Flat Tire
2	2	Stalled Engine
3	3	Disabling Vehicle Failure (e.g., wheel fell off)
4	4	Non-Disabling Vehicle Problem (e.g., hood flew up)
5	--	Poor Road Conditions (Puddle, Pothole, Ice, etc.)
--	5	Suddenly Encountered Poor Road Conditions (puddle, pothole, ice, etc.)
6	--	Traveling Too Fast for Conditions
--	6	Traveling Too Fast for Conditions or Road Configuration
8	8	Other Cause of Control Loss
9	9	Unknown Cause of Control Loss

This Vehicle Traveling:

2016-2018	2019-Later	
10	10	Over the Lane Line on Left Side of Travel Lane
11	11	Over the Lane Line on Right Side of Travel Lane
12	12	Off the Edge of the Road on the Left Side
13	13	Off the Edge of the Road on the Right Side

2016-2018	2019-Later	
14	14	End Departure
15	15	Turning Left
16	16	Turning Right
17	17	Crossing Over (Passing Through) Junction
18	18	This Vehicle Decelerating
19	19	Unknown Travel Direction
20	20	Backing
21	21	Making a U-Turn

Other Motor Vehicle In Lane

2016-2018	2019-Later	
50	50	Other Vehicle Stopped
51	51	Traveling in Same Direction with Lower Steady Speed
52	52	Traveling in Same Direction While Decelerating
53	53	Traveling in Same Direction with Higher Speed
54	54	Traveling in Opposite Direction
55	55	In Crossover
56	56	Backing
59	--	Unknown Travel Direction of the Other Motor Vehicle in Lane
--	59	Unknown Travel Direction/Speed of the Other Motor Vehicle in Lane

Other Motor Vehicle Encroaching Into Lane

2016-2018	2019-Later	
60	60	From Adjacent Lane (Same Direction)-Over Left Lane Line
61	61	From Adjacent Lane (Same Direction)-Over Right Lane Line
62	62	From Opposite Direction Over Left Lane Line
63	63	From Opposite Direction Over Right Lane Line
64	64	From Parking Lane/Shoulder, Median/Crossover, Roadside
65	65	From Crossing Street, Turning Into Same Direction
66	66	From Crossing Street, Across Path
67	67	From Crossing Street, Turning Into Opposite Direction
68	68	From Crossing Street, Intended Path Not Known
70	70	From Driveway, Turning Into Same Direction
71	71	From Driveway, Across Path
72	72	From Driveway, Turning Into Opposite Direction
73	73	From Driveway, Intended Path Not Known

2016-2018	2019-Later	
74	74	From Entrance to Limited Access Highway
78	78	Encroaching By Other Vehicle – Details Unknown

Pedestrian, Pedalcyclist Or Other Non-Motorist

2016-2018	2019-Later	
80	80	Pedestrian in Road
81	81	Pedestrian Approaching Road
82	82	Pedestrian Unknown Location
83	83	Pedalcyclist/Other Non-Motorist in Road
84	84	Pedalcyclist/Other Non-Motorist Approaching Road
85	85	Pedalcyclist or Other Non-Motorist Unknown Location

Object or Animal

2016-2018	2019-Later	
87	87	Animal in Road
88	88	Animal Approaching Road
89	89	Animal Unknown Location
90	90	Object in Road
91	91	Object Approaching Road
92	92	Object Unknown Location

Other

2016-2018	2019-Later	
98	98	Other Critical Precrash Event

Unknown

2016-2018	2019-Later	
99	99	Unknown

PC20. Attempted Avoidance Maneuver

Definition

This data element identifies the attribute that best describes the movements/actions taken by this driver, within a critical crash envelope, in response to the “Critical Precrash Event.”

Additional Information

This data element identifies the actions taken by the driver in response to the impending danger. Because this data element focuses upon the driver’s action just prior to the first harmful event it is coded independently of any maneuvers associated with this vehicle’s “Crash Type.”

These data elements were designed to identify: (1) what the vehicle was doing just prior to the critical precrash event, (2) what made the vehicle’s situation critical, (3) what was the corrective action made, if any, to this critical situation, and what was the (4) location and (5) stability of the vehicle just prior to impact.

SAS Name

P_CRASH3

Attribute Codes

2016-Later	
0	No Driver Present/Unknown if Driver Present
1	No Avoidance Maneuver
5	Releasing Brakes
6	Steering Left
7	Steering Right
8	Braking and Steering Left
9	Braking and Steering Right
10	Accelerated
11	Accelerating and Steering Left
12	Accelerating and Steering Right
15	Braking and Unknown Steering Direction
16	Braking
98	Other Actions
99	Unknown/Not Reported

PC21. Pre-Impact Stability

Definition

This data element identifies the attribute that best describes the stability of this vehicle after the “Critical Precrash Event,” but before the impact.

Additional Information

These data elements were designed to identify: (1) what the vehicle was doing just prior to the critical precrash event, (2) what made the vehicle’s situation critical, (3) what was the corrective action made, if any, to this critical situation, and what was the (4) location and (5) stability of the vehicle just prior to impact.

SAS Name

PCRASH4

Attribute Codes

2016-Later	
0	No Driver Present/Unknown if Driver Present
1	Tracking
2	Skidding Longitudinally – Rotation Less Than 30 Degrees
3	Skidding Laterally – Clockwise Rotation
4	Skidding Laterally – Counterclockwise Rotation
5	Skidding Laterally – Rotation Direction Unknown
7	Other Vehicle Loss-of-Control
9	Precrash Stability Unknown

PC22. Pre-Impact Location

Definition

This data element identifies the attribute that best describes the location of this vehicle after the “Critical Precrash Event,” but before the impact.

Additional Information

These data elements were designed to identify: (1) what the vehicle was doing just prior to the critical precrash event, (2) what made the vehicle’s situation critical, (3) what was the corrective action made, if any, to this critical situation, and what was the (4) location and (5) stability of the vehicle just prior to impact.

SAS Name

PCRASH5

Attribute Codes

2016-Later	
0	No Driver Present/Unknown if Driver Present
1	Stayed in Original Travel Lane
2	Stayed on Roadway But Left Original Travel Lane
3	Stayed on Roadway, Not Known if Left Original Travel Lane
4	Departed Roadway
5	Remained off Roadway
6	Returned to Roadway
7	Entered Roadway
9	Unknown

PC23. Crash Type

Definition

This data element identifies the attribute that best describes the type of crash this vehicle was involved in based on the “First Harmful Event” and the precrash circumstances. For graphic descriptions of possible values see [Appendix A: Crash Type Diagram](#).

Additional Information

In 2023, the data element “Crash Type” was condensed and made available as “Crash Type Configuration” in addition to the existing “Crash Type.” For graphic descriptions of the condensed possible values see [Appendix A: PC23A. Crash Type Configuration](#).

SAS Name

ACC_TYPE

Attribute Codes

2010-Later

2010-Later	
0	No Impact

Category I: Single Driver

Configuration A: Right Roadside Departure

1	Drive off Road
2	Control/Traction Loss
3	Avoid Collision With Vehicle, Pedestrian, Animal
4	Specifics Other
5	Specifics Unknown

Configuration B: Left Roadside Departure

6	Drive off Road
7	Control/Traction Loss
8	Avoid Collision With Vehicle, Pedestrian, Animal
9	Specifics Other
10	Specifics Unknown

Configuration C: Forward Impact

11	Parked Vehicle
12	Stationary Object
13	Pedestrian/Animal
14	End Departure
15	Specifics Other
16	Specifics Unknown

Category II: Same Trafficway, Same Direction***Configuration D: Rear End***

20	Stopped
21	Stopped, Straight
22	Stopped, Left
23	Stopped, Right
24	Slower
25	Slower, Going Straight
26	Slower, Going Left
27	Slower, Going Right
28	Decelerating (Slowing)
29	Decelerating (Slowing), Going Straight
30	Decelerating (Slowing), Going Left
31	Decelerating (Slowing), Going Right
32	Specifics Other
33	Specifics Unknown

Configuration E: Forward Impact

34	Control/Traction Loss, Avoiding Non-Contact Vehicle- Vehicle's Frontal Area Impacts Another Vehicle
35	Control/Traction Loss, Avoiding Non-Contact Vehicle- Vehicle Is Impacted by Frontal Area of Another Vehicle
36	Control/Traction Loss, Avoiding Non-Fixed Object- Vehicle's Frontal Area Impacts Another Vehicle
37	Control/Traction Loss, Avoiding Non-Fixed Object- Vehicle Is Impacted by Frontal Area of Another Vehicle
38	Avoiding Non-Contact Vehicle- Vehicle's Frontal Area Impacts Another Vehicle
39	Avoiding Non-Contact Vehicle- Vehicle Is Impacted by Frontal Area of Another Vehicle

34	Control/Traction Loss, Avoiding Non-Contact Vehicle- Vehicle's Frontal Area Impacts Another Vehicle
40	Avoiding Non-Fixed Object- Vehicle's Frontal Area Impacts Another Vehicle
41	Avoiding Non-Fixed Object- Vehicle Is Impacted by Frontal Area of Another Vehicle
42	Specifics Other
43	Specifics Unknown

Configuration F: Sideswipe/Angle

44	Straight Ahead on Left
45	Straight Ahead on Left/Right
46	Changing Lanes to the Right
47	Changing Lanes to the Left
48	Specifics Other
49	Specifics Unknown

Category III: Same Trafficway, Opposite Direction*Configuration G: Head-On*

50	Lateral Move (Left/Right)
51	Lateral Move (Going Straight)
52	Specifics Other
53	Specifics Unknown

Configuration H: Forward Impact

54	Control/Traction Loss, Avoiding Non-Contact Vehicle- Vehicle's Frontal Area Impacts Another Vehicle
55	Control/Traction Loss, Avoiding Non-Contact Vehicle- Vehicle Is Impacted by Frontal Area of Another Vehicle
56	Control/Traction Loss, Avoiding Non-Fixed Object- Vehicle's Frontal Area Impacts Another Vehicle
57	Control/Traction Loss, Avoiding Non-Fixed Object- Vehicle Is Impacted by Frontal Area of Another Vehicle
58	Avoiding Non-Contact Vehicle- Vehicle's Frontal Area Impacts Another Vehicle
59	Avoiding Non-Contact Vehicle- Vehicle Is Impacted by Frontal Area of Another Vehicle
60	Avoiding Non-Fixed Object- Vehicle's Frontal Area Impacts Another Vehicle

54	Control/Traction Loss, Avoiding Non-Contact Vehicle- Vehicle's Frontal Area Impacts Another Vehicle
61	Avoiding Non-Fixed Object- Vehicle Is Impacted by Frontal Area of Another Vehicle
62	Specifics Other
63	Specifics Unknown

Configuration I: Sideswipe/Angle

64	Lateral Move (Left/Right)
65	Lateral Move (Going Straight)
66	Specifics Other
67	Specifics Unknown

Category IV: Changing Trafficway, Vehicle Turning*Configuration J: Turn Across Path*

68	Initial Opposite Directions (Left/Right)
69	Initial Opposite Directions (Going Straight)
70	Initial Same Directions (Turning Right)
71	Initial Same Directions (Going Straight)
72	Initial Same Directions (Turning Left)
73	Initial Same Directions (Going Straight)
74	Specifics Other
75	Specifics Unknown

Configuration K: Turn Into Path

76	Turn Into Same Direction (Turning Left)
77	Turn Into Same Direction (Going Straight)
78	Turn Into Same Direction (Turning Right)
79	Turn Into Same Direction (Going Straight)
80	Turn Into Opposite Directions (Turning Right)
81	Turn Into Opposite Directions (Going Straight)
82	Turn Into Opposite Directions (Turning Left)
83	Turn Into Opposite Directions (Going Straight)
84	Specifics Other
85	Specifics Unknown

Category V: Intersecting Paths (Vehicle Damage)*Configuration L: Straight Paths*

86	Striking From the Right
87	Struck on the Right
88	Striking From the Left
89	Struck on the Left
90	Specifics Other
91	Specifics Unknown

Category VI: Miscellaneous*Configuration M: Backing, Etc.*

92	Backing Vehicle
93	Other Vehicle or Object (2010-2012)
93	Other Vehicle (2013-Later)
98	Other Crash Type
99	Unknown Crash Type

PC23A. Crash Type Configuration

Definition

This data element identifies the attribute that best describes the type of crash this vehicle was involved in based on the “First Harmful Event” and the precrash circumstances. For graphic descriptions of possible values see [Appendix A: PC23 Crash Type Configuration Diagram](#).

Additional Information

This is a derived element of “Crash Type.” In 2023, the data element “Crash Type” was condensed and made available as “Crash Type Configuration” in addition to “Crash Type.”

SAS Name

ACC_CONFIG 2023

Attribute Codes

2023	
0	No Impact

Category I: Single Driver

2023	
101	Right Roadside Departure
102	Left Roadside Departure
103	Struck Object While Moving Forward

Category II: Same Trafficway, Same Direction

2023	
201	Rear End, Trailing Vehicle
202	Rear End, Lead Vehicle
203	Rear End, Other or Unknown
204	Forward Impact, Frontal Impact After Maneuver
205	Forward Impact, Rear End Impact After Maneuver
206	Forward Impact, Other or Unknown
207	Sideswipe, Angle, Vehicle on Left
208	Sideswipe, Angle, Vehicle on Right
209	Sideswipe, Angle, Other or Unknown

Category III: Same Trafficway, Opposite Direction

2023	
301	Lateral Move [Left/Right], Head-On, Sideswipe, or Angle
302	Lateral Move [Going Straight], Head-On, Sideswipe, or Angle
303	Lateral Move, Other or Unknown
304	Forward Impact After Maneuver, Departed Lane
305	Forward Impact After Maneuver, Remained in Lane
306	Foward Impact After Maneuver, Other or Unknown

Category IV: Changing Trafficway, Vehicle Turning

2023	
401	Turn Across Path, Initial Opposite Directions [Left/Right]
402	Turn Across Path, Initial Opposite Directions [Going Straight]
403	Turn Across Path, Initial Same Directions [Turning Right]
404	Turn Across Path, Initial Same Directions [Going Straight, Other Vehicle Turning Right]
405	Turn Across Path, Initial Same Directions [Turning Left]
406	Turn Across Path, Initial Same Directions [Going Straight, Other Vehicle Turning Left]
407	Turn Across Path, Other or Unknown
408	Turn Into Path, Turn into Same Direction [Turning Left]
409	Turn Into Path, Turn into Same Direction [Going Straight, Other Vehicle Turning Left]
410	Turn Into Path, Turn into Same Direction [Turning Right]
411	Turn Into Path, Turn into Same Direction [Going Straight, Other Vehicle Turning Right]
412	Turn Into Path, Turn into Opposite Directions [Turning Right]
413	Turn Into Path, Turn into Opposite Directions [Going Straight, Other Vehicle Turning Right]
414	Turn Into Path, Turn into Opposite Directions [Turning Left]
415	Turn Into Path, Turn into Opposite Directions [Going Straight, Other Vehicle Turning Left]
416	Turn Into Path, Other or Unknown

Category V: Intersecting Paths (Vehicle Damage)

2023	
501	Straight Paths, Striking from the Right
502	Straight Paths, Struck on the Right
503	Straight Paths, Striking from the Left
504	Straight Paths, Struck on the Left
505	Straight Paths, Other or Unknown

Category VI: Miscellaneous

2023	
000	No Impact
992	Backing Vehicle
993	Other Vehicle
998	Other Crash Type
999	Unknown Crash Type

Discontinued VEHICLE Data Elements

Gross Vehicle Weight Rating (discontinued)

Definition

This data element identifies the gross vehicle weight rating of this vehicle if applicable.

Additional Information

The Gross Vehicle Weight Rating (GVWR) or Gross Combination Weight Rating (GCWR) is a value specified by the manufacturer for a single-unit truck, truck tractor, or trailer. In the absence of a gross vehicle weight rating, an estimate of the gross weight of a fully loaded unit can be substituted.

This data element is the gross vehicle weight of the power unit only. The weight of trailers is not added. Beginning in 2020 this data element is replaced by two data elements, Power Unit GVWR and Trailer GVWR, which are derived from their VINs.

This data element also appears in the Parkwork data file as PGVWR.

SAS Name

GVWR

Attribute Codes

2016-2017	2018-2019	
0	0	Not Applicable
1	1	10,000 lbs or Less
2	2	10,001 lbs -26,000 lbs
3	3	26,001 lbs or More
8	8	Not Reported
9	--	Unknown
--	9	Reported as Unknown

Imputed Hit-and-Run (discontinued)

Definition

This imputed data element has the same definition and data element values as “**Hit-and-Run**,” excluding value 9 for unknown hit-and-run.

Additional Information

See the [CRSS Imputation](#) section of this manual.

SAS Name**HITRUN_IM** (2016-2019)***Imputed NCSA Body Type (discontinued)*****Definition**

The attributes for this imputed data element have changed over the years to mirror the values for “NCSA **Body Type**,” excluding values 49, 79, and 99 for unknown light vehicle type, unknown truck type (light/medium/heavy), and unknown body type, respectively, and value 98 for not reported body type.

Additional Information

Prior to 2020 this data element’s name was "Imputed Body Type" and the Data Element ID was V11I.

See the [CRSS Imputation](#) section of this manual.

SAS Name**BDYTYP_IM** (2016-2020)***Related Factors—Driver Level (discontinued)*****Definition**

This data element records factors related to this driver expressed in the case materials.

Additional Information

There are also crash level related factors in the Accident data file, CF1, CF2, and CF3; vehicle-related factors, namely VEH_SC1 and VEH_SC2 in the Vehicle data file; and person-related factors P_SF1, P_SF2, and P_SF3 in the person data file.

The CRSS coder may have used any of the four data elements to code a related factor. One must test all four data elements to insure that the selected related factor is included.

The person-related factors P_SF1, P_SF2, and P_SF3 are all set to 0 for drivers.

Beginning in 2020 this data element was no longer collected at the Vehicle level. It is now collected in the Driverrf data file as DRIVERRF.

SAS Name**DR_SF1, DR_SF2, DR_SF3, DR_SF4**

Attribute Codes

2016	2017	2018	2019	
0	0	0	0	None
6	6	6	6	Careless Driving
8	8	8	8	Road Rage/Aggressive Driving
--	--	9	--	Emergency Services Personnel
--	--	10	10	Looked But Did Not See
16	16	16	16	Police or Law Enforcement Officer
18	18	18	18	Traveling on Prohibited Trafficways
20	20	20	20	Leaving Vehicle Unattended with Engine Running; Leaving Vehicle Unattended in Roadway
21	21	21	21	Overloading or Improper Loading of Vehicle with Passenger or Cargo
22	22	22	22	Towing or Pushing Vehicle Improperly
23	23	23	23	Failing to Dim Lights or to Have Lights on When Required
24	24	24	24	Operating Without Required Equipment
--	--	--	29	Intentional Illegal Driving off the Roadway
32	32	32	32	Opening Vehicle Closure into Moving Traffic or Vehicle Is in Motion or Operating at Erratic or Suddenly Changing Speeds
36	36	36	36	Operating the Vehicle in an Erratic, Reckless, Careless or Negligent Manner
37	37	37	37	Police Pursuing this Driver or Police Officer in Pursuit
50	50	50	50	Driving Wrong Way on One-Way Trafficway
51	51	51	51	Driving on Wrong Side of Two-Way Trafficway (Intentionally or Unintentionally)
54	54	54	54	Stopping in Roadway (Vehicle Not Abandoned)
--	55	55	55	Improper Management of Vehicle Controls
--	56	56	56	Object Interference with Vehicle Controls
--	57	57	57	Driving with Tire-Related Problems
58	58	58	58	Over Correcting
59	59	59	59	Getting off/out of a Vehicle
--	60	60	60	Alcohol and/or Drug Test Refused
91	91	91	91	Non-Traffic Violation Charged (Manslaughter, Homicide or Other Assault Offense Committed Without Malice)
--	--	--	94	Emergency Medical Service Personnel
--	--	--	95	Fire Personnel
--	--	--	96	Tow Operator

2016	2017	2018	2019	
--	--	--	97	Transportation (Maintenance Workers, Safety Service Patrol Operators, etc.)
99	99	--	--	Unknown
--	--	99	99	Reported as Unknown

Related Factors—Vehicle Level (discontinued)

Definition

This data element records factors related to this vehicle expressed in the case materials.

Additional Information

There are also crash level related factors in the Accident data file, CF1, CF2, and CF3; driver-related factors in the Vehicle data file, namely DR_SF1, DR_SF2, DR_SF3, and DR_SF4; and person-related factors P_SF1, P_SF2, and P_SF3 in the Person data file.

The CRSS coder may have used either of the two data elements to code a related factor. One must test both data elements to insure that the selected related factor is included.

These data elements also appear in the Parkwork data file as PVEH_SC1 and PVEH_SC2.

Prior to 2019 the Data Element ID was V34. Beginning in 2020 this data element was no longer collected at the Vehicle level. It is now collected in the Vehiclesf data file as VEHICLESF.

SAS Name

VEH_SC1, VEH_SC2

Attribute Codes

2016-2017	2018	2019	
0	0	0	None
--	--	29	Default Code Used for Vehicle Numbering
30	30	30	Multi-Wheeled Motorcycle Conversion
33	33	33	Vehicle Being Pushed by Non-Motorist
35	35	35	Reconstructed/Altered Vehicle
39	39	39	Highway Construction, Maintenance or Utility Vehicle, In-Transport (Inside or Outside Work Zone)
40	40	--	Highway Incident Response Vehicle
41	41	41	Police Fire or EMS Vehicle Working at the Scene of an Emergency or Performing Other Traffic Control Activities

2016-2017	2018	2019	
42	42	42	Other Working Vehicle (Not Construction, Maintenance, Utility, Police, Fire, or EMS Vehicle)
44	44	44	Adaptive Equipment
--	45	45	Slide-in Camper
99	--	--	Unknown
--	99	99	Reported as Unknown

The PERSON Data File

The Person data file includes motorist and non-motorist data. It contains the data elements CASENUM, PSU, PJ, STRATUM, PSUSTRAT, PSU_VAR, REGION, URBANICITY, WEIGHT, VEH_NO, and PER_NO, which are described in the Key Data Elements at the beginning of the Data Element Definitions and Codes section. The Person data file also contains the data elements on the following pages.

CASENUM, VEH_NO, and PER_NO are the unique identifiers for each record. CASENUM should be used to merge the Person data file with the Accident data file for a set of all motorists and non-motorists. CASENUM and VEH_NO should be used to merge the Person data file with the Vehicle and Parkwork data files for a set of all motor vehicle occupants. CASENUM and PER_NO should be used to merge the Person data file with non-motorist person level data files.

In the Person data file, motor vehicle occupants are PER_TYPE = 1, 2, 3, 9. Motor vehicle occupants have assigned vehicle numbers starting with 1. When PER_TYPE = 3, the occupied vehicle will be found in the PARKWORK data file. Non-motor vehicle occupants are PER_TYPE = 4, 5, 6, 7, 8, 10, 11, 12, 13, or 19. VEH_NO = 0 for non-motor vehicle occupants.

P5/NM5. Age**Definition**

This data element identifies the person's age in years on the date of the crash.

Additional Information**SAS Name****AGE****Attribute Codes**

2016-2017	2018-Later	
0	0	Less Than One Year
1-120	1-120	Age in Years
998	998	Not Reported
999	--	Unknown
--	999	Reported as Unknown

P5/NM5I. Imputed Age**Definition**

This imputed data element has the same definition and data element values as Age, excluding the value 999 for unknown age and value 998 for not reported age.

Additional Information

See the [CRSS Imputation](#) section of this manual.

SAS Name**AGE_IM**

P6/NM6. Sex**Definition**

This data element identifies the sex of this person involved in the crash.

Additional Information**SAS Name****SEX****Attribute Codes**

2016-2017	2018-Later	
1	1	Male
2	2	Female
8	8	Not Reported
9	--	Unknown
--	9	Reported as Unknown

P6/NM6I. Imputed Sex**Definition**

This imputed data element has the same definition and data element values as Sex, excluding value 9 for unknown sex and value 8 for not reported sex.

Additional Information

See the [CRSS Imputation](#) section of this manual.

SAS Name**SEX_IM**

P7/NM7. Person Type**Definition**

This data element describes the role of this person or nonmotorist at the time they became involved in the crash.

Additional Information

See [Appendix C: Analytical Classification of Select CRSS Data Elements](#) for the standard NCSA classifications for this data element.

SAS Name**PER_TYP****Attribute Codes***Motorists*

2016-2019	2020-2021	2022-Later	
1	1	1	Driver of a Motor Vehicle In-Transport
2	2	2	Passenger of a Motor Vehicle In-Transport
9	9	9	Unknown Occupant Type in a Motor Vehicle In-Transport

Non-Motorists-Occupant

2016-2019	2020-2021	2022-Later	
3	3	3	Occupant of a Motor Vehicle Not In-Transport
4	4	4	Occupant of a Non-Motor Vehicle Transport Device

Non-Motorists-Non-Occupant

2016-2019	2020-2021	2022-Later	
5	5	5	Pedestrian
6	6	6	Bicyclist
7	7	--	Other Cyclist
--	--	7	Other Pedalcyclist
8	--	--	Persons on Personal Conveyances
--	--	8	Person on a Personal Conveyance
10	10	--	Persons in/on Buildings
--	--	10	Person In/On a Building
--	11	--	Person on Motorized Personal Conveyance

2016-2019	2020-2021	2022-Later	
--	12	--	Person on Non-Motorized Personal Conveyance
--	13	--	Person on Personal Conveyance, Unknown if Motorized or Non-Motorized
19	19	19	Unknown Type of Non-Motorist

P8/NM10. Injury Severity**Definition**

This data element describes the severity of the injury to this person in the crash using the KABCO scale.

Additional Information

See the Accident data file for C90 Maximum Injury Severity in Crash and the Vehicle data file for V90 Maximum Injury Severity in Vehicle, both of which are derived from this data element.

See [Appendix C: Analytical Classification of Select CRSS Data Elements](#) for the standard NCSA classifications for this data element.

Prior to 2022 the Data Element ID was P8/NM8.

SAS Name

INJ_SEV

Attribute Codes

2016-Later	
0	No Apparent Injury (O)
1	Possible Injury (C)
2	Suspected Minor Injury (B)
3	Suspected Serious Injury (A)
4	Fatal Injury (K)
5	Injured, Severity Unknown (U)
6	Died Prior to Crash
9	Unknown/Not Reported

P8/NM10I. Imputed Injury Severity**Definition**

This imputed data element has the same definition and data element values as Injury Severity, excluding value 9 for unknown if injured or not reported if injured.

Additional Information

See the [CRSS Imputation](#) section of this manual.

SAS Name

INJSEV_IM

P9. Seating Position

Definition

This data element identifies the location of this person in or on the vehicle.

Additional Information

More than one person can be assigned the same seat position, however this is coded only when a person is sitting on someone's lap.

SAS Name

SEAT_POS

Attribute Codes

2016-2018	2019-Later	
0	0	Not a Motor Vehicle Occupant
11	11	Front Seat, Left Side (Driver's Side)
12	12	Front Seat, Middle
13	13	Front Seat, Right Side
18	18	Front Seat, Other
19	19	Front Seat, Unknown
21	21	Second Seat, Left Side
22	22	Second Seat, Middle
23	23	Second Seat, Right Side
28	28	Second Seat, Other
29	29	Second Seat, Unknown
31	31	Third Seat, Left Side
32	32	Third Seat, Middle
33	33	Third Seat, Right Side
38	38	Third Seat, Other
39	39	Third Seat, Unknown
41	41	Fourth Seat, Left Side
42	42	Fourth Seat, Middle
43	43	Fourth Seat, Right Side
48	48	Fourth Seat, Other
49	49	Fourth Seat, Unknown
50	50	Sleeper Section of Cab (Truck)
51	51	Other Passenger in Enclosed Passenger or Cargo Area
52	52	Other Passenger in Unenclosed Passenger or Cargo Area

2016-2018	2019-Later	
53	53	Other Passenger in Passenger or Cargo Area, Unknown Whether or Not Enclosed
54	54	Trailing Unit
55	55	Riding on Exterior of Vehicle
--	56	Appended to a Motor Vehicle for Motion
98	98	Not Reported
99	99	Unknown/Reported as Unknown (Since 2018)

P9I. Imputed Seating Position.

Definition

This imputed data element has the same definition and data element values as Seating Position, excluding values 19, 29, 39, 49, and 99 for unknown seating position and values 98 for not reported seating position.

Additional Information

See the [CRSS Imputation](#) section of this manual.

SAS Name

SEAT_IM

P10A. Restraint System Use

Definition

This data element records the restraint equipment in use by this occupant at the time of the crash.

Additional Information

Prior to 2019 this data element's name was "Restraint System/Helmet Use" that included helmet use, and the Data Element ID was P10. Starting in 2019 helmet use is captured as part of the data element "Helmet Use."

See [Appendix C: Analytical Classification of Select CRSS Data Elements](#) for the standard NCSA classifications for this data element.

SAS Name

REST_USE

Attribute Codes

2016	2017-2018	2019-Later	
0	--	--	Not Applicable
1	1	1	Shoulder Belt Only Used
2	2	2	Lap Belt Only Used
3	3	3	Shoulder and Lap Belt Used
4	4	4	Child Restraint – Type Unknown
5	5	--	DOT-Compliant Motorcycle Helmet
--	--	6	Racing-Style Harness Used
7	--	--	None Used
8	8	8	Restraint Used – Type Unknown
10	10	10	Child Restraint System – Forward Facing
11	11	11	Child Restraint System – Rear Facing
12	12	12	Booster Seat
16	16	--	Helmet, Other Than DOT-Compliant Motorcycle Helmet
17	17	--	No Helmet
19	19	--	Helmet, Unknown if DOT-Compliant
--	20	20	None Used/Not Applicable
29	29	--	Unknown if Helmet Worn
96	96	96	Not a Motor Vehicle Occupant
97	97	97	Other
98	98	98	Not Reported
99	99	99	Unknown/Reported as Unknown (Since 2018)

P10B. Indication of Restraint System Misuse**Definition**

This data element identifies any misuse of the available restraint system used by this person.

Additional Information

Prior to 2019 this data element's name was "Indication of Misuse of Restraint System/Helmet" that included helmet misuse, and the Data Element ID was P11. Starting in 2019 helmet misuse is captured as part of the data element "Indication of Helmet Misuse."

SAS Name

REST_MIS

Attribute Codes

2016-2018	2019-Later	
0	--	No
--	0	No Indication of Misuse
1	--	Yes
--	1	Yes, Indication of Misuse
--	7	None Used/Not Applicable
8	8	Not a Motor Vehicle Occupant

P11A. Helmet Use

Definition

This data element records the helmet use by this occupant at the time of the crash.

Additional Information

This data element is applicable to occupants of body types 80-91, 96, and 97. (See [NCSA Body Type](#))

See [Appendix C: Analytical Classification of Select CRSS Data Elements](#) for the standard NCSA classifications for this data element.

Prior to 2019 this data was collected as part of the data element “Restraint System/Helmet Use,” and the Data Element ID was P10.

SAS Name

HELM_USE

Attribute Codes

2019-Later	
5	DOT-Compliant Motorcycle Helmet
16	Helmet, Other Than DOT-Compliant Motorcycle Helmet
17	No Helmet
19	Helmet, Unknown if DOT-Compliant
20	Not Applicable
96	Not a Motor Vehicle Occupant
98	Not Reported
99	Unknown/Reported as Unknown

P11B. Indication of Helmet Misuse**Definition**

This data element identifies any misuse of the helmet used by this person.

Additional Information

This data element is applicable to occupants of body types 80-91, 96, and 97.

Prior to 2019 this data was collected as part of the data element “Indication of Misuse of Restraint System/Helmet,” and the Data Element ID was P11.

SAS Name**HELM_MIS****Attribute Codes**

2019-Later	
0	No Indication of Misuse
1	Yes, Indication of Misuse
7	None Used/Not Applicable
8	Not a Motor Vehicle Occupant

P12. Air Bag Deployed

Definition

This data element records air bag availability and deployment for this person as reported in the police crash report.

Additional Information

This data element is designed to collect both air bag availability and deployment for each occupied seat position. Variation in the presentation of the source data on the State crash report forms and the selections coded on the police report may produce unlikely combinations or missing data. For example:

- If the seat position does not have an air bag at the time of manufacture, but the information on the police report indicates an air bag was available or deployed, the information on the police report takes precedence.
- If the seat position has an air bag installed at the time of manufacture and the police report indicates there is no air bag available, then the police report information takes precedence.

SAS Name

AIR_BAG

Attribute Codes

2016	2017	2018-Later	
0	--	--	Not Applicable
1	1	1	Deployed – Front
2	2	2	Deployed – Side (Door, Seat Back)
3	3	3	Deployed – Curtain (Roof)
7	7	7	Deployed – Other (Knee, Air Belt, etc.)
8	8	8	Deployed – Combination
9	9	9	Deployment – Unknown Location
20	20	20	Not Deployed
28	--	--	Switched off
97	97	97	Not a Motor Vehicle Occupant
98	98	98	Not Reported
99	99	--	Deployment Unknown
--	--	99	Reported as Deployment Unknown

P13. Ejection**Definition**

This data element describes the ejection status and the degree of ejection for this person, excluding motorcycle occupants.

Additional Information**SAS Name****EJECTION****Attribute Codes**

2016-2017	2018-Later	
0	0	Not Ejected
1	1	Totally Ejected
2	2	Partially Ejected
3	3	Ejected – Unknown Degree
7	7	Not Reported
8	8	Not Applicable
9	--	Unknown
--	9	Reported as Unknown if Ejected

P13I. Imputed Ejection**Definition**

This imputed data element had the same definition and data element values as Ejection, excluding 9 (Unknown) and 7 (Not Reported).

Additional Information

See the [CRSS Imputation](#) section of this manual.

SAS Name**EJECT_IM**

P16/NM18. Police Reported Alcohol Involvement**Definition**

This data element records whether alcohol was involved for this person and reflects the judgment of law enforcement.

Additional Information

This data element does not indicate that alcohol was a cause of the crash. If a police crash report indicates that opened or unopened alcohol bottles were found in the vehicle, then this information does not by itself constitute involvement.

Prior to 2019 the Data Element ID was P16/NM15. From 2019-2021 the Data Element ID was P16/NM16.

SAS Name**DRINKING****Attribute Codes**

2016-2017	2018-Later	
0	0	No (Alcohol Not Involved)
1	1	Yes (Alcohol Involved)
8	8	Not Reported
9	--	Unknown (Police Reported)
--	9	Reported as Unknown

P16/NM18I. Imputed Police Reported Alcohol Involvement**Definition**

The definition and data element values are the same as Police Reported Alcohol Involvement, excluding 8 for not reported and 9 for unknown alcohol involvement.

Additional Information

See the [CRSS Imputation](#) section of this manual. From 2019-2021 the Data Element ID was P16/NM16.

SAS Name**PERALCH_IM**

P17/NM19. Alcohol Test**P17A/NM19A. Alcohol Test Status****Definition**

This data element identifies whether an alcohol test was given to this person.

Additional Information

Prior to 2019 the Data Element ID was P18A/NM17A. From 2019-2021 the Data Element ID was P18A/NM18A.

SAS Name

ALC_STATUS

Attribute Codes

2016	2017	2018-Later	
0	0	0	Test Not Given
1	--	--	Test Refused
2	2	2	Test Given
8	8	8	Not Reported
9	9	--	Unknown if Tested
--	--	9	Reported as Unknown if Tested

P17B/NM19B. Alcohol Test Type**Definition**

This data element identifies the type of alcohol test that was given to this person.

Additional Information

If a valid blood test is administered along with another type of test then blood test is coded.

Prior to 2019 the Data Element ID was P18B/NM17B. From 2019-2021 the Data Element ID was P18B/NM18B.

SAS Name

ATST_TYP

Attribute Codes

2016-2017	2018-Later	
0	0	Test Not Given
1	1	Blood
2	2	Breath Test (AC)
3	3	Urine
8	8	Other Test Type
10	10	Preliminary Breath Test (PBT)
--	11	Breath Test, Unknown Type
95	95	Not Reported
98	98	Unknown Test Type
99	--	Unknown if Tested
--	99	Reported as Unknown if Tested

P17C/NM19C. Alcohol Test Result**Definition**

This data element identifies the alcohol test result for this person.

Additional Information

A BAC of .10 is coded as 100. The decimal is implied. The BAC is expressed in grams per deciliter (g/dL) or a clinical evaluation of the same.

See [Appendix C: Analytical Classification of Select CRSS Data Elements](#) for the standard NCSA classifications for this data element.

Prior to 2019 the Data Element ID was P18C/NM18C. From 2019-2021 the Data Element ID was P18C/NM18C.

SAS Name

ALC_RES

Attribute Codes

2016-2017	2018-Later	
0-939	0-939	Actual Value
940	940	0.94 or Greater
995	995	Not Reported
996	996	Test Not Given
997	997	Test Performed, Results Unknown
998	998	Positive Reading With No Actual Value
999	--	Unknown if Tested
--	999	Reported as Unknown if Tested

P18/NM20. Police Reported Drug Involvement**Definition**

This data element records whether drugs were involved for this person and reflects the judgment of law enforcement.

Additional Information

Involvement is not an indication that drugs were in any way cause of the crash, even though it may have been. If the police crash report indicates that drugs were found in the vehicle, then this information does not by itself constitute involvement.

Prior to 2019 the Data Element ID was P19/NM18. From 2019-2021 the Data Element ID was P19/NM19.

SAS Name**DRUGS****Attribute Codes**

2016-2017	2018-Later	
0	0	No (Drugs Not Involved)
1	1	Yes (Drugs Involved)
8	8	Not Reported
9	--	Unknown (Police Reported)
--	9	Reported as Unknown

P20/NM22. Transported to First Medical Facility By**Definition**

This data element identifies the mode of transportation to a hospital or medical facility provided for this person.

Additional Information

Prior to 2019 the Data Element ID was P22/NM21. From 2019-2021 the Data Element ID was P22/NM22.

SAS Name

HOSPITAL

Attribute Codes

2016-2017	2018-2019	2020-2021	
0	0	--	Not Transported
--	--	0	Not Transported for Treatment
1	1	1	EMS Air
2	2	2	Law Enforcement
3	3	3	EMS Unknown Mode
4	4	4	Transported Unknown Source
5	5	5	EMS Ground
6	6	6	Other
8	8	8	Not Reported
9	--	--	Unknown
--	9	9	Reported as Unknown

NM4. Vehicle Number of Motor Vehicle Striking Non-Motorist**Definition**

This data element identifies the “Vehicle Number” (VEH_NO) of the motor vehicle in-transport that made contact with this non-motorist.

Additional Information

This data element applies only to non-motorists/non-occupants and reflects the vehicle that made contact with the non-motorist/non-occupant being coded.

The number must match the vehicle number of the striking vehicle. This number is similar to VEH_NO, except that the non-motorist/non-occupant was struck by the vehicle, rather than being within the vehicle.

SAS Name**STR_VEH****Attribute Codes**

2016-2017	2018-Later	
0	0	Occupant of Motor Vehicle
1-998	1-998	Vehicle Number of Striking Vehicle
999	--	Unknown

NM8. Non-Motorist Device Type**Definition**

This element describes the type of transport device operated by the non-motorist.

Additional Information:**SAS Name****DEVTYPE****Attribute Codes**

2022-Later	
0	Not Applicable
1	Ridden Animal, Animal Drawn Conveyance, or Trailer
2	Railway Vehicle or Road Vehicle on Rails
3	Bicycle
4	Other Pedalcycle
5	Mobility Aid Device
6	Skates
7	Non-Self-Balancing Board (Skateboard)
8	Self-Balancing Board
9	Standing or Seated Scooter
97	Personal Conveyance, Other
98	Personal Conveyance, Unknown Type
99	Unknown Type of Non-Motorist

NM9. Non-Motorist Device Motorization**Definition**

This element describes the motorization of the device operated by the non-motorist.

Additional Information**SAS Name****DEVMOTOR****Attribute Codes**

2022-Later	
0	Not Applicable
1	Not Motorized
2	Motorized
3	Unknown/Not Reported if Motorized
9	Unknown Type of Non-Motorist

NM12. Non-Motorist Location at Time of Crash

Definition

This data element identifies the attribute that best describes the location of this non-motorist with respect to the roadway at the time of the crash.

Additional Information

Non-motorists who are occupants of motor vehicles not in-transport are coded with respect to the location of the vehicle. Prior to 2022 the Data Element ID was NM10.

SAS Name

LOCATION

Attribute Codes

2016-2017	2018-Later	
0	0	Not Applicable-Motor Vehicle Occupant
1	1	At Intersection-In Marked Crosswalk
2	2	At Intersection-Unmarked/Unknown if Marked Crosswalk
3	3	At Intersection-Not in Crosswalk
9	9	At Intersection-Unknown Location
10	10	Not at Intersection-In Marked Crosswalk
11	11	Not at Intersection-On Roadway, Not in Marked Crosswalk Unknown
13	13	Not at Intersection-On Roadway, Crosswalk Availability Unknown
14	14	Parking Lane/Zone
16	16	Bicycle Lane
20	20	Shoulder/Roadside
21	21	Sidewalk
22	22	Median/Crossing Island
23	23	Driveway Access
24	24	Shared-Use Path
25	25	Non-Trafficway Area
28	28	Other
98	98	Not Reported
99	--	Unknown Location
--	99	Reported as Unknown Location

See [Analysis of Pedestrian and Bicycle Crashes Around Intersections](#) for guidance on analyzing Pedestrian/Bicyclist crash locations.

Discontinued PERSON Data Elements***Drug Test Result (discontinued)*****Definition**

This data element identifies the drug test result for this person.

SAS Name

DRUGRES1, DRUGRES2, DRUGRES3

Attribute Codes

2016-2017	
0	Test Not Given
1	Tested for Drugs, No Drugs Found/Negative
95	Not Reported
997	Tested for Drugs, Result Unknown
998	Tested for Drugs, Drugs Found, Type Unknown/Positive
999	Unknown if Tested

Drug Test Status (discontinued)**Definition**

This data element identifies whether a drug test was given to this person.

Additional Information**SAS Name**

DSTATUS

Attribute Codes

2016	2017	
0	0	Test Not Given
1	--	Test Refused
2	2	Test Given
8	8	Not Reported
9	9	Unknown if Tested

Drug Test Type (discontinued)**Definition**

This data element identifies the type of drug test that was given to this person.

Additional Information**SAS Name**

DRUGTST1, DRUGTST2, DRUGTST3

Attribute Codes

2016-2017	
0	Test Not Given
1	Blood
2	Urine
3	Both Blood and Urine Tests
6	Not Reported
7	Unknown Test Type
8	Other Test Type
9	Unknown if Tested

Related Factors—Person Level (discontinued)**Definition**

This data element records factors related to motor vehicle occupants other than drivers and people not in motor vehicles as expressed in the case materials.

Additional Information

There are also vehicle level related factors in the Vehicle data file, VEH_SC1 and VEH_SC2 and driver-related factors, also in the Vehicle data file, namely DR_SF1, DR_SF2, DR_SF3, and DR_SF4. There are also crash-related factors CF1, CF2, and CF3 in the Accident data file.

Any of the three data elements may have been used to code a related factor. One must test all three data elements to insure that the selected related factor is included.

Person-related factors for all drivers are coded 0. Person-related factors for non-drivers can have non-zero values as listed below.

Prior to 2019 the Data Element ID was P26/NM25. Beginning in 2020 this data element was no longer collected at the Person level. It is now collected in the Personrf data file as PERSONRF.

SAS Name**P_SF1, P_SF2, P_SF3****Attribute Codes**

2016	2017	2018	2019	
0	0	0	0	None/Not Applicable-Driver
5	5	5	5	Interfering With Driver*
9	9	9	9	Construction/Maintenance/Utility Worker/Highway Department, Contractor, Utility Company Personnel, etc.
--	10	10	10	Alcohol and/or Drug Test Refused
13	13	13	13	Motorized Wheelchair Rider**
21	21	21	21	Overloading or Improper Loading of Vehicle with Passengers or Cargo
--	--	31	31	Default Code Used for Vehicle Numbering**
32	32	32	32	Opening Vehicle Closure into Moving Traffic or While Vehicle Is in Motion*
--	--	--	53	Non-Motorist Previously Used a Motor Vehicle for Motion**
--	--	--	54	Non-Motorist Attempting to Use a Motor Vehicle for Motion**
--	--	--	55	Non-Motorist Attempting to Use or Previously Used a Motor Vehicle for Motion, Details Not Reported**
56	56	56	56	Non-Driver Flees Scene
86	86	86	--	Emergency Services Personnel
87	87	87	87	Police or Law Enforcement Officer
89	89	89	89	Parked Motor Vehicle With Equipment Extending into the Travel Lane*
90	90	90	90	Non-Motorist Pushing a Vehicle**
91	91	91	91	Portable Electronic Devices
92	92	92	92	Person in Ambulance Treatment Compartment*
93	93	93	93	Non-Motorist Wearing Motorcycle Helmet**
--	--	--	94	Emergency Medical Services Personnel
--	--	--	95	Fire Personnel
--	--	--	96	Tow Operator
--	--	--	97	Transportation (Maintenance Workers, Safety Service Patrol Operators, etc.)
99	99	--	--	Unknown
--	--	99	99	Reported as Unknown

* Attribute is only applicable to occupants (other than drivers) of motor vehicles.

** Attribute is only applicable to people not in motor vehicles.

The PARKWORK Data File

The Parkwork data file includes Vehicle data elements applicable to Parked and Working Vehicles. It contains the data elements CASENUM, PSU, PJ, STRATUM, PSUSTRAT, PSU_VAR, REGION, URBANICITY, WEIGHT, and VEH_NO, which are described in the Key Data Elements at the beginning of the Data Element Definitions and Codes section. The Parkwork data file also contains the data elements on the following pages.

CASENUM and VEH_NO are the unique identifiers for each record. CASENUM should be used to merge the Parkwork data file with the Accident data file. CASENUM and VEH_NO should be used to merge the Parkwork data file with the Person data file.

C4A. Number of Motor Vehicles In-Transport (MVIT)**Definition**

This data element is a count of the number of motor vehicles in-transport involved in the crash. Legally parked vehicles are not included.

Additional Information

See this data element in the Accident data file section for more information.

SAS Name**PVE_FORMS****Attribute Codes**

2016-Later	
1-100	Number of Vehicle Forms

C8. Crash Date**C8A. Month of Crash****Definition**

This data element records the month in which the crash occurred.

Additional Information

See this data element in the Accident data file section for more information.

SAS Name**PMONTH****Attribute Codes**

2016-Later	
1	January
2	February
3	March
4	April
5	May
6	June
7	July
8	August
9	September
10	October
11	November
12	December

C9. Crash Time**C9A. Hour of Crash****Definition**

This data element records the hour at which the crash occurred.

Additional Information

See this data element in the Accident data file section for more information.

SAS Name**PHOUR****Attribute Codes**

2016-Later	
0-23	Hour
99	Unknown

C9B. Minute of Crash**Definition**

This data element records the minutes after the hour at which the crash occurred.

Additional Information

See this data element in the Accident data file section for more information.

SAS Name**PMINUTE****Attribute Codes**

2016-Later	
0-59	Minute
99	Unknown

C19. First Harmful Event**Definition**

This data element describes the first injury or damage producing event of the crash.

Additional Information

See this data element in the Accident data file section for more information.

SAS Name

PHARM_EV

Attribute Codes*Non-collision*

2016	2017	2018-Later	
1	1	1	Rollover/Overturn
2	2	2	Fire/Explosion
3	3	3	Immersion or Partial Immersion
4	4	4	Gas Inhalation
5	5	5	Fell/Jumped From Vehicle
6	6	6	Injured in Vehicle (Non-Collision)
7	7	7	Other Non-Collision
16	16	16	Thrown or Falling Object
44	44	44	Pavement Surface Irregularity (Ruts, Potholes, Grates, etc.)
51	51	51	Jackknife (Harmful to This Vehicle)
72	72	--	Cargo/Equipment Loss or Shift (Harmful to This Vehicle)
--	--	72	Cargo/Equipment Loss, Shift, or Damage (Harmful)

Collision with Motor Vehicle In-Transport

2016	2017	2018-Later	
12	12	12	Motor Vehicle In-Transport
54	54	54	Motor Vehicle In-Transport Strikes or Is Struck by Cargo, Persons or Objects Set-in-Motion From/By Another Motor Vehicle In-Transport
55	55	88	Motor Vehicle in Motion Outside the Trafficway

Collision with Object Not Fixed

2016	2017	2018-Later	
8	8	8	Pedestrian
9	9	9	Pedalcyclist
10	10	10	Railway Vehicle
11	11	11	Live Animal
14	14	14	Parked Motor Vehicle
15	15	15	Non-Motorist on Personal Conveyance
18	18	18	Other Object (Not Fixed)
45	45	45	Working Motor Vehicle
49	49	49	Ridden Animal or Animal Drawn Conveyance
73	73	73	Object That Had Fallen From Motor Vehicle In-Transport
74	74	74	Road Vehicle on Rails
--	91	91	Unknown Object Not Fixed

Collision with Fixed Object

2016	2017	2018-Later	
17	17	17	Boulder
19	19	19	Building
20	20	20	Impact Attenuator/Crash Cushion
21	21	21	Bridge Pier or Support
23	23	23	Bridge Rail (Includes Parapet)
24	24	24	Guardrail Face
25	25	25	Concrete Traffic Barrier
26	26	26	Other Traffic Barrier
30	30	30	Utility Pole/Light Support
31	31	31	Post, Pole or Other Support
32	32	32	Culvert
33	33	33	Curb
34	34	34	Ditch
35	35	35	Embankment
38	38	38	Fence
39	39	39	Wall
40	40	40	Fire Hydrant
41	41	41	Shrubbery
42	42	42	Tree (Standing Only)
43	43	43	Other Fixed Object

2016	2017	2018-Later	
46	46	46	Traffic Signal Support
48	48	48	Snow Bank
50	50	50	Bridge Overhead Structure
52	52	52	Guardrail End
53	53	53	Mail Box
57	57	57	Cable Barrier
58	58	58	Ground
59	59	59	Traffic Sign Support
--	93	93	Unknown Fixed Object
--	--	98	Harmful Event, Details Not Reported (Since 2019)
99	99	--	Unknown
--	--	99	Reported as Unknown

C20. Manner of Collision of the First Harmful Event**Definition**

This data element describes the orientation of two motor vehicles in-transport when they are involved in the “First Harmful Event” of a collision crash. If the “First Harmful Event” is not a collision between two motor vehicles in-transport, it is classified as such.

Additional Information

See this data element in the Accident data file section for more information.

SAS Name

PMAN_COLL

Attribute Codes

2016-2017	2018	2019-Later	
0	0	--	Not Collision With Motor Vehicle In-Transport
--	--	0	First Harmful Event Was Not a Collision With Motor Vehicle In-Transport
1	1	1	Front-to-Rear
2	2	2	Front-to-Front
6	6	6	Angle
7	7	7	Sideswipe – Same Direction
8	8	8	Sideswipe – Opposite Direction
9	9	9	Rear-to-Side
10	10	10	Rear-to-Rear
11	11	11	Other
98	98	98	Not Reported
99	--	--	Unknown
--	99	99	Reported as Unknown

V4. Number of Occupants**Definition**

This data element is a count of the number of occupants in this vehicle.

Additional Information

See this data element in the Vehicle data file section for more information.

SAS Name**PNUMOCCS****Attribute Codes**

2016-Later	
0	None
1-98	Number of Occupants
99	Unknown

V5. Unit Type

Definition

This data element identifies the type of unit that applies to this motor vehicle at the time it became an involved vehicle in the crash and was reported as a unit on the police crash report.

Additional Information

This data element also appears in the Vehicle data file as UNITTYPE. The only valid attribute for UNITTYPE is 1 (Motor Vehicle In-Transport [Inside or Outside the Trafficway]).

SAS Name

PTYPE

Attribute Codes

2016-Later	
2	Motor Vehicle Not In-Transport Within the Trafficway
3	Motor Vehicle Not In-Transport Outside the Trafficway
4	Working Motor Vehicle (Highway Construction, Maintenance, Utility Only)

V6. Hit-and-Run**Definition**

This data element identifies whether this vehicle was a contact vehicle in the crash that did not stop to render aid (this can include drivers who flee the scene on foot). Hit-and-run is coded when a motor vehicle in-transport, or its driver, departs from the scene; motor vehicles not in-transport are excluded. It does not matter whether the hit-and-run vehicle was striking or struck.

Additional Information

See this data element in the Vehicle data file section for more information.

SAS Name**PHIT_RUN****Attribute Codes**

2016-2017	2018-2019	2020-Later	
0	0	0	No
1	1	1	Yes
9	--	--	Unknown
--	9	--	Reported as Unknown

V9. Vehicle Identification Number (VIN)

Definition

This data element records the vehicle identification number (VIN) of this vehicle assigned by the vehicle manufacturer. The VIN contains information on the vehicle such as: manufacturer, model year, model, body type, restraint type, etc.

Additional Information

See this data element in the Vehicle data file section for more information.

SAS Name

PVIN

Attribute Codes

2016-2017	2018-2020	2021-Later	
000000000000	000000000000	--	No VIN Required
--	--	000000000000	No VIN Required, Not a Vehicle for Road Use
xxxxxxxxxxxx	xxxxxxxxxxxx	xxxxxxxxxxxx	First 12 Characters of the VIN
888888888888	888888888888	888888888888	Not Reported
999999999999	--	--	Unknown
--	999999999999	999999999999	Reported as Unknown
--	*	*	VIN Character Missing or Not Decipherable

V10. Vehicle Model Year**Definition**

This data element identifies the manufacturer's model year of this vehicle.

Additional Information

See this data element in the Vehicle data file section for more information.

SAS Name**PMODYEAR****Attribute Codes**

2016-Later	
xxxx	Actual Model Year
9998	Not Reported
9999	Unknown

V11. vPIC Make**Definition**

This element identifies the make (manufacturer brand name) of this vehicle as per NHTSA vPIC submissions.

Additional Information

Refer to [Addition of VIN-Decoded Data](#) for more details. See this data element in the Vehicle data file section for more information.

SAS Name**PVPICMAKE****Attribute Codes**

2020-Later	
xxxxx	Actual Make (up to five digits)
99997	Other
99998	Not Reported
99999	Unknown

V12. vPIC Model**Definition**

This element identifies the model of this vehicle using NHTSA's VIN decoder application, vPIC.

Additional Information

Refer to [Addition of VIN-Decoded Data](#) for more details. See this data element in the Vehicle data file section for more information.

SAS Name**PVPICMODEL****Attribute Codes**

2020-Later	
xxxxx	Actual Model (up to five digits)
99997	Other
99998	Not Reported
99999	Unknown

V13. vPIC Body Class

Definition

This element identifies a classification of this vehicle based on its general body configuration, size, shape, doors, etc., as defined by the manufacturer.

Additional Information

Refer to [Addition of VIN-Decoded Data](#) for more details. See this data element in the Vehicle data file section for more information.

Attributes with an asterisk (*) must have a finished body class for an incomplete vehicle captured under Final Stage Body Class. Other attributes may have a Final Stage Body Class if VIN decoding indicates that the vehicle is manufactured as an incomplete vehicle.

SAS Name

PVPICBODYCLASS

Attribute Codes

2020	2021-Later	
1	1	Convertible/Cabriolet
2	2	Minivan
3	3	Coupe
4	4	Low Speed Vehicle (LSV)/Neighborhood Electric Vehicle (NEV)
5	5	Hatchback/Liftback/Notchback
6	6	Motorcycle -Standard
7	7	Sport Utility Vehicle (SUV)/Multi-Purpose Vehicle (MPV)
8	8	Crossover Utility Vehicle (CUV)
9	9	Van
10	10	Roadster
11	11	Truck
12	12	Motorcycle -Scooter
13	13	Sedan/Saloon
15	15	Wagon
16	16	Bus
60	60	Pickup
62	62	Incomplete -Cutaway*
63	63	Incomplete -Chassis Cab (Single Cab)*
64	64	Incomplete -Glider*
65	65	Incomplete*
66	66	Truck-Tractor

2020	2021-Later	
67	67	Incomplete -Stripped Chassis*
68	68	Streetcar/Trolley
69	69	Off-Road Vehicle -All Terrain Vehicle (ATV) (Motorcycle-Style)
70	70	Incomplete -Chassis Cab (Double Cab)*
71	71	Incomplete -School Bus Chassis*
72	72	Incomplete -Commercial Bus Chassis*
73	73	Bus -School Bus
74	74	Incomplete -Chassis Cab (Number of Cab Unknown)*
75	75	Incomplete -Transit Bus Chassis*
76	76	Incomplete -Motor Coach Chassis*
77	77	Incomplete -Shuttle Bus Chassis*
78	78	Incomplete -Motor Home Chassis*
80	80	Motorcycle -Sport
81	81	Motorcycle -Touring/Sport Touring
82	82	Motorcycle -Cruiser
83	83	Motorcycle -Trike
84	84	Off-Road Vehicle -Dirt Bike/Off-Road
85	85	Motorcycle -Dual Sport/Adventure/Supermoto/On/Off-Road
86	86	Off-Road Vehicle -Enduro (off-road long-distance racing)
87	87	Motorcycle -Small/Minibike
88	88	Off-Road Vehicle -Go Kart
90	90	Motorcycle -Side Car
94	94	Motorcycle -Custom
95	95	Cargo Van
97	97	Off-Road Vehicle -Snowmobile
98	98	Motorcycle -Street
100	100	Motorcycle -Enclosed Three Wheeled/Enclosed Autocycle
103	103	Motorcycle -Unenclosed Three Wheeled/Open Autocycle
104	104	Motorcycle -Moped
105	105	Off-Road Vehicle -Recreational Off-Road Vehicle (ROV)
107	107	Incomplete -Bus Chassis*
108	108	Motorhome
109	109	Motorcycle -Cross Country
110	110	Motorcycle -Underbone
111	111	Step Van/Walk-in Van
112	112	Incomplete -Commercial Chassis*
113	113	Off-Road Vehicle -Motocross (Off-Road Short-Distance, Closed-Track Racing)

2020	2021-Later	
114	114	Motorcycle -Competition
117	117	Limousine
119	119	Sport Utility Truck (SUT)
124	124	Off-Road Vehicle -Golf Cart
125	125	Motorcycle -Unknown Body Type
126	126	Off-Road Vehicle -Farm Equipment
127	127	Off-Road Vehicle -Construction Equipment
--	128	Ambulance
--	129	Street Sweeper
--	130	Fire Apparatus
996	996	Motorized Bicycle (discontinued in 2022)
997	997	Other
998	998	Not Reported
999	999	Unknown

V14. NCSA Make**Definition**

This data element identifies the make (manufacturer) of this vehicle by NCSA historically.

Additional Information

See this data element in the Vehicle data file section for more information.

SAS Name

PMAKE

Attribute Codes

2016-Later	
1	American Motors
2	Jeep/Kaiser-Jeep/Willys-Jeep
3	AM General
6	Chrysler
7	Dodge
8	Imperial
9	Plymouth
10	Eagle
12	Ford
13	Lincoln
14	Mercury
18	Buick/Opel
19	Cadillac
20	Chevrolet
21	Oldsmobile
22	Pontiac
23	GMC
24	Saturn
25	Grumman
26	Coda
29	Other Domestic Manufacturers
	Avanti
	Checker
	DeSoto
	Excalibur
	Hudson

2016-Later	
	Packard
	Panoz
	Saleen
	Studebaker
	Stutz
	Tesla
30	Volkswagen
31	Alfa Romeo
32	Audi
33	Austin/Austin-Healey
34	BMW
35	Datsun/Nissan
36	Fiat
37	Honda
38	Isuzu
39	Jaguar
40	Lancia
41	Mazda
42	Mercedes-Benz
43	MG
44	Peugeot
45	Porsche
46	Renault
47	Saab
48	Subaru
49	Toyota
50	Triumph
51	Volvo
52	Mitsubishi
53	Suzuki
54	Acura
55	Hyundai
56	Merkur
57	Yugo
58	Infiniti
59	Lexus
60	Daihatsu
61	Sterling

2016-Later		
62	Land Rover	
63	Kia	
64	Daewoo	
65	Smart	
67	Scion	
69	Other Import	
		Aston Martin
		Bentley
		Bertone
		Bricklin
		Bugatti
		Caterham
		Citroen
		DeLorean
		Desta
		Ferrari
		Fisker
		Gazelle
		Hillman
		Jensen
		Koenigsegg
		Lada
		Lamborghini
		Lotus
		Mahindra
		Maserati
		Maybach
		McLaren
		Mini Cooper
		Morgan
		Morris
		Reliant (British)
		Rolls-Royce
		Simca
		Singer
		Spyker
		Sunbeam
		TVR

2016-Later	
70	BSA
71	Ducati
72	Harley-Davidson
73	Kawasaki
74	Moto-Guzzi
75	Norton
76	Yamaha
78	Other Make Moped
79	Other Make Motored Cycle
80	Brockway
81	Diamond Reo/Reo
82	Freightliner/White
83	FWD
84	International Harvester/Navistar
85	Kenworth
86	Mack
87	Peterbilt
88	Iveco/Magirus
89	White/Autocar, White/GMC
90	Bluebird
91	Eagle Coach
92	Gillig
93	MCI
94	Thomas Built
97	Not Reported
	Auto-Union-DKW
	Carpenter
	Collins Bus
	DINA
	Divco
	Hino
	Meyers Motors
	Mid Bus
	Neoplan
	Orion
	Oshkosh
	Scania
	Sterling

2016-Later		
		Think
		UD
		Van Hool
		Western Star

V15. NCSA Model**Definition**

This data element identifies the NCSA model of this vehicle within a given NCSA make.

Additional Information

See this data element in the Vehicle data file section for more information.

SAS Name**PMODEL****Attribute Codes**

2016-Later	
	See the current FARS/CRSS Coding and Validation Manual for vehicle model codes.

V16. NCSA Body Type

Definition

This data element identifies a classification of this vehicle based on its general body configuration, size, shape, doors, etc., as defined by NCSA.

Additional Information

See this data element in the Vehicle data file section for more information.

SAS Name

PBODYTYP

Attribute Codes

Automobiles

2016	2017-2019	2020-Later	
1	1	1	Convertible (Excludes Sun-Roof, T-Bar)
2	2	2	2-Door Sedan, Hardtop, Coupe
3	3	3	3-Door/2-Door Hatchback
4	4	4	4-Door Sedan, Hardtop
5	5	5	5-Door/4-Door Hatchback
6	6	6	Station Wagon (Excluding Van-and Truck-Based)
7	7	7	Hatchback, Number of Doors Unknown
8	8	8	Sedan/Hardtop, Number of Doors Unknown
9	9	9	Other or Unknown Automobile Type
17	17	17	3-Door Coupe

Automobile Derivatives

2016	2017-2019	2020-Later	
10	10	10	Auto Based Pickup (Includes El Camino, Caballero, Ranchero, SSR, G8-ST, Baha, Brat, and Rabbit Pickup)
11	11	11	Auto Based Panel (Cargo Station Wagon, Auto-Based Ambulance/Hearse)
12	12	12	Large Limousine (More Than 4 Side Doors or Stretched Chassis)
13	13	13	Three Wheel Automobile or Automobile Derivative

Utility Vehicles

2016	2017-2019	2020-Later	
14	14	14	Compact Utility (ANSI D.16 Utility Vehicle Categories “Small” and “Midsize”)
15	15	15	Large Utility (ANSI D.16 Utility Vehicle Categories “Full Size” and “Large”)
16	16	16	Utility Station Wagon
19	19	17	Utility Vehicle, Unknown Body Type

Van-Based Light Trucks (GVWR ≤ 10,000 lbs)

2016	2017-2019	2020-Later	
20	20	20	Minivan
21	21	21	Large Van – Includes Van-Based Buses
22	22	22	Step Van or Walk-in Van (GVWR ≤ 10,000 lbs)
28	28	28	Other Van Type
29	29	29	Unknown Van Type

Light Conventional Trucks (Pickup style cab, GVWR ≤10,000 lbs)

2016	2017-2019	2020-Later	
30	--	--	Compact Pickup (S-10, LUV, Ram 50, Rampage, Courier, Ranger, S-5, Pup, Mazda Pickup, Mitsubishi Truck, Datsun/Nissan Pickup, Arrow Pickup, Scamp, Toyota Pickup, VW Pickup, D50, Colt P/U, T-10, S-15, T-15, Ram 100, Dakota, Sonoma)
31	--	--	Standard Pickup (C10-C35, Jeep P/U, Comanche, Ram P/U, K10-K35, D100-D350, W100-350, F100-F350, R100-500, R10-R35, V10-35, Silverado, Sierra, T100)
32	32	32	Pickup With Slide-in Camper (2016-2017 Only)
33	33	33	Convertible Pickup
--	34	34	Light Pickup
39	39	39	Unknown (Pickup Style) Light Conventional Truck

Other Light Trucks (GVWR ≤10,000 lbs)

2016	2017-2019	2020-Later	
40	40	40	Cab Chassis Based (Included Rescue Vehicle, Light Stake, Dump, and Tow Truck)

2016	2017-2019	2020-Later	
41	41	41	Truck-Based Panel
45	45	45	Other Light Conventional Truck Type
48	48	48	Unknown Light Truck Type
49	49	49	Unknown Light Vehicle Type (Automobile, Utility, Van, or Light Truck)

Buses (excludes van based buses with a GVWR $\leq$ 10,000 lbs)

2016	2017-2019	2020-Later	
50	50	50	School Bus (Designed to Carry Students, Not Cross Country or Transit)
51	51	51	Cross Country/Intercity Bus (i.e., Greyhound)
52	52	52	Transit Bus (City Bus)
55	55	55	Van-Based Bus (GVWR > 10,000 lbs)
58	58	58	Other Bus Type
59	59	59	Unknown Bus Type

Medium/Heavy Trucks (GVWR > 10,000 lbs)

2016	2017-2019	2020-Later	
60	60	60	Step Van (GVWR > 10,000 lbs)
61	61	61	Single-Unit Straight Truck or Cab-Chassis (GVWR Range 10,001 to 19,500 lbs)
62	62	62	Single-Unit Straight Truck or Cab-Chassis (GVWR Range 19,501 to 26,000 lbs)
63	63	63	Single-Unit Straight Truck or Cab-Chassis (GVWR > 26,000 lbs)
64	64	64	Single Unit Straight Truck or Cab-Chassis (GVWR Unknown)
66	66	66	Truck-Tractor (Cab Only, or With Any Number of Trailing Units; Any Weight)
67	67	67	Medium/Heavy Pickup (GVWR > 10,000 lbs)
71	71	71	Unknown if Single-Unit or Combination-Unit Medium Truck (GVWR Range 10,001 to 26,000 lbs)
72	72	72	Unknown if Single-Unit or Combination-Unit Heavy Truck (GVWR > 26,000 lbs)
78	78	78	Unknown Medium/Heavy Truck Type
79	79	79	Unknown Truck Type (Light/Medium/Heavy)

Motor Homes

2016	2017-2019	2020-Later	
42	42	--	Light Truck-Based Motor Home (Chassis Mounted)
--	--	42	Light Vehicle-Based Motor Home (Chassis Mounted)
65	65	--	Medium/Heavy Truck-Based Motor Home
--	--	65	Medium/Heavy Vehicle-Based Motor Home
73	73	--	Camper or Motor Home, Unknown Truck Type
--	--	73	Camper or Motor Home, Unknown GVWR

Motored Cycles, Mopeds, All-Terrain Vehicles, All-Terrain cycles

2016	2017-2019	2020-Later	
80	--	--	Motorcycle
--	80	80	Two-Wheel Motorcycle (Excluding Motor Scooters)
81	--	--	Moped (Motorized Bicycle)
--	81	81	Moped (Since 2022)
82	--	--	Three-Wheeled Motorcycle or Moped
--	82	82	Three-Wheel Motorcycle (2 Rear Wheels)
83	--	--	Off-Road Motorcycle (2 Wheels)
--	83	83	Off-Road Motorcycle
--	84	84	Motor Scooter
--	85	85	Unenclosed Three-Wheel Motorcycle/Unenclosed Autocycle (1 Rear Wheel)
--	86	86	Enclosed Three-Wheel Motorcycle/Enclosed Autocycle (1 Rear Wheel)
--	87	87	Unknown Three-Wheel Motorcycle Type
88	--	--	Other Motored Cycle Type (Minibike, Motor Scooter, Pocket Motorcycles, Pocket Bikes)
--	88	88	Other Motored Cycle Type (Mini-bikes, Pocket Motorcycles, "Pocket Bikes")
89	89	89	Unknown Motored Cycle Type
90	90	90	ATV (All-Terrain Vehicle)

Other Vehicles

2016	2017-2019	2020-Later	
91	91	91	Snowmobile
92	92	92	Farm Equipment Other Than Trucks

2016	2017-2019	2020-Later	
93	93	93	Construction Equipment Other Than Trucks (Includes Graders)
94	94	94	Low-Speed Vehicle (LSV)/Neighborhood Electric Vehicle (NEV)
95	95	95	Golf Cart
--	96	96	Recreational Off-Highway Vehicle (ROV)
97	97	97	Other Vehicle Type (Includes Go-Cart, Fork-Lift, City Street Sweeper)
98	98	98	Not Reported
99	99	99	Unknown Body Type

V17. Final Stage Body Class

Definition

This element captures the completed/finished body class for an Incomplete Vehicle. An incomplete vehicle is completed by a final stage manufacturer. The intent of this data element is to capture the body class for incomplete vehicles when they are finished for road-use.

Additional Information

This data element is only applicable to incomplete vehicles, and the attributes are a subset of the vPIC Body Class attributes. Information captured in this data element is based on the police crash report. See this data element in the Vehicle data file section for more information.

SAS Name

PICFINALBODY

Attribute Codes

2020	2021-Later	
0	0	Not Applicable
2	2	Minivan
4	4	Low-Speed Vehicle (LSV)
7	7	Sport Utility Vehicle (SUV)/Multi-Purpose Vehicle (MPV)
8	8	Crossover Utility Vehicle (CUV)
9	9	Van
11	11	Truck
15	15	Wagon
16	16	Bus
60	60	Pickup
66	66	Truck-Tractor
68	68	Streetcar/Trolley
73	73	Bus-School Bus
95	95	Cargo Van
108	108	Motorhome
111	111	Step Van/Walk-in Van
117	117	Limousine
119	119	Sport Utility Truck
--	128	Ambulance
--	129	Street Sweeper
--	130	Fire Apparatus
997	997	Other

2020	2021-Later	
998	998	Not Reported
999	999	Unknown

V18. Power Unit Gross Vehicle Weight Rating (GVWR)

Definition

This element identifies the range of gross vehicle weight rating of the power unit as identified by the manufacturer through the vehicle's VIN submission. GVWR_FROM defines the lowest value and GVWR_TO defines the highest value for the range of the GVWR specified by the manufacturer as the recommended loaded weight for a vehicle.

Additional Information

Refer to [Addition of VIN-Decoded Data](#) for more details. See this data element in the Vehicle data file section for more information.

SAS Name

PGVWR_FROM, PGVWR_TO

Attribute Codes

2020-Later	
11	Class 1: 6,000 lbs or less (2,722 kg or less)
12	Class 2: 6,001 - 10,000 lbs (2,722 - 4,536 kg)
13	Class 3: 10,001 - 14,000 lbs (4,536 - 6,350 kg)
14	Class 4: 14,001 - 16,000 lbs (6,350 - 7,258 kg)
15	Class 5: 16,001 - 19,500 lbs (7,258 - 8,845 kg)
16	Class 6: 19,501 - 26,000 lbs (8,845 - 11,794 kg)
17	Class 7: 26,001 - 33,000 lbs (11,794 - 14,969 kg)
18	Class 8: 33,001 lbs and above (14,969 kg and above)
98	Not Reported
99	Reported as Unknown

V19. Vehicle Trailing

Definition

This data element identifies whether this vehicle had any attached trailing units or was towing another motor vehicle.

Additional Information

Trailing unit applies to any device connected to a motor vehicle by a hitch, including tractor-trailer combinations, a single-unit truck pulling a trailer (truck trailer), a boat trailer hitched onto a motor vehicle, etc.

See this data element in the Vehicle data file section for more information.

SAS Name

PTRAILER

Attribute Codes

2016-2021	2022-Later	
0	--	No Trailing Units
--	0	No Trailers
1	--	Yes, One Trailing Unit
--	1	One Trailer
2	--	Yes, Two Trailing Units
--	2	Two Trailers
3	--	Yes, Three or More Trailing Units
--	3	Three or More Trailers
4	--	Yes, Number of Trailing Units Unknown
--	4	Yes, Number of Trailers Unknown
5	5	Vehicle Towing Another Motor Vehicle – Fixed Linkage
6	6	Vehicle Towing Another Motor Vehicle – Non-Fixed Linkage
--	7	Trailing Unit Other than a Trailer or Another Motor Vehicle
9	9	Unknown

V20. Trailer Vehicle Identification Number

Definition

This data element records the vehicle identification number (VIN) of any trailing units of a combination vehicle.

Additional Information

See this data element in the Vehicle data file section for more information.

SAS Name

PTRLR1VIN, PTRLR2VIN, PTRLR3VIN

Attribute Codes

2016-2017	2018-2020	2021-Later	
000000000000	000000000000	--	No VIN Required
--	--	000000000000	No VIN Required, Not a Vehicle for Road Use
xxxxxxxxxxxx	xxxxxxxxxxxx	xxxxxxxxxxxx	First 12 Characters of the VIN
777777777777	777777777777	777777777777	No Trailing Units
888888888888	888888888888	888888888888	Not Reported
999999999999	--	--	Unknown
--	999999999999	999999999999	Reported as Unknown
--	*	*	VIN Character Missing or Not Decipherable

V21. Trailer Gross Vehicle Weight Rating (GVWR)

Definition

This element identifies the Gross Vehicle Weight Rating of any trailing units as identified by the manufacturer in the vehicle's VIN.

Additional Information

Refer to [Addition of VIN-Decoded Data](#) for more details. See this data element in the Vehicle data file section for more information.

SAS Name

PTRLR1GVWR, PTRLR2GVWR, PTRLR3GVWR

Attribute Codes

2020-Later	
0	No Trailer GVWR Required
11	Class 1: 6,000 lbs or less (2,722 kg or less)
12	Class 2: 6,001 - 10,000 lbs (2,722 - 4,536 kg)
13	Class 3: 10,001 - 14,000 lbs (4,536 - 6,350 kg)
14	Class 4: 14,001 - 16,000 lbs (6,350 - 7,258 kg)
15	Class 5: 16,001 - 19,500 lbs (7,258 - 8,845 kg)
16	Class 6: 19,501 - 26,000 lbs (8,845 - 11,794 kg)
17	Class 7: 26,001 - 33,000 lbs (11,794 - 14,969 kg)
18	Class 8: 33,001 lbs and above (14,969 kg and above)
77	No Trailing Units
98	Not Reported
99	Reported as Unknown

V23. Motor Carrier Identification Number (MCID)**Definition**

This data element records the issuing authority and motor carrier identification number if applicable to this vehicle.

Additional Information

This 11-character data element is the combination of two data elements, the two-digit Motor Carrier Issuing Authority code (MCARR_I1) followed by the nine-character Identification Number (MCARR_I2).

Additional Information

See this data element in the Vehicle data file section for more information.

SAS Name

PMCARR_ID

Attribute Codes

2016-Later	
00000000000	Not Applicable
xxxxxxxxxxx	11-Character (Combination of MCARR_I1 followed by MCARR_I2)
77777777777	Not Reported
88888888888	None
99999999999	Unknown (Reported as Unknown, 2018-2019)

V23A. MCID Issuing Authority**Definition**

This data element records the issuing authority if applicable to this vehicle.

Additional Information

See this data element in the Vehicle data file section for more information.

SAS Name

PMCARR_I1

Attribute Codes

2016-Later	
0	Not Applicable
1-56	State Code
57	U.S. DOT
58	MC/MX (ICC)
77	Not Reported
88	None
95	Canada
96	Mexico
99	Unknown (Reported as Unknown, 2018-2019)

V23B. MCID Identification Number**Definition**

This data element records the motor carrier identification number if applicable to this vehicle.

Additional Information

See this data element in the Vehicle data file section for more information.

SAS Name**PMCARR_I2****Attribute Codes**

2016-Later	
000000000	Not Applicable
xxxxxxxxx	Actual 9-Digit Number
777777777	Not Reported
888888888	None
999999999	Unknown (Reported as Unknown, 2018-2019)

V24. Vehicle Configuration

Definition

This data element describes the general configuration of this vehicle if applicable.

Additional Information

See this data element in the Vehicle data file section for more information.

SAS Name

PV_CONFIG

Attribute Codes

2016-2020	2021-Later	
0	0	Not Applicable
1	1	Single-Unit Truck (2 Axles and GVWR More Than 10,000 lbs)
2	2	Single-Unit Truck (3 or More Axles)
4	4	Truck Pulling Trailer(s)
5	5	Truck Tractor (Bobtail)
6	6	Truck Tractor/Semi-Trailer
7	7	Truck Tractor/Double
8	8	Truck Tractor/Triple
10	10	Vehicle 10,000 lbs or Less Placarded for Hazardous Materials
19	--	Truck More Than 10,000 lbs, Cannot Classify
--	19	Vehicle More Than 10,000 lbs, Other
20	20	Bus/Large Van (Seats for 9-15 Occupants, Including Driver)
21	21	Bus (Seats for More Than 15 Occupants, Including Driver)
--	88	Qualifying Vehicle, Unknown Configuration
99	99	Unknown (Reported as Unknown, 2018-2019)

V25. Cargo Body Type

Definition

This data element identifies the primary cargo carrying capability of this vehicle if applicable.

Additional Information

See this data element in the Vehicle data file section for more information.

SAS Name

PCARGTYP

Attribute Codes

2016-Later	
0	Not Applicable
1	Van/Enclosed Box
2	Cargo Tank
3	Flatbed
4	Dump
5	Concrete Mixer
6	Auto Transporter
7	Garbage/Refuse
8	Grain/Chips/Gravel
9	Pole-Trailer
10	Log
11	Intermodal Container Chassis
12	Vehicle Towing Another Vehicle
22	Bus
96	No Cargo Body
97	Other
98	Unknown Cargo Body Type
99	Unknown (Reported as Unknown, 2018-2019)

V26A/HM1. Hazardous Materials Involvement**Definition**

This data element identifies whether this vehicle was carrying hazardous materials.

Additional Information

See this data element in the Vehicle data file section for more information.

SAS Name**PHAZ_INV****Attribute Codes**

2016-Later	
1	No
2	Yes

V26B/HM2. Hazardous Materials Placard**Definition**

This data element identifies the presence of hazardous materials for this vehicle and whether this vehicle displayed a hazardous materials placard.

Additional Information

See this data element in the Vehicle data file section for more information.

SAS Name**PHAZPLAC****Attribute Codes**

2016-Later	
0	Not Applicable
1	No
2	Yes
8	Not Reported

V26C/HM3. Hazardous Material Identification Number**Definition**

This data element identifies the four-digit hazardous material identification number for this vehicle.

Additional Information

In 2018 this data element was changed to alphanumeric to retain all four digits.

See this data element in the Vehicle data file section for more information.

SAS Name**PHAZ_ID****Attribute Codes**

2016-Later	
0	Not Applicable
xxxx	Actual 4-Digit Number
8888	Not Reported

V26D/HM4. Hazardous Material Class Number**Definition**

This data element identifies the single-digit hazardous material class number for this vehicle.

Additional Information

See this data element in the Vehicle data file section for more information.

SAS Name

PHAZ_CNO

Attribute Codes

2016-Later	
0	Not Applicable
1	Explosives
2	Gases
3	Flammable/Combustible Liquid
4	Flammable Solid, Spontaneously Combustible, and Dangerous When Wet
5	Oxidizer and Organic Peroxide
6	Poison and Poison Inhalation Hazard
7	Radioactive
8	Corrosive
9	Miscellaneous
88	Not Reported

V26E/HM5. Release of Hazardous Material from the Cargo Compartment**Definition**

This data element identifies whether any hazardous cargo was released from the cargo tank or compartment of this vehicle.

Additional Information

See this data element in the Vehicle data file section for more information.

SAS Name**PHAZ_REL****Attribute Codes**

2016-Later	
0	Not Applicable
1	No
2	Yes
8	Not Reported

V27. Bus Use**Definition**

This data element describes the common type of bus service this vehicle was being used as at the time of the crash or the primary use for the bus if not in service at the time of the crash.

Additional Information

See this data element in the Vehicle data file section for more information.

SAS Name

PBUS_USE

Attribute Codes

2016-2017	2018-2021	2022-Later	
0	0	0	Not a Bus
1	1	1	School
4	4	4	Intercity
5	5	5	Charter/Tour
6	6	6	Transit/Commuter
7	7	7	Shuttle
8	8	8	Modified for Personal/Private Use
--	--	97	Bus, Unknown Use
98	98	98	Not Reported
99	--	--	Unknown
--	99	99	Reported as Unknown

V28. Special Use**Definition**

This data element identifies any special use associated with this vehicle at the time of the crash.

Additional Information

See this data element in the Vehicle data file section for more information.

SAS Name

PSP_USE

Attribute Codes

2016-2018	2019	2020	2021-Later	
0	0	0	--	No Special Use
--	--	--	0	No Special Use Noted
1	1	1	1	Taxi
2	2	2	2	Vehicle Used for School Transport
3	3	3	3	Vehicle Used as Other Bus
4	4	4	4	Military
5	5	5	5	Police
6	6	6	6	Ambulance
7	7	7	7	Fire Truck
8	8	8	8	Non-Transport Emergency Services Vehicle
--	10	10	10	Safety Service Patrols – Incident Response
--	11	11	11	Other Incident Response
--	12	12	12	Towing – Incident Response
13	--	--	--	Incident Response
--	--	19	19	Motor Vehicle Used for Vehicle Sharing Mobility
--	20	--	--	Vehicle Used for Electronic Ride-Hailing (Transportation Network Company)
--	--	20	20	Motor Vehicle Used for Electronic Ride-Hailing
--	21	21	21	Mail Carrier
--	22	22	22	Public Utility
--	23	23	23	Rental Truck Over 10,000 lbs
--	24	24	24	Truck Operating With Crash Attenuator Equipment
98	98	98	--	Not Reported
99	--	--	--	Unknown
99	99	99	99	Reported as Unknown (Since 2018)

V29. Emergency Motor Vehicle Use

Definition

This data element identifies whether this vehicle was engaged in emergency use. Emergency Use indicates operation of any motor vehicle that is legally authorized by a government authority to respond to emergencies with or without the use of emergency warning equipment, such as a police vehicle, fire truck or ambulance while actually engaged in such response.

Additional Information

See this data element in the Vehicle data file section for more information.

SAS Name

PEM_USE

Attribute Codes

2016-2017	2018-Later	
0	0	Not Applicable
2	2	Non-Emergency, Non-Transport
3	3	Non-Emergency Transport
4	4	Emergency Operation, Emergency Warning Equipment Not in Use
5	5	Emergency Operation, Emergency Warning Equipment in Use
6	6	Emergency Operation, Emergency Warning Equipment in Use Unknown
8	8	Not Reported
9	--	Unknown
--	9	Reported as Unknown

V31. Vehicle Underride/Override**Definition**

This element indicates whether this vehicle experienced an underride or override with another vehicle during the crash.

Additional Information

See [Appendix H: Notable Changes](#) for more information on this data element.

SAS Name**PUNDEROVERRIDE****Attribute Codes**

2021-Later	
0	No Underride or Override
1	Underride
2	Override
7	Not Applicable
8	Not Reported
9	Reported as Unknown

V34A. Areas of Impact—Initial Contact Point**Definition**

This data element identifies the area on this vehicle that produced the first instance of injury to non-motorists or occupants of this vehicle, or that resulted in the first instance of damage to other property or to this vehicle.

Additional Information

See this data element in the Vehicle data file section for more information.

SAS Name**PIMPACT1****Attribute Codes**

2016	2017	2018	2019-Later	
0	0	0	0	Non-Collision
1-12	1-12	1-12	1-12	Clock Points
13	13	13	13	Top
14	14	14	14	Undercarriage
18	18	18	18	Cargo/Vehicle Parts Set-in-Motion
19	19	19	--	Other Objects Set-in-Motion
--	--	--	19	Other Objects or Person Set-in-Motion
--	20	20	20	Object Set in Motion, Unknown if Cargo/Vehicle Parts or Other
61	61	61	61	Left
62	62	62	62	Left-Front Side
63	63	63	63	Left-Back Side
81	81	81	81	Right
82	82	82	82	Right-Front Side
83	83	83	83	Right-Back Side
98	98	98	98	Not Reported
99	99	--	--	Unknown
--	--	99	99	Reported as Unknown

V35. Extent of Damage**Definition**

This data element records the amount of damage sustained by this vehicle as indicated on the police crash report based on an operational damage scale.

Additional Information

See this data element in the Vehicle data file section for more information.

SAS Name

PVEH_SEV

Attribute Codes

2016-2017	2018-2021	2022-Later	
0	0	0	No Damage
2	2	2	Minor Damage
4	4	4	Functional Damage
6	6	6	Disabling Damage
--	--	7	Damage Reported, Extent Unknown
8	8	8	Not Reported
9	--	--	Unknown
--	9	9	Reported as Unknown

V36. Vehicle Towed

Definition

This data element identifies whether the vehicle was towed from the scene of the crash.

Additional Information

Prior to 2022 this data element's name was "Vehicle Removal." In 2022 the attribute structure was revised to remove the multiple constructs (i.e., tow status and damage) and to simply indicate whether or not the vehicle was towed. See this data element in the Parkwork data file section for more information.

Prior to 2020 the Data Element ID was V31.

SAS Name

PTOWED

Attribute Codes

2016-2017	2018-2019	2020-2021	2022-Later	
2	2	2	--	Towed Due to Disabling Damage
3	3	--	--	Towed Not Due to Disabling Damage
--	--	3	--	Towed but Not Due to Disabling Damage
5	5	5	5	Not Towed
--	--	--	6	Towed
--	7	7	--	Towed, Unknown Reason
8	8	8	8	Not Reported
9	--	--	--	Unknown
--	9	9	9	Reported as Unknown

V38. Most Harmful Event**Definition**

This data element describes the event that resulted in the most severe injury or, if no injury, the greatest property damage involving this vehicle.

Additional Information

See this data element in the Vehicle data file section for more information.

SAS Name

PM_HARM

Attribute Codes*Non-collision*

2016	2017	2018-Later	
1	1	1	Rollover/Overturn
2	2	2	Fire/Explosion
3	3	3	Immersion or Partial Immersion
4	4	4	Gas Inhalation
5	5	5	Fell/Jumped From Vehicle
6	6	6	Injured in Vehicle (Non-Collision)
7	7	7	Other Non-Collision
16	16	16	Thrown or Falling Object
44	44	44	Pavement Surface Irregularity (Ruts, Potholes, Grates, etc.)
51	51	51	Jackknife (Harmful to This Vehicle)
72	72	--	Cargo/Equipment Loss or Shift (Harmful to This Vehicle)
--	--	72	Cargo/Equipment Loss, Shift, or Damage (Harmful)

Collision with Motor Vehicle In-Transport

2016	2017	2018-Later	
12	12	12	Motor Vehicle In-Transport
54	54	54	Motor Vehicle In-Transport Strikes or Is Struck by Cargo, Persons or Objects Set-in-Motion From/by Another Motor Vehicle In-Transport
55	55	55	Motor Vehicle in Motion Outside the Trafficway

Collision with Object Not Fixed

2016	2017	2018-Later	
8	8	8	Pedestrian
9	9	9	Pedalcyclist
10	10	10	Railway Vehicle
11	11	11	Live Animal
14	14	14	Parked Motor Vehicle
15	15	15	Non-Motorist on Personal Conveyance
18	18	18	Other Object Not Fixed
45	45	45	Working Motor Vehicle
49	49	49	Ridden Animal or Animal Drawn Conveyance
73	73	73	Object That Had Fallen From Motor Vehicle In-Transport
74	74	74	Road Vehicle on Rails
--	91	91	Unknown Object Not Fixed

Collision with Fixed Object

2016	2017	2018-Later	
17	17	17	Boulder
19	19	19	Building
20	20	20	Impact Attenuator/Crash Cushion
21	21	21	Bridge Pier or Support
23	23	23	Bridge Rail (Includes Parapet)
24	24	24	Guardrail Face
25	25	25	Concrete Traffic Barrier
26	26	26	Other Traffic Barrier
30	30	30	Utility Pole/Light Support
31	31	31	Post, Pole or Other Support
32	32	32	Culvert
33	33	33	Curb
34	34	34	Ditch
35	35	35	Embankment
38	38	38	Fence
39	39	39	Wall
40	40	40	Fire Hydrant
41	41	41	Shrubbery
42	42	42	Tree (Standing Only)
43	43	43	Other Fixed Object

2016	2017	2018-Later	
46	46	46	Traffic Signal Support
48	48	48	Snow Bank
50	50	50	Bridge Overhead Structure
52	52	52	Guardrail End
53	53	53	Mail Box
57	57	57	Cable Barrier
58	58	58	Ground
59	59	59	Traffic Sign Support
--	93	93	Unknown Fixed Object

--	--	98	Harmful Event, Details Not Reported (Since 2019)
99	99	--	Unknown
--	--	99	Reported as Unknown

V39. Fire Occurrence**Definition**

This data element identifies whether a fire in any way related to the crash occurred in this vehicle.

Additional Information

See this data element in the Vehicle data file section for more information.

SAS Name**PFIRE****Attribute Codes**

2016-Later	
0	No or Not Reported
1	Yes

V100. NCSA Make Model Combined**Definition**

This derived data element represents the five-digit combination of two data elements, the two-digit “NCSA Make” code (MAKE) followed by the three-digit “NCSA Model” code (MODEL).

Additional Information

Prior to 2020 this data element’s name was “Make Model Combined.”

See this data element in the Vehicle data file section for more information.

SAS Name**PMAK_MOD****Attribute Codes**

2016-Later	
	See the current FARS/CRSS Coding and Validation Manual for vehicle make and model codes.

Discontinued PARKWORK Data Elements***Gross Vehicle Weight Rating (discontinued)*****Definition**

This data element identifies the gross vehicle weight rating of this vehicle if applicable.

Additional Information

See this data element in the Vehicle data file section for more information.

SAS Name

PGVWR

Attribute Codes

2016-2017	2018-2019	
0	0	Not Applicable
1	1	10,000 lbs or Less
2	2	10,001 lbs -26,000 lbs
3	3	26,001 lbs or More
8	8	Not Reported
9	--	Unknown
--	9	Reported as Unknown

Related Factors—Vehicle Level (discontinued)**Definition**

This data element records factors related to this vehicle expressed in the case materials.

Additional Information

Beginning in 2020 this data element was no longer collected at the Vehicle level. It is now collected in the Pvehiclesf data file as PVEHICLESF.

SAS Name

PVEH_SC1, PVEH_SC2

Attribute Codes

2016-2017	2018	2019	
0	0	0	None

2016-2017	2018	2019	
--	--	29	Default Code Used for Vehicle Numbering
30	30	30	Multi-Wheeled Motorcycle Conversion
33	33	33	Vehicle Being Pushed by Non-Motorist
35	35	35	Reconstructed/Altered Vehicle
39	39	39	Highway Construction, Maintenance or Utility Vehicle, In-Transport (Inside or Outside Work Zone)
40	40	--	Highway Incident Response Vehicle
41	41	41	Police Fire or EMS Vehicle Working at the Scene of an Emergency or Performing Other Traffic Control Activities
42	42	42	Other Working Vehicle (Not Construction, Maintenance, Utility, Police, Fire, or EMS Vehicle)
44	44	44	Adaptive Equipment
--	45	45	Slide-in Camper
99	--	--	Unknown
--	99	99	Reported as Unknown

The PBTYPEData File

The Pdtype data file includes data on pedestrians, bicyclists, and people on personal conveyances. It contains the data elements CASENUM, PSU, PJ, STRATUM, PSUSTRAT, PSU_VAR, REGION, URBANICITY, WEIGHT, VEH_NO, and PER_NO, which are described in the Key Data Elements at the beginning of the Data Element Definitions and Codes section. The Pdtype data file also contains the data elements on the following pages.

CASENUM, VEH_NO, and PER_NO are the unique identifiers. CASENUM should be used to merge the Pdtype data file with the Accident data file.

P5/NM5. Age**Definition**

This data element identifies the person's age in years on the date of the crash.

Additional Information**SAS Name****PBAGE****Attribute Codes**

2016-2017	2018-Later	
0	0	Less Than One Year
1-120	1-120	Age in Years
998	998	Not Reported
999	--	Unknown
--	999	Reported as Unknown

P6/NM6. Sex**Definition**

This data element identifies the sex of the person involved in the crash.

Additional Information**SAS Name****PBSEX****Attribute Codes**

2016-2017	2018-Later	
1	1	Male
2	2	Female
8	8	Not Reported
9	--	Unknown
--	9	Reported as Unknown

P7/NM7. Person Type**Definition**

This data element describes the role of this person involved in the crash.

Additional Information**SAS Name****PBTYPE****Attribute Codes**

2016-2019	2020-2021	2022-Later	
5	5	5	Pedestrian
6	6	6	Bicyclist
7	7	7	Other Pedalcyclist
8	--	8	Person on a Personal Conveyance
--	11	--	Person on Motorized Personal Conveyance
--	12	--	Person on Non-Motorized Personal Conveyance
--	13	--	Person on Personal Conveyance, Unknown if Motorized or Non-Motorized

NM11-PB27. Marked Crosswalk Present**Definition**

This data element indicates if a marked crosswalk was present at the crash site.

Additional Information

This data element is applicable to both pedestrians and bicyclists. Prior to 2022 the Data Element ID was NM9-PB27.

SAS Name**PBCWALK****Attribute Codes**

2016-Later	
0	None Noted
1	Yes
9	Unknown

NM11-PB28. Sidewalk Present**Definition**

This data element indicates if a sidewalk was present at the crash site.

Additional Information

This data element is applicable to both pedestrians and bicyclists. Prior to 2022 the Data Element ID was NM9-PB28.

SAS Name**PBSWALK****Attribute Codes**

2016-Later	
0	None Noted
1	Yes
9	Unknown

NM11-PB29. School Zone**Definition**

This data element indicates if the crash occurred in a school zone.

Additional Information

This data element is applicable to both pedestrians and bicyclists. Prior to 2022 the Data Element ID was NM9-PB29.

SAS Name**PBSZONE****Attribute Codes**

2016-Later	
0	None Noted
1	Yes
9	Unknown

NM11-PB30. Crash Type – Pedestrian**Definition**

This data element summarizes the circumstances of the crash for this pedestrian.

Additional Information

This data element is applicable to pedestrians only. Prior to 2022 the Data Element ID was NM9-PB30.

SAS Name**PEDCTYPE****Attribute Codes**

2016	2017-2019	2020-Later	
0	0	0	Not a Pedestrian
120	120	120	Dispute-Related
130	130	130	Pedestrian on Vehicle
140	140	140	Vehicle Into Vehicle or Vehicle Into Object
150	150	150	Motor Vehicle Loss of Control
160	160	160	Pedestrian Loss of Control
190	190	190	Other Unusual Circumstances
211	211	211	Backing Vehicle – Non-Trafficway – Driveway
212	212	212	Backing Vehicle – Driveway Access
213	213	213	Backing Vehicle – Trafficway
214	214	214	Backing Vehicle – Non-Trafficway – Parking Lot
219	219	219	Backing Vehicle – Other/Unknown
220	220	220	Driverless Vehicle
230	230	230	Disabled Vehicle-Related
240	240	240	Emergency Vehicle-Related
250	250	250	Play Vehicle-Related
311	311	311	Working in Roadway
312	312	312	Playing in Roadway
313	313	313	Lying in Roadway
320	320	320	Entering/Exiting Parked or Stopped Vehicle
330	330	330	Mailbox-Related
341	--	--	Transit Bus-Related
--	341	341	Transit Bus Stop-Related
342	342	342	School Bus Stop-Related
360	360	360	Ice Cream/Vendor Truck-Related

2016	2017-2019	2020-Later	
410	410	410	Walking/Running Along Roadway With Traffic – From Behind
420	420	420	Walking/Running Along Roadway With Traffic – From Front
430	430	430	Walking/Running Along Roadway Against Traffic – From Behind
440	440	440	Walking/Running Along Roadway Against Traffic – From Front
459	459	459	Walking/Running Along Roadway – Direction/Position Unknown
461	461	461	Motorist Entering Driveway
465	465	465	Motorist Exiting Driveway
469	469	469	Driveway Access – Other/Unknown
510	510	510	Waiting to Cross – Vehicle Turning
520	520	520	Waiting to Cross – Vehicle Not Turning
590	590	590	Waiting to Cross – Vehicle Action Unknown
610	610	610	Standing in Roadway
620	620	620	Walking in Roadway
680	680	680	Not at Intersection – Other/Unknown
690	690	690	At Intersection – Other/Unknown
710	710	710	Multiple Threat
730	730	730	Trapped
741	741	--	Dash
--	--	741	Dash – Run, No Visual Obstruction Noted
742	742	--	Dart-out
--	--	742	Dash out – Visual Obstruction Noted
760	760	760	Pedestrian Failed to Yield
770	770	770	Motorist Failed to Yield
781	781	781	Motorist Left Turn – Parallel Paths
782	782	782	Motorist Left Turn – Perpendicular Paths
791	791	791	Motorist Right Turn – Parallel Paths
792	792	792	Motorist Right Turn on Red – Parallel Paths
794	794	794	Motorist Right Turn on Red – Perpendicular Paths
795	795	795	Motorist Right Turn – Perpendicular Paths
799	799	799	Motorist Turn/Merge – Other/Unknown
830	830	830	Non-Trafficway – Parking Lot
890	890	890	Non-Trafficway – Other/Unknown
900	900	900	Other – Unknown Location
910	910	910	Crossing an Expressway

NM11-PB30B. Crash Type – Bicycle**Definition**

This data element summarizes the circumstances of the crash for this bicyclist.

Additional Information

This data element is applicable to bicyclists only. Prior to 2022 the Data Element ID was NM9-PB30B.

SAS Name

BIKECTYPE

Attribute Codes

2016-Later	
0	Not a Cyclist
111	Motorist Turning Error – Left Turn
112	Motorist Turning Error – Right Turn
113	Motorist Turning Error – Other
114	Bicyclist Turning Error – Left Turn
115	Bicyclist Turning Error – Right Turn
116	Bicyclist Turning Error – Other
121	Bicyclist Lost Control – Mechanical Problems
122	Bicyclist Lost Control – Oversteering, Improper Braking, Speed
123	Bicyclist Lost Control – Alcohol/Drug Impairment
124	Bicyclist Lost Control – Surface Conditions
129	Bicyclist Lost Control – Other/Unknown
131	Motorist Lost Control – Mechanical Problems
132	Motorist Lost Control – Oversteering, Improper Braking, Speed
133	Motorist Lost Control – Alcohol/Drug Impairment
134	Motorist Lost Control – Surface Conditions
139	Motorist Lost Control – Other/Unknown
141	Motorist Drive-out – Sign-Controlled Intersection
142	Bicyclist Ride-out – Sign-Controlled Intersection
143	Motorist Drive-Through – Sign-Controlled Intersection
144	Bicyclist Ride-Through – Sign-Controlled Intersection
147	Multiple Threat – Sign-Controlled Intersection
148	Sign-Controlled Intersection – Other/Unknown
151	Motorist Drive-out – Right Turn on Red
152	Motorist Drive-out – Signalized Intersection

2016-Later	
153	Bicyclist – Ride-out – Signalized Intersection
154	Motorist Drive-Through – Signalized Intersection
155	Bicyclist Ride-Through – Signalized Intersection
156	Bicyclist Failed to Clear – Trapped
157	Bicyclist Failed to Clear – Multiple Threat
158	Signalized Intersection – Other/Unknown
159	Bicyclist Failed to Clear – Unknown
160	Crossing Paths – Uncontrolled Intersection
180	Crossing Paths – Intersection – Other/Unknown
211	Motorist Left Turn – Same Direction
212	Motorist Left Turn – Opposite Direction
213	Motorist Right Turn – Same Direction
214	Motorist Right Turn – Opposite Direction
215	Motorist Drive-in/out – Parking
216	Bus/Delivery Vehicle Pullover
217	Motorist Right Turn on Red – Same Direction
218	Motorist Right Turn on Red – Opposite Direction
219	Motorist Turn/Merge – Other/Unknown
221	Bicyclist Left Turn – Same Direction
222	Bicyclist Left Turn – Opposite Direction
223	Bicyclist Right Turn – Same Direction
224	Bicyclist Right Turn – Opposite Direction
225	Bicyclist Ride-out – Parallel Path
231	Motorist Overtaking – Undetected Bicyclist
232	Motorist Overtaking – Misjudged Space
235	Motorist Overtaking – Bicyclist Swerved
239	Motorist Overtaking – Other/Unknown
241	Bicyclist Overtaking – Passing on Right
242	Bicyclist Overtaking – Passing on Left
243	Bicyclist Overtaking – Parked Vehicle
244	Bicyclist Overtaking – Extended Door
249	Bicyclist Overtaking – Other/Unknown
250	Wrong-Way/Wrong-Side – Bicyclist
255	Wrong-Way/Wrong-Side – Motorist
259	Wrong-Way/Wrong-Side – Unknown
280	Parallel Paths – Other/Unknown
311	Bicyclist Ride-out – Residential Driveway
312	Bicyclist Ride-out – Commercial Driveway

2016-Later	
313	Bicyclist Ride-out – Driveway, Unknown Type
318	Bicyclist Ride-out – Other Midblock
319	Bicyclist Ride-out – Unknown
321	Motorist Drive-out – Residential Driveway
322	Motorist Drive-out – Commercial Driveway
323	Motorist Drive-out – Driveway, Unknown Type
328	Motorist Drive-out – Other Midblock
329	Motorist Drive-out – Midblock – Unknown
357	Multiple Threat – Midblock
380	Crossing Paths – Midblock – Other/Unknown
610	Backing Vehicle
700	Play Vehicle-Related
800	Unusual Circumstances
910	Non-Trafficway
970	Unknown Approach Paths
980	Unknown Location

NM11-PB31. Crash Location – Pedestrian**Definition**

This data element identifies where the pedestrian crash occurred with respect to an intersection.

Additional Information

This data element is applicable to pedestrians only. Prior to 2022 the Data Element ID was NM9-PB31.

SAS Name**PEDLOC****Attribute Codes**

2016-Later	
1	At Intersection
2	Intersection-Related
3	Not at Intersection
4	Non-Trafficway Location
7	Not a Pedestrian
9	Unknown/Insufficient Information

See [Analysis of Pedestrian and Bicycle Crashes Around Intersections](#) for guidance on analyzing Pedestrian/Bicyclist crash locations.

NM11-PB31B. Crash Location – Bicycle**Definition**

This data element identifies where the bicyclist crash occurred with respect to an intersection.

Additional Information

This data element is applicable to bicyclists only. Prior to 2022 the Data Element ID was NM9-PB31B.

SAS Name**BIKELOC****Attribute Codes**

2016-Later	
1	At Intersection
2	Intersection-Related
3	Not at Intersection
4	Non-Trafficway Location
7	Not a Cyclist
9	Unknown/Insufficient Information

See [Analysis of Pedestrian and Bicycle Crashes Around Intersections](#) for guidance on analyzing Pedestrian/Bicyclist crash locations.

NM11-PB32. Pedestrian Position**Definition**

This data element identifies the position/location of the pedestrian with respect to the trafficway when contacted.

Additional Information

This data element is applicable to pedestrians only. Prior to 2022 the Data Element ID was NM9-PB32.

SAS Name

PEDPOS

Attribute Codes

2016-Later	
1	Intersection Area
2	Crosswalk Area
3	Travel Lane
4	Paved Shoulder/Bicycle Lane/Parking Lane
5	Sidewalk/Shared-Use Path/Driveway Access
6	Unpaved Right-of-Way
7	Non-Trafficway – Driveway
8	Non-Trafficway – Parking Lot/Other
9	Other/Unknown
77	Not a Pedestrian

See [Analysis of Pedestrian and Bicycle Crashes Around Intersections](#) for guidance on analyzing Pedestrian/Bicyclist crash locations.

NM11-PB32B. Bicyclist Position**Definition**

This data element identifies the position/location of the bicyclist with respect to the trafficway when contacted.

Additional Information

This data element is applicable to bicyclists only. Prior to 2022 the Data Element ID was NM9-PB32B.

SAS Name

BIKEPOS

Attribute Codes

2016-Later	
1	Travel Lane
2	Bicycle Lane/Paved Shoulder/Parking Lane
3	Sidewalk/Crosswalk/Driveway Access
4	Shared-Use Path
5	Non-Trafficway – Driveway
6	Non-Trafficway – Parking Lot/Other
7	Not a Cyclist
8	Other
9	Unknown

See [Analysis of Pedestrian and Bicycle Crashes Around Intersections](#) for guidance on analyzing Pedestrian/Bicyclist crash locations.

NM11-PB33. Pedestrian Initial Direction of Travel**Definition**

This data element identifies the initial direction of travel of the pedestrian prior to being contacted in the crash.

Additional Information

This data element is applicable to pedestrians only. Prior to 2022 the Data Element ID was NM9-PB33.

SAS Name

PEDDIR

Attribute Codes

2016	2017-Later	
1	1	Northbound
2	2	Eastbound
3	3	Southbound
4	4	Westbound
7	7	Not a Pedestrian
8	8	Not Applicable
9	--	Unknown Initial Direction of Travel
--	9	Not Derived/Unknown Initial Direction of Travel

NM11-PB33B. Bicyclist Initial Direction of Travel**Definition**

This data element identifies the initial travel direction of the bicyclist with respect to the flow of traffic prior to being contacted in the crash.

Additional Information

This data element is applicable to bicyclists only. Prior to 2022 the Data Element ID was NM9-PB33B.

SAS Name**BIKEDIR****Attribute Codes**

2016-Later	
1	With Traffic
2	Facing Traffic
3	Not Applicable
7	Not a Cyclist
9	Unknown

NM11-PB34. Motorist Initial Direction of Travel**Definition**

This data element identifies the initial direction of travel of the motorist prior to being involved in a pedestrian crash.

Additional Information

This data element is applicable to pedestrians only. Prior to 2022 the Data Element ID was NM9-PB34.

SAS Name**MOTDIR****Attribute Codes**

2016-Later	
1	Northbound
2	Eastbound
3	Southbound
4	Westbound
7	Not a Pedestrian
8	Not Applicable
9	Unknown Initial Direction of Travel

NM11-PB35. Motorist Maneuver**Definition**

This data element identifies if the motorist was engaged in a turning maneuver at an intersection prior to being involved in a pedestrian crash. The data element indicates the maneuver being made by the motorist at the time of a pedestrian collision.

Additional Information

This data element is applicable to pedestrians only. Prior to 2022 the Data Element ID was NM9-PB35.

SAS Name**MOTMAN****Attribute Codes**

2016-Later	
1	Left Turn
2	Right Turn
3	Straight Through
7	Not a Pedestrian
8	Not Applicable
9	Unknown Motorist Maneuver

NM11-PB36. Intersection Leg**Definition**

The data element identifies the leg of the intersection where the pedestrian crash occurred.

Additional Information

This data element is applicable to pedestrians only. Prior to 2022 the Data Element ID was NM9-PB36.

SAS Name**PEDLEG****Attribute Codes**

2016-Later	
1	Nearside
2	Farside
7	Not a Pedestrian
8	Not Applicable
9	Unknown/None of the Above

NM11-PB37. Pedestrian Scenario**Definition**

This data element identifies the location and travel directions of the motorist and pedestrian for those crashes that occur at intersections. This data element summarizes the movements of the pedestrian and motorist in an intersection area.

Additional Information

This data element is applicable to pedestrians only. Prior to 2022 the Data Element ID was NM9-PB37.

SAS Name**PEDSNR****Attribute Codes**

Motorist traveling straight through – Crash Occurred on Near (Approach) Side of Intersection

2016	2017-Later	
1a	1a	Pedestrian Within Crosswalk Area, Traveled From Motorist's Left.
1b	1b	Pedestrian Within Crosswalk Area, Traveled From Motorist's Right.
1c	1c	Pedestrian Within Crosswalk Area, Approach Direction Unknown.
--	1d	Pedestrian Within Crosswalk Area, Other
2a	2a	Pedestrian Outside Crosswalk Area, Traveled From Motorist's Left.
2b	2b	Pedestrian Outside Crosswalk Area, Traveled From Motorist's Right.
2c	2c	Pedestrian Outside Crosswalk Area, Approach Direction Unknown.
--	2d	Pedestrian Outside Crosswalk Area, Other

Motorist traveling straight through – Crash Occurred on Far Side of Intersection

2016	2017-Later	
3a	3a	Pedestrian Within Crosswalk Area, Traveled From Motorist's Left.
3b	3b	Pedestrian Within Crosswalk Area, Traveled From Motorist's Right.

2016	2017-Later	
3c	3c	Pedestrian Within Crosswalk Area, Approach Direction Unknown.
--	3d	Pedestrian Within Crosswalk Area, Other
4a	4a	Pedestrian Outside Crosswalk Area, Traveled From Motorist's Left.
4b	4b	Pedestrian Outside Crosswalk Area, Traveled From Motorist's Right.
4c	4c	Pedestrian Outside Crosswalk Area, Approach Direction Unknown.
--	4d	Pedestrian Outside Crosswalk Area, Other

Motorist turning right – Crash Occurred on Near (Approach) Side of Intersection

2016	2017-Later	
5a	5a	Pedestrian Within Crosswalk Area, Traveled From Motorist's Left.
5b	5b	Pedestrian Within Crosswalk Area, Traveled From Motorist's Right.
5c	5c	Pedestrian Within Crosswalk Area, Approach Direction Unknown.
--	5d	Pedestrian Within Crosswalk Area, Other
6a	6a	Pedestrian Outside Crosswalk Area, Traveled From Motorist's Left.
6b	6b	Pedestrian Outside Crosswalk Area, Traveled From Motorist's Right.
6c	6c	Pedestrian Outside Crosswalk Area, Approach Direction Unknown.
--	6d	Pedestrian Outside Crosswalk Area, Other

Motorist turning right – Crash Occurred on Far Side of Intersection

2016	2017-Later	
7a	7a	Pedestrian Within Crosswalk Area, Approach Direction Same as Motorist's.
7b	7b	Pedestrian Within Crosswalk Area, Approach Direction Opposite Motorist's.
7c	7c	Pedestrian Within Crosswalk Area, Approach Direction Unknown.
--	7d	Pedestrian Within Crosswalk Area, Other
8a	8a	Pedestrian Outside Crosswalk Area, Approach Direction Same as Motorist's.

2016	2017-Later	
8b	8b	Pedestrian Outside Crosswalk Area, Approach Direction Opposite Motorist's.
8c	8c	Pedestrian Outside Crosswalk Area, Approach Direction Unknown.
--	8d	Pedestrian Outside Crosswalk Area, Other

Motorist turning left – Crash Occurred on Near (Approach) Side of Intersection

2016	2017-Later	
9a	9a	Pedestrian Within Crosswalk Area, Traveled From Motorist's Left.
9b	9b	Pedestrian Within Crosswalk Area, Traveled From Motorist's Right.
9c	9c	Pedestrian Within Crosswalk Area, Approach Direction Unknown.
--	9d	Pedestrian Within Crosswalk Area, Other
10a	10a	Pedestrian Outside Crosswalk Area, Traveled From Motorist's Left.
10b	10b	Pedestrian Outside Crosswalk Area, Traveled From Motorist's Right.
10c	10c	Pedestrian Outside Crosswalk Area, Approach Direction Unknown.
--	10d	Pedestrian Outside Crosswalk Area, Other

Motorist turning left – Crash Occurred on Far Side of Intersection

2016	2017-Later	
11a	11a	Pedestrian Within Crosswalk Area, Approach Direction Same as Motorist's.
11b	11b	Pedestrian Within Crosswalk Area, Approach Direction Opposite Motorist's.
11c	11c	Pedestrian Within Crosswalk Area, Approach Direction Unknown.
--	11d	Pedestrian Within Crosswalk Area, Other
12a	12a	Pedestrian Outside Crosswalk Area, Approach Direction Same as Motorist's.
12b	12b	Pedestrian Outside Crosswalk Area, Approach Direction Opposite Motorist's.
12c	12c	Pedestrian Outside Crosswalk Area, Approach Direction Unknown.
--	12d	Pedestrian Outside Crosswalk Area, Other

2016	2017-Later	
7	7	Not a Pedestrian
8	8	Not Applicable
--	99	Unknown/Insufficient Information

NM11-PB38. Crash Group – Pedestrian**Definition**

This data element provides general groupings of the more specific individual Pedestrian Crash Types.

Additional Information

This data element is applicable to pedestrians only. Prior to 2022 the Data Element ID was NM9-PB38.

SAS Name**PEDCGP****Attribute Codes**

2016	2017-2019	2020-Later	
0	0	0	Not a Pedestrian
100	100	100	Unusual Circumstances
200	200	200	Backing Vehicle
310	310	310	Working or Playing in Roadway
340	--	--	Bus-Related
--	340	340	Bus Stop-Related
350	350	350	Unique Midblock
400	400	400	Walking/Running Along Roadway
460	460	460	Driveway Access/Driveway Access Related
500	500	500	Waiting to Cross
600	600	600	Pedestrian in Roadway – Circumstances Unknown
720	720	720	Multiple Threat/Trapped
740	740	--	Dash/Dart-out
--	--	740	Dash – Run, No Visual Obstruction Noted/ Dart-out – Visual Obstruction Noted
750	750	750	Crossing Roadway – Vehicle Not Turning
790	790	790	Crossing Roadway – Vehicle Turning
800	800	800	Non-Trafficway
910	910	910	Crossing Expressway
990	990	990	Other/Unknown – Insufficient Details

NM11-PB38B. Crash Group – Bicycle**Definition**

This data element provides general groupings of the more specific individual Bicyclist Crash Types.

Additional Information

This data element is applicable to bicyclists only. Prior to 2022 the Data Element ID was NM9-PB38B.

SAS Name

BIKECGP

Attribute Codes

2016-Later	
0	Not a Cyclist
110	Loss of Control/Turning Error
140	Motorist Failed to Yield – Sign-Controlled Intersection
145	Bicyclist Failed to Yield – Sign-Controlled Intersection
150	Motorist Failed to Yield – Signalized Intersection
158	Bicyclist Failed to Yield – Signalized Intersection
190	Crossing Paths – Other Circumstances
210	Motorist Left Turn/Merge
215	Motorist Right Turn/Merge
219	Parking/Bus-Related
220	Bicyclist Left Turn/Merge
225	Bicyclist Right Turn/Merge
230	Motorist Overtaking Bicyclist
240	Bicyclist Overtaking Motorist
258	Wrong-Way/Wrong-Side
290	Parallel Paths – Other Circumstances
310	Bicyclist Failed to Yield – Midblock
320	Motorist Failed to Yield – Midblock
600	Backing Vehicle
850	Other/Unusual Circumstances
910	Non-Trafficway
990	Other/Unknown – Insufficient Details

The CEVENT Data File

The Cevent data file includes harmful and non-harmful events in the crash. It contains the data elements CASENUM, PSU, PJ, STRATUM, PSUSTRAT, PSU_VAR, REGION, URBANICITY, WEIGHT, and EVENTNUM, which are described in the Key Data Elements at the beginning of the Data Element Definitions and Codes section. The Cevent data file also contains the data elements on the following pages.

CASENUM and EVENTNUM are the unique identifiers for each record. CASENUM should be used to merge the Cevent data file with the Accident data file.

C18A. Vehicle Number (This Vehicle)**Definition**

This data element identifies the “Vehicle Number” (VEH_NO) of this motor vehicle in-transport described in this event.

Additional Information

This is the vehicle described in “Sequence of Events” for this event.

SAS Name**VNUMBER1****Attribute Codes**

2016-Later	
1-999	Vehicle Number

C18B. Area of Impact (This Vehicle)**Definition**

This data element identifies the impact point, if any, on this motor vehicle in-transport that produced property damage or personal injury in this event.

Additional Information

This is the impact area of the vehicle recorded in “Vehicle Number (This Vehicle)” and described in “Sequence of Events.”

SAS Name**AOI1****Attribute Codes**

2016	2017	2018	2019-Later	
0	0	0	0	Non-Collision
1-12	1-12	1-12	1-12	Clock Points
13	13	13	13	Top
14	14	14	14	Undercarriage
18	18	18	18	Cargo/Vehicle Parts Set-in-Motion
19	19	19	--	Other Objects Set-in-Motion
--	--	--	19	Other Objects or Person Set-in-Motion
--	20	20	20	Object Set in Motion, Unknown if Cargo/Vehicle Parts or Other
55	55	55	55	Non-Harmful Event
61	61	61	61	Left
62	62	62	62	Left-Front Side
63	63	63	63	Left-Back Side
81	81	81	81	Right
82	82	82	82	Right-Front Side
83	83	83	83	Right-Back Side
98	98	98	98	Not Reported
99	99	--	--	Unknown
--	--	99	99	Reported as Unknown

V37. Sequence of Events

Definition

This data element describes this event. A motor vehicle traffic crash is a series of events resulting from an unstabilized situation. This series of harmful and non-harmful events is recorded in chronological order based on the police crash report narrative and diagram.

Additional Information

“First Harmful Event, Most Harmful Event,” and the “Sequence of Events” data elements have the same harmful event attributes. “Sequence of Events” also has non-harmful event attributes. Prior to 2020 the Data Element ID was V32.

SAS Name

SOE

Attribute Codes

Non-Harmful Events

2016	2017	2018-Later	
60	60	60	Cargo/Equipment Loss or Shift (Non-Harmful)
61	61	61	Equipment Failure (Blown Tire, Brake Failure, etc.)
62	62	62	Separation of Units
63	63	63	Ran off Roadway-Right
64	64	64	Ran off Roadway-Left
65	65	65	Cross Median
66	66	66	Downhill Runaway
67	67	67	Vehicle Went Airborne
68	68	68	Cross Centerline
69	69	69	Re-entering Roadway
70	70	70	Non-harmful, Swaying Trailer/Jackknife
71	71	71	End Departure
79	79	79	Ran off Roadway -Direction Unknown

Non-collision Harmful Events

2016	2017	2018-Later	
1	1	1	Rollover/Overtake
2	2	2	Fire/Explosion
3	3	3	Immersion or Partial Immersion
4	4	4	Gas Inhalation

2016	2017	2018-Later	
5	5	5	Fell/Jumped From Vehicle
6	6	6	Injured in Vehicle (Non-Collision)
7	7	7	Other Non-Collision
16	16	16	Thrown or Falling Object
44	44	44	Pavement Surface Irregularity (Ruts, Potholes, Grates, etc.)
51	51	51	Jackknife (Harmful to This Vehicle)
72	72	--	Cargo/Equipment Loss or Shift (Harmful to This Vehicle)
--	--	72	Cargo/Equipment Loss, Shift, or Damage (Harmful)

Collision with Motor Vehicle In-Transport

2016	2017	2018-Later	
12	12	12	Motor Vehicle In-Transport
54	54	54	Motor Vehicle In-Transport Strikes or Is Struck by Cargo, Persons or Objects Set-in-Motion From/by Another Motor Vehicle In-Transport
55	55	55	Motor Vehicle in Motion Outside the Trafficway

Collision with Object Not Fixed

2016	2017	2018-Later	
8	8	8	Pedestrian
9	9	9	Pedalcyclist
10	10	10	Railway Vehicle
11	11	11	Live Animal
14	14	14	Parked Motor Vehicle
15	15	15	Non-Motorist on Personal Conveyance
18	18	18	Other Object (Not Fixed)
45	45	45	Working Motor Vehicle
49	49	49	Ridden Animal or Animal Drawn Conveyance
73	73	73	Object That Had Fallen From Motor Vehicle In-Transport
74	74	74	Road Vehicle on Rails
--	91	91	Unknown Object Not Fixed

Collision with Fixed Object

2016	2017	2018-Later	
17	17	17	Boulder
19	19	19	Building
20	20	20	Impact Attenuator/Crash Cushion
21	21	21	Bridge Pier or Support
23	23	23	Bridge Rail (Includes Parapet)
24	24	24	Guardrail Face
25	25	25	Concrete Traffic Barrier
26	26	26	Other Traffic Barrier
30	30	30	Utility Pole/Light Support
31	31	31	Post, Pole or Other Support
32	32	32	Culvert
33	33	33	Curb
34	34	34	Ditch
35	35	35	Embankment
38	38	38	Fence
39	39	39	Wall
40	40	40	Fire Hydrant
41	41	41	Shrubbery
42	42	42	Tree (Standing Only)
43	43	43	Other Fixed Object
46	46	46	Traffic Signal Support
48	48	48	Snow Bank
50	50	50	Bridge Overhead Structure
52	52	52	Guardrail End
53	53	53	Mail Box
57	57	57	Cable Barrier
58	58	58	Ground
59	59	59	Traffic Sign Support
--	93	93	Unknown Fixed Object
--	--	98	Harmful Event, Details Not Reported (Since 2019)
99	99	--	Unknown
--	--	99	Reported as Unknown

C18C. Vehicle Number (Other Vehicle)**Definition**

This data element identifies the “Vehicle Number” (VEH_NO) of the other motor vehicle, if any, in this event.

Additional Information

This is the vehicle contacted by the motor vehicle in-transport recorded in “Vehicle Number (This Vehicle).” Another vehicle must have been involved in this event for this data element to be a valid vehicle number (i.e., “Sequence of Events” for this event must be 12, 14, 45, 54, or 55).

SAS Name**VNUMBER2****Attribute Codes**

2016-Later	
1-999	Vehicle Number
5555	Non-Harmful Event
9999	Not a Motor Vehicle

C18D. Area of Impact (Other Vehicle)**Definition**

This data element identifies the impact point on the other motor vehicle, if any, in this event.

Additional Information

This is the impact area of the vehicle recorded in “Vehicle Number (Other Vehicle).” Another vehicle must have been involved in this event for this data element to be a valid impact location (i.e., “Sequence of Events” for this event must be 12, 14, 45, 54, or 55).

SAS Name**AOI2****Attribute Codes**

2016	2017	2018	2019-Later	
0	0	0	0	Non-Collision
1-12	1-12	1-12	1-12	Clock Points
13	13	13	13	Top
14	14	14	14	Undercarriage
18	18	18	18	Cargo/Vehicle Parts Set-in-Motion
19	19	19	--	Other Objects Set-in-Motion
--	--	--	19	Other Objects or Person Set-in-Motion
--	20	20	20	Object Set in Motion, Unknown if Cargo/Vehicle Parts or Other
55	55	55	55	Non-Harmful Event
61	61	61	61	Left
62	62	62	62	Left-Front Side
63	63	63	63	Left-Back Side
77	77	77	77	Not a Motor Vehicle
81	81	81	81	Right
82	82	82	82	Right-Front Side
83	83	83	83	Right-Back Side
98	98	98	98	Not Reported
99	99	--	--	Unknown
--	--	99	99	Reported as Unknown

The VEVENT Data File

The Vevent data file includes harmful and non-harmful events for each motor vehicle in-transport. It contains the data elements CASENUM, PSU, PJ, STRATUM, PSUSTRAT, PSU_VAR, REGION, URBANICITY, WEIGHT, VEH_NO, EVENTNUM, and VEVENTNUM, which are described in the Key Data Elements at the beginning of the Data Element Definitions and Codes section. The Vevent data file also contains the data elements on the following pages.

CASENUM, VEH_NO, and VEVENTNUM are the unique identifiers for each record. CASENUM and VEH_NO should be used to merge the Vevent data file with the Vehicle data file.

C18A. Vehicle Number (This Vehicle)**Definition**

This data element identifies the “Vehicle Number” (VEH_NO) of the motor vehicle in-transport described in this event.

Additional Information

This is the vehicle described in “Sequence of Events” for this event.

If Vehicle #1 (V1) impacts Vehicle #2 (V2) then we have at least 2 Vevent records.

Example:

VEH_NO	EVENTNUM	VNUMBER1	SOE	VNUMBER2
1	1	1	12	2
2	1	1	12	2

The explanation of these 2 records is as follows:

V1 was involved in event 1 where V1 impacts V2.

V2 was involved in event 1 where V1 impacts V2.

SAS Name

VNUMBER1

Attribute Codes

2016-Later	
1-999	Vehicle Number

C18B. Area of Impact (This Vehicle)**Definition**

This data element identifies the impact point, if any, on this motor vehicle in-transport that produced property damage or personal injury in this event.

Additional Information

This is the impact area of the vehicle recorded in “Vehicle Number (This Vehicle)” and described in “Sequence of Events.”

SAS Name**AOI1****Attribute Codes**

2016	2017	2018	2019-Later	
0	0	0	0	Non-Collision
1-12	1-12	1-12	1-12	Clock Points
13	13	13	13	Top
14	14	14	14	Undercarriage
18	18	18	18	Cargo/Vehicle Parts Set-in-Motion
19	19	19	--	Other Objects Set-in-Motion
--	--	--	19	Other Objects or Person Set-in-Motion
--	20	20	20	Object Set in Motion, Unknown if Cargo/ Vehicle Parts or Other
55	55	55	55	Non-Harmful Event
61	61	61	61	Left
62	62	62	62	Left-Front Side
63	63	63	63	Left-Back Side
81	81	81	81	Right
82	82	82	82	Right-Front Side
83	83	83	83	Right-Back Side
98	98	98	98	Not Reported
99	99	--	--	Unknown
--	--	99	99	Reported as Unknown

V37. Sequence of Events

Definition

This data element describes this event. A motor vehicle traffic crash is a series of events resulting from an unstabilized situation. This series of harmful and non-harmful events is recorded in chronological order based on the police crash report narrative and diagram.

Additional Information

“First Harmful Event, Most Harmful Event,” and the “Sequence of Events” data elements have the same harmful event attributes. “Sequence of Events” also has non-harmful event attributes. Prior to 2020 the Data Element ID was V32.

SAS Name

SOE

Attribute Codes

Non-Harmful Events

2016	2017	2018-Later	
60	60	60	Cargo/Equipment Loss or Shift (Non-Harmful)
61	61	61	Equipment Failure (Blown Tire, Brake Failure, etc.)
62	62	62	Separation of Units
63	63	63	Ran off Roadway-Right
64	64	64	Ran off Roadway-Left
65	65	65	Cross Median
66	66	66	Downhill Runaway
67	67	67	Vehicle Went Airborne
68	68	68	Cross Centerline
69	69	69	Re-entering Roadway
70	70	70	Non-harmful, Swaying Trailer/Jackknife
71	71	71	End Departure
79	79	79	Ran off Roadway -Direction Unknown

Non-collision Harmful Events

2016	2017	2018-Later	
1	1	1	Rollover/Overturn
2	2	2	Fire/Explosion
3	3	3	Immersion or Partial Immersion
4	4	4	Gas Inhalation

2016	2017	2018-Later	
5	5	5	Fell/Jumped From Vehicle
6	6	6	Injured in Vehicle (Non-Collision)
7	7	7	Other Non-Collision
16	16	16	Thrown or Falling Object
44	44	44	Pavement Surface Irregularity (Ruts, Potholes, Grates, etc.)
51	51	51	Jackknife (Harmful to This Vehicle)
72	72	--	Cargo/Equipment Loss or Shift (Harmful to This Vehicle)
--	--	72	Cargo/Equipment Loss, Shift, or Damage (Harmful)

Collision with Motor Vehicle In-Transport

2016	2017	2018-Later	
12	12	12	Motor Vehicle In-Transport
54	54	54	Motor Vehicle In-Transport Strikes or Is Struck by Cargo, Persons or Objects Set-in-Motion From/by Another Motor Vehicle In-Transport
55	55	55	Motor Vehicle in Motion Outside the Trafficway

Collision with Object Not Fixed

2016	2017	2018-Later	
8	8	8	Pedestrian
9	9	9	Pedalcyclist
10	10	10	Railway Vehicle
11	11	11	Live Animal
14	14	14	Parked Motor Vehicle
15	15	15	Non-Motorist on Personal Conveyance
18	18	18	Other Object (Not Fixed)
45	45	45	Working Motor Vehicle
49	49	49	Ridden Animal or Animal Drawn Conveyance
73	73	73	Object That Had Fallen From Motor Vehicle In-Transport
74	74	74	Road Vehicle on Rails
--	91	91	Unknown Object Not Fixed

Collision with Fixed Object

2016	2017	2018-Later	
17	17	17	Boulder
19	19	19	Building
20	20	20	Impact Attenuator/Crash Cushion
21	21	21	Bridge Pier or Support
23	23	23	Bridge Rail (Includes Parapet)
24	24	24	Guardrail Face
25	25	25	Concrete Traffic Barrier
26	26	26	Other Traffic Barrier
30	30	30	Utility Pole/Light Support
31	31	31	Post, Pole, or Other Support
32	32	32	Culvert
33	33	33	Curb
34	34	34	Ditch
35	35	35	Embankment
38	38	38	Fence
39	39	39	Wall
40	40	40	Fire Hydrant
41	41	41	Shrubbery
42	42	42	Tree (Standing Only)
43	43	43	Other Fixed Object
46	46	46	Traffic Signal Support
48	48	48	Snow Bank
50	50	50	Bridge Overhead Structure
52	52	52	Guardrail End
53	53	53	Mail Box
57	57	57	Cable Barrier
58	58	58	Ground
59	59	59	Traffic Sign Support
--	93	93	Unknown Fixed Object
--	--	98	Harmful Event, Details Not Reported (Since 2019)
99	99	--	Unknown
--	--	99	Reported as Unknown

C18C. Vehicle Number (Other Vehicle)**Definition**

This data element identifies the “Vehicle Number” (VEH_NO) of the other motor vehicle, if any, in this event.

Additional Information

This is the vehicle contacted by the motor vehicle in-transport recorded in “Vehicle Number (This Vehicle).” Another vehicle must have been involved in this event for this data element to be a valid vehicle number (i.e., “Sequence of Events” for this event must be 12, 14, 45, 54, or 55).

SAS Name**VNUMBER2****Attribute Codes**

2016-Later	
1-999	Vehicle Number
5555	Non-Harmful Event
9999	Not a Motor Vehicle

C18D. Area of Impact (Other Vehicle)**Definition**

This data element identifies the impact point on the other motor vehicle, if any, in this event.

Additional Information

This is the impact area of the vehicle recorded in “Vehicle Number (Other Vehicle).” Another vehicle must have been involved in this event for this data element to be a valid impact location (i.e., “Sequence of Events” for this event must be 12, 14, 45, 54, or 55).

SAS Name**AOI2****Attribute Codes**

2016	2017	2018	2019-Later	
0	0	0	0	Non-Collision
1-12	1-12	1-12	1-12	Clock Points
13	13	13	13	Top
14	14	14	14	Undercarriage
18	18	18	18	Cargo/Vehicle Parts Set-in-Motion
19	19	19	--	Other Objects Set-in-Motion
--	--	--	19	Other Objects or Person Set-in-Motion
--	20	20	20	Object Set in Motion, Unknown if Cargo/Vehicle Parts or Other
55	55	55	55	Non-Harmful Event
61	61	61	61	Left
62	62	62	62	Left-Front Side
63	63	63	63	Left-Back Side
77	77	77	77	Not a Motor Vehicle
81	81	81	81	Right
82	82	82	82	Right-Front Side
83	83	83	83	Right-Back Side
98	98	98	98	Not Reported
99	99	--	--	Unknown
--	--	99	99	Reported as Unknown

The VSOE Data File

The Vsoe data file includes harmful and non-harmful events for each motor vehicle in-transport. It contains the data elements CASENUM, PSU, PJ, STRATUM, PSUSTRAT, PSU_VAR, REGION, URBANICITY, WEIGHT, VEH_NO, and VEVENTNUM, which are described in the Key Data Elements at the beginning of the Data Element Definitions and Codes section. The Vsoe data file also contains the data elements on the following pages.

CASENUM, VEH_NO, and VEVENTNUM are the unique identifiers for each record. CASENUM and VEH_NO should be used to merge the Vsoe data file with the Vehicle data file.

C18E. Area of Impact

Definition

This data element identifies the impact point, if any, on this motor vehicle in-transport that produced property damage or personal injury in this event.

Additional Information

This is the impact area of the vehicle recorded as “Vehicle Number (This Vehicle)” or “Vehicle Number (Other Vehicle)” in the crash events.

SAS Name

AOI

Attribute Codes

2016	2017	2018	2019-Later	
0	0	0	0	Non-Collision
1-12	1-12	1-12	1-12	Clock Points
13	13	13	13	Top
14	14	14	14	Undercarriage
18	18	18	18	Cargo/Vehicle Parts Set-in-Motion
19	19	19	--	Other Objects Set-in-Motion
--	--	--	19	Other Objects or Person Set-in-Motion
--	20	20	20	Object Set in Motion, Unknown if Cargo/ Vehicle Parts or Other
55	55	55	55	Non-Harmful Event
61	61	61	61	Left
62	62	62	62	Left-Front Side
63	63	63	63	Left-Back Side
81	81	81	81	Right
82	82	82	82	Right-Front Side
83	83	83	83	Right-Back Side
98	98	98	98	Not Reported
99	99	--	--	Unknown
--	--	99	99	Reported as Unknown

V37. Sequence of Events

Definition

The events in sequence related to this motor vehicle, regardless of injury and/or property damage. Events for the vehicle are recorded in the order in which they occur, time-wise, from the police crash report narrative and diagram.

Additional Information

“First Harmful Event,” “Most Harmful Event,” and the “Sequence of Events” data elements have the same harmful event attributes. “Sequence of Events” also has non-harmful event attributes. Prior to 2020 the Data Element ID was V32.

SAS Name

SOE

Attribute Codes

Non-Harmful Events

2016	2017	2018-Later	
60	60	60	Cargo/Equipment Loss or Shift (Non-Harmful)
61	61	61	Equipment Failure (Blown Tire, Brake Failure, etc.)
62	62	62	Separation of Units
63	63	63	Ran off Roadway-Right
64	64	64	Ran off Roadway-Left
65	65	65	Cross Median
66	66	66	Downhill Runaway
67	67	67	Vehicle Went Airborne
68	68	68	Cross Centerline
69	69	69	Re-entering Roadway
70	70	70	Non-harmful, Swaying Trailer/Jackknife
71	71	71	End Departure
79	79	79	Ran off Roadway -Direction Unknown

Non-collision Harmful Events

2016	2017	2018-Later	
1	1	1	Rollover/Overturn
2	2	2	Fire/Explosion
3	3	3	Immersion or Partial Immersion
4	4	4	Gas Inhalation

2016	2017	2018-Later	
5	5	5	Fell/Jumped From Vehicle
6	6	6	Injured in Vehicle (Non-Collision)
7	7	7	Other Non-Collision
16	16	16	Thrown or Falling Object
44	44	44	Pavement Surface Irregularity (Ruts, Potholes, Grates, etc.)
51	51	51	Jackknife (Harmful to This Vehicle)
72	72	--	Cargo/Equipment Loss or Shift (Harmful to This Vehicle)
--	--	72	Cargo/Equipment Loss, Shift, or Damage (Harmful)

Collision with Motor Vehicle In-Transport

2016	2017	2018-Later	
12	12	12	Motor Vehicle In-Transport
54	54	54	Motor Vehicle In-Transport Strikes or Is Struck by Cargo, Persons or Objects Set-in-Motion From/by Another Motor Vehicle In-Transport
55	55	55	Motor Vehicle in Motion Outside the Trafficway

Collision with Object Not Fixed

2016	2017	2018-Later	
8	8	8	Pedestrian
9	9	9	Pedalcyclist
10	10	10	Railway Vehicle
11	11	11	Live Animal
14	14	14	Parked Motor Vehicle
15	15	15	Non-Motorist on Personal Conveyance
18	18	18	Other Object (Not Fixed)
45	45	45	Working Motor Vehicle
49	49	49	Ridden Animal or Animal Drawn Conveyance
73	73	73	Object That Had Fallen From Motor Vehicle In-Transport
74	74	74	Road Vehicle on Rails
--	91	91	Unknown Object Not Fixed

Collision with Fixed Object

2016	2017	2018-Later	
17	17	17	Boulder
19	19	19	Building
20	20	20	Impact Attenuator/Crash Cushion
21	21	21	Bridge Pier or Support
23	23	23	Bridge Rail (Includes Parapet)
24	24	24	Guardrail Face
25	25	25	Concrete Traffic Barrier
26	26	26	Other Traffic Barrier
30	30	30	Utility Pole/Light Support
31	31	31	Post, Pole or Other Support
32	32	32	Culvert
33	33	33	Curb
34	34	34	Ditch
35	35	35	Embankment
38	38	38	Fence
39	39	39	Wall
40	40	40	Fire Hydrant
41	41	41	Shrubbery
42	42	42	Tree (Standing Only)
43	43	43	Other Fixed Object
46	46	46	Traffic Signal Support
48	48	48	Snow Bank
50	50	50	Bridge Overhead Structure
52	52	52	Guardrail End
53	53	53	Mail Box
57	57	57	Cable Barrier
58	58	58	Ground
59	59	59	Traffic Sign Support
--	93	93	Unknown Fixed Object
--	--	98	Harmful Event, Details Not Reported (Since 2019)
99	99	--	Unknown
--	--	99	Reported as Unknown

The CRASHRF Data File

The Crashrf data file identifies each crash related factor as a separate record. That is, there can be more than one record for each crash. It contains the data elements CASENUM, PSU, PJ, STRATUM, PSUSTRAT, PSU_VAR, REGION, URBANICITY, and WEIGHT, which are described in the Key Data Elements at the beginning of the Data Element Definitions and Codes section. The data file also contains CRASHRF that is described below.

CASENUM and CRASHRF are the unique identifiers for each record. CASENUM should be used to merge the Crashrf data file with the Accident data file.

C32. Related Factors—Crash Level

Definition

This data element records factors related to the crash expressed in the case materials.

Additional Information

There are also vehicle related factors in the Vehiclesf and Pvehiclesf data files, driver related factors in the Driverrf data file, and person related factors in the Personrf data file.

Prior to 2020 this data element was collected at the Crash level and up to three factors could be selected. These three elements were discontinued and moved to the Discontinued Accident Data Elements at the end of the Accident Data File section. Refer to the discontinued element for a history of this data element's attributes.

SAS Name

CRASHRF

Attribute Codes

2020	2021	2022	2023-Later	
0	0	--	--	None
--	--	0	0	None Noted
3	3	3	3	Other Maintenance or Construction-Created Condition
5	5	5	5	Surface Under Water
7	7	7	7	Surface Washed out (Caved in, Road Slippage)
--	10	10	10	Emergency Vehicle Related
12	12	12	12	Distracted Driver of a Non-Contact Vehicle
13	13	13	--	Aggressive Driving/Road Rage by Non-Contact Vehicle Driver
14	14	14	14	Motor Vehicle Struck by Falling Cargo or Something That Came Loose From or Something That Was Set in Motion by a Vehicle
15	15	15	15	Non-Occupant Struck by Falling Cargo, or Something Came Loose From or Something That Was Set in Motion by a Vehicle
16	16	16	16	Non-Occupant Struck Vehicle
17	17	--	--	Vehicle Set in Motion by Non-Driver
--	--	17	17	Stopped Vehicle Set in Motion by Non-Driver
19	19	19	19	Recent Previous Crash Scene Nearby
20	20	20	20	Police-Pursuit-Involved
21	21	21	21	Within Designated School Zone

2020	2021	2022	2023- Later	
23	23	23	23	Indication of a Stalled/Disabled Vehicle
24	24	24	24	Unstabilized Situation Began and All Harmful Events Occurred off of the Roadway
25	25	25	25	Toll Booth/Plaza Related
26	26	26	26	Prior Non-Recurring Incident
27	27	27	27	Backup Due to Prior Crash
28	28	28	28	Regular Congestion
30	30	30	30	Obstructed Crosswalks
31	31	31	31	Related to a Bus Stop
--	--	--	102	Aggressive Driving by Non-Contact Vehicle Driver
--	--	--	103	Road Rage by Non-Contact Vehicle Driver
999	999	--	--	Reported as Unknown

The WEATHER Data File

The Weather data file identifies each atmospheric condition as a separate record. That is, there can be more than one record for each crash. It contains the data elements CASENUM, PSU, PJ, STRATUM, PSUSTRAT, PSU_VAR, REGION, URBANICITY, and WEIGHT, which are described in the Key Data Elements at the beginning of the Data Element Definitions and Codes section. The data file also contains WEATHER that is described below.

CASENUM and WEATHER are the unique identifiers for each record. CASENUM should be used to merge the Weather data file with the Accident data file.

C26. Atmospheric Conditions

Definition

This data element records the prevailing atmospheric conditions that existed at the time of the crash as indicated in the police crash report.

Additional Information

Prior to 2020 this data element identified up to two values. If more than two atmospheric conditions were reported, the two conditions that most affect visibility were selected. Accident.WEATHER1 and Accident.WEATHER2 were the coded data elements, and Accident.WEATHER was derived from these two. The two coded data elements were discontinued after 2019 and moved to the Discontinued Accident Data Elements at the end of the Accident Data File section.

Beginning in 2020 all applicable atmospheric conditions are selected and stored in this data file. Only the derived data element WEATHER is still stored in the Accident data file and is now derived from the multiple responses in this data file using the same hierarchy.

SAS Name

WEATHER

Attribute Codes

2020-Later	
1	Clear
2	Rain
3	Sleet or Hail
4	Snow
5	Fog, Smog, Smoke
6	Severe Crosswinds
7	Blowing Sand, Soil, Dirt
8	Other
10	Cloudy
11	Blowing Snow
12	Freezing Rain or Drizzle
98	Not Reported
99	Reported as Unknown

The VEHICLESF Data File

The Vehiclesf data file identifies each vehicle related factor for a motor vehicle in-transport as a separate record. That is, there can be more than one record for each vehicle. It contains the data elements CASENUM, PSU, PJ, STRATUM, PSUSTRAT, PSU_VAR, REGION, URBANICITY, WEIGHT, and VEH_NO, which are described in the Key Data Elements at the beginning of the Data Element Definitions and Codes section. The data file also contains VEHICLESF that is described below.

CASENUM, VEH_NO, and VEHICLESF are the unique identifiers for each record. CASENUM and VEH_NO should be used to merge the Vehiclesf data file with vehicles from the Vehicle data file.

V41. Related Factors—Vehicle Level (Motor Vehicles In-Transport)

Definition

This data element records factors related to this motor vehicle in-transport expressed in the case materials.

Additional Information

There are also crash related factors in the Crashrf data file, vehicle related factors in the Pvehiclesf data file (for parked/working vehicles), driver related factors in the Driverrf data file, and person related factors in the Personrf data file.

Pre-existing vehicle defects are captured in the data element “Contributing Circumstances, Motor Vehicle” (Factor.MFACTOR).

Prior to 2020 this data element’s ID was V36 and it was collected at the Vehicle level with up to two factors being selected. These two elements were discontinued and moved to the Discontinued Vehicle Data Elements at the end of the Vehicle Data File section. Refer to the discontinued element for a history of this data element’s attributes.

SAS Name

VEHICLESF

Attribute Codes

2020-2021	2022-Later	
0	--	None
--	0	None Noted
29	29	Default Code Used for Vehicle Numbering
30	30	Multi-Wheeled Motorcycle Conversion
33	33	Vehicle Being Pushed by Non-Motorist
35	35	Reconstructed/Altered Vehicle
39	39	Highway Construction, Maintenance or Utility Vehicle, In-Transport (Inside or Outside Work Zone)
41	41	Police Fire or EMS Vehicle Working at the Scene of an Emergency or Performing Other Traffic Control Activities
42	42	Other Working Vehicle (Not Construction, Maintenance, Utility, Police, Fire, or EMS Vehicle)
44	44	Adaptive Equipment
45	45	Slide-in Camper
999	--	Reported as Unknown

The PVEHICLESF Data File

The Pvehiclesf data file identifies each vehicle related factor for a parked/working motor vehicle as a separate record. That is, there can be more than one record for each vehicle. It contains the data elements CASENUM, PSU, PJ, STRATUM, PSUSTRAT, PSU_VAR, REGION, URBANICITY, WEIGHT, and VEH_NO, which are described in the Key Data Elements at the beginning of the Data Element Definitions and Codes section. The data file also contains PVEHICLESF that is described below.

CASENUM, VEH_NO, and PVEHICLESF are the unique identifiers for each record. CASENUM and VEH_NO should be used to merge the Pvehiclesf data file with vehicles from the Vehicle data file.

V41. Related Factors—Vehicle Level (Parked/Working Vehicles)

Definition

This data element records factors related to this parked/working motor vehicle expressed in the case materials.

Additional Information

There are also crash related factors in the Crashrf data file, vehicle related factors in the Vehiclesf data file (for motor vehicles in-transport), driver related factors in the Driverrf data file, and person related factors in the Personrf data file.

Prior to 2020 this data element's ID was V36 and it was collected at the Vehicle level with up to two factors being selected. These two elements were discontinued and moved to the Discontinued Parkwork Data Elements at the end of the Parkwork Data File section. Refer to the discontinued element for a history of this data element's attributes.

SAS Name

PVEHICLESF

Attribute Codes

2020-2021	2022-Later	
0	--	None
--	0	None Noted
29	29	Default Code Used for Vehicle Numbering
30	30	Multi-Wheeled Motorcycle Conversion
33	33	Vehicle Being Pushed by Non-Motorist
35	35	Reconstructed/Altered Vehicle
39	39	Highway Construction, Maintenance or Utility Vehicle, In-Transport (Inside or Outside Work Zone)
41	41	Police Fire or EMS Vehicle Working at the Scene of an Emergency or Performing Other Traffic Control Activities
42	42	Other Working Vehicle (Not Construction, Maintenance, Utility, Police, Fire, or EMS Vehicle)
44	44	Adaptive Equipment
45	45	Slide-in Camper
999	--	Reported as Unknown

The DRIVERRF Data File

The Driverrf data file identifies each driver related factor as a separate record. That is, there can be more than one record for each driver. It contains the data elements CASENUM, PSU, PJ, STRATUM, PSUSTRAT, PSU_VAR, REGION, URBANICITY, WEIGHT, and VEH_NO, which are described in the Key Data Elements at the beginning of the Data Element Definitions and Codes section. The data file also contains DRIVERRF that is described below.

CASENUM, VEH_NO, and DRIVERRF are the unique identifiers for each record. CASENUM and VEH_NO should be used to merge the Driverrf data file with drivers from the Vehicle data file.

D24. Related Factors—Driver Level

Definition

This data element records factors related to this driver expressed in the case materials.

Additional Information

There are also crash related factors in the Crashrf data file, vehicle related factors in the Vehiclesf and Pvehiclesf data files, and person related factors in the Personrf data file.

Person related factors are all set to 0 for drivers.

Prior to 2020 this data element was collected at the Vehicle level and up to four factors could be selected. These four elements were discontinued and moved to the Discontinued Vehicle Data Elements at the end of the Vehicle Data File section. Refer to the discontinued element for a history of this data element's attributes.

SAS Name

DRIVERRF

Attribute Codes

2020	2021	2022	2023-Later	
0	0	--	--	None
--	--	0	0	None Noted
6	6	6	6	Careless Driving, Inattentive Operation, Improper Driving, Driving Without Due Care
8	8	8	--	Road Rage/Aggressive Driving
10	10	10	10	Looked but Did Not See
16	16	16	16	Police or Law Enforcement Officer
18	18	18	18	Traveling on Prohibited Trafficways
20	20	20	20	Leaving Vehicle Unattended With Engine Running; Leaving Vehicle Unattended in Roadway
21	21	21	21	Overloading or Improper Loading of Vehicle With Passenger or Cargo
22	22	22	22	Towing or Pushing Vehicle Improperly
23	23	23	23	Failing to Dim Lights or to Have Lights on When Required
24	24	24	24	Operating Without Required Equipment
29	29	29	29	Intentional Illegal Driving off the Roadway
32	32	32	32	Opening Vehicle Closure Into Moving Traffic or Vehicle Is in Motion

2020	2021	2022	2023- Later	
36	36	36	36	Operating the Vehicle in an Erratic, Reckless, Careless or Negligent Manner
37	37	37	--	Police Pursuing This Driver or Police Officer in Pursuit (See Police Pursuits in Appendix C: Additional Data Element Information)
50	50	50	50	Driving Wrong Way on One-Way Trafficway
51	51	51	51	Driving on Wrong Side of Two-way Trafficway (Intentionally or Unintentionally)
54	54	54	54	Stopping in Roadway (Vehicle Not Abandoned)
55	55	55	55	Improper Management of Vehicle Controls
56	56	56	56	Object Interference With Vehicle Controls
57	57	57	57	Driving With Tire-Related Problems
58	58	58	58	Over Correcting
59	59	59	59	Getting off/out of a Vehicle
60	60	60	60	Alcohol and/or Drug Test Refused
91	--	--	--	Non-Traffic Violation Charged (Manslaughter, Homicide or Other Assault Offense Committed Without Malice)
94	94	94	94	Emergency Medical Service Personnel
95	95	95	95	Fire Personnel
96	96	96	96	Tow Operator
97	97	97	97	Transportation (i.e., Maintenance Workers, Safety Service Patrol Operators, etc.)
--	--	--	100	Using a Belt-Positioning Device or Other
--	--	--	102	Aggressive Driving
--	--	--	103	Road Rage
--	--	--	104	Police Pursuing This Driver (See Police Pursuits in Appendix C: Additional Data Element Information)
--	--	--	105	Police Officer in Pursuit (See Police Pursuits in Appendix C: Additional Data Element Information)
999	999	--	--	Reported as Unknown

The DAMAGE Data File

The Damage data file identifies each area of damage as a separate record. That is, there can be more than one damage record for each vehicle. It contains the data elements CASENUM, PSU, PJ, STRATUM, PSUSTRAT, PSU_VAR, REGION, URBANICITY, WEIGHT, and VEH_NO, which are described in the Key Data Elements at the beginning of the Data Element Definitions and Codes section. The data file also contains DAMAGE that is described below.

CASENUM, VEH_NO, and DAMAGE are the unique identifiers for each record. CASENUM and VEH_NO should be used to merge the Damage data file with vehicles from the Vehicle data file.

V34B. Areas of Impact—Damaged Areas**Definition**

This data element identifies all the areas on this vehicle that were damaged in the crash as reflected in the case materials.

Additional Information

Prior to 2020 the Data Element ID was V29B.

SAS Name

MDAREAS 2016-2019

DAMAGE 2020-Later

Attribute Codes

2016-Later	
1-12	Clock Points
13	Top
14	Undercarriage
15	No Damage
99	Damage Areas Unknown

The DISTRACT Data File

The Distract data file identifies each driver distraction as a separate record. That is, there can be more than one distraction record for each driver. It contains the data elements CASENUM, PSU, PJ, STRATUM, PSUSTRAT, PSU_VAR, REGION, URBANICITY, WEIGHT, and VEH_NO, which are described in the Key Data Elements at the beginning of the Data Element Definitions and Codes section. The data file also contains DRDISTRACT that is described below.

CASENUM, VEH_NO, and DRDISTRACT are the unique identifiers for each record. CASENUM and VEH_NO should be used to merge the Distract data file with drivers from the Vehicle data file.

PC16. Driver Distracted By

Definition

This data element identifies the attributes that best describe this driver's attention to driving prior to the driver's realization of an impending critical event or just prior to impact if realization of an impending critical event does not occur. This element reports on the presence of any distractions that may or may not have contributed to the crash.

Additional Information

Distraction from the primary task of driving occurs when drivers divert their attention from the driving task to some other activity. Also, driving while daydreaming or lost in thought is identified as distracted driving by NHTSA. Physical conditions/impairments (fatigue, alcohol, medical condition, etc.) or psychological states (anger, emotional, depressed, etc.) are not identified as distractions by NHTSA.

SAS Name

MDRDSTRD 2016-2019

DRDISTRACT 2020-Later

Attribute Codes

2016-2017	2018-Later	
0	0	Not Distracted
1	--	Looked But Did Not See
3	3	By Other Occupant(s)
4	4	By a Moving Object in Vehicle
5	5	While Talking or Listening to Mobile Phone
6	6	While Manipulating Mobile Phone
7	7	While Adjusting Audio or Climate Controls
9	9	While Using Other Component/Controls Integral to Vehicle
10	10	While Using or Reaching for Device/Object Brought Into Vehicle
12	12	Distracted By Outside Person, Object or Event
13	13	Eating or Drinking
14	14	Smoking Related
15	15	Other Mobile Phone Related
16	16	No Driver Present/Unknown if Driver Present
17	17	Distraction/Inattention
18	18	Distraction/Careless
19	19	Careless/Inattentive
92	92	Distraction (Distracted), Details Unknown

2016-2017	2018-Later	
93	93	Inattention (Inattentive), Details Unknown
96	96	Not Reported
97	97	Lost in Thought/Daydreaming
98	98	Other Distraction
99	--	Unknown if Distracted
--	99	Reported as Unknown if Distracted

The DRIMPAIR Data File

The Drimpair data file identifies each driver impairment as a separate record. That is, there can be more than one impairment record for each driver. It contains the data elements CASENUM, PSU, PJ, STRATUM, PSUSTRAT, PSU_VAR, REGION, URBANICITY, WEIGHT, and VEH_NO, which are described in the Key Data Elements at the beginning of the Data Element Definitions and Codes section. The data file also contains DRIMPAIR that is described below.

CASENUM, VEH_NO, and DRIMPAIR are the unique identifiers for each record. CASENUM and VEH_NO should be used to merge the Drimpair data file with drivers from the Vehicle data file.

D23. Condition (Impairment) at Time of Crash—Driver**Definition**

This data element identifies physical impairments to this driver that may have contributed to the crash as identified by law enforcement.

Additional Information**SAS Name****DRIMPAIR****Attribute Codes**

2016	2017	2018-2020	2021-Later	
0	0	0	0	None/Apparently Normal
1	1	1	1	Ill, Blackout
2	2	2	2	Asleep or Fatigued
3	3	3	3	Walking With a Cane or Crutches, etc.
4	--	--	--	Paraplegic or Restricted to Wheelchair
--	4	4	4	Paraplegic or in a Wheelchair
5	5	5	5	Impaired Due to Previous Injury
6	6	6	--	Deaf
--	--	--	6	Deaf/Hard of Hearing
7	7	7	--	Blind
--	--	--	7	Blind/Low Vision
8	8	8	8	Emotional (Depressed, Angry, Disturbed, etc.)
9	9	9	9	Under the Influence of Alcohol, Drugs, or Medication
10	10	10	10	Physical Impairment – No Details
95	95	95	95	No Driver Present/Unknown if Driver Present
96	96	96	96	Other Physical Impairment
98	98	98	98	Not Reported
99	99	--	--	Unknown if Impaired
--	--	99	99	Reported as Unknown if Impaired

The FACTOR Data File

The Factor data file identifies each vehicle factor as a separate record. That is, there can be more than one factor record for each vehicle. It contains the data elements CASENUM, PSU, PJ, STRATUM, PSUSTRAT, PSU_VAR, REGION, URBANICITY, WEIGHT, and VEH_NO, which are described in the Key Data Elements at the beginning of the Data Element Definitions and Codes section. The data file also contains VEHICLECC that is described below.

CASENUM, VEH_NO, and VEHICLECC are the unique identifiers for each record. CASENUM and VEH_NO should be used to merge the Factor data file with vehicles from the Vehicle data file.

PC4. Contributing Circumstances, Motor Vehicle**Definition**

This data element describes this vehicle's possible pre-existing defects or maintenance conditions that may have contributed to the crash.

Additional Information**SAS Name**

MFACTOR 2016-2019

VEHICLECC 2020-Later

Attribute Codes

2016-2017	2018-2019	2020-Later	
0	0	--	None
--	--	0	None Noted
1	1	1	Tires
2	2	2	Brake System
3	3	3	Steering System
4	4	4	Suspension
5	5	5	Power Train
6	6	6	Exhaust System
7	7	7	Headlights
8	8	8	Signal Lights
9	9	9	Other Lights
10	10	10	Wipers
11	11	11	Wheels
12	12	12	Mirrors
13	13	13	Windows/Windshield
14	14	14	Body, Doors
15	15	15	Truck Coupling/Trailer Hitch/Safety Chains
16	16	16	Safety Systems
17	17	17	Vehicle Contributing Factors-No Details
97	97	97	Other
98	98	--	Not Reported
99	--	--	Unknown
--	99	99	Reported as Unknown

The MANEUVER Data File

The Maneuver data file identifies each avoidance attempt as a separate record. That is, there can be more than one maneuver record for each driver. It contains the data elements CASENUM, PSU, PJ, STRATUM, PSUSTRAT, PSU_VAR, REGION, URBANICITY, WEIGHT, and VEH_NO, which are described in the Key Data Elements at the beginning of the Data Element Definitions and Codes section. The data file also contains MANEUVER that is described below.

CASENUM, VEH_NO, and MANEUVER are the unique identifiers for each record. CASENUM and VEH_NO should be used to merge the Maneuver data file with vehicles from the Vehicle data file.

PC15. Driver Maneuvered to Avoid**Definition**

This data element identifies the things this driver attempted to avoid while the vehicle was on the road portion of the trafficway, just prior to the first harmful event for this vehicle.

Additional Information**SAS Name**

MDRMANAV 2016-2019

MANEUVER 2020-Later

Attribute Codes

2016-2017	2018-2019	2020-Later	
0	0	0	Driver Did Not Maneuver to Avoid
1	1	1	Object
2	2	2	Poor Road Conditions (Puddle, Ice, Pot Hole, etc.)
3	3	3	Live Animal
4	4	--	Motor Vehicle
--	--	4	Contact Motor Vehicle (in this Crash)
5	5	5	Pedestrian, Pedalcyclist, or Other Non-Motorist
92	92	92	Phantom/Non-Contact Motor Vehicle
95	95	95	No Driver Present/Unknown if Driver Present
98	98	98	Not Reported
99	--	--	Unknown
--	99	99	Reported as Unknown

The VIOLATN Data File

The Violatn data file identifies each violation as a separate record. That is, there can be more than one violation record for each driver. It contains the data elements CASENUM, PSU, PJ, STRATUM, PSUSTRAT, PSU_VAR, REGION, URBANICITY, WEIGHT, and VEH_NO, which are described in the Key Data Elements at the beginning of the Data Element Definitions and Codes section. The data file also contains VIOLATION that is described below.

CASENUM, VEH_NO, and VIOLATION are the unique identifiers for each record. CASENUM and VEH_NO should be used to merge the Violatn data file with drivers from the Vehicle data file.

D21. Violations Charged

Definition

This element documents the violations, citations, or infractions of the Vehicle Code issued on the police crash report for this driver in this crash, regardless of whether the driver survived the crash.

Additional Information

SAS Name

MVIOLATN 2016-2019

VIOLATION 2020-Later

Attribute Codes

2016-2019	2020-Later	
0	0	None

Reckless/Careless/Hit-and-Run Type Offenses

2016-2019	2020-Later	
1	1	Manslaughter or Homicide
2	2	Willful Reckless Driving; Driving to Endanger; Negligent Driving
3	3	Unsafe Reckless (Not Willful, Wanton Reckless) Driving
4	--	Inattentive, Careless, Improper Driving
--	4	Inattentive, Careless, Improper Driving, Driving Without Due Care
5	5	Fleeing or Eluding Police
6	6	Fail to Obey Police, Fireman, Authorized Person Directing Traffic
7	7	Hit-And-Run, Fail to Stop After Crash
8	8	Fail to Give Aid, Information, Wait for Police After Crash
9	9	Serious Violation Resulting in Death
10	10	Use of Telecommunications Device

Impairment Offenses

2016-2019	2020-Later	
11	11	Driving While Intoxicated (Alcohol or Drugs) or BAC Above Limit (Any Detectable BAC for CDLs)
12	12	Driving While Impaired
13	13	Driving Under Influence of Substance Not Intended to Intoxicate

2016-2019	2020-Later	
14	14	Drinking While Operating
15	15	Illegal Possession of Alcohol or Drugs
16	16	Driving With Detectable Alcohol
18	18	Refusal to Submit to Chemical Test
19	19	Alcohol, Drug or Impairment Violations Generally

Speed-Related Offenses

2016-2019	2020-Later	
21	21	Racing
22	22	Speeding (Above the Speed Limit)
23	23	Speed Greater Than Reasonable and Prudent (Not Necessarily Over the Limit)
24	24	Exceeding Special Limit (for Trucks, Buses, Cycles, or on Bridge, in School Zone, etc.)
25	25	Energy Speed (Exceeding 55 mph, Non-Pointable)
26	26	Driving Too Slowly
29	29	Speed Related Violations, Generally

Rules of The Road – Traffic Sign and Signals

2016-2019	2020-Later	
31	31	Fail to Stop for Red Signal
32	32	Fail to Stop for Flashing Red
33	33	Violation of Turn on Red (Fail to Stop and Yield, Yield to Pedestrians Before Turning)
34	34	Fail to Obey Flashing Signal (Yellow or Red)
35	35	Fail to Obey Signal, Generally
36	36	Violate RR Grade Crossing Device/Regulations
37	37	Fail to Obey Stop Sign
38	38	Fail to Obey Yield Sign
39	39	Fail to Obey Traffic Control Device

Rules of The Road – Turning, Yielding, Signaling

2016-2019	2020-Later	
41	41	Turn in Violation of Traffic Control (Disobey Signs, Turn Arrow or Pavement Markings; This Is Not a Right-on-Red Violation)
42	42	Improper Method and Position of Turn (Too Wide, Wrong Lane)
43	43	Fail to Signal for Turn or Stop

2016-2019	2020-Later	
45	45	Fail to Yield to Emergency Vehicle
46	46	Fail to Yield, Generally
48	48	Enter Intersection When Space Insufficient
49	49	Turn, Yield, Signaling Violations, Generally

Rules of The Road – Wrong Side, Passing and Following

2016-2019	2020-Later	
51	51	Driving Wrong Way on One-Way Road
52	52	Driving on Left, Wrong Side of Road, Generally
53	53	Improper, Unsafe Passing
54	54	Pass on Right (Drive off Pavement to Pass)
55	55	Pass Stopped School Bus
56	56	Fail to Give Way When Overtaken
58	58	Following Too Closely
59	59	Wrong Side, Passing, Following Violations, Generally

Rules of The Road – Lane Usage

2016-2019	2020-Later	
61	61	Unsafe or Prohibited Lane Change
62	62	Improper Use of Lane (Enter of 3-Lane Road, HOV Designated Lane)
63	63	Certain Traffic to Use Right Lane (Trucks, Slow-Moving, etc.)
66	66	Motorcycle Lane Violations (More Than Two per Lane, Riding Between Lanes, etc.)
67	67	Motorcyclist Attached to Another Vehicle
69	69	Lane Violations, Generally

Non-Moving – License And Registration Violations

2016-2019	2020-Later	
71	71	Driving While License Withdrawn
72	72	Other Driver License Violations
73	73	Commercial Driver Violations (Log Book, Hours, Permits Carried)
74	74	Vehicle Registration Violations
75	78	Fail to Carry Insurance Card
76	76	Driving Uninsured Vehicle
79	79	Non-Moving Violations, Generally

Equipment

2016-2019	2020-Later	
81	81	Lamp Violations
82	82	Brake Violations
83	83	Failure to Require Restraint Use (By Self or Passengers)
84	84	Motorcycle Equipment Violations (Helmet, Special Equipment)
85	85	Violation of Hazardous Cargo Regulations
86	86	Size, Weight, Load Violations
89	89	Equipment Violations, Generally

License, Registration and Other Violations

2016-2019	2020-Later	
91	91	Parking
92	92	Theft, Unauthorized Use of Motor Vehicle
93	93	Driving Where Prohibited (Sidewalk, Limited Access, off Truck Route)
95	95	No Driver Present/Unknown if Driver Present
97	97	Not Reported
98	98	Other Moving Violation (Coasting, Backing, Opening Door)
99	99	Unknown Violations

The VISION Data File

The Vision data file identifies each visual obstruction as a separate record. That is, there can be more than one vision record for each driver. It contains the data elements CASENUM, PSU, PJ, STRATUM, PSUSTRAT, PSU_VAR, REGION, URBANICITY, WEIGHT, and VEH_NO, which are described in the Key Data Elements at the beginning of the Data Element Definitions and Codes section. The data file also contains VISION that is described below.

CASENUM, VEH_NO, and VISION are the unique identifiers for each record. CASENUM and VEH_NO should be used to merge the Vision data file with drivers from the Vehicle data file.

PC14. Driver's Vision Obscured By**Definition**

This data element records impediments to this driver's visual field that were noted in the case materials.

Additional Information**SAS Name**

MVISOBSC 2016-2019

VISIO 2020-Late

Attribute Codes

2016-2017	2018-Later	
0	0	No Obstruction Noted
1	1	Rain, Snow, Fog, Smoke, Sand, Dust
2	2	Reflected Glare, Bright Sunlight, Headlights
3	3	Curve, Hill, or Other Roadway Design Features
4	4	Building, Billboard, or Other Structure
5	5	Trees, Crops, Vegetation
6	6	In-Transport Motor Vehicle (Including Load)
7	7	Not In-Transport Motor Vehicle (Parked, Working)
8	8	Splash or Spray of Passing Vehicle
9	9	Inadequate Defrost or Defog System
10	10	Inadequate Vehicle Lighting System
11	11	Obstructing Interior to the Vehicle
12	12	External Mirrors
13	13	Broken or Improperly Cleaned Windshield
14	14	Obstructing Angles on Vehicle
95	95	No Driver Present/Unknown if Driver Present
97	97	Vision Obscured – No Details
98	98	Other Visual Obstruction
99	--	Unknown
--	99	Reported as Unknown

The PERSONRF Data File

The Personrf data file identifies each person related factor for motorists and non-motorists as a separate record. That is, there can be more than one record for each person. It contains the data elements CASENUM, PSU, PJ, STRATUM, PSUSTRAT, PSU_VAR, REGION, URBANICITY, WEIGHT, VEH_NO, and PER_NO, which are described in the Key Data Elements at the beginning of the Data Element Definitions and Codes section. The data file also contains PERSONRF that is described below.

CASENUM, VEH_NO, PER_NO, and PERSONRF are the unique identifiers for each record. CASENUM, VEH_NO, and PER_NO should be used to merge the Personrf data file with motorists and non-motorists from the Person data file. VEH_NO equals 0 for non-motorists in this data file.

P24/NM26. Related Factors—Person Level**Definition**

This data element records factors related to motor vehicle occupants (other than drivers) and people not in motor vehicles as expressed in the case materials.

Additional Information

Person related factors are all set to 0 for drivers.

There are also crash related factors in the Crashrf data file, vehicle related factors in the Vehiclesf and Pvehiclesf data files, and driver related factors in the Driverrf data file.

Attributes with a single asterisk (*) are only applicable to occupants (other than drivers) of motor vehicles. Attributes with a double asterisk (**) are only applicable to people not in motor vehicles.

Prior to 2020 this data element was collected at the Person level, and up to three factors could be selected. These three elements were discontinued and moved to the Discontinued Person Data Elements at the end of the Person Data File section. Refer to the discontinued element for a history of this data element's attributes.

Prior to 2022 this data element ID was P26/NM26.

SAS Name**PERSONRF****Attribute Codes**

2020	2021	2022	2023-Later	
0	0	--	--	None/Not Applicable-Driver
--	--	0	0	None Noted
5	5	5	5	Interfering With Driver*
9	9	9	9	Construction/Maintenance/Utility Worker
10	10	10	10	Alcohol and/or Drug Test Refused
13	13	13	13	Motorized Wheelchair Rider**
21	21	21	21	Overloading or Improper Loading of Vehicle With Passengers or Cargo
31	31	31	31	Default Code Used for Vehicle Numbering**
32	32	32	32	Opening Vehicle Closure Into Moving Traffic or While Vehicle Is in Motion*
53	53	53	53	Non-Motorist Previously Used a Motor Vehicle for Motion**
54	54	54	54	Non-Motorist Attempting to Use a Motor Vehicle for Motion**

2020	2021	2022	2023- Later	
55	55	55	55	Non-Motorist Attempting to Use or Previously Used a Motor Vehicle for Motion, Details Not Reported**
56	--	--	--	Non-Driver Flees Scene
--	56	56	56	Non-Operator Flees Scene
87	87	87	87	Police or Law Enforcement Officer
89	89	89	89	Parked Motor Vehicle With Equipment Extending Into the Travel Lane
90	90	90	90	Non-Motorist Pushing a Vehicle**
91	91	91	91	Portable Electronic Devices
92	92	92	92	Person in Ambulance Treatment Compartment*
93	93	93	93	Non-Motorist Wearing Motorcycle Helmet**
94	94	94	94	Emergency Medical Services Personnel
95	95	95	95	Fire Personnel
96	96	96	96	Tow Operator
97	97	97	97	Transportation (Maintenance Workers, Safety Service Patrol Operators, etc.)
100	100	100	100	Using a Shared Micromobility Device**
101	101	101	101	Obstructed Sidewalk (for this Person)**
--	--	--	102	Motor Vehicle Occupant in Prior Crash**
--	--	--	103	Road Rage**
--	--	--	104	Using a Belt-Positioning Device*
--	--	--	105	Paraplegic or in a Wheelchair*
999	999	--	--	Reported as Unknown

The NMCRASH Data File

The Nmcraash data file identifies each non-motorist action or circumstance that may have contributed to the crash as a separate record. That is, there can be more than one contributing circumstance record for each non-motorist. It contains the data elements CASENUM, PSU, PJ, STRATUM, PSUSTRAT, PSU_VAR, REGION, URBANICITY, WEIGHT, VEH_NO, and PER_NO, which are described in the Key Data Elements at the beginning of the Data Element Definitions and Codes section. The data file also contains NMCC that is described below.

CASENUM, PER_NO, and NMCC are the unique identifiers for each record. CASENUM, VEH_NO, and PER_NO should be used to merge the Nmcraash data file with non-motorists from the Person data file. VEH_NO equals 0 for all records in this data file.

NM14. Non-Motorist Contributing Circumstances

Definition

This data element describes the actions and/or circumstances of this non-motorist that law enforcement indicated may have contributed to the crash.

Additional Information

Prior to 2022 the Data Element ID was NM12.

SAS Name

MTM_CRSH 2016-2019

NMCC 2020-Later

Attribute Codes

2016-2017	2018	2019-2020	2021-Later	
0	0	0	0	None Noted
1	1	--	--	Dart-out
--	--	1	1	Dart-out – Visual Obstruction Noted
2	2	2	2	Failure to Yield Right-of-Way
3	3	3	3	Failure to Obey Traffic Signs, Signals, or Officer
4	4	4	4	In Roadway Improperly (Standing, Lying, Working, Playing)
5	5	5	5	Entering/Exiting Parked or Stopped Vehicle
6	6	6	6	Inattentive (Talking, Eating, etc.)
7	7	7	7	Improper Turn/Merge
8	8	8	8	Improper Passing
9	9	9	9	Wrong-Way Riding or Walking
10	10	10	10	Riding on Wrong Side of Road
11	11	--	--	Dash
--	--	11	11	Dash – Run, No Visual Obstruction Noted
12	12	12	12	Improper Crossing of Roadway or Intersection (Jaywalking)
13	13	13	13	Failing to Have Lights on When Required
14	14	14	14	Operating Without Required Equipment
15	15	15	15	Improper or Erratic Lane Changing
16	16	16	16	Failure to Keep in Proper Lane or Running off Road
17	17	17	17	Making Improper Entry to or Exit From Trafficway

2016-2017	2018	2019-2020	2021-Later	
18	18	18	18	Operating in Other Erratic, Reckless, Careless or Negligent Manner
19	19	19	19	Not Visible (Dark Clothing, No Lighting, etc.)
20	20	20	20	Passing With Insufficient Distance or Inadequate Visibility or Failing to Yield to Overtaking Vehicle
21	21	21	21	Other
--	--	--	92	Contributing Circumstance -No Details
99	--	--	--	Unknown
--	99	99	99	Reported as Unknown

The NMDISTRACT Data File

The Nmdistract data file identifies each non-motorist distraction as a separate record. That is, there can be more than one distraction record for each non-motorist. It contains the data elements CASENUM, PSU, PJ, STRATUM, PSUSTRAT, PSU_VAR, REGION, URBANICITY, WEIGHT, VEH_NO, and PER_NO, which are described in the Key Data Elements at the beginning of the Data Element Definitions and Codes section. The data file also contains NMDISTRACT that is described below.

CASENUM, PER_NO, and NMDISTRACT are the unique identifiers for each record. CASENUM, VEH_NO, and PER_NO should be used to merge the Nmdistract data file with non-motorists from the Person data file. VEH_NO equals 0 for all records in this data file.

NM15. Non-Motorist Distracted By

Definition

This data element identifies the attributes that best describe this non-motorist's attention prior to the non-motorist's involvement in this crash. This element reports on the presence of any distractions that may or may not have contributed to the crash.

Additional Information

Distraction, for a non-motorist, occurs when a non-motorist's attention is diverted from the task of navigating in public to some other activity. Also, daydreaming or lost in thought are identified as distractions by NHTSA. Physical conditions/impairments (fatigue, alcohol, medical condition, etc.) or psychological states (anger, emotional, depressed, etc.) are not identified as distractions by NHTSA.

Prior to 2022 the Data Element ID was NM13.

SAS Name

MNMDSTRD 2019

NMDISTRACT 2020-Later

Attribute Codes

2019-Later	
0	Not Distracted
2	By Other Non-Motorist(s)
3	By a Driver or Occupant of a Motor Vehicle
5	While Talking or Listening to Mobile Phone
6	While Manipulating Mobile Phone
7	Adjusting or Listening to Portable Audio Device (Other Than on a Mobile Phone)
8	Adjusting, Talking to, or Manipulating Other Portable Electronic Device
12	Distracted by Animal, Other Object, Event, or Activity
13	Eating or Drinking
14	Smoking Related
15	Other Mobile Phone Related
17	Distraction/Inattention
18	Distraction/Careless
19	Careless/Inattentive
92	Distraction (Distracted), Details Unknown
93	Inattention (Inattentive), Details Unknown
97	Lost in Thought/Day Dreaming

2019-Later	
98	Other Distraction
96	Not Reported
99	Reported as Unknown if Distracted

The NMIMPAIR Data File

The Nmimpair data file identifies each non-motorist impairment as a separate record. That is, there can be more than one impairment record for each non-motorist. It contains the data elements CASENUM, PSU, PJ, STRATUM, PSUSTRAT, PSU_VAR, REGION, URBANICITY, WEIGHT, VEH_NO, and PER_NO, which are described in the Key Data Elements at the beginning of the Data Element Definitions and Codes section. The data file also contains NMIMPAIR that is described below.

CASENUM, PER_NO, and NMIMPAIR are the unique identifiers for each record. CASENUM, VEH_NO, and PER_NO should be used to merge the Nmimpair data file with non-motorists from the Person data file. VEH_NO equals 0 for all records in this data file.

NM17. Condition (Impairment) at Time of Crash—Non-Motorist**Definition**

This data element identifies physical impairments to this non-motorist that may have contributed to the crash as identified by law enforcement.

Additional Information

Prior to 2019 the Data Element ID was NM14. From 2019-2021 the Element Data ID was NM15.

SAS Name

NMIMPAIR

Attribute Codes

2016	2017	2018-2020	2021-Later	
0	0	0	0	None/Apparently Normal
1	1	1	1	Ill, Blackout
2	2	2	2	Asleep or Fatigued
3	3	3	3	Walking With a Cane or Crutches, etc.
4	--	--	--	Paraplegic or Restricted to Wheelchair
--	4	4	4	Paraplegic or in a Wheelchair
5	5	5	5	Impaired Due to Previous Injury
6	6	6	--	Deaf
--	--	--	6	Deaf/Hard of Hearing
7	7	7	--	Blind
--	--	--	7	Blind/Low Vision
8	8	8	8	Emotional (Depressed, Angry, Disturbed, etc.)
9	9	9	9	Under the Influence of Alcohol, Drugs or Medication
10	10	10	10	Physical Impairment – No Details
96	96	96	96	Other Physical Impairment
98	98	98	98	Not Reported
99	99	--	--	Unknown if Impaired
--	--	99	99	Reported as Unknown if Impaired

The NMPRIOR Data File

The Nmprior data file identifies each non-motorist action at the time of their involvement in the crash as a separate record. That is, there can be more than one action record for each non-motorist. It contains the data elements CASENUM, PSU, PJ, STRATUM, PSUSTRAT, PSU_VAR, REGION, URBANICITY, WEIGHT, VEH_NO, and PER_NO, which are described in the Key Data Elements at the beginning of the Data Element Definitions and Codes section. The data file also contains NMACTION that is described below.

CASENUM, PER_NO, and NMACTION are the unique identifiers for each record. CASENUM, VEH_NO, and PER_NO should be used to merge the Nmprior data file with non-motorists from the Person data file. VEH_NO equals 0 for all records in this data file.

NM13. Non-Motorist Action/Circumstances

Definition

This data element describes the actions of the non-motorist immediately prior to their involvement in the crash.

Additional Information

It is also an indication of whether the non-motorist was walking/cycling to/from school in addition to the action of the non-motorist immediately prior to their involvement in the crash. Prior to 2022 the Data Element ID was NM11.

SAS Name

MPR_ACT 2016-2019

NMACTION 2020-Later

Attribute Codes

2016-2017	2018-2022	2023-Later	
1	1	--	Going to or From School (K-12)
--	--	1	Going To or From School [Pre-K-12]
2	2	2	Waiting to Cross Roadway
3	3	3	Crossing Roadway
4	4	4	Jogging/Running
5	5	5	Movement Along Roadway with Traffic (In or Adjacent to Travel Lane)
6	6	6	Movement Along Roadway Against Traffic (In or Adjacent to Travel Lane)
8	8	8	In Roadway-Other [Working, Playing, etc.]
9	--	--	Adjacent to Roadway (e.g., Shoulder, Median)
--	9	9	Stationary and Adjacent to Roadway (e.g., Shoulder, Median, Sidewalk)
10	10	10	Working in Trafficway [Incident Response]
11	11	11	Entering/Exiting a Parked or Stopped Vehicle
12	12	12	Disabled Vehicle Related (Working on, Pushing, Leaving/Approaching)
14	14	14	Other
16	16	16	Movement Along Roadway – Direction Unknown
98	98	98	Not Reported
99	--	--	Unknown
--	99	99	Reported as Unknown

The SAFETYEQ Data File

The Safetyeq data file includes non-motorist safety equipment. It contains the data elements CASENUM, PSU, PJ, STRATUM, PSU_VAR, REGION, URBANICITY, WEIGHT, VEH_NO, and PER_NO, which are described in the Key Data Elements at the beginning of the Data Element Definitions and Codes section. The data file also contains the data elements on the following pages.

CASENUM and PER_NO are the unique identifiers for each record. CASENUM, VEH_NO, and PER_NO should be used to merge the Safetyeq data file with non-motorists from the Person data file. VEH_NO equals 0 for all records in this data file.

Prior to 2017 the Safetyeq data file identified each item of safety equipment as a separate record. That is, there could be more than one safety equipment record for each non-motorist. The data element that captured each item of safety equipment is MSAFEQMT. This element has been moved to the Discontinued Safetyeq Data Elements.

NM16. Non-Motorist Safety Equipment Use**NM16A. Non-Motorist Helmet Use****Definition**

This data element indicates if the non-motorist was wearing a safety helmet.

Additional Information

This includes all helmets (e.g., bicycle helmets, motorcycle helmets, racing helmets).

Prior to 2019 the Data Element ID was NM13A. From 2019-2021 the Element Data ID was NM14A.

SAS Name

NMHELMET

Attribute Codes

2017	2018-Later	
1	1	No
2	2	Yes
8	8	Not Reported
9	--	Unknown
--	9	Reported as Unknown

NM16B. Non-Motorist Use of Protective Pads**Definition**

This data element indicates if the non-motorist was wearing padded, shaped attachments to protect specific areas of the body (e.g., elbows, knees, shins) from injury.

Additional Information

Prior to 2019 the Data Element ID was NM13B. From 2019-2021 the Element Data ID was NM14B.

SAS Name**NMPROPAD****Attribute Codes**

2017	2018-Later	
1	1	No
2	2	Yes
8	8	Not Reported
9	--	Unknown
--	9	Reported as Unknown

NM16C. Non-Motorist Use of Other Protective Safety Equipment**Definition**

This data element indicates if the non-motorist was using protective safety equipment other than a helmet or pads (e.g., eye wear/face shields, gloves, wrist guards).

Additional Information

Prior to 2019 the Data Element ID was NM13C. From 2019-2021 the Element Data ID was NM14C.

SAS Name**NMOTHPRO****Attribute Codes**

2017	2018-Later	
1	1	No
2	2	Yes
8	8	Not Reported
9	--	Unknown
--	9	Reported as Unknown

NM16D. Non-Motorist Use of Reflective Clothing/Carried Item**Definition**

This data element indicates if the non-motorist was wearing or carrying some type of reflective item (e.g., jacket, backpack, vest).

Additional Information

Prior to 2019 the Data Element ID was NM13D. From 2019-2021 the Element Data ID was NM14D.

SAS Name**NMREFCLO****Attribute Codes**

2017	2018-Later	
1	1	No
2	2	Yes
8	8	Not Reported
9	--	Unknown
--	9	Reported as Unknown

NM16E. Non-Motorist Use of Lighting**Definition**

This data element indicates if the non-motorist was using a light on his/her person or on a pedalcycle or personal conveyance for safety purposes, to include flashlights.

Additional Information

Prior to 2019 the Data Element ID was NM13E. From 2019-2021 the Element Data ID was NM16E.

SAS Name**NMLIGHT****Attribute Codes**

2017	2018-Later	
1	1	No
2	2	Yes
8	8	Not Reported
9	--	Unknown
--	9	Reported as Unknown

NM16F. Non-Motorist Use of Other Preventive Safety Equipment**Definition**

This data element indicates if the non-motorist was using preventive safety equipment other than a reflective clothing/carried item or light (e.g., bicycle reflectors and flags, reflectors and triangles on a buggy, hi-glo orange clothing, rollerblade stoppers).

Additional Information

Prior to 2019 the Data Element ID was NM13F. From 2019-2021 the Element Data ID was NM16F.

SAS Name**NMOTHPRE****Attribute Codes**

2017	2018-Later	
1	1	No
2	2	Yes
8	8	Not Reported
9	--	Unknown
--	9	Reported as Unknown

Discontinued SAFETYEQ Data Elements

Non-Motorist Safety Equipment Use (discontinued)

Definition

This data element indicates the safety equipment that was used by this non-motorist involved in the crash.

Additional Information

There can be one or more safety equipment responses for each non-motorist.

SAS Name

MSAFEQMT

Attribute Codes

2016	
1	None Used
2	Helmet
3	Reflective Clothing (Jacket, Backpack, etc.)
4	Protective Pads (Elbows, Knees, Shins, etc.)
5	Lighting
7	Other Safety Equipment
8	Not Reported
9	Unknown if Used

The VPICDECODE Data File

The Vpicdecode data file provides specification data for all vehicles derived from the VIN. It contains the data elements CASENUM and VEH_NO, which are described in the Key Data Elements at the beginning of the Data Element Definitions and Codes section. CASENUM and VEH_NO are the unique identifiers for each record. CASENUM and VEH_NO should be used to merge the Vpicdecode data file with the Vehicle or Parkwork data file.

The Vpicdecode data file contains approximately 200 data elements derived from the VIN using NHTSA's Product Information Catalog and Vehicle Listing, known as vPIC. There is one record for each VIN that can be cleanly decoded. If a VIN has issues and cannot be decoded cleanly, there will not be a record. For the definition of clean decoding, and descriptions of the data elements, see the *Product Information Catalog and Vehicle Listing (vPIC) Analytical User's Manual* found in the [NCSA Publications — Manuals and Documentation](#) section of NHTSA's website.

1. The VINs decoded in CRSS Vpicdecode file are based on two sources:
2. The VINs from the police crash report that are coded by CRSS coding staff;
3. The VINs that NHTSA obtains from third parties: each year, after CRSS cases are coded, NHTSA uses the license plate number and vehicle registration information to find missing VINs or incomplete VINs in the vehicle records. Once these additional VINs are obtained, the vPICDecode file is created using both sets of VINs.

The VPICTRAILERDECODE Data File

The Vpictrailerdecode data file provides specification data for all trailers derived from the VIN. It contains the data elements CASENUM and VEH_NO, which are described in the Key Data Elements at the beginning of the Data Element Definitions and Codes section. CASENUM, VEH_NO, and TRAILER_NO are the unique identifiers for each record. CASENUM and VEH_NO should be used to merge the Vpictrailerdecode data file with the Vehicle or Parkwork data file.

The Vpictrailerdecode data file contains approximately forty data elements derived from the VIN using NHTSA's Product Information Catalog and Vehicle Listing, known as vPIC. There is one record for each trailer VIN that can be cleanly decoded. If a VIN has issues and cannot be decoded cleanly, there will not be a record. For the definition of clean decoding, and descriptions of the data elements, see the *Product Information Catalog and Vehicle Listing (vPIC) Analytical User's Manual* found in the [NCSA Publications- Manuals and Documentation](#) section of NHTSA's website.

References

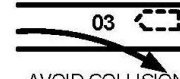
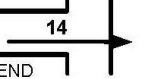
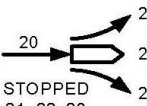
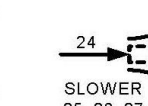
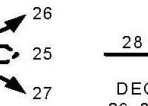
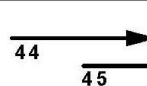
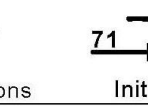
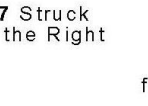
- Association of Transportation Safety Information Professionals. (2017, December 18). *Manual on classification of motor vehicle traffic crashes*, 8th edition (ANSI D16.1-2017). www.transportation.gov/sites/dot.gov/files/docs/resources/government/traffic-records/304331/ansid16-2017.pdf
- Federal Highway Administration. (2009). *Manual on uniform traffic control devices for streets and highways*. <https://mutcd.fhwa.dot.gov/index.htm>
- Herbert, G. C. (2019, September). *Crash report sampling system: Imputation* (Report No. DOT HS 812 795). National Highway Traffic Safety Administration. <https://crashstats.nhtsa.dot.gov/Api/Public/ViewPublication/812795>
- MMUCC Expert Panel. (2017). *MMUCC guideline: Model minimum uniform crash criteria*, 5th edition. National Highway Traffic Safety Administration. <https://crashstats.nhtsa.dot.gov/Api/Public/Publication/812433>
- National Center for Statistics and Analysis. (2024, April). *Product information catalog and vehicle listing (vPIC) analytical user's manual 2022* (Report No. DOT HS 813 547). National Highway Traffic Safety Administration. <https://crashstats.nhtsa.dot.gov/Api/Public/ViewPublication/813547>
- National Highway Traffic Safety Administration. (2025, February, Revised). 2022 FARS/CRSS coding and validation manual (Report No. DOT HS 813 545). <https://crashstats.nhtsa.dot.gov/Api/Public/Publication/813545>
- National Highway Traffic Safety Administration. (2019). *National Automotive Sampling System (NASS) General Estimates System (GES) Analytical User's Manual 1988-2015*. <https://crashstats.nhtsa.dot.gov/Api/Public/Publication/812320>
- Robinson, B. W., Rodegerdts, L., Scarborough, W., Kittelson, W., Troutbeck, R., Brilon, W., Bondzio, L., Courage, K., Kyle, M., Mason, J., Flannery, A., Myers, E., Bunker, J., & Jacquemart, G. (2000, June). *Roundabouts: An informational guide* (Report No. FHWA-RD-00-067). Federal Highway Administration. www.fhwa.dot.gov/publications/research/safety/00067/00067.pdf
- SAE International. (2021). *SAE J3016 Levels of driving automation*. https://www.sae.org/binaries/content/assets/cm/content/blog/sae-j3016-visual-chart_5.3.21.pdf
- Zhang, F., & Acevedo-Diaz, E. (2020, December). *Crash Report Sampling System: Generalized variance functions* (Report No. DOT HS 813 041). National Highway Traffic Safety Administration. <https://crashstats.nhtsa.dot.gov/Api/Public/ViewPublication/813041>
- Zhang, F., Noh, E. Y., Subramanian, R., & Chen, C.-L. (2019, May). *Crash Report Sampling System: Sample design and weighting* (Report No. DOT HS 812 706). National Highway Traffic Safety Administration. <https://crashstats.nhtsa.dot.gov/Api/Public/ViewPublication/812706>

References

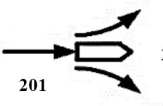
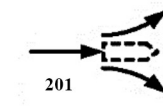
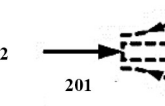
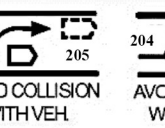
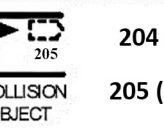
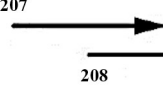
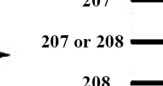
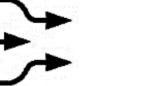
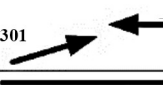
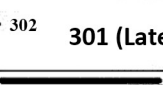
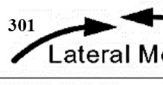
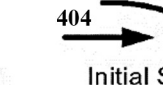
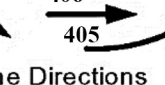
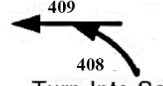
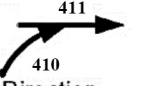
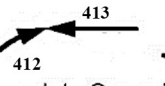
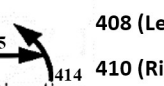
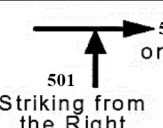
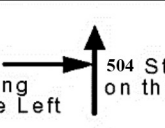
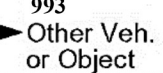
- Zhang, F., Subramanian, R., Chen, C.-L., & Noh, E. Y. (2019, April). *Crash Report Sampling System: Design overview, analytical guidance, and FAQs* (Report No. DOT HS 812 688). National Highway Traffic Safety Administration.
<https://crashstats.nhtsa.dot.gov/Api/Public/ViewPublication/812688>
- Vehicle Identification Number (VIN) Requirements, Fed. Reg 73, No. 84 / Wednesday, April 30, 2008, Rules and Regulations, 23379. <https://www.govinfo.gov/content/pkg/FR-2008-04-30/pdf/08-1197.pdf>

Appendix A: Crash Type Diagrams

Crash Type Diagram

Category	Configuration	CRASH TYPES (includes intent)					
I Single Driver	A Right Roadside Departure	 01 DRIVE OFF ROAD	 02 CONTROL/ TRACTION LOSS	 03 AVOID COLLISION WITH VEH., PED., ANIM.	04 SPECIFICS OTHER	05 SPECIFICS UNKNOWN	
	B Left Roadside Departure	 06 DRIVE OFF ROAD	 07 CONTROL/ TRACTION LOSS	 08 AVOID COLLISION WITH VEH., PED., ANIM.	09 SPECIFICS OTHER	10 SPECIFICS UNKNOWN	
	C Forward Impact	 11 PARKED VEH	 12 STA OBJECT	 13 PEDESTRIAN/ ANIMAL	 14 END DEPARTURE	15 SPECIFICS OTHER	16 SPECIFICS UNKNOWN
II Same Trafficway Same Direction	D Rear End	 20 STOPPED 21, 22, 23	 24 SLOWER 25, 26, 27	 28 DECEL. 29, 30, 31	(EACH - 32) SPECIFICS OTHER	(EACH - 33) SPECIFICS UNKNOWN	
	E Forward Impact	 34 CONTROL/ TRACTION LOSS	 36 CONTROL/ TRACTION LOSS	 38 AVOID COLLISION WITH VEH.	 40 AVOID COLLISION WITH OBJECT	(EACH - 42) SPECIFICS OTHER	(EACH - 43) SPECIFICS UNKNOWN
	F Angle, Sideswipe	 44 45	 46 47	(EACH - 48) SPECIFICS OTHER	(EACH - 49) SPECIFICS UNKNOWN		
III Same Trafficway Opposite Direction	G Head-On	 50 51	(EACH - 52) SPECIFICS OTHER	(EACH - 53) SPECIFICS UNKNOWN			
	H Forward Impact	 54 CONTROL/ TRACTION LOSS	 56 CONTROL/ TRACTION LOSS	 58 AVOID COLLISION WITH VEH.	 60 AVOID COLLISION WITH OBJECT	(EACH - 62) SPECIFICS OTHER	(EACH - 63) SPECIFICS UNKNOWN
	I Angle, Sideswipe	 64 65 Lateral Moves	(EACH - 66) SPECIFICS OTHER	(EACH - 67) SPECIFICS UNKNOWN			
IV Change Trafficway Vehicle Turning	J Turn Across Path	 68 69 Initial Opposite Directions	 70 71 72 73 Initial Same Directions	(EACH - 74) SPECIFICS OTHER	(EACH - 75) SPECIFICS UNKNOWN		
	K Turn Into Path	 76 77 78 79 Turn Into Same Direction	 80 81 82 83 Turn Into Opposite Direction	(EACH - 84) SPECIFICS OTHER	(EACH - 85) SPECIFICS UNKNOWN		
V Intersect Paths	L Straight Paths	 86 87 Struck on the Right Striking from the Right	 88 89 Struck on the left Striking from the Left	(EACH - 90) SPECIFICS OTHER	(EACH - 91) SPECIFICS UNKNOWN		
VI Misc.	M Backing, Etc.	 92 Backing Veh.	93 Other Veh. or Object	98 Other Accident Type 99 Unknown Accident Type 00 No Impact			

Crash Type Configuration Diagram (2023)

Category	Configuration	CRASH TYPES (includes intent)							
I Single Driver	Right Roadside Departure	 DRIVE OFF ROAD	 CONTROL/ TRACTION LOSS	 AVOID COLLISION WITH VEH., PED., ANIM.	101				
	Left Roadside Departure	 DRIVE OFF ROAD	 CONTROL/ TRACTION LOSS	 AVOID COLLISION WITH VEH., PED., ANIM.	102				
	Forward Impact	 PARKED VEH.	 STA OBJECT	 PEDESTRIAN/ ANIMAL	 END DEPARTURE	103			
II Same Trafficway Same Direction	Rear End	 201			 202	 202	201 (Rear End, Trailing) 202 (Rear End, Lead)		
	Forward Impact	 CONTROL/ TRACTION LOSS	 CONTROL/ TRACTION LOSS	 AVOID COLLISION WITH VEH.	 AVOID COLLISION WITH OBJECT	204 (Frontal Impact) 205 (Rear End Impact)			
	Angle, Sideswipe	 207			 207 or 208	 208	207 (Vehicle on Left) 208 (Vehicle on Right)		
III Same Trafficway Opposite Direction	Head-On	 301					 302	301 (Lateral Move Left/Right), 302 (Lateral Move Going Straight)	
	Forward Impact	 CONTROL/ TRACTION LOSS	 CONTROL/ TRACTION LOSS	 AVOID COLLISION WITH VEH.	 AVOID COLLISION WITH OBJECT	304 (Departed Lane) 305 (Remained in Lane)			
	Angle, Sideswipe	 301					 302	301 (Lateral Move Left/Right), 302 (Lateral Move Going Straight)	
IV Change Trafficway Vehicle Turning	Turn Across Path	 401			 402	 403	 406	401 (Left/Right) /402 (Going Straight) 403 (Right) /404 (Going Straight) 405 (Left) /406 (Going Straight)	
	Turn Into Path	 409			 411	 413	 415	408 (Left)/409 (Going Straight) 410 (Right)/411 (Going Straight) 412 (Right)/413 (Going Straight) 414 (Left)/415 (Going Straight)	
V Intersect Paths	Straight Paths	 501 Striking from the Right					 503 Striking from the Left	 504 Struck on the left	
VI Misc.	Backing, Etc.	 992 Backing Veh.					 993 Other Veh. or Object	998 OTHER CRASH TYPE 999 UNKNOWN CRASH TYPE 000 NO IMPACT	

Appendix B: Rules for Derived Data Elements

Appendix B: Rules for Derived Data Elements

Several derived data elements are included in the data files. A derived data element is any element that is not coded (i.e., not directly entered into the system) but translated from existing data. Derived data elements include:

- translations from coded data elements (e.g., “Driver Drinking in Vehicle”),
- translations from collected information (e.g., “Urbanicity”),
- records counted from vehicle and person levels as crash level counters (e.g., “Number of Parked/Working Vehicles”),
- data extracted across several records (e.g., “First Harmful Event,” “Rollover”), and
- element combinations (e.g., “Motor Carrier Issuing Authority and ID Number”).

The derived data elements are provided to facilitate analyses and as a common platform for presenting findings. These elements and the translations used to derive them are described in this Appendix.

Crash Level Counts

Number of Motor Vehicles In-Transport (MVIT)

Accident. VE_FORMS

(also provided as Vehicle.VE_FORMS, Parkwork.PVE_FORMS, Person.VE_FORMS)

Logic of Derivation

All Vehicle records linked to the crash are used. This data element is derived as the count of all vehicles in the crash where “Unit Type” = 1. It is the number of records in the Vehicle data file.

Number of Parked/Working Vehicles

Accident. PVH_INVL

Logic of Derivation

All Vehicle records linked to the crash are used. This data element is derived as the count of all vehicles in the crash where “Unit Type” is in (2, 3, or 4). It is the number of records in the Parkwork data file.

Number of Persons in Motor Vehicles In-Transport (MVIT)

Accident. PERMVIT

Logic of Derivation

All Person records linked to the crash are used. This data element is derived as the count of all people in the crash where “Person Type” is in (1, 2, or 9).

Number of Persons Not in Motor Vehicles In-Transport (MVIT)

Accident. PERNOTMVIT

Logic of Derivation

All Person records linked to the crash are used. Prior to 2020 this data element is derived as the count of all people in the crash where “Person Type” is in (3, 4, 5, 6, 7, 8, 10, or 19). Starting in 2020 the attributes are in (3, 4, 5, 6, 7, 10, 11, 12, 13, or 19).

Crash and Vehicle Level Derived Data Elements

Maximum Injury Severity in Crash

Accident.MAX_SEV

Attribute Labels	2016-Later
No Apparent Injury	0
Possible Injury	1
Suspected Minor Injury	2
Suspected Serious Injury	3
Fatal	4
Injured, Severity Unknown	5
Died Prior to Crash	6
No person involved in the Crash	8
Unknown if Injured/Not Reported	9

Logic of Derivation

All Person records linked to the crash are used. If there are no records then the value 8 is assigned. If there is a single record then the SAS code for Person.INJ_SEV is used. If there are multiple records, all SAS codes for Person.INJ_SEV are obtained and prioritized. Follow the priority ranking of each attribute as follows: 4, 3, 2, 1, 5, 0, 6, 9.

Maximum Injury Severity in Vehicle

Vehicle.MAX_VSEV

Attribute Labels	2016-Later
No Apparent Injury	0
Possible Injury	1
Suspected Minor Injury	2
Suspected Serious Injury	3
Fatal	4
Injured, Severity Unknown	5
Died Prior to Crash	6
No person in Vehicle	8
Unknown if Injured/Not Reported	9

Logic of Derivation

All Person records linked to the vehicle are used. If there are no records then the value 8 is assigned. If there is a single record then the SAS code for Person.INJ_SEV is used. If there are multiple records, all SAS codes for Person.INJ_SEV are obtained and prioritized. Follow the priority ranking of each attribute as follows: 4, 3, 2, 1, 5, 0, 6, 9.

Number Injured in Crash

Accident.NUM_INJ

Attribute Labels	2016-Later
No Person Injured/Property Damage Only Crash	0
Number of Known Injured	x
No Person involved in the Crash	98
All Persons in Crash Are Unknown if Injured	99

Logic of Derivation

All Person records linked to the crash are used. If there are no records then the value 98 is assigned. If the SAS code for Person.INJ_SEV is 9 for all people in the crash then the value is 99. If not, the value assigned is the number (count) of Person records where the SAS code for Person.INJ_SEV is between 1 and 5.

Number Injured in Vehicle

Vehicle.NUM_INJV

Attribute Labels	2016-Later
No Person Injured in Vehicle	0
Number of Known Injured	1-97
No Person involved in the Vehicle	98
All Persons in Vehicle Are Unknown if Injured	99

Logic of Derivation

All Person records linked to the vehicle are used. If there are no records then the value 98 is assigned. If the SAS code for Person.INJ_SEV is 9 for all people in the vehicle then the value is 99. If not, the value assigned is the number (count) of Person records where the SAS code for Person.INJ_SEV is between 1 and 5.

Alcohol Involved in Crash

Accident.ALCOHOL

Attribute Labels	2016-Later
Alcohol Involved	1
No Alcohol involved	2
No applicable person	8
Unknown	9

Logic of Derivation

Alcohol Involved in Crash is calculated based on drivers and non-motorists (except occupants of motor vehicles not in-transport) in the crash and are referred to here as “involved active participants.” This translates to Person Type NOT in 2, 3, or 9.

The following order of alcohol involvement is used. The SAS value for the case was determined by:

- 1 (Alcohol Involved)
 - If “Police Reported Alcohol Involvement” is Yes for any of the involved active participants in the crash,
 - Then Alcohol Involved in Crash should be 1 (Alcohol Involved).
- 2 (No Alcohol Involved)
 - If “Police Reported Alcohol Involvement” is No for ALL of the involved active participants in the crash,

Appendix B: Rules for Derived Data Elements

Then Alcohol Involvement in Crash should be 2 (No Alcohol Involved).

- 9 (Unknown)

If NOT #1 (Alcohol Involved) and “Police Reported Alcohol Involvement” is Unknown or Not Reported for ANY of the involved active participants,

Then Alcohol Involvement in Crash should be 9 (Unknown).

- 8 (No Applicable Person)

Default value if no active participants coded for this case.

Examples:

Case 1: V1 Driver- alcohol is No, V2 Driver- alcohol is Unknown, one non-motorist- alcohol is No, V3 with the situation that three unknown occupants with none coded the role of driver- alcohol for occ1 is Yes, alcohol for occ2 is No, alcohol for occ3 is Unknown.

Alcohol Involved in Crash is 9 (Unknown).

Case 2: V1 driver, alcohol is Unknown, one non-motorist, alcohol is No,

Alcohol Involved in Crash is 9 (Unknown).

Case 3: V1 driver, alcohol is No, one non-motorist, alcohol is unknown,

Alcohol Involved in Crash is 9 (Unknown).

Note: For a single vehicle crash, if a motor vehicle in-transport is listed as having a driver present, but no occupant is coded with the role of driver, then Alcohol Involved in Crash equals 9 (Unknown) unless all occupants are coded No (Alcohol Not involved) or all the occupants are coded Yes (Alcohol Involved). In the case where all occupants are coded No (Alcohol Not Involved) then Alcohol Involved in Crash is 2 (No Alcohol Involved). In the case where all occupants are coded Yes (Alcohol Involved), then Alcohol Involved in Crash is 1 (Alcohol Involved). In the case where not all occupants are coded Yes or No, then Alcohol Involved in Crash equals 9 (Unknown). For a multi-vehicle crash or a crash having non-motorists, the highest priority alcohol value in each vehicle in the case and each applicable non-motorist is taken.

Driver Drinking in Vehicle

Vehicle.VEH_ALCH

Attribute Labels	2016-Later
Alcohol Involved	1
No Alcohol involved	2
No Driver Present/Unknown if Driver Present	8
Unknown	9

Logic of Derivation

- If “Driver Presence” equals 0 (No Driver Present/Not Applicable) or 9 (Unknown), Then “Driver Drinking in Vehicle” is set to 8 (No Driver Present/Unknown if Driver Present).
- If “Driver Presence” equals 1 (Yes) and there is a person in the vehicle where “Person Type” equals 1 (Driver of a Motor Vehicle In-Transport), Then “Police Reported Alcohol Involvement” for the driver is used for the derivation of “Driver Drinking in Vehicle” as follows:

Police Reported Alcohol InvolvementDriver Drinking in Vehicle

- | | | |
|-------------------------------|---|-----------------------|
| ▪ 0 No (Alcohol Not Involved) | → | 2 No Alcohol Involved |
| • 1 Yes (Alcohol Involved) | → | 1 Alcohol Involved |
| ▪ 8 Not Reported | → | 9 Unknown |
| ▪ 9 Unknown (Police Reported) | → | 9 Unknown |

- If “Driver Presence” equals 1 (Yes) and there is **not** a person in the vehicle where “Person Type” equals 1 (Driver of a Motor Vehicle In-Transport), Then
 - If “Police Reported Alcohol Involvement” is the same for the occupants of the vehicle where “Person Type” equals 9 (Unknown Occupant Type in a Motor Vehicle In-Transport), Then “Driver Drinking in Vehicle” is derived from “Police Reported Alcohol Involvement” as shown above,
 - Else “Driver Drinking in Vehicle” is set to 9 (Unknown).

Example:

V1 Driver- alcohol is No, V2 Driver- alcohol is unknown, one non-motorist- alcohol is No, V3 (driver present) with the situation that three unknown occupants with none coded the role of driver- alcohol for occ1 is Yes, alcohol for occ2 is No, occ3 for alcohol is Unknown.

Driver Drinking in Vehicle for V1 is 2 (No Alcohol Involved), for V2 is 9 (Unknown), for V3 is 9 (Unknown).

Note: If a motor vehicle in-transport is listed as having a driver present, but no occupant is coded with the role of driver, then Driver Drinking in Vehicle equals 9 (Unknown) unless all the

unknown occupant types (PER_TYP=9) are coded No (Alcohol Not Involved) or all the unknown occupant types are coded Yes (Alcohol Involved). In the case where all the unknown occupant types are coded Yes (Alcohol Involved) then Driver Drinking in Vehicle is 2 (No Alcohol Involved). In the case where all the unknown occupant types are coded Yes (Alcohol Involved) then Driver Drinking in Vehicle is 1 (Alcohol Involved). For example, if there is a vehicle where there is a driver present and there are two unknown occupant types, both coded Yes (Alcohol Involved) but neither is coded as the driver; then Driver Drinking in Vehicle equals 1 (Alcohol Involved). Another example: if there is a vehicle where there is a driver present and there are two unknown occupant types (neither coded as the driver--that is, the police report indicates it is unknown who was actually driving), and one is coded Yes (Alcohol Involved) and the other is coded No (Alcohol Not Involved); then Driver Drinking in Vehicle equals 9 (Unknown).

Atmospheric Conditions

Accident.WEATHER

Attribute Labels	2016-2019	2020-Later
No Additional Atmospheric Conditions	0	--
Clear	1	1
Rain	2	2
Sleet or Hail	3	3
Snow	4	4
Fog, Smog, Smoke	5	5
Severe Crosswinds	6	6
Blowing Sand, Soil, Dirt	7	7
Other	8	8
Cloudy	10	10
Blowing Snow	11	11
Freezing Rain or Drizzle	12	12
Not Reported	98	98
Unknown/Reported as Unknown	99	99

Logic of Derivation

Prior to 2020 this data element is derived from the coded data elements, Accident.WEATHER1 and Accident.WEATHER2. Beginning in 2020 this data element is derived from Weather.WEATHER that allows the coding of all applicable attributes.

Appendix B: Rules for Derived Data Elements

The following priority ranking of the attributes is used to derive Accident.WEATHER:

- Snow
 - Blowing Snow
 - Sleet or Hail
 - Freezing Rain or Drizzle
 - Rain
 - Fog, Smog, Smoke
 - Severe Crosswinds
 - Blowing Sand, Soil, Dirt
 - Other
 - Cloudy
 - Clear
 - Not Reported
 - Unknown
 - No Additional Atmospheric Conditions
-

Region of the Country

Accident.REGION

Logic of Derivation

This element is derived from the data element “Primary Sampling Unit (PSU)” where the crash occurred. The country is divided into four regions with each of the 50 States and the District of Columbia falling into one of the regions. Region of the Country, therefore, is based on the State in which the Primary Sampling Unit is located.

Urbanicity

Accident.URBANICITY

Logic of Derivation

This element is derived from the data element “Primary Sampling Unit (PSU)” where the crash occurred. A PSU is considered Urban if the county (or counties) in the PSU has a population of 250,000 or greater, otherwise it is Rural.

Primary Sampling Unit for Variance Estimation

Accident.PSU_VAR

Logic of Derivation

This element is derived from the data elements “Primary Sampling Unit (PSU)” and “Police Jurisdiction” where the crash occurred.

First Harmful Event

Accident.HARM_EV

(also provided as Vehicle.HARM_EV, Parkwork.PHARM_EV, Person.HARM_EV)

Logic of Derivation

This data element is derived from the set of all crash events. Each event in a crash is recorded in chronological order. The data element that records the event is “Sequence of Events” and includes both harmful and non-harmful events. First Harmful Event, therefore, is the first “Sequence of Events” value that is not between codes 60 and 79 (non-harmful events).

Rollover

Vehicle.ROLLOVER

(also provided as Person.ROLLOVER)

Logic of Derivation

Since 2022 this data element is derived from the set of all crash events for a vehicle. Each event in a crash is recorded in chronological order. If any "Sequence of Events" associated with "Vehicle Number (This Vehicle) is 1 (Rollover/Overturn) then Rollover for that vehicle is 3 (Rollover), otherwise it is 0 (No Rollover). The exception is any two-wheeled motorcycles which are 8 (Not Applicable). The data element that records the event is “Sequence of Events” and includes both harmful and non-harmful events. “Rollover” is the “Sequence of Events” value 01 that is associated with “This Vehicle” in the crash event.

Initial Contact Point

Vehicle. IMPACT1, Parkwork.PIMPACT1

(also provided as Person.IMPACT1)

Logic of Derivation

This data element is derived from the set of all crash events for a vehicle. Each event in a crash is recorded in chronological order. The data element that records each impact for a vehicle is “Area of Impact (This Vehicle)” or “Area of Impact (Other Vehicle).” The area of impact is only coded for harmful events, that is “Sequence of Events” values that are not between codes 60 and 79. Initial Contact Point, therefore, is the vehicle’s first recorded Area of Impact value for a harmful event. Note that the vehicle may be “This Vehicle” or the “Other Vehicle” in the crash event.

NCSA Make Model Combined

Vehicle. MAK_MOD, Parkwork. PMAK_MOD

(also provided as Person. MAK_MOD)

Logic of Derivation

This five-digit data element is the combination of two data elements, the two-digit “Vehicle Make” code followed by the three-digit “Vehicle Model” code.

Motor Carrier Identification Number

Vehicle. MCARR_ID, Parkwork. PMCARR_ID

Logic of Derivation

This 11-character data element is the combination of two data elements, the two-digit “Motor Carrier Issuing Authority” code followed by the nine-character “Identification Number.”

Appendix C: Analytical Classification of Select CRSS Data Elements

Appendix C: Analytical Classification of Select CRSS Data Elements

Several data elements in CRSS are classified or collapsed according to analytical needs. In various NCSA's published reports and analysis, select CRSS data elements have been given a standard classification. This section shows how CRSS data elements are classified, assisting users in understanding and duplicating statistics presented in NCSA's published reports.

For analytical purposes, fatal crashes and fatalities are extracted from the Fatality Analysis Reporting System (FARS), not CRSS. FARS contains data on a census of fatal traffic crashes within the 50 States, the District of Columbia, and Puerto Rico. To be included in FARS, a crash must involve a motor vehicle traveling on a trafficway customarily open to the public and result in the death of a person (occupant of a vehicle or non-motorist) within 30 days of the crash. Since FARS contains records on all fatal crashes, it is a more accurate representation of fatal crashes and fatalities than the sample contained in CRSS.

It is important to note that these are NCSA's classifications and are subject to modification.

The following tables show the specific coding schemes of select CRSS data elements that are used in NCSA's publications and analysis:

Time of Day/Day of Week

Classification	Data Year and Code
	2016-Later
Time of Day	HOURL (Military)
Daytime (6:00 a.m. – 5:59 p.m.)	6-17
Nighttime (6:00 p.m. – 5:59 a.m.)	0-5, 18-23
Unknown	99
Day of Week	WKDY_I with HOURL_I
Weekday 6 a.m. Monday thru 5:59 p.m. Friday	(WKDY_I =2 and 6<=HOURL_I<=23) or (WKDY_I in (3,4,5)) or (WKDY_I =6 and (0<= HOURL_I <=17)
Weekend 6 p.m. Friday thru 5:59 a.m. Monday	(WKDY_I =6 and 18<= HOURL_I <=23) or (WKDY_I in (1,7)) or (WKDY_I =2 and (0<= HOURL_I <=5))
Unknown	NA

[Return](#)

Vehicle Classification by vPIC Data Elements

Classification	Description	2020	2021-Later
Passenger Cars	Vehicles with VPIC Body Class in the following list: • 1 (Convertible/Cabriolet) • 3 (Coupe) • 5 (Hatchback/Liftback/Notchback) • 10 (Roadster) • 13 (Sedan/Saloon) • 15 (Wagon) Use vPIC Body Class only when the Final Stage Body Class⁽¹⁾ is 0 (Not Applicable), 998 (Not Reported) or 999 (Unknown).	[VPICBODYCLASS] IN (1, 3, 5, 10, 13, 15) AND [ICFINALBODY] IN (0, 998, 999)	[VPICBODYCLASS] IN (1, 3, 5, 10, 13, 15) AND [ICFINALBODY] IN (0, 998, 999)

Appendix C: Analytical Classification of Select CRSS Data Elements

Classification	Description	2020	2021-Later
Light Trucks, Vans, and Multi-Purpose Vehicle	Vehicles with VPIC Body Class or Final Stage Body Class in the following list with a GVWR range in Class 1 or 2 (GVWR of 10K lbs or less): • 2 (Minivan) • 7 (Sport Utility Vehicle/Multi-Purpose Vehicle) • 8 (Crossover Utility Vehicle) • 9 (Van) • 11 (Truck) • 60 (Pickup) • 95 (Cargo Van) • 111 (Step Van/Walk-in Van) • 119 (Sport Utility Truck) • 128 (Ambulance) [since 2021] Use vPIC Body Class only when the Final Stage Body Class is 0 (Not Applicable), 998 (Not Reported) or 999 (Unknown).	([VPICBODYCLASS] IN (2, 7, 8, 9, 11, 60, 95, 111, 119) AND [ICFINALBODY] IN (0, 998, 999)) OR [ICFINALBODY] IN (2, 7, 8, 9, 11, 60, 95, 111, 119, 128)) AND ([GVWR_FROM] IN (11, 12) AND [GVWR_TO] IN (11, 12))	([VPICBODYCLASS] IN (2, 7, 8, 9, 11, 60, 95, 111, 119, 128) AND [ICFINALBODY] IN (0, 998, 999)) OR [ICFINALBODY] IN (2, 7, 8, 9, 11, 60, 95, 111, 119, 128)) AND ([GVWR_FROM] IN (11, 12) AND [GVWR_TO] IN (11, 12))

Appendix C: Analytical Classification of Select CRSS Data Elements

Classification	Description	2020	2021-Later
Light Utility Vehicles	Vehicles with VPIC Body Class or Final Stage Body Class in the following list with a GVWR range in Class 1 or 2 (GVWR of 10K lbs or less): • 7 (Sport Utility Vehicle/Multi-Purpose Vehicle) • 8 (Crossover Utility Vehicle) Use vPIC Body Class only when the Final Stage Body Class is 0 (Not Applicable), 998 (Not Reported) or 999 (Unknown).	([VPICBODYCLASS] IN (7, 8) AND [ICFINALBODY] IN (0, 998, 999)) OR [ICFINALBODY] IN (7, 8)) AND ([GVWR_FROM] IN (11, 12) AND [GVWR_TO] IN (11, 12))	([VPICBODYCLASS] IN (7, 8) AND [ICFINALBODY] IN (0, 998, 999)) OR [ICFINALBODY] IN (7, 8)) AND ([GVWR_FROM] IN (11, 12) AND [GVWR_TO] IN (11, 12))
Light Pickups	Vehicles with VPIC Body Class or Final Stage Body Class in the following list with a GVWR range in Class 1 or 2 (GVWR of 10K lbs or less): • 60 (Pickup) • 119 (Sport Utility Truck) Use vPIC Body Class only when the Final Stage Body Class is 0 (Not Applicable), 998 (Not Reported) or 999 (Unknown).	([VPICBODYCLASS] IN (60, 119) AND [ICFINALBODY] IN (0, 998, 999)) OR [ICFINALBODY] IN (60, 119)) AND ([GVWR_FROM] IN (11, 12) AND [GVWR_TO] IN (11, 12))	([VPICBODYCLASS] IN (60, 119) AND [ICFINALBODY] IN (0, 998, 999)) OR [ICFINALBODY] IN (60, 119)) AND ([GVWR_FROM] IN (11, 12) AND [GVWR_TO] IN (11, 12))

Appendix C: Analytical Classification of Select CRSS Data Elements

Classification	Description	2020	2021-Later
Light Vans	Vehicles with VPIC Body Class or Final Stage Body Class in the following list with a GVWR range in Class 1 or 2 (GVWR of 10K lbs or less): • 2 (Minivan) • 9 (Van) • 95 (Cargo Van) • 111 (Step Van/Walk-in Van) Use vPIC Body Class only when the Final Stage Body Class is 0 (Not Applicable), 998 (Not Reported) or 999 (Unknown).	(([VPICBODYCLASS] IN (2, 9, 95, 111) AND [ICFINALBODY] IN (0, 998, 999)) OR [ICFINALBODY] IN (2, 9, 95, 111)) AND ([GVWR_FROM] IN (11, 12) AND [GVWR_TO] IN (11, 12)))	(([VPICBODYCLASS] IN (2, 9, 95, 111) AND [ICFINALBODY] IN (0, 998, 999)) OR [ICFINALBODY] IN (2, 9, 95, 111)) AND ([GVWR_FROM] IN (11, 12) AND [GVWR_TO] IN (11, 12)))

Appendix C: Analytical Classification of Select CRSS Data Elements

Classification	Description	2020	2021-Later
Other Light Trucks	Vehicles with VPIC Body Class or Final Stage Body Class in the following list with a GVWR range in Class 1 or 2 (GVWR of 10K lbs. or less): • 11 (Truck) • 128 (Ambulance) [since 2021] Use vPIC Body Class only when the Final Stage Body Class is 0 (Not Applicable), 998 (Not Reported) or 999 (Unknown).	(([VPICBODYCLASS] IN (11) AND [ICFINALBODY] IN (0, 998, 999)) OR [ICFINALBODY] IN (11)) AND ([GVWR_FROM] IN (11, 12) AND [GVWR_TO] IN (11, 12))	(([VPICBODYCLASS] IN (11, 128) AND [ICFINALBODY] IN (0, 998, 999)) OR [ICFINALBODY] IN (11, 128)) AND ([GVWR_FROM] IN (11, 12) AND [GVWR_TO] IN (11, 12))

Appendix C: Analytical Classification of Select CRSS Data Elements

Classification	Description	2020	2021-Later
Large Trucks	Vehicles with VPIC Body Class or Final Stage Body Class in the following list with a GVWR range of Class 3 or higher (GVWR greater than 10K lbs) : • 7 (Sport Utility Vehicle/Multi-Purpose Vehicle) • 8 (Crossover Utility Vehicle) • 9 (Van) • 11 (Truck) • 60 (Pickup) • 66 (Truck-Tractor) • 95 (Cargo Van) • 111 (Step Van/Walk-in Van) • 119 (Sport Utility Truck) • 128 (Ambulance) [since 2021] • 130 (Fire Apparatus) [since 2021] Use vPIC Body Class only when the Final Stage Body Class is 0 (Not Applicable), 998 (Not Reported) or 999 (Unknown).	(([VPICBODYCLASS] IN (7, 8, 9, 11, 60, 66, 95, 111, 119) AND [ICFINALBODY] IN (0, 998, 999)) OR [ICFINALBODY] IN (7, 8, 9, 11, 60, 66, 95, 111, 119)) AND ([GVWR_FROM] IN (13, 14, 15, 16, 17, 18) AND [GVWR_TO] IN (13, 14, 15, 16, 17, 18, 98, 99))	(([VPICBODYCLASS] IN (7, 8, 9, 11, 60, 66, 95, 111, 119, 128, 130) AND [ICFINALBODY] IN (0, 998, 999)) OR [ICFINALBODY] IN (7, 8, 9, 11, 60, 66, 95, 111, 119, 128, 130)) AND ([GVWR_FROM] IN (13, 14, 15, 16, 17, 18) AND [GVWR_TO] IN (13, 14, 15, 16, 17, 18, 98, 99))

Appendix C: Analytical Classification of Select CRSS Data Elements

Classification	Description	2020	2021-Later
Medium-Duty Trucks⁽²⁾	Vehicles with VPIC Body Class or Final Stage Body Class in the following list with a GVWR range in Classes 3 to 6 (GVWR between 10K and 26K lbs): • 7 (Sport Utility Vehicle/Multi-Purpose Vehicle) • 8 (Crossover Utility Vehicle) • 9 (Van) • 11 (Truck) • 60 (Pickup) • 66 (Truck-Tractor) • 95 (Cargo Van) • 111 (Step Van/Walk-in Van) • 119 (Sport Utility Truck) • 128 (Ambulance) [since 2021] • 130 (Fire Apparatus) [since 2021] Use vPIC Body Class only when the Final Stage Body Class is 0 (Not Applicable), 998 (Not Reported) or 999 (Unknown).	(([VPICBODYCLASS] IN (7, 8, 9, 11, 60, 66, 95, 111, 119) AND [ICFINALBODY] IN (0, 998, 999)) OR [ICFINALBODY] IN (7, 8, 9, 11, 60, 66, 95, 111, 119)) AND ([GVWR_FROM] IN (13, 14, 15, 16) AND [GVWR_TO] IN (13, 14, 15, 16)))	(([VPICBODYCLASS] IN (7, 8, 9, 11, 60, 66, 95, 111, 119, 128, 130) AND [ICFINALBODY] IN (0, 998, 999)) OR [ICFINALBODY] IN (7, 8, 9, 11, 60, 66, 95, 111, 119, 128, 130)) AND ([GVWR_FROM] IN (13, 14, 15, 16) AND [GVWR_TO] IN (13, 14, 15, 16)))

Appendix C: Analytical Classification of Select CRSS Data Elements

Classification	Description	2020	2021-Later
Heavy-Duty Trucks⁽²⁾	Vehicles with VPIC Body Class or Final Stage Body Class in the following list with a GVWR range in Class 7 or 8 (GVWR greater than 26K lbs): • 7 (Sport Utility Vehicle/Multi-Purpose Vehicle) • 8 (Crossover Utility Vehicle) • 9 (Van) • 11 (Truck) • 60 (Pickup) • 66 (Truck-Tractor) • 95 (Cargo Van) • 111 (Step Van/Walk-in Van) • 119 (Sport Utility Truck) • 128 (Ambulance) [since 2021] • 130 (Fire Apparatus) [since 2021] Use vPIC Body Class only when the Final Stage Body Class is 0 (Not Applicable), 998 (Not Reported) or 999 (Unknown).	(([VPICBODYCLASS] IN (7, 8, 9, 11, 60, 66, 95, 111, 119) AND [ICFINALBODY] IN (0, 998, 999)) OR [ICFINALBODY] IN (7, 8, 9, 11, 60, 66, 95, 111, 119) AND ([GVWR_FROM] IN (17, 18) AND [GVWR_TO] IN (17, 18)))	(([VPICBODYCLASS] IN (7, 8, 9, 11, 60, 66, 95, 111, 119, 128, 129, 130) AND [ICFINALBODY] IN (0, 998, 999)) OR [ICFINALBODY] IN (7, 8, 9, 11, 60, 66, 95, 111, 119, 128, 130)) AND ([GVWR_FROM] IN (17, 18) AND [GVWR_TO] IN (17, 18)))

Appendix C: Analytical Classification of Select CRSS Data Elements

Classification	Description	2020	2021-Later
Buses	Vehicles with VPIC Body Class or Final Stage Body Class in the following list: • 16 (Bus) • 68 (Streetcar/ Trolley) • 73 (Bus - School Bus) Use vPIC Body Class only when the Final Stage Body Class is 0 (Not Applicable), 998 (Not Reported) or 999 (Unknown).	([VPICBODYCLASS] IN (16, 68, 73) AND [ICFINALBODY] IN (0, 998, 999)) OR [ICFINALBODY] IN (16, 68, 73)	([VPICBODYCLASS] IN (16, 68, 73) AND [ICFINALBODY] IN (0, 998, 999)) OR [ICFINALBODY] IN (16, 68, 73)

Appendix C: Analytical Classification of Select CRSS Data Elements

Motorcycles⁽³⁾	Vehicles with VPIC Body Class in the following list: • 6 (Motorcycle – Standard) • 12 (Motorcycle – Scooter) • 80 (Motorcycle – Sport) • 81 (Motorcycle – Touring/Sport Touring) • 82 (Motorcycle – Cruiser) • 83 (Motorcycle – Trike) • 84 (Off-road Vehicle - Dirt Bike / Off-Road) • 85 (Motorcycle – Dual Sport/ Adventure/Supernatural/ On/Off-Road) • 86 (Off-road Vehicle - Enduro (Off-road long distance racing)) • 87 (Motorcycle – Small/ Minibike) • 90 (Motorcycle – Side Car) • 94 (Motorcycle – Custom) • 98 (Motorcycle – Street) • 100 (Motorcycle – Enclosed Three Wheeled/Enclosed Autocycle) • 103 (Motorcycle – Unenclosed Three Wheeled/Open Autocycle)	[VPICBODYCLASS] IN (6, 12, 80, 81, 82, 83, 84, 85, 86, 87, 90, 94, 98, 100, 103, 104, 109, 110, 113, 114, 125, 996)	[VPICBODYCLASS] IN (6, 12, 80, 81, 82, 83, 84, 85, 86, 87, 90, 94, 98, 100, 103, 104, 109, 110, 113, 114, 125, 996 [deleted for 2022 and later])
----------------------------------	--	--	---

Appendix C: Analytical Classification of Select CRSS Data Elements

Classification	Description	2020	2021-Later
	• 104 (Motorcycle – Moped) • 109 (Motorcycle – Cross County) • 110 (Motorcycle – Underbone) • 113 (Off-road Vehicle - Motocross (Off-road short distance, closed track racing)) • 114 (Motorcycle – Competition) • 125 (Motorcycle – Unknown Body Class) • 996 (Motorized Bicycle) [deleted for 2022 and later])		

Appendix C: Analytical Classification of Select CRSS Data Elements

Classification	Description	2020	2021-Later
Off-Road Vehicles	Vehicles with VPIC Body Class in the following list: • 69 (Off-Road Vehicle – All Terrain Vehicle (ATV) [Motorcycle-style]) • 88 (Off-Road Vehicle – Go Kart) • 97 (Off-Road Vehicle – Snowmobile) • 105 (Off-Road Vehicle – Recreational Off-Road Vehicle [ROV]) • 124 (Off-Road Vehicle – Golf Cart) • 126 (Off-Road Vehicle – Farm Equipment) • 127 (Off-Road Vehicle – Construction Equipment)	[VPICBODYCLASS] IN (69, 88, 97, 105, 124, 126, 127)	[VPICBODYCLASS] IN (69, 88, 97, 105, 124, 126, 127)
Low-Speed Vehicles	Vehicles with VPIC Body Class or Final Stage Body Class as 4 (Low-Speed Vehicle) Use vPIC Body Class only when the Final Stage Body Class is 0 (Not Applicable), 998 (Not Reported) or 999 (Unknown).	([VPICBODYCLASS]=4 AND [ICFINALBODY] IN (0, 998, 999)) OR [ICFINALBODY]=4	([VPICBODYCLASS]=4 AND [ICFINALBODY] IN (0, 998, 999)) OR [ICFINALBODY]=4

Appendix C: Analytical Classification of Select CRSS Data Elements

Classification	Description	2020	2021-Later
Other/Unknown	Vehicles not meeting the criteria specified above. Includes vehicles with VPIC Body Class or Final Stage Body Class in the following list: • 108 (Motorhome) • 117 (Limousine) • 129 (Street Sweeper) [since 2021] • 997 (Other, Specify) [Final Stage Body Class Only] • 999 (Unknown)		

- (1) Final Stage Body Class is only applicable to incomplete vehicles.
- (2) The "Medium-Duty Trucks" and "Heavy-Duty Trucks" groups will not include vehicles with a GVWR range overlapping the medium-duty (GVWR classes 3-6) and heavy-duty (GVWR classes 7-8) categories.
- (3) In 2022 996 (Motorized Bicycle) was removed from the motorcycle range. Motorized bicycles were no longer considered motor vehicles and crashes involving only motorized bicycles were no longer collected.

2020-2022 Changes:

- The groups now confirm that Final Stage Body Class would not result in a different classification for the vehicle.
- Off-road motorcycles (84, 86, 113) are now grouped within the "Motorcycles" group for ease of comparison to crash data in 2019 and earlier.
- Ambulance (128) and Trucks (11) with a GVWR of 10K lbs. or less are now grouped within a new group of "Other Light Trucks".
- The "Other" and "Unknown" groups are combined into one group.
- Street Sweeper (129) is now grouped within the "Other/Unknown" group.

Vehicle Classification by NCSA Data Elements

Classification	Data Year and Code
	2016-Later
	(BODY_TYP)
Passenger Cars	01-11, 17
Light Trucks & Vans	14-16, 19-22, 28-41, 45-49, or (79 and TOW_VEH =0 or 9)
Large Trucks	60-64, 66, 67, 71, 72, 78, or (79 and TOW_VEH in 1-4)
Motorcycles ¹	80-89
Buses	50-59
Other/Unknown Vehicles	12, 13, 42, 65, 73, 90-97, 98, 99, or (79 and TOW_VEH=5 or 6)
Passenger Vehicles	01-11, 14-22, 28-41, 45-49, or (79 and TOW_VEH=0 or 9)
Utility Vehicles (a.k.a. On/Off Road)	14-16, 19
Pickups ²	30-39
Vans	20, 21, 22, 28, 29
Medium Trucks	60-62, 64, 67, 71
Heavy Trucks	63, 66, 72, 78, or (79 and TOW_VEH in 1-4)
Combination Trucks	(60-63, 64, 67, 71, 72, 78 and TOW_VEH in (1-4)) or 66
Single Unit Trucks	60-63, 64, 67, 71, 72, 78, 79 and TOW_VEH in (0, 5, 6, 9)
Unknown (not in Imputed Body Type)	98, 99

⁽¹⁾ In 2017 new attributes were added to the motorcycle range: motor scooter (84); unenclosed three wheel motorcycle/unenclosed autocycle (1 rear wheel) (85); enclosed three wheel motorcycle/enclosed autocycle (1 rear wheel) (86); unknown three wheel motorcycle type (87). In 2022 motorized bicycles were no longer considered motor vehicles and attribute (81) was modified to reflect that. Single vehicle crashes involving motorized bicycles were no longer collected.

⁽²⁾ In 2017 attributes compact pickup (30) and standard pickup (31) were deleted and replaced with attribute light pickup (34). In 2018 attribute pickup with slide in camper (32) was deleted.

[Return](#)

Injury Severity

CRSS Description	Data Year and Code	Classification
	2016-Later	
No Apparent Injury (O)	0	Not Injured
Died Prior	6	
Possible Injury (C)	1	Injured
Suspected Minor Injury (B)	2	
Suspected Serious Injury (A)	3	
Unknown Injury Severity (U)	5	
Fatal (K)*	4	Killed

* Fatality counts from the FARS are used in NCSA's publications and analysis.

[Return](#)

Person Type

CRSS Description	Data Year and Code			Classification
	2016-2019	2020-2021	2022-Later	
<i>Occupants</i>				
Driver of a motor vehicle in-transport	1	1	1	Driver
Passenger of a motor vehicle in-transport	2	2	2	Passenger
Unknown occupant type of a motor vehicle in-transport ⁽¹⁾	9	9	9	
<i>Non-occupants</i>				
Occupant of a motor vehicle not in-transport ⁽²⁾	3	3	3	Other non-occupant
Occupant of a non-motor vehicle transport device ⁽³⁾	4	4	4	
Pedestrian	5	5	5	Pedestrian
Bicyclist ⁽⁴⁾	6	6	6	Pedalcyclist
Other Cyclist	7	7	--	
Other Pedalcyclist ⁽⁴⁾	--	--	7	

Appendix C: Analytical Classification of Select CRSS Data Elements

CRSS Description	Data Year and Code			Classification
	2016-2019	2020-2021	2022-Later	
Persons on personal conveyances	8	--	--	Other non-occupant
Person on a personal conveyance	--	--	8	
Persons in/on buildings	10	10	--	
Person in/on a building	--	--	10	
Person on motorized personal conveyance	-	11	--	
Person on non-motorized personal conveyance	-	12	--	
Person on personal conveyance, unknown if motorized or non-motorized	-	13	--	Unknown non-occupant type
Unknown type of non-motorist	19	19	19	

⁽¹⁾ Customarily, “Unknown Occupant” is placed in the “Passenger” category, unless they need to be distinguished from “Passengers.”

⁽²⁾ “Occupant of motor vehicle not in-transport” refers to occupants of parked motor vehicles (any motor vehicle stopped off the roadway). This includes occupants of motor vehicles in motion outside the trafficway boundaries.

⁽³⁾ “Occupant of non-motor vehicle transport device” refers to people riding in an animal-drawn conveyance, on an animal, or injured occupants of railway trains, etc.

⁽⁴⁾ Prior to 2022 motorize bicyclists were considered motor vehicle occupants. After 2022 motorized bicycles were no longer collected as motor vehicles and the occupants are now considered non-motorists, “bicyclist (06)” and “Other pedalcyclist (07).”

[Return](#)

Restraint System Use

The restraint use classification should be used for all vehicle occupants, except for motorcyclists. However, most restraint use analysis focuses on child safety seat use or belt use for passenger vehicle occupants. Be sure to include the appropriate vehicle body type occupied in your selection criteria—see the section on [Vehicle Body Type Classification](#).

CRSS Description	Data Year and Code			Classification
	2016	2017-2018	2019-Later	
Not Applicable	0	--	--	Not Used
None Used – Motor Vehicle Occupant	7	--	--	
None Used/Not Applicable	--	20	20	
No Helmet	17	17	--	
DOT-Compliant Motorcycle Helmet	5	5	--	
Helmet, Other Than DOT-Compliant Motorcycle Helmet	16	16	--	
Helmet, Unknown if DOT-Compliant	19	19	--	
Shoulder and Lap Belt Used	3	3	3	Used
Shoulder Belt Only Used	1	1	1	
Lap Belt Only Used	2	2	2	
Racing-Style Harness Used	--	--	6	
Child Restraint System – Forward Facing	10	10	10	
Child Restraint System – Rear Facing	11	11	11	
Booster Seat	12	12	12	
Child Restraint – Type Unknown	4	4	4	
Other Restraint/ Safety Equipment Used	97	97	97	
Restraint Used – Type Unknown	8	8	8	
Not Reported	98	98	98	Unknown
Unknown if Helmet Worn	29	29	--	
Unknown if Used/ Reported as Unknown if Used	99	99	99	

[Return](#)

Helmet Use

The helmet use classification should be used for motorcyclists only. Be sure to include the appropriate vehicle body type occupied in your selection criteria—see the section on [Vehicle Body Type Classification](#).

CRSS Description	Data Year and Code			Classification
	2016	2017-2018	2019-Later	
Not Applicable	0	--	--	Not Helmeted
None Used – Motor Vehicle Occupant	7	--	--	
None Used/Not Applicable	--	20	20	
Shoulder and Lap Belt Used	3	3	--	
Shoulder Belt Only	1	1	--	
Lap Belt Only	2	2	--	
Child Restraint System – Forward Facing	10	10	--	
Child Restraint System – Rear Facing	11	11	--	
Booster Seat	12	12	--	
Child Restraint – Type Unknown	4	4	--	
No Helmet	17	17	17	
Helmet Used Improperly	(5, 16, 19) and REST_MIS=1	(5, 16, 19) and REST_MIS=1	(5, 16, 19) and HELM_MIS=1	Helmeted
Restraint Used – Other or Type Unknown	(8, 97) and REST_MIS=1	(8, 97) and REST_MIS=1	--	
DOT-Compliant Motorcycle Helmet	5 and REST_MIS=0	5 and REST_MIS=0	5 and HELM_MIS=0	
Other/Unknown Helmet	(16, 19) and REST_MIS=0	(16, 19) and REST_MIS=0	(16, 19) and HELM_MIS=0	
Other Restraint/ Safety Equipment Used	97 and REST_MIS=0	97 and REST_MIS=0	--	
Restraint Used – Type Unknown	8 and REST_MIS=0	8 and REST_MIS=0	--	

Appendix C: Analytical Classification of Select CRSS Data Elements

CRSS Description	Data Year and Code			Classification
	2016	2017-2018	2019-Later	
Not Reported	98	98	98	Unknown
Unknown if Helmet Worn	29	29	--	
Unknown if Used/ Reported as Unknown if Used	99	99	99	

[Return](#)

Alcohol Test Result

CRSS Description	Data Year and Code	Classification	
	2016-Later		
.00 - Actual Value	0-9	No Alcohol	Tested with Known Results
.01-.93 – Actual Value	10-939	Positive BAC	
.94 or Greater	940		
Positive Reading with No Actual Value	998		
None Given	996	Not Tested	Unknown BAC
AC Test Performed, Results Unknown	997	Tested, with Unknown Results	
Unknown if Tested/ Not Reported	-	Unknown if Tested	
Unknown if Tested/ Reported as Unknown if Tested	999		
Not Reported	995		

[Return](#)

Police Pursuits

A pursuit is an event that is initiated when a law enforcement officer, operating an authorized emergency vehicle, gives notice to stop (either through the use of visual or audible emergency signals or a combination of emergency devices) to a motorist who the officer is attempting to apprehend and that motorist fails to comply with the signal by either maintaining his/her speed, increasing speed or taking other evasive action to elude the officer's continued attempts to stop the motorist.

Note: CRSS data does not include crashes that were the result of Legal Intervention, as defined in the ANSI D16.1-2017 Manual as:

2.4.3 legal intervention: Legal intervention is a category of deliberate intent in which the person who acts or refrains from acting is a law-enforcing agent or other official.

Examples:

1. If a lawbreaker crashes either intentionally or unintentionally into a road block set up by police to stop him, the crash is considered a result of legal intervention. If a driver other than the lawbreaker crashes into the road block, the crash is not considered to be a result of legal intervention.
2. If a police car is intentionally driven into another vehicle, the crash is considered to result from legal intervention. If a lawbreaker being pursued by the police loses control of his vehicle and crashes, the crash is not considered to result from legal intervention unless the police intended that the lawbreaker crash.
3. If, during the pursuit, the police vehicle strikes a road vehicle other than the subject of the pursuit, a non-motorist or property, then that harmful event is not legal intervention.

Police Pursuits			
Classification	Data Year and Codes		
	2016-2019	2020-2022	2023-Later
Related Factors-Crash Level	Accident.CF1, CF2, CF3	Crashrf.CRASHRF	
Police Pursuit Involved	20	20	20
Related Factors-Driver Level	Vehicle.DR_SF1, DR_SF2, DR_SF3, DR_SF4	Driverrf.DRIVERRF	
Police Pursuing This Driver or Police Officer in Pursuit	37	37	-

Police Pursuits			
Classification	Data Year and Codes		
	2016-2019	2020-2022	2023-Later
Police Pursuing This Driver	-	-	104
Police Officer in Pursuit	-	-	105

CRSS 2016-2019

If a crash has a “Related Factor—Crash Level” of 20 (Police Pursuit Involved) or a driver in the crash has a “Related Factor—Driver Level” of 37 (Police Pursuing This Driver or Police Officer in Pursuit), then that crash is considered a “police pursuit crash.”

(CF1=20) or (CF2=20) or (CF3=20) or (DR_SF1=37) or (DR_SF2=37) or (DR_SF3=37) or (DR_SF4=37)

Specific person types can be identified as follows:

1. Occupant of police vehicle – all occupants (PER_TYP in (1,2,9)) of special use vehicle police (SPEC_USE=5)
2. Occupant of chased vehicle – all occupants (PER_TYP in (1,2,9)) of vehicle with a driver having a “Related Factor – Driver Level” of police pursuing this driver or police officer in pursuit (DR_SF1=37 or DR_SF2=37 or DR_SF3=37 or DR_SF4=37)
3. Occupant of other vehicle – all other occupants (PER_TYP IN (1,2,9)) – excludes occupant of police vehicle and chased vehicle
4. Non-occupant – pedestrians, pedalcyclists, and other non-occupants (PER_TYP IN (3,4,5,6,7,8,10,19))

CRSS 2020-2022

Starting in 2020, related factors at each level (e.g., Accident level, Vehicle level) are stored in their own tables and as many factors as apply are included. For example, “Related Factors—Crash Level” are stored in the Crashrf table and the data element name is now CRASHRF. Similarly, “Related Factors—Driver Level” are stored in the Driverrf table and the data element name is now DRIVERRF. However, the logic for identifying crashes involving police pursuit remains the same.

If a crash has a “Related Factor—Crash Level” of 20 (Police Pursuit Involved) or a driver in the crash has a “Related Factor—Driver Level” of 37 (Police Pursuing This Driver or Police Officer in Pursuit), then that crash is considered a “police pursuit crash.”

(CRASHRF=20) or (DRIVERRF=37)

Specific person types can be identified as follows:

Appendix C: Analytical Classification of Select CRSS Data Elements

1. Occupant of police vehicle – all occupants (PER_TYP IN (1,2,9)) of “Special Use” vehicle police (SPEC_USE=5)
2. Occupant of chased vehicle – all occupants (PER_TYP IN (1,2,9)) of vehicle with a driver having a “Related Factor – Driver Level” of police pursuing this driver or police officer in pursuit (DRIVERRF=37)
3. Occupant of other vehicle - all other occupants (PER_TYP IN (1,2,9)) – excludes occupant of police vehicle and chased vehicle
4. Nonoccupant - pedestrians, pedalcyclists, and other nonoccupants (PER_TYP IN (3,4,5,6,7,8,10,11,12,13,19)) – 11, 12, and 13 replaced 8 in 2020. In 2022, 11, 12, and 13 were removed and 8 was reinstated.

CRSS 2023 and Later

Starting in 2023, a “Related Factor—Driver Level” of 37 (Police Pursuing This Driver or Police Officer in Pursuit) was discontinued and replaced with separate attributes 104 (Police Pursuing This Driver) and 105 (Police Officer in Pursuit). The logic for identifying crashes involving police pursuit remains the same.

If a crash has a “Related Factor—Crash Level” of 20 (Police Pursuit Involved) or a driver in the crash has a “Related Factor—Driver Level” of 104 (Police Pursuing This Driver) or 105 (Police Officer in Pursuit), then that crash is considered a “police pursuit crash.”

(CRASHRF=20) or (DRIVERRF in (104, 105))

Specific person types can be identified as follows:

1. Occupant of police vehicle - all occupants (PER_TYP IN (1,2,9)) of
 - a. “Special Use” vehicle police (SPEC_USE=5) OR
 - b. vehicle with a driver having a “Driver Related Factor” of police officer in pursuit (DRIVERRF=105).
2. Occupant of chased vehicle - all occupants (PER_TYP IN (1,2,9)) of vehicle with a driver having a “Driver Related Factor” of police pursuing this driver (DRIVERRF=104).
3. Occupant of other vehicle - all other occupants (PER_TYP IN (1,2,9)) - excludes occupant of police vehicle and chased vehicle
4. Nonoccupant - pedestrians, pedalcyclists, and other nonoccupants (PER_TYP IN (3,4,5,6,7,8,10,19)).

Appendix D: Auxiliary Data Files

Appendix D: Auxiliary Data Files

A set of auxiliary files contains elements derived from the CRSS datasets to make it easier to extract certain data classifications and topical areas, such as commonly used age groups, speeding involved crashes, and distraction involved crashes. There is an Accident (acc_aux), Vehicle (veh_aux), and Person (per_aux) level auxiliary file for each year of data. Univariates for each derived data element can be obtained from the [Motor Vehicle Crash Data Systems Data-Book Application](#). A listing of data elements in each file follows:

Accident Data File (acc_aux)

Variable	Description
YEAR	Crash Year
CASENUM	Case Number
WEIGHT	Case Weight
A_CRAINJ	Crash Injury Type
A_CT	Crash Type
A_D15_19	Crashes Involving a Young Driver (Aged 15-19)
A_D15_20	Crashes Involving a Young Driver (Aged 15-20)
A_D16_19	Crashes Involving a Young Driver (Aged 16-19)
A_D16_20	Crashes Involving a Young Driver (Aged 16-20)
A_D16_24	Crashes Involving a Young Driver (Aged 16-24)
A_D21_24	Crashes Involving a Young Driver (Aged 21-24)
A_D65PLS	Crashes Involving an Older Driver (Aged 65+)
A_DIST	Involving a Distracted Driver
A_DOW	Day of Week
A_HR	Involving a Hit-and-Run
A_INTER	Interstate
A_INTSEC	Intersection
A_JUNC	Junction
A_LT	Involving a Large Truck
A_MANCOL	Manner of Collision
A_MC	Involving a Motorcycle
A_PED	Involving a Pedestrian
A_PEDAL	Involving a Pedalcyclist
A_RELRD	Relationship to the Trafficway

Appendix D: Auxiliary Data Files

Variable	Description
A_ROLL	Involving a Rollover
A_SPCRA	Involving Speeding
A_TOD	Time of Day
A_WEATHER	Atmospheric Conditions

Vehicle Data File (veh_aux)

Variable	Description
YEAR	Crash Year
CASENUM	Case Number
VEH_NO	Vehicle Number
WEIGHT	Case Weight
A_BODY	Vehicle Type
A_DRDIS	Distracted Driver
A_IMP1	Initial Impact Point
A_MOD_YR	Vehicle Model Year (4-digit model year for all data years)
A_SPVEH	Speeding Vehicle
A_VROLL	Rollover

Person Data File (per_aux)

Variable	Description
YEAR	Crash Year
CASENUM	Case Number
VEH_NO	Vehicle Number
PER_NO	Person Number
WEIGHT	Case Weight
A_AGE1	Age Group 1
A_AGE2	Age Group 2
A_AGE3	Age Group 3
A_AGE4	Age Group 4
A_AGE5	Age Group 5
A_AGE6	Age Group 6

Appendix D: Auxiliary Data Files

Variable	Description
A_AGE7	Age Group 7
A_AGE8	Age Group 8
A_AGE9	Age Group 9
A_EJECT	Ejection
A_HELMUSE*	Helmet Use (use for motorcyclists only)*
A_LOC	Non-Motorist Location
A_PERINJ	Injury Type
A_PTYPE	Person Type
A_RESTUSE*	Restraint Use (use for all vehicle occupants except motorcyclists)*

***A_RESTUSE** focuses on belts and child seats and should be used when doing restraint use analysis on motor vehicle occupants except for motorcyclists. **A_HELMUSE** focuses on motorcycle helmet use and should be used when doing helmet use analysis for motorcyclists. When using these variables, be sure to include the appropriate body types in your selection criteria as well (see [Vehicle Classification by NCSA Data Elements](#)). For the specific type of restraint system used—child seat, lap belt, shoulder belt, DOT-compliant motorcycle helmet, etc.—refer to the [Restraint System Use](#) (REST_USE) and [Helmet Use](#) (HELM_USE) in the Person data file.

Important: Although autocycles are considered motorcycles, they are usually equipped with belts. As an exception, both **A_RESTUSE** (belt use) and **A_HELMUSE** (helmet use) are captured for occupants of autocycles.

Appendix E: Summary Statistics

Appendix E: Summary Statistics

The following two tables provide a summary of descriptive statistics from the CRSS data files. Table 1: Unweighted Sample represents the actual number of records and Table 2: Weighted Sample represents the national estimates. These statistics provide the analyst a benchmark to compare against numbers obtained from the analytical data files.

Table E-1. Unweighted Sample

Year	Crashes	Vehicles (In-Transport)	People	Drivers	Occupants	Pedestrians	Pedalcyclists
2016	46,511	82,149	117,759	82,000	113,405	2,257	1,576
2017	54,969	97,625	138,913	97,388	133,408	2,881	1,946
2018	48,443	86,105	120,230	85,916	115,774	2,444	1,436
2019	54,409	96,717	135,410	96,488	129,980	2,949	1,802
2020	54,745	94,718	131,962	94,500	126,460	2,882	1,923
2021	54,200	95,785	133,734	95,551	128,315	2,886	1,820
2022	53,955	94,756	132,175	94,510	126,442	2,967	1,870
2023	50,103	87,461	122,388	87,262	116,597	2,907	2,050

Table E-2. Weighted Sample

Year	Crashes	Vehicles (In-Transport)	People	Drivers	Occupants	Pedestrians	Pedalcyclists
2016	6,821,129	12,094,306	16,617,091	12,074,087	16,386,624	95,492	69,929
2017	6,452,285	11,547,079	15,758,853	11,521,902	15,557,000	78,671	55,067
2018	6,734,416	12,049,038	16,208,490	12,024,889	15,997,232	81,573	51,286
2019	6,755,841	12,144,348	16,279,944	12,118,123	16,069,748	83,296	54,219
2020	5,250,837	9,145,835	12,156,001	9,126,739	11,997,178	61,799	43,195
2021	6,102,936	10,842,675	14,439,132	10,820,045	14,264,521	68,423	45,842
2022	5,930,496	10,527,209	14,045,582	10,501,979	13,840,598	75,666	50,827
2023	6,138,359	10,906,228	14,567,906	10,884,875	14,360,683	77,264	57,101

Drivers: PERSON TYPE = 1

Occupants: PERSON TYPE IN (1,2,9)

Pedestrians: PERSON TYPE = 5

Pedalcyclists: PERSON TYPE IN (6, 7)

Appendix F: Standard Errors

Appendix F: Standard Errors

The estimates generated using CRSS data are subject to sampling errors because they are based on a probability sample of crashes instead of all crashes. The sampling error is a measure of the variability of an estimator from its mean under repeated sample selections. The magnitude of sampling error depends on the study variable, the estimator used, and the CRSS sample design.

For various reasons, it is necessary to use design features such as stratification, clustering, and unequal selection probabilities to select the CRSS probability sample. As a result, the CRSS sample is not a simple random sample. Failing to consider these design features in estimation can cause bias to both CRSS point estimates and the associated standard error estimates.

Estimation methods and computer software have been developed to make estimates from complex survey data like CRSS. Specialized procedures for complex survey data analysis, such as SAS PROC SURVEY procedures and SUDAAN procedures, should be used for CRSS data analysis along with proper design statements. A SAS PROC SURVEY procedure and a SUDAAN procedure are provided below as examples of CRSS estimation. See the NHTSA Technical Report [Crash Report Sampling System: Design Overview, Analytic Guidance, and FAQs](#) for some basic concepts of complex survey data analysis and more examples.

SAS and SUDAAN Examples for Single Year CRSS Estimation

/*SAS Example*/

```
PROC SURVEYFREQ DATA=IMPUTED.ACCIDENT VARMETHOD=JK;  
    STRATA PSUSTRAT;  
    CLUSTER PSU_VAR;  
    TABLES MAXSEV_IM;  
    WEIGHT WEIGHT;  
    RUN;
```

/*SUDAAN Example*/

```
PROC CROSSTAB DATA=IMPUTED.ACCIDENT DESIGN=JACKKNIFE NOTSORTED;  
    NEST PSUSTRAT PSU_VAR;  
    WEIGHT WEIGHT;  
    TABLES MAXSEV_IM;  
    CLASS MAXSEV_IM;  
    PRINT NSUM="SAMSIZE" WSUM="POPSIZE" SEWGT;  
    RUN;
```

For readers who do not have access to the specialized software, the generalized variance function (GVF) method can be used to generate ballpark standard error estimates for a large quantity of estimates in a simpler way. In this approach, it is assumed that in CRSS, the standard error (Ste)

Appendix F: Standard Errors

of a point estimate X can be approximated by a known generalized variance function f of X indexed by estimated parameters, say, a , b , and c :

$$Ste \approx f(X; a, b, c)$$

The survey statisticians normally provide the estimated parameters a , b , c and specify the GVF form $f(X; a, b, c)$. To have a quick estimate of the standard error of X , the data user simply first estimates X and plugs X into $f(X; a, b, c)$ to calculate Ste .

In 2020 NHTSA conducted a study to determine the GVFs for CRSS. The CRSS GVF was determined as:

$$ste(X) = e^{a+b*\ln(X)+c*\ln^2(x)}$$

For more detailed information about NHTSA's CRSS GVF study, see Zhang and Diaz¹ (Report No. DOT HS813 041).

The following table lists estimated coefficients for crash, vehicle, and person level GVFs since 2016.

Estimated coefficients of CRSS GVFs since 2016

Year	Crash Level Coefficients	Vehicle Level Coefficients	Person Level Coefficients
2016	a = 1.92772 b = 0.38750 c = 0.01947	a = 1.17146 b = 0.53866 c = 0.01425	a = 1.79032 b = 0.40622 c = 0.01930
2017	a = 2.33171 b = 0.30826 c = 0.02344	a = 1.43152 b = 0.48824 c = 0.01629	a = 2.05394 b = 0.35287 c = 0.02119
2018	a = 2.33242 b = 0.31521 c = 0.02258	a = 1.69299 b = 0.44262 c = 0.01787	a = 2.02774 b = 0.35777 c = 0.02075
2019	a = 2.19494 b = 0.33465 c = 0.02185	a = 1.70176 b = 0.43713 c = 0.01826	a = 2.14416 b = 0.32619 c = 0.02238
2020	a = 1.81266 b = 0.38881 c = 0.01959	a = 1.69637 b = 0.42507 c = 0.01877	a = 1.88630 b = 0.36439 c = 0.02074
2021	a = 2.16680 b = 0.33005 c = 0.02222	a = 1.58125 b = 0.45204 c = 0.01775	a = 1.95844 b = 0.35704 c = 0.02129

¹ <https://crashstats.nhtsa.dot.gov/Api/Public/ViewPublication/813041>

Year	Crash Level Coefficients	Vehicle Level Coefficients	Person Level Coefficients
2022	a = 1.82248 b = 0.38876 c = 0.02057	a = 1.06573 b = 0.52025 c = 0.01622	a = 1.71565 b = 0.39384 c = 0.02069
2023	a = 1.66293 b = 0.42425 c = 0.01866	a = 0.67458 b = 0.60142 c = 0.01233	a = 1.30776 b = 0.47508 c = 0.01702

Example 1: To use these GVF's to estimate the standard errors, use the vehicle level “hit-and-run” indicator variable HITRUN_IM as an example. The total number of “hit-and-run” vehicles in 2018 CRSS is estimated as $X = 817,573$ (the summation of the weights of all “hit-and-run” vehicles). Using the 2018 vehicle level model coefficients listed in the table above, the corresponding GVF standard error estimate is:

$$ste(X) = e^{1.69299 + 0.44262 \cdot \ln(817,573) + 0.01787 \cdot (\ln(817,573))^2} = 61,756$$

At the end of this section, tables of pre-calculated point and GVF standard error estimates are also provided from 2016. Alternatively, these tables can also be used to estimate standard errors.

Example 2: For the “hit-and-run” point estimate $X = 817,573$, there is no standard error estimate in the 2018 CRSS GVF Standard Error Estimate table below. We need to make approximation by interpolation. The following is an excerpt of the 2018 CRSS GVF Standard Error Estimate table around estimate $X = 817,573$ at vehicle level:

2018 CRSS GVF Standard Error Estimates	
Vehicle	
Estimate (X)	Standard Error*
800,000	60,500
900,000	67,500

$X = 817,573$ is between $X = 800,000$ and $X = 900,000$. We approximate the standard error for estimate $X = 817,573$ by interpolation as the following:

$$ste(X) = 60,500 + \frac{817,573 - 800,000}{900,000 - 800,000} * (67,500 - 60,500) = 61,730$$

Example 3: GVF's may also be used to estimate the standard error of a proportion estimate. Here, the proportion estimate is referred to as the ratio of two total estimates:

$$\hat{R} = \frac{\hat{X}_d}{\hat{X}_p}$$

where $\hat{X}_p$ is the total estimate of variable X (numeric or categorical) for population p , $\hat{X}_d$ is the total estimate of variable X for domain d within population p . So, in general $\hat{X}_d \leq \hat{X}_p$.

Appendix F: Standard Errors

The GVF for the standard error estimate of the above proportion estimate is:

$$ste(\hat{R}) = \hat{R} \sqrt{\frac{var(\hat{X}_d)}{\hat{X}_d^2} - \frac{var(\hat{X}_p)}{\hat{X}_p^2}}$$

For more discussion on this formula, see Zhang and Diaz (2022).

In Example 1, it is estimated $\hat{X}_d = 817,573$ “hit-and-run” vehicles in 2018. This comprises 6.7854% of total vehicles involved in police reported crashes ($X_p=12,049,038$ —the summation of the weights of all sampled vehicles). To estimate the associated standard error of this proportion estimate $\hat{R} = 6.7854\%$, notice:

$$var(\hat{X}_d) = ste^2(X_d) = 61,756^2$$

$$var(\hat{X}_p) = [e^{1.69299+0.44262*\ln(12,049,038)+0.01787*(\ln(12,049,038))^2}]^2 = 856,137^2$$

$$ste(\hat{R}) = 6.7854\% * \sqrt{\left(\frac{61,756}{817,573}\right)^2 - \left(\frac{856,137}{12,049,038}\right)^2} \approx 0.17\%$$

In some cases, this method still generates imaginary number standard error estimate due to approximation and rounding. It should be noted that the CRSS GVF is mainly for the standard error of the total estimates. For estimates that are non-linear in totals such as percentage, ratio, and regression estimates, special software should be used in general.

Appendix F: Standard Errors

2016 CRSS Estimates and GVF Standard Error Estimates					
Crash		Vehicle		Person	
Estimate (X)	Standard Error*	Estimate (X)	Standard Error*	Estimate (X)	Standard Error*
1,000	300	1,000	300	1,000	200
5,000	800	5,000	900	5,000	800
6,000	900	10,000	1,500	10,000	1,300
7,000	1,000	20,000	2,700	20,000	2,200
8,000	1,100	30,000	3,800	30,000	3,100
9,000	1,200	40,000	4,800	40,000	3,900
10,000	1,300	50,000	5,800	50,000	4,700
20,000	2,200	60,000	6,800	60,000	5,400
30,000	3,000	70,000	7,700	70,000	6,200
40,000	3,700	80,000	8,700	80,000	6,900
50,000	4,400	90,000	9,600	90,000	7,600
60,000	5,200	100,000	10,500	100,000	8,300
70,000	5,800	200,000	19,300	200,000	15,100
80,000	6,500	300,000	27,800	300,000	21,700
90,000	7,200	400,000	36,000	400,000	28,000
100,000	7,900	500,000	44,100	500,000	34,300
200,000	14,200	600,000	52,100	600,000	40,600
300,000	20,200	700,000	60,000	700,000	46,800
400,000	26,000	800,000	67,900	800,000	53,000
500,000	31,700	900,000	75,700	900,000	59,100
600,000	37,400	1,000,000	83,500	1,000,000	65,300
700,000	43,000	2,000,000	160,500	2,000,000	126,300
800,000	48,600	3,000,000	236,700	3,000,000	187,500
900,000	54,200	4,000,000	312,800	4,000,000	249,100
1,000,000	59,700	5,000,000	388,800	5,000,000	311,200
2,000,000	114,500	6,000,000	464,900	6,000,000	373,800
3,000,000	169,000	7,000,000	541,200	7,000,000	436,900
4,000,000	223,600	8,000,000	617,700	8,000,000	500,500
5,000,000	278,600	9,000,000	694,300	9,000,000	564,500
6,000,000	333,800	10,000,000	771,200	10,000,000	629,000
6,500,000	361,500	11,000,000	848,300	11,000,000	693,800
7,000,000	389,300	12,000,000	925,500	12,000,000	759,200
*: $ste(X) = e^{a+b\ln(X)+c\ln(X)^2}$					
a = 1.92772 b = 0.38750 c = 0.01947		a = 1.17146 b = 0.53866 c = 0.01425		a = 1.79032 b = 0.40622 c = 0.01930	

Appendix F: Standard Errors

2017 CRSS Estimates and GVF Standard Error Estimates					
Crash		Vehicle		Person	
Estimate (X)	Standard Error*	Estimate (X)	Standard Error*	Estimate (X)	Standard Error*
1,000	300	1,000	300	1,000	200
5,000	800	5,000	900	5,000	700
6,000	900	10,000	1,500	10,000	1,200
7,000	1,000	20,000	2,600	20,000	2,100
8,000	1,100	30,000	3,600	30,000	2,800
9,000	1,200	40,000	4,600	40,000	3,500
10,000	1,300	50,000	5,500	50,000	4,200
20,000	2,200	60,000	6,500	60,000	4,900
30,000	3,000	70,000	7,400	70,000	5,600
40,000	3,800	80,000	8,300	80,000	6,200
50,000	4,500	90,000	9,100	90,000	6,900
60,000	5,200	100,000	10,000	100,000	7,500
70,000	5,900	200,000	18,400	200,000	13,600
80,000	6,600	300,000	26,400	300,000	19,400
90,000	7,300	400,000	34,200	400,000	25,100
100,000	8,000	500,000	41,900	500,000	30,700
200,000	14,600	600,000	49,600	600,000	36,300
300,000	20,900	700,000	57,200	700,000	41,800
400,000	27,100	800,000	64,700	800,000	47,300
500,000	33,300	900,000	72,200	900,000	52,800
600,000	39,400	1,000,000	79,700	1,000,000	58,300
700,000	45,500	2,000,000	153,900	2,000,000	112,900
800,000	51,700	3,000,000	227,900	3,000,000	167,700
900,000	57,800	4,000,000	302,000	4,000,000	223,000
1,000,000	63,900	5,000,000	376,400	5,000,000	278,900
2,000,000	125,300	6,000,000	451,200	6,000,000	335,300
3,000,000	187,800	7,000,000	526,300	7,000,000	392,300
4,000,000	251,400	8,000,000	601,800	8,000,000	449,700
5,000,000	316,100	9,000,000	677,700	9,000,000	507,700
6,000,000	381,700	10,000,000	753,900	10,000,000	566,100
6,500,000	414,900	11,000,000	830,500	11,000,000	625,000
7,000,000	448,400	12,000,000	907,400	12,000,000	684,300
*: $ste(X) = e^{a+b\ln(X)+c\ln(X)^2}$					
a = 2.33171 b = 0.30826 c = 0.02344		a = 1.43152 b = 0.48824 c = 0.01629		a = 2.05394 b = 0.35287 c = 0.02119	

Appendix F: Standard Errors

2018 CRSS Estimates and GVF Standard Error Estimates					
Crash		Vehicle		Person	
Estimate (X)	Standard Error*	Estimate (X)	Standard Error*	Estimate (X)	Standard Error*
1,000	300	1,000	300	1,000	200
5,000	800	5,000	900	5,000	700
6,000	900	10,000	1,500	10,000	1,200
7,000	1,000	20,000	2,500	20,000	2,000
8,000	1,100	30,000	3,500	30,000	2,800
9,000	1,200	40,000	4,400	40,000	3,500
10,000	1,300	50,000	5,300	50,000	4,100
20,000	2,100	60,000	6,200	60,000	4,800
30,000	2,900	70,000	7,000	70,000	5,400
40,000	3,700	80,000	7,800	80,000	6,100
50,000	4,400	90,000	8,700	90,000	6,700
60,000	5,100	100,000	9,500	100,000	7,300
70,000	5,800	200,000	17,300	200,000	13,200
80,000	6,400	300,000	24,800	300,000	18,800
90,000	7,100	400,000	32,100	400,000	24,200
100,000	7,700	500,000	39,300	500,000	29,600
200,000	14,000	600,000	46,400	600,000	34,900
300,000	19,900	700,000	53,500	700,000	40,200
400,000	25,700	800,000	60,500	800,000	45,400
500,000	31,500	900,000	67,500	900,000	50,700
600,000	37,200	1,000,000	74,500	1,000,000	55,900
700,000	42,800	2,000,000	143,800	2,000,000	107,600
800,000	48,500	3,000,000	213,000	3,000,000	159,400
900,000	54,100	4,000,000	282,500	4,000,000	211,400
1,000,000	59,700	5,000,000	352,300	5,000,000	263,900
2,000,000	115,700	6,000,000	422,500	6,000,000	316,800
3,000,000	172,100	7,000,000	493,200	7,000,000	370,100
4,000,000	229,200	8,000,000	564,300	8,000,000	423,800
5,000,000	286,900	9,000,000	635,700	9,000,000	477,900
6,000,000	345,300	10,000,000	707,600	10,000,000	532,300
6,500,000	374,700	11,000,000	779,900	11,000,000	587,200
7,000,000	404,300	12,000,000	852,600	12,000,000	642,400
*: $ste(X) = e^{a+b\ln(X)+c\ln(X)^2}$					
a = 2.33242 b = 0.31521 c = 0.02258		a = 1.69299 b = 0.44262 c = 0.01787		a = 2.02774 b = 0.35777 c = 0.02075	

Appendix F: Standard Errors

2019 CRSS Estimates and GVF Standard Error Estimates					
Crash		Vehicle		Person	
Estimate (X)	Standard Error*	Estimate (X)	Standard Error*	Estimate (X)	Standard Error*
1,000	300	1,000	300	1,000	200
5,000	800	5,000	900	5,000	700
6,000	900	10,000	1,400	10,000	1,100
7,000	1,000	20,000	2,500	20,000	1,900
8,000	1,100	30,000	3,500	30,000	2,700
9,000	1,200	40,000	4,400	40,000	3,300
10,000	1,200	50,000	5,300	50,000	4,000
20,000	2,100	60,000	6,100	60,000	4,600
30,000	2,900	70,000	7,000	70,000	5,300
40,000	3,600	80,000	7,800	80,000	5,900
50,000	4,300	90,000	8,600	90,000	6,500
60,000	5,000	100,000	9,500	100,000	7,100
70,000	5,700	200,000	17,300	200,000	12,800
80,000	6,400	300,000	24,800	300,000	18,400
90,000	7,000	400,000	32,200	400,000	23,800
100,000	7,700	500,000	39,400	500,000	29,100
200,000	13,800	600,000	46,600	600,000	34,400
300,000	19,700	700,000	53,800	700,000	39,700
400,000	25,500	800,000	60,900	800,000	44,900
500,000	31,200	900,000	68,000	900,000	50,200
600,000	36,900	1,000,000	75,100	1,000,000	55,400
700,000	42,500	2,000,000	145,500	2,000,000	107,800
800,000	48,100	3,000,000	215,900	3,000,000	160,700
900,000	53,600	4,000,000	286,900	4,000,000	214,200
1,000,000	59,200	5,000,000	358,300	5,000,000	268,500
2,000,000	114,700	6,000,000	430,200	6,000,000	323,400
3,000,000	170,400	7,000,000	502,700	7,000,000	378,900
4,000,000	226,800	8,000,000	575,700	8,000,000	435,100
5,000,000	283,700	9,000,000	649,100	9,000,000	491,800
6,000,000	341,200	10,000,000	723,100	10,000,000	549,000
6,500,000	370,200	11,000,000	797,500	11,000,000	606,800
7,000,000	399,300	12,000,000	872,300	12,000,000	665,100
*: $ste(X) = e^{a+b\ln(X)+c\ln(X)^2}$					
a = 2.19494 b = 0.33465 c = 0.02185		a = 1.70176 b = 0.43713 c = 0.01826		a = 2.14416 b = 0.32619 c = 0.02238	

Appendix F: Standard Errors

2020 CRSS Estimates and GVF Standard Error Estimates					
Crash		Vehicle		Person	
Estimate (X)	Standard Error*	Estimate (X)	Standard Error*	Estimate (X)	Standard Error*
1,000	200	1,000	300	1,000	200
5,000	700	5,000	800	5,000	700
6,000	800	10,000	1,300	10,000	1,100
7,000	900	20,000	2,300	20,000	1,900
8,000	1,000	30,000	3,200	30,000	2,600
9,000	1,100	40,000	4,100	40,000	3,200
10,000	1,200	50,000	4,900	50,000	3,900
20,000	2,000	60,000	5,700	60,000	4,500
30,000	2,700	70,000	6,500	70,000	5,100
40,000	3,400	80,000	7,200	80,000	5,700
50,000	4,100	90,000	8,000	90,000	6,300
60,000	4,700	100,000	8,800	100,000	6,800
70,000	5,400	200,000	16,000	200,000	12,400
80,000	6,000	300,000	23,000	300,000	17,700
90,000	6,600	400,000	29,800	400,000	22,900
100,000	7,200	500,000	36,500	500,000	28,000
200,000	13,100	600,000	43,200	600,000	33,000
300,000	18,600	700,000	49,900	700,000	38,100
400,000	24,000	800,000	56,500	800,000	43,100
500,000	29,400	900,000	63,100	900,000	48,100
600,000	34,700	1,000,000	69,700	1,000,000	53,100
700,000	39,900	2,000,000	135,200	2,000,000	102,600
800,000	45,100	3,000,000	201,000	3,000,000	152,400
900,000	50,300	4,000,000	267,200	4,000,000	202,500
1,000,000	55,500	5,000,000	334,000	5,000,000	253,100
2,000,000	106,700	6,000,000	401,400	6,000,000	304,200
3,000,000	157,800	7,000,000	469,300	7,000,000	355,700
4,000,000	209,100	8,000,000	537,700	8,000,000	407,700
5,000,000	260,700	9,000,000	606,600	9,000,000	460,000
6,000,000	312,600	10,000,000	676,000	10,000,000	512,800
6,500,000	338,700	11,000,000	745,900	11,000,000	566,000
7,000,000	364,900	12,000,000	816,200	12,000,000	619,500
*: $ste(X) = e^{a+b\ln(X)+c\ln(X)^2}$					
a = 1.81266 b = 0.38881 c = 0.01959		a = 1.69637 b = 0.42507 c = 0.01877		a = 1.88630 b = 0.36439 c = 0.02074	

Appendix F: Standard Errors

2021 CRSS Estimates and GVF Standard Error Estimates					
Crash		Vehicle		Person	
Estimate (X)	Standard Error*	Estimate (X)	Standard Error*	Estimate (X)	Standard Error*
1,000	200	1,000	300	1,000	200
5,000	700	5,000	800	5,000	700
6,000	800	10,000	1,400	10,000	1,200
7,000	900	20,000	2,400	20,000	2,000
8,000	1,000	30,000	3,400	30,000	2,700
9,000	1,100	40,000	4,300	40,000	3,400
10,000	1,200	50,000	5,200	50,000	4,100
20,000	2,000	60,000	6,000	60,000	4,700
30,000	2,800	70,000	6,900	70,000	5,400
40,000	3,500	80,000	7,700	80,000	6,000
50,000	4,200	90,000	8,500	90,000	6,600
60,000	4,900	100,000	9,300	100,000	7,300
70,000	5,500	200,000	17,000	200,000	13,200
80,000	6,200	300,000	24,500	300,000	18,900
90,000	6,800	400,000	31,700	400,000	24,500
100,000	7,400	500,000	38,900	500,000	30,000
200,000	13,400	600,000	46,100	600,000	35,500
300,000	19,200	700,000	53,100	700,000	41,000
400,000	24,900	800,000	60,200	800,000	46,400
500,000	30,500	900,000	67,200	900,000	51,800
600,000	36,000	1,000,000	74,200	1,000,000	57,200
700,000	41,500	2,000,000	143,800	2,000,000	111,300
800,000	47,000	3,000,000	213,500	3,000,000	165,900
900,000	52,500	4,000,000	283,500	4,000,000	221,000
1,000,000	58,000	5,000,000	354,000	5,000,000	276,900
2,000,000	112,700	6,000,000	425,100	6,000,000	333,300
3,000,000	168,000	7,000,000	496,600	7,000,000	390,400
4,000,000	223,900	8,000,000	568,600	8,000,000	448,000
5,000,000	280,600	9,000,000	641,100	9,000,000	506,100
6,000,000	337,900	10,000,000	714,000	10,000,000	564,800
6,500,000	366,800	11,000,000	787,300	11,000,000	624,000
7,000,000	395,900	12,000,000	861,100	12,000,000	683,600
*: $ste(X) = e^{a+b\ln(X)+c\ln(X)^2}$					
a = 2.16680 b = 0.33005 c = 0.02222		a = 1.58125 b = 0.45204 c = 0.01775		a = 1.95844 b = 0.35704 c = 0.02129	

Appendix F: Standard Errors

2022 CRSS Estimates and GVF Standard Error Estimates					
Crash		Vehicle		Person	
Estimate (X)	Standard Error*	Estimate (X)	Standard Error*	Estimate (X)	Standard Error*
1,000	200	1,000	200	1,000	200
5,000	800	5,000	800	5,000	700
6,000	900	10,000	1,400	10,000	1,200
7,000	1,000	20,000	2,500	20,000	2,100
8,000	1,100	30,000	3,500	30,000	2,900
9,000	1,200	40,000	4,400	40,000	3,700
10,000	1,300	50,000	5,400	50,000	4,400
20,000	2,200	60,000	6,300	60,000	5,200
30,000	3,000	70,000	7,200	70,000	5,900
40,000	3,800	80,000	8,200	80,000	6,600
50,000	4,600	90,000	9,100	90,000	7,300
60,000	5,400	100,000	9,900	100,000	8,000
70,000	6,100	200,000	18,600	200,000	14,800
80,000	6,900	300,000	27,100	300,000	21,400
90,000	7,600	400,000	35,400	400,000	28,000
100,000	8,300	500,000	43,700	500,000	34,400
200,000	15,300	600,000	52,000	600,000	40,900
300,000	22,000	700,000	60,200	700,000	47,300
400,000	28,600	800,000	68,400	800,000	53,700
500,000	35,100	900,000	76,700	900,000	60,100
600,000	41,600	1,000,000	84,900	1,000,000	66,500
700,000	48,100	2,000,000	167,400	2,000,000	131,100
800,000	54,600	3,000,000	250,900	3,000,000	196,800
900,000	61,000	4,000,000	335,300	4,000,000	263,700
1,000,000	67,500	5,000,000	420,700	5,000,000	331,500
2,000,000	132,300	6,000,000	507,100	6,000,000	400,400
3,000,000	198,000	7,000,000	594,200	7,000,000	470,200
4,000,000	264,000	8,000,000	682,200	8,000,000	540,800
5,000,000	332,100	9,000,000	770,900	9,000,000	612,200
6,000,000	400,500	10,000,000	860,300	10,000,000	684,400
6,500,000	435,000	11,000,000	950,300	11,000,000	757,200
7,000,000	469,700	12,000,000	1,041,000	12,000,000	830,800
*: $ste(X) = e^{a+b*ln(X)+c*ln(X)^2}$					
a = 1.82248 b = 0.38876 c = 0.02057		a = 1.06573 b = 0.52025 c = 0.01622		a = 1.71055 b = 0.39513 c = 0.02062	

Appendix F: Standard Errors

2023 CRSS Estimates and GVF Standard Error Estimates					
Crash		Vehicle		Person	
Estimate (X)	Standard Error*	Estimate (X)	Standard Error*	Estimate (X)	Standard Error*
1,000	200	1,000	200	1,000	200
5,000	800	5,000	800	5,000	700
6,000	900	10,000	1,400	10,000	1,200
7,000	1,000	20,000	2,500	20,000	2,200
8,000	1,100	30,000	3,600	30,000	3,000
9,000	1,200	40,000	4,600	40,000	3,800
10,000	1,300	50,000	5,600	50,000	4,600
20,000	2,200	60,000	6,500	60,000	5,400
30,000	3,000	70,000	7,500	70,000	6,200
40,000	3,800	80,000	8,400	80,000	6,900
50,000	4,600	90,000	9,300	90,000	7,600
60,000	5,400	100,000	10,200	100,000	8,400
70,000	6,100	200,000	19,000	200,000	15,400
80,000	6,800	300,000	27,500	300,000	22,200
90,000	7,600	400,000	35,700	400,000	28,800
100,000	8,300	500,000	43,900	500,000	35,300
200,000	15,100	600,000	52,000	600,000	41,800
300,000	21,600	700,000	60,000	700,000	48,300
400,000	28,000	800,000	68,000	800,000	54,700
500,000	34,300	900,000	75,900	900,000	61,100
600,000	40,600	1,000,000	83,900	1,000,000	67,500
700,000	46,800	2,000,000	162,100	2,000,000	131,000
800,000	52,900	3,000,000	239,600	3,000,000	194,600
900,000	59,100	4,000,000	317,000	4,000,000	258,600
1,000,000	65,200	5,000,000	394,400	5,000,000	323,000
2,000,000	126,300	6,000,000	471,900	6,000,000	387,800
3,000,000	187,400	7,000,000	549,600	7,000,000	453,100
4,000,000	248,800	8,000,000	627,400	8,000,000	518,800
5,000,000	310,700	9,000,000	705,400	9,000,000	584,900
6,000,000	373,100	10,000,000	783,500	10,000,000	651,400
6,500,000	404,500	11,000,000	861,900	11,000,000	718,200
7,000,000	435,900	12,000,000	940,400	12,000,000	785,500
*: $ste(X) = e^{a+b*ln(X)+c*ln(X)^2}$					
a = 1.66293 b = 0.42425 c = 0.01866		a = 0.67458 b = 0.60142 c = 0.01233		a = 1.30776 b = 0.47508 c = 0.01702	

Appendix G: Special Notes for Analysts

The Effect of the COVID-19 Pandemic on CRSS Data Collection

Beginning in the first quarter of 2020, the COVID-19 pandemic presented several challenges to CRSS data collection. As States implemented shutdowns and closures to limit exposure to COVID-19, several police jurisdictions closed their doors to non-essential personnel, thus the CRSS Samplers were unable to access crash reports for varying amounts of time. Additionally, some jurisdictions stopped responding to Property Damage Only (PDO) crashes and opted to complete citizen reports or desk reports for PDO motor vehicle crashes. CRSS only samples from police reported crashes on official crash reports. Thus, there was a decline in CRSS applicable cases to sample.

COVID-19 also had an impact on the transmission of crash report data. As agencies needed time to coordinate working remotely, the processes to key in and upload electronic crash reports were delayed across various sample sites ranging from weeks to months before NHTSA was able to receive data. The pandemic has impacted CRSS data collection, but NHTSA has adapted to the new safety protocols and changes in sampling schedules with police jurisdictions and has extended listings of crashes to incorporate the delay of receiving crash reports. The situation brought on by the current crisis is unlike anything our administration has experienced before. Nevertheless, NHTSA will continue to explore traffic safety during the COVID-19 pandemic.

2022 CRSS Generalized Variance Function Study

In 2022 NHTSA's Mathematical Analysis Division completed a study to establish the General Variance Functions for CRSS. As for any probability-based sample, the estimates generated from CRSS data are subject to sampling errors. The magnitude of sampling error depends on the study variable, the estimator used, and the sample design. CRSS data is collected under a complex survey design with features such as multistage sampling, stratification, and unequal selection probabilities to ensure it is a nationally representative sample. CRSS case weights are derived corresponding to its complex design features in order to produce unbiased and robust estimates.

Estimation methods and computer software have been developed to make estimates from complex survey data. Specialized procedures for complex survey data analysis, such as SAS PROC SURVEY procedures and SUDAAN procedures, can be used in CRSS data analysis along with proper design statements to take the complex survey design into account. (See more on these methods in [Appendix F: Standard Errors](#)) However, for users who do not have access to specialized software and wish to have a quick assessment of the magnitude of the standard errors of CRSS estimates, the generalized variance functions can be used to generate ballpark standard error estimates for a large quantity of estimates. The study is published in the NHTSA Technical Report [Crash Report Sampling System: Generalized Variance Functions](#).

Analysis of Police Reported Alcohol Involvement Elements

In recent years the percentage of "Not Reported" cases for *Police Reported Alcohol Involvement* (DRINKING) increased to over 40 percent. In 2020 the percentage of unknown or not reported cases increased to over 45 percent. This percentage increase can be attributed to NHTSA's improved quality control over the data collection year. This process included but was not limited to establishing state specific coding instructions to help coders with the interpretation of the police crash report fields. In addition, NHTSA implemented new quality assurance plans to

improve coding accuracy across the data collection system. Two other data elements, *Alcohol Involved in Crash* (ALCOHOL) and *Driver Drinking in Vehicle* (VEH_ALCH), are derived from the *Police Reported Alcohol Involvement* data element. Additionally, *Police Reported Alcohol Involvement* is the basis of the imputed data element, *Imputed Police Reported Involvement* (PERALCH_IM), which is used to derive *Imputed Alcohol Involved in Crash* (ALCHL_IM) and *Imputed Driver Drinking in Vehicle* (V_ALCH_IM). NHTSA continues to impute the drinking related data elements, but analysts should be aware when using these data elements for analysis.

Removal of Automated Driving Systems (ADS) Data Elements in CRSS

In 2019 three Motor Vehicle Automated Driving Systems (ADS) data elements were added to the CRSS data collection. These elements were added in response to the inclusion of ADS in the [Model Minimum Uniform Crash Criteria 5th ed.](#) (MMUCC) released in 2017. The concepts and definitions in MMUCC were adopted from the [Society of Automotive Engineers \(SAE\) J3016 Levels of Driving Automation](#) and were applied to both the MMUCC and CRSS elements. The data are intended for crash avoidance and countermeasure research and development.

In 2020 NHTSA continued to collect the ADS data that were added in 2019; however, collection proved to be difficult. The source for CRSS to collect ADS data is the police crash report and this information is limited on crash reports. Few States have crash reports with ADS-related fields and only a small number of those are compatible with the CRSS ADS definitions and attributes. Most States do not have an ADS field on their crash report and therefore the identification of vehicle automation is only possible through the crash report narrative. At this time the CRSS ADS data elements are largely coded as “Not Reported.”

Extensive quality control checks and analyses were performed using the 2019 and 2020 data. The results of the analyses highlighted inconsistencies in collecting and accurately identifying specifics with these elements that can lead to varying or misleading results. Consequently, NHTSA has removed the ADS data elements from the 2019 and following CRSS files while additional research is conducted on how improvements can be made. NHTSA will continue to collect these data for internal quality control, review, and analysis purposes only. The following data elements have been removed from the 2019 and following CRSS files:

Automation System or Systems Present in Vehicle (Vehicle.ADS_PRES)

Highest Automation System Level Present in Vehicle (Vehicle.ADS_LEV)

Highest Automation System Level Engaged at Time of Crash (Vehicle.ADS_ENG)

2018 Increase in National Estimate of Property Damage Only (PDO) Crashes Due to Change in Police Reporting Procedures

Motor vehicle crashes that include fatalities, injuries, or property damage in excess of a pre-determined dollar amount are considered reportable crashes. Previously, at five CRSS sampled data collection sites, police officers could report the property damage costs to vehicles involved in a crash as “Unknown.” Due to updates to the State’s reporting criteria for these data collection sites, “unknown” property damage cost was removed as an option. This change converted many PDO crashes with unknown property damage costs that were previously non-reportable crashes

into reportable crashes in 2018. This in turn caused the increase of the national PDO crash estimate.

Analysis of Pedestrian and Bicycle Crashes Around Intersections

When using the Accident, Person, and Pdtype data files to study pedestrian and cyclist crashes, care must be taken when describing their locations in and around intersections.

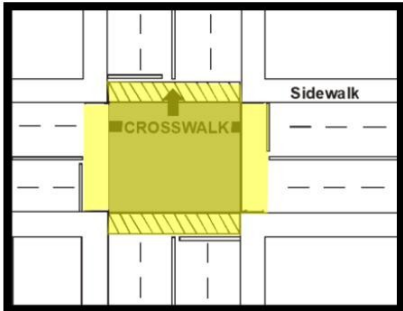
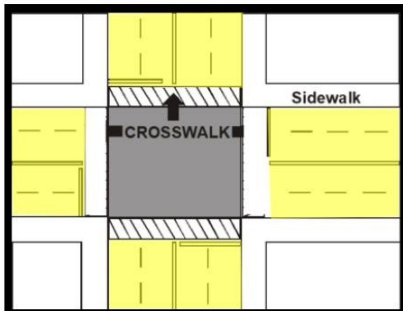
The Accident data file contains the data element, “Relation to Junction-Specific Location.” This element identifies the location of the “First Harmful Event” of the crash and not necessarily the location of any pedestrian or bicyclist involved. In addition, this element’s attributes have specific definitions for *Intersection* (in the intersection) and *Intersection-Related*.

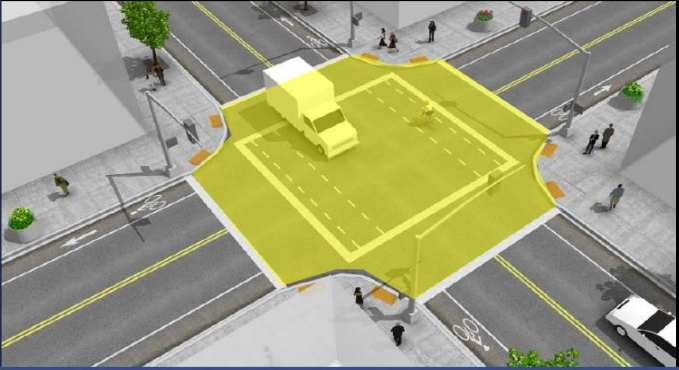
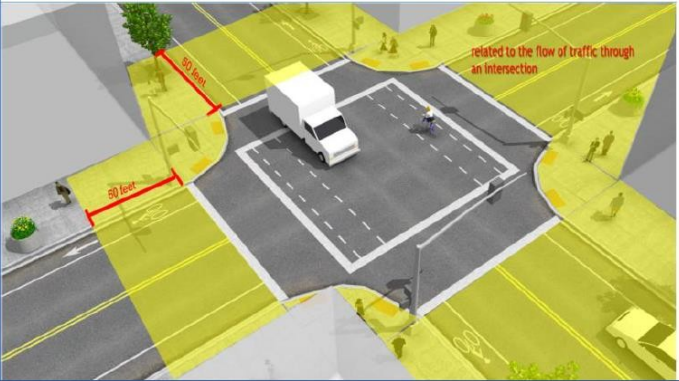
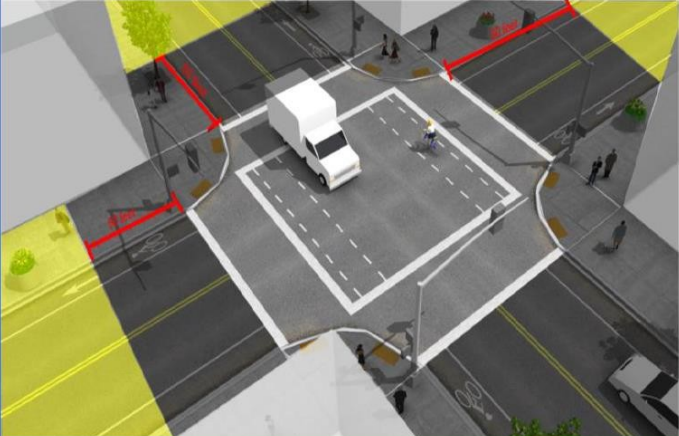
The Person data file contains the data element, “Non-Motorist Location at Time of Crash.” This element employs the defined concepts of *At Intersection* and *Not at Intersection*, but does not include the concept of *Intersection-Related*.

Finally, the Pdtype data file contains the data elements, “Crash Location – Pedestrian,” “Crash Location – Bicycle,” “Pedestrian Position,” and “Bicyclist Position.” These elements employ the defined concepts of *At Intersection*, *Not at Intersection*, and *Intersection Related* (defined somewhat differently from the Accident file concept).

The following graphics may be helpful aids in conjunction with the FARS/CRSS Coding and Validation Manual and the Pedestrian-Bicyclist Crash Typing Manual:

 C21b RELATION TO JUNCTION 	
02 (Intersection)  02 (Intersection) is used when the FIRST HARMFUL EVENT occurs in an area which: (1) contains a crossing or connection of two or more roadways not classified as a driveway access, and (2) is embraced within the prolongation of the lateral curb lines or, if none, the lateral boundary lines of the roadways.	03 (Intersection-Related)  03 (Intersection-Related) means that the FIRST HARMFUL EVENT: (1) occurs on an approach to or exit from an intersection and (2) results from an activity, behavior, or control related to the movement of traffic units through the intersection.

 NM10 NON-MOTORIST LOCATION AT TIME OF CRASH 	
AT INTERSECTION  “At intersection” means: The person is on a roadway (travel lanes) either: (1) in the intersection, (2) in an area between a crosswalk and the perimeter of the intersection, or (3) in a crosswalk (whether marked or unmarked) adjacent to an intersection. If there are no crosswalks, “at intersection” means only the intersection, which is the area embraced within the prolongation of the lateral curb lines or, if none, the lateral boundary lines of the roadways.	NOT AT INTERSECTION  The person is on a roadway (travel lanes), but not “At Intersection.”

  PB31/PB31b Pedestrian/Bicycle Crash Location  
AT INTERSECTION  1 (At Intersection) is used when a person is on a roadway (travel lanes): (1) in the intersection, (2) in an area between a crosswalk and the perimeter of the intersection, OR (3) in a crosswalk (whether marked or unmarked) adjacent to an intersection.
INTERSECTION RELATED  2 (Intersection-Related) is used when a person is: • within the trafficway <u>50 feet</u> out from the perimeter of an “At intersection” area including the entire cross section of the trafficway (e.g., medians, turn lanes, bike lanes, parking lanes, shoulders, sidewalks, etc.) OR • the crash is related to the flow of traffic through an intersection (e.g., the result of queuing traffic).
NOT AT INTERSECTION  3 (Not At Intersection) is used when a person is: • within the trafficway <u>more than 50 feet out</u> from the perimeter of an “At Intersection” area AND • the crash is not identified as related to the movement of the traffic units through an intersection. This includes the entire cross section of the trafficway (e.g., medians, turn lanes, bike lanes, parking lanes, shoulders, sidewalks, etc.). This attribute is the default when the case materials give no indication that the crash is within 50 feet of an intersection.

Appendix H: Notable Changes

Non-responding CRSS Primary Sampling Unit

For 2022 one PSU was deemed non-responding due to significant delays in receiving PCRs. Thus, there were 59 responding PSUs across the country in 2022 compared to 60 responding PSUs in 2021. A PSU non-response adjustment was applied to mitigate potential non-response bias. For more information on the PSU non-response adjustment, please refer to [Crash Report Sampling System: Sample Design and Weighting](#). It is important to note 2022 published data only includes information from the 59 responding PSUs.

Change in Categorization for Motorized Bicycles

For 2022 CRSS is no longer collecting motorized/motor assisted bicycles as motor vehicles. Consequently, the operators of motorized/motor assisted bicycles will be captured as non-motorists when involved in a motor vehicle traffic crash. Single-vehicle crashes involving motorized/motor assisted bicycles will no longer be captured. In addition, to address this change and the range of non-motorist devices appearing on the nation's roadways two new data elements, [Non-Motorist Device Type](#) and [Non-Motorist Device Motorization](#), have been added to the Person (Not a MV Occupant) Level.

Addition of VIN-Decoded Data

Prior to 2020 the descriptive vehicle information in Vehicle Make, Vehicle Model, and Body Type were coded from information in the police crash reports and based on a Vehicle Make/Model/Body Type table maintained by NCSA for this purpose. Starting in 2020 this table will no longer be updated and a new set of data elements has been added to the Vehicle and Parkwork data files. These new data elements are the following.

- [vPIC Make](#)
- [vPIC Model](#)
- [vPIC Body Class](#)
- [Final Stage Body Class](#)
- [Power Unit Gross Vehicle Weight Rating – From](#)
- [Power Unit Gross Vehicle Weight Rating – To](#)
- [Trailer Gross Vehicle Weight Rating](#) (data collected up to three trailers)

Elements *vPIC Make*, *vPIC Model*, *vPIC Body Class*, and *Final Stage Body Class* are also added to Person data file.

These data elements are mostly derived from VIN decoding using NHTSA's tool, Product Information Catalog and Vehicle Listing (vPIC), which is based on the vehicle manufacturer submissions to NHTSA mandated by Federal Motor Vehicle Safety Standard (FMVSS) 49 Code of Federal Regulation (CFR) 565. If a vehicle VIN or trailer VIN can be decoded cleanly, such as with no errors or minor issues, *vPIC Make*, *vPIC Model*, *vPIC Body Class*, *Power Unit* or *Trailer Gross Vehicle Weight Rating (From and To)* are coded using information derived from vPIC VIN decoder. If a VIN cannot be decoded cleanly or there is no VIN reported in the police crash report, these elements are coded by analysts using the information on the crash report. *Final Stage Body Class* is applicable only to incomplete vehicles and always coded using the information from police crash report.

Appendix H: Notable Changes

To further differentiate between these new data elements and the historic NCSA descriptions for Make, Model, and Body Type, the following data elements have been renamed the following.

- Vehicle Make → NCSA Make
- Vehicle Model → NCSA Model
- Body Type → NCSA Body Type

Also, *Gross Vehicle Weight Ratio/Gross Vehicle Combination Ratio (GVWR/GCWR)* has been discontinued in response to the new vPIC data elements that collect GVWR for the power unit (upper and lower limits) and any trailers separately. The attributes represent vehicle Class 1 to Class 8.

It is important to note that the new VIN-derived data elements will eventually replace the NCSA ones and result in new body class designations that will differ from NCSA's historic body type classifications. See [Appendix C: Analytical Classification of Select CRSS Data Elements](#) for new classifications based on vPIC Body Class.

In addition to the data elements added to the existing data files, two additional data files are available with many data elements decoded from the VIN, one for vehicles (Vpicdecode) and one for trailers (Vpictrailerdecode). These data files have their own user manual, the *Product Information Catalog and Vehicle Listing (vPIC) Analytical User's Manual*, found in the [NCSA Publications- Manuals and Documentation](#) section of NHTSA's website.

For more information on NHTSA's Product Information Catalog and Vehicle Listing (vPIC), go to <https://vpic.nhtsa.dot.gov/>.

Addition of Non-Motorist Person Types

The data element [Person Type](#) has expanded to collect more specific types of non-motorists on motorized or non-motorized personal conveyances. A personal conveyance is a device, other than a transport device, used by a pedestrian for personal mobility assistance or recreation. These devices can be motorized or human powered, but not propelled by pedaling. Examples include rideable toys, roller skates, motorized and non-motorized skateboards, scooters and wheelchairs. The new attributes replace attribute 8 (Persons on Personal Conveyances) and include the following.

- 11 (Person on Motorized Personal Conveyance)
- 12 (Person on Non-Motorized Personal Conveyance)
- 13 (Person on Personal Conveyance, Unknown if Motorized or Non-Motorized)

These additions were necessitated by the growing variety and use of these devices. This allows these devices to be more clearly identified and targeted in analyses.

In addition, the NCSA [Person Type Classifications](#) in *Appendix C: Analytical Classification of Select CRSS Data Elements* were updated accordingly.

Change from Multiple Elements to Single Elements that Allow Selection of Multiple Values

Prior to 2020 *Atmospheric Conditions* and the “*Related Factor*” data elements were comprised of more than one element to allow the selection of more than one attribute. For example, Crash Related Factors was made up of three elements (i.e., CF1, CF2, CF3) allowing up to three selections. This format, however, limited the number of selections to the available number of elements. Beginning in 2020 these elements have been changed to a single element that allows for the selection of all attributes that apply.

Changes to SAS Names

In 2020 the conversion of six more data elements to allow the coding of more than one attribute brought the total to 17 data files that store these “select all that apply” elements. With this many data files and elements, it was an appropriate time to standardize the SAS names for this type of element. It was also an opportunity to update the SAS names for two of these elements where the element name had changed but the SAS name had not (i.e., Non Motorist Action/Circumstances and Non-Motorist Contributing Circumstances). Nine SAS names were updated and are identified in the Summary of the SAS Naming Changes.

Changes to Imputed Elements

The imputed data element for *Relation to Junction-Within Interchange Area* (RELJCT1_IM) was added back to CRSS in 2020 after being removed in 2019. During the 2019 data collection year, the data element’s unknown rate increased to over 60 percent due to additional quality control measures during the data collection process. For the 2020 data collection year, NHTSA incorporated a geolocator tool to assist in coding *Relation to Junction- Within Interchange Area*. A little over 7 percent of the data element was unknown or not reported in 2020 and NHTSA decided to reinstate *Relation to Junction – Within Interchange Area* as a candidate for imputation.

Hit-and-Run (HITRUN_IM) was removed from the imputation process due to the removal of the “Reported as Unknown” attribute (0.01% in 2019).

NCSA Body Type was removed from the imputation process in 2021. Prior to 2021, imputed *NCSA Body Type* was the basis of vehicle classification. However, starting in 2021, the basis of vehicle classification will transition to vPIC data elements. Currently, NHTSA does not plan to impute *vPIC Body Class*.

More details on the imputation process can be found in the [CRSS Imputation](#) section.

Addition of Automated Driving System Data Elements

Automated Driving System (ADS) data elements were added to CRSS to start collecting information on autonomous vehicles in 2019. Motor vehicle Automated Driving Systems are defined by the Model Minimum Uniform Crash Criteria (MMUCC), 5th ed., as "The hardware and software that are collectively capable of performing part or all of the dynamic driving task on a sustained basis; this term is used generically to describe any system capable of level 1-5

driving automation." The automation level refers to the SAE International standard (SAE J3016). For details on Automated Driving Systems, see NHTSA's website.

Three ADS data elements were added: one to capture the presence of an Automation System or Systems in the vehicle (ADS_PRES); a second to capture the highest level of automation present in the vehicle (ADS_LEV); and a third to capture the highest level of automation that was known to have been engaged in this vehicle at the time of the crash (ADS_ENG). Currently, information on ADS is not available on most crash reports and is limited in the data decoded from VINs, but States are beginning to update crash reports to collect information on autonomous vehicles. The addition of these data elements to CRSS prepares for future enhanced collection of ADS in vehicles involved in crashes.

However, at this time this data is not publicly available for analysis. For more information, see [Appendix G: Special Notes for Analysts - Removal of Automated Driving Systems \(ADS\) Data Elements in CRSS](#).

Separation of Restraint System/Helmet Use into Two Data Elements

The 2019 change to *Restraint System/Helmet Use* is in response to more vehicle types where the use of both helmets and belt restraints are possible (e.g., three-wheel motorcycles and ROVs). Splitting the data element into two data elements, *Restraint System Use* and *Helmet Use*, allows both pieces of information to be captured. Analysts will be able to examine the varying State safety equipment laws for both seat belt and helmet use, and will no longer need to rely on focus groups and observational studies on use. *Restraint System Use* retained the SAS name REST_USE and the new SAS name for *Helmet Use* is HELM_USE.

A similar change to *Indication of Misuse of Restraint System/Helmet* was made to correspond to the change in *Restraint System/Helmet Use*. This data element was also split into two new data elements, *Restraint System Misuse* and *Helmet Misuse*. *Restraint System Misuse* retained the SAS name REST_MIS and the new SAS name for *Helmet Misuse* is HELM_MIS.

Addition of Attributes for Incident Responders

The *Related Factors–Driver Level* and *Related Factors–Person Level* data elements were modified in 2019 to capture information on specific types of emergency services personnel, tow operators, and transportation workers involved in crashes. This may provide more detail for analyses and evaluation of "move over" laws, which require other drivers to slow down and move over for emergency vehicles and hazard vehicles. Specifically, attribute 86 (Emergency Services Personnel) was replaced with 94 (Emergency Medical Services Personnel), 95 (Fire Personnel), 96 (Tow Operator), and 97 (Transportation [maintenance workers, safety service patrol operators, etc.]). Existing attribute 87 (Police or Law Enforcement Officer) remains unchanged.

Addition of the Nmdistract Data File and Non-Motorist Distracted By Data Element

The data element *Non-Motorist Distracted By* was added to CRSS in 2019 to begin capturing non-motorist distractions. Previously CRSS only captured distractions for drivers of motor vehicles in-transport. The data element is defined as identifying the attributes that best describe

the non-motorist's attention prior to their involvement in the crash. Distraction, for a non-motorist, occurs when a non-motorist's attention is diverted from the task of navigating in public to some other activity. Also, daydreaming or lost in thought are identified as distractions by NHTSA. Physical conditions/impairments (fatigue, alcohol, medical condition, etc.) or psychological states (anger, emotional, depressed, etc.) are not identified as distractions by NHTSA.

Non-Motorist Distracted By is structured the same as the current *Driver Distracted By* data element, both of which allow all applicable attributes to be recorded. Therefore, a separate Nmdistract data file is necessary to store (potentially) multiple distraction records for each non-motorist. Details on this new data element and data file can be found in [The Nmdistract Data File](#) section.

New Vehicle Underride/Override Data Element

The 2020 FARS-only data element *Underride/Override* was retired. This element was replaced in 2021 by a completely new element that is applicable to CRSS titled [Vehicle Underride/Override](#). This new element and data collection structure is intended to support NHTSA rulemaking activities. The element is designed to indicate whether a vehicle experienced an underride or override with another vehicle during the crash.

Vehicle Underride/Override is assessed for each vehicle in every vehicle-to-vehicle collision. If this vehicle is identified in the case materials as going under another vehicle during the events of the crash, then this vehicle is coded as UNDERRIDE while the other vehicle is coded as OVERRIDE. This data element is not applicable to:

1. Single vehicle crashes (i.e., underride or override events require two vehicles),
2. Any vehicle in a multi-vehicle crash that has no vehicle-to-vehicle collision events,
3. All vehicle-to-vehicle collisions involving motor vehicle types for which this data is not collected; specifically, motorcycles, all-terrain cycles, and snowmobiles, but excluding "autocycles."

DOT HS 813 707
August 2025



U.S. Department
of Transportation
**National Highway
Traffic Safety
Administration**

